



Smithsonian Institution

Office of Planning, Design & Construction

SPECIFICATIONS

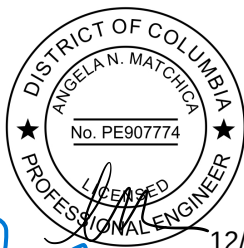
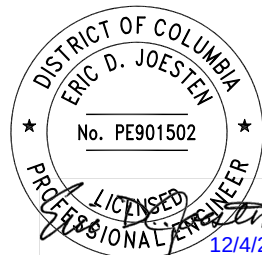
VOLUME 2 - DIVISION 21-33

PROJECT NO.:

PROJECT TITLE:

FACILITY:

DATE:



This project is approved as being in conformance with applicable provisions of the Smithsonian Directive (SD) 410.

Michael J. Carrancho, P.E., Deputy Director

1/15/26

Date

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PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.
- B. The following sections apply to the work of this section:
 - 1. 078413 Penetration Firestopping
 - 2. 238111 Fire Detection and Alarm Systems

1.2 SUMMARY

- A. Scope: Provide a complete, code compliant wet pipe sprinkler system, and associated equipment, ready for operation.
- B. The work includes designing and modifying an automatic wet-pipe fire extinguishing sprinkler system for Ordinary Hazard II and uniform distribution of water by hydraulic design to afford complete fire protection coverage throughout the contract area.
- C. Existing Sprinkler Equipment: Existing sprinkler equipment shall be maintained fully operational until the new equipment has been tested and accepted by the COTR and OSHEM fire protection engineer. After acceptance of the new equipment, all existing equipment so indicated shall be removed.
- D. Authority Having Jurisdiction: Any reference in the specifications or applicable codes to the "authority having jurisdiction" shall be interpreted to mean the OSHEM Fire Protection Engineer.

1.3 DEFINITIONS

- A. COTR: Contracting Officer Technical Representative
- B. FM: FM Global (Factory Mutual)
- C. FPE: Fire Protection Engineer
- D. Furnish: To supply the stated equipment or materials
- E. Install: To set in position and connect or adjust for use
- F. NFPA: National Fire Protection Association
- G. NICET: National Institute for Certification in Engineering Technologies

- H. OSHM: Office of Safety Health and Environmental Management
- I. Provide: To furnish and install the stated equipment or materials
- J. UL: Underwriters Laboratories

1.4 SYSTEM DESCRIPTION:

- A. The design shall conform to NFPA 13 and the requirements specified herein. Design of the automatic wet pipe sprinkler system shall be for Ordinary Hazard II occupancy by hydraulic calculation. The design, equipment, materials, installation, workmanship, examination, inspection and testing shall be in strict accordance with NFPA 13, except as modified herein.
- B. The system shall include all materials, accessories, and equipment inside and outside the building to provide a system complete and ready for use. Refer to Section 211100 for Installation of Exterior Water Supply Mains for Fire Protection Service.
- C. Design and provide each system giving full consideration to obstructions, blind spaces, piping, electrical equipment, duct work and other construction and equipment in accordance with detailed drawings to be submitted for approval.
- D. Equipment for fire protection service shall be UL listed or FM approved for use in wet pipe sprinkler systems.
- E. In the NFPA publications referred to herein, the advisory provisions shall be considered mandatory, as though the word "shall" had been substituted for "should," wherever it appears.

1.5 PERFORMANCE REQUIREMENTS

- A. Water Distribution: Distribution shall be uniform throughout the area which it is assumed the sprinkler heads will open. Variation in discharge from individual heads in the hydraulically most remote area shall be between 100 and 120 percent of the specified density.
- B. Piping Restrictions: Piping is prohibited in the following areas, except when solely supplying sprinklers in such areas:
 - 1. Transformer, switchgear, or similar rooms
 - 2. Elevator machine rooms and elevator shafts.
 - 3. IT Computer spaces (mission critical).
 - 4. Collection storage rooms.
- C. Clearance From Electrical Equipment: Piping and automatic sprinklers are prohibited directly over:
 - 1. Transformers.
 - 2. Substations.
 - 3. Switchboards.
 - 4. Motor control centers.

5. Emergency generators.
 6. Bus Ducts.
 7. Electrical panels.
- D. If installing pipe over electrical equipment is unavoidable, provide drip pans under piping to protect electrical equipment.
- E. Location of Sprinkler Heads: Heads in relation to the ceiling and the spacing of sprinkler heads shall not exceed their listed area of coverage for Ordinary Hazard II. Uniformly space sprinklers on the branch piping. Locate sprinkler heads in a consistent pattern with ceiling grid, lights, and supply air diffusers.
- F. Sprinkler Discharge Area: The sprinkler discharge area shall be the hydraulically most remote areas as defined in NFPA 13. Remote area reductions permitted per NFPA 13 shall not apply.
- G. Hose Allowances: System design shall include an allowance of 100 gpm (6.31 l/s) for inside hose stream, and 250 gpm (15.77 l/s) for total combined inside and outside hose streams.
- H. Hydraulic Calculations: Hydraulic calculations shall be in conformance with NFPA 13 and the following requirements:
1. Water Supply: Fire Pump test reports as follows: Main Building: 1000 GPM @ 100 PSI, Max suction pressure: 131 PSI, West Wing: 750 GPM @ 100 PSI, Max suction pressure: 105 PSI, East Wing: 750 GPM @ 100 PSI, Max suction pressure: 103 PSI. Provide a copy of the most recent waterflow test with the hydraulic calculations. The water supply test may be no more than 10 months old when used for hydraulic calculations.
 2. Margin of Safety: The sprinkler system shall be hydraulically designed so that the total waterflow and pressure demand is at least 10% less than the available water flow and pressure at the site.
 3. Friction Losses: Calculate losses in piping in accordance with the Hazen-Williams formula with "C" value of 100 for unlined ductile iron piping, 120 for steel piping, 140 for cement-lined ductile-iron piping, 150 for copper tubing and plastic piping.
 4. Flow Velocity: Piping shall be sized so that the water flow velocity does not exceed 20 ft/sec (6.1 m/s) at any point in the system during maximum water flow (including inside hose demand). Flow velocity in each pipe segment is to be provided in the calculations.
 5. Test Point: Calculations shall be brought back to the flow test point.
 6. Area of Coverage and Density: The area of coverage and density for each sprinkler shall be provided in the calculations.
 7. Equivalent Lengths: The equivalent lengths for all types of fittings and valves used shall be provided.
 8. Supply and Demand Graph(s): A graph comparing the water supply and the sprinkler and hose stream demand for each remote area shall be provided. Pressure and flow values for the supply and demand curves are to be provided on $N^{1.85}$ graph paper.

1.6 SUBMITTALS

- A. Partial submittals will not be acceptable. Any installation work performed prior to the approval of the submittal shall be at the Contractor's own risk.

- B. Before any work is commenced, submit manufacturer's data (with listing or approval), system calculations, water supply data, and complete sets of working drawings.
- C. The OSHM Fire Protection Engineer shall review and approve submittals.
- D. Manufacturers' Data: Annotate descriptive data to show the specific model, type and size of each item the Contractor proposes to furnish. Include data for proper installation of each system including:
 - 1. Pipe and fittings
 - 2. Gate and check valves
 - 3. Globe valves
 - 4. Sprinkler heads
 - 5. Pipe hangers and supports
 - 6. Lubricating compound/PTFE tape.
 - 7. Signs
 - 8. Caps, chains
 - 9. Cabinets
 - 10. Hose valves
- E. Shop Drawings: Prepare working drawings on sheets not smaller than 24 in by 36 in (610 mm by 914 mm), in accordance with the requirements for "Working Plans" as specified in NFPA 13. A scaled site plan, with the location and elevation of the water flow test, shall be provided on the drawings. Drawings are to include isometric diagram of sprinkler risers and feed mains, including all control valves.
- F. As-Built (Record) Working Drawings: On a weekly basis, the Contractor Superintendent, in conjunction with the COTR, shall review and record as-built conditions on a set of drawings maintained at the job site. After completion, but before final acceptance of the work, furnish a complete set of as-built drawings for review and approval by OSHM. Make all necessary corrections to the drawings and furnish four sets of as-built drawings for record purposes. All deviations from the approved shop drawings shall be highlighted on the as-built drawings; if required by OSHM the Contractor shall also provide hydraulic calculations justifying deviations. The drawings shall not be smaller than 24 in by 36 in (610 mm by 914 mm) on reproducible media with title block similar to full size contract documents. Provide 1 compact disc containing CAD based drawings in DXF and PDF based format of all as-built drawings and schematics.
- G. Operation and Maintenance Manuals: Furnish four (4) instruction manuals containing complete operation and maintenance instructions for the specific make and model of all check valves, detector check valves, alarm valves, waterflow and tamper switches, backflow preventers, and other trim furnished. Serial numbers and ordering information shall be provided. Place one copy of each instruction manual in a flexible, oil-resistant protective binder and mount in an accessible location in the vicinity of each alarm control valve. Furnish three additional copies of each instruction manual.

1.7 QUALITY ASSURANCE

- A. The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only. All publications listed below refer to the most current edition.
- B. Manufacturers Qualifications
 - 1. American Society for Testing and Materials (ASTM) Publications:
 - a. A53 Pipe, Steel, Black and Hot Dipped, Zinc-Coated, Welded and Seamless
 - b. A135 Welded and Seamless Steel Pipe
 - 2. Factory Mutual System (FM) Publications
 - a. Approval Guide
 - 3. National Fire Protection Association (NFPA) Publications
 - a. 13 Standard for the Installation of Sprinkler Systems
 - b. 70 National Electrical Code
 - c. 72 National Fire Alarm and Signaling Code
 - d. 101 Life Safety Code
 - 4. International Code Council (ICC)
 - a. International Building Code
 - 5. Underwriters Laboratories, Inc. (UL) Publications:
 - a. Fire Protection Equipment Directory
 - b. 262 Gate Valves for Fire Protection Service
 - c. 789 Indicator Posts for Fire Protection Service
 - 6. American Society of Mechanical Engineers (ASME)
 - a. A17.1 Safety Code for Elevators and Escalators
- C. Qualifications Of Installer:
 - 1. Prior to installation, submit data for approval by OSHM, showing that the Contractor has successfully installed automatic wet pipe fire extinguishing sprinkler systems of at least 200 sprinkler heads each, or there is a firm contractual agreement with a subcontractor having such required experience. These systems shall be the same type and design which have been retrofitted into existing museums and historical buildings as specified herein. The data shall include the names and locations of at least two installations where the Contractor, or the subcontractor referred to above, has installed such systems. The Contractor shall indicate the type and design of each system and certify that each system has performed satisfactorily in the manner intended for a period of not less than 18 months.

2. Design (including hydraulic calculations) shall be by a NICET Level III or IV Technician (in automatic sprinkler system design) or a Professional Engineer (P.E.), licensed in Fire Protection Engineering. Qualifications of the designer must be submitted to OSHM for approval. The designer's NICET or P.E. stamp shall be present on each sheet of the working drawings.

- D. Service Organization: The Contractor shall furnish, to the COTR, evidence that there is an experienced and effective service organization which carries a stock of repair parts for the system in order to readily conduct repairs throughout the warranty period. Should the Contractor fail to comply with the service requirements of this section, the Government will then have the option to make the necessary repairs and back charge the Contractor without any loss of warranty or guarantee as provided by the contract documents.

1.8 DELIVERY STORAGE AND HANDLING

- A. Deliver products to project site in original, unopened packages with intact and legible manufacturers' labels identifying product and manufacturer, date of manufacture, and shelf life if applicable.
- B. Store materials inside, under cover, above ground, kept dry and protected from physical damage until ready for use. Remove from site and discard wet or damaged materials.
- C. Automatic sprinklers must be kept in original packaging until they are installed. Loose carrying or storage is not permitted. Loose sprinklers shall be discarded immediately and replaced at Contractor's expense.

1.9 COORDINATION

- A. Coordinate sprinkler head layout with reflected ceiling plan and all ceiling – mounted equipment, including diffusers, lights, security cameras, fire alarm devices, exit signs, and other devices.
- B. Coordinate major equipment and piping layouts with other trades to avoid obstructions and excessive changes in direction for piping.

1.10 WARRANTY

- A. The Contractor shall guarantee labor, materials, and equipment provided under this contract against defects for a period of one year after the date of final acceptance of this work by the Government.
- B. Final acceptance includes, but is not limited to, the receipt and OSHM approval of, as-built drawings and operation and maintenance manuals.

1.11 EXTRA MATERIALS

- A. Provide 6 extra sprinkler heads of each type installed and wrenches suitable for each head type.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. All products are subject to the following listed acceptable manufacturers. If the product is not addressed herein the product shall be from a US based manufacturer and listed for fire protection use.
- B. All products shall be FM approved/UL listed.
- C. Fire Protection Valves and Drain Assemblies
 - 1. Tyco Fire & Building Products
 - 2. Victaulic
 - 3. Stockham
 - 4. Kennedy
 - 5. Jenkins
 - 6. Reliable Automatic Sprinkler Co.
 - 7. Viking Corporation
- D. Sprinklers
 - 1. Tyco Fire & Building Products
 - 2. Reliable Automatic Sprinkler Co.
 - 3. Viking Corporation
 - 4. Victaulic Corporation
- E. Valve Supervisory Switches
 - 1. Potter Electric Signal
 - 2. System Sensor
 - 3. McWane, In. Kennedy Valve Div.

2.2 ABOVEGROUND PIPING SYSTEMS

- A. Provide fittings for changes in direction of piping and for all connections. Arrange piping so that it can be drained at the main riser. Make changes in piping sizes through standard tapered, reducing pipe fittings; the use of bushings is not permitted. Perform welding in the shop; field welding is not permitted.
- B. Jointing compound for pipe threads shall be polytetrafluoroethylene (PTFE) pipe thread tape only, applied on the male threads.
- C. Use of pipe dope is not permitted.
- D. Lubricant used on gaskets for mechanical fittings must be non-petroleum based and approved by OSHM.
- E. Run piping concealed in areas with suspended ceilings, except as noted on the drawings.

- F. Sprinkler Pipe and Fittings: Provide in accordance with NFPA 13, except as modified herein. Steel piping shall be Schedule 40 per ASTM A53 for sizes less than 4 inches (101.6 mm) and Schedule 10 or 40 per ASTM A53 for sizes 4 inches (101.6 mm) and larger. ASTM A135, Schedule 40 piping may be used for pipe sizes less than 2 ½ inches (63.5 mm).

1. Standard Installation:

- a. Nominal pipe sizes 4 inches or larger: Schedule 10 or 40 Pipe meeting ASTM A-53, A-135 or A-795 with factory- or field-formed, roll-grooved for Schedule 10 or 40 ends, or cut-grooved for Schedule 40 ends.
 - b. Nominal pipe sizes smaller than 4-inches: Schedule 40 Pipe meeting ASTM A-53, A-135, and A-795 with factory- or field-formed threaded ends.
 - c. For connections between 4 inch and larger pipes on risers, feed mains, cross-mains and drain lines, the requirements are as follows:
 - 1) Grooved pipe couplings and fittings for grooved pipe.
 - 2) Outlet coupling with screwed connection for grooved pipe.
 - d. For connections from risers, feed mains, cross-mains, or drains to branch lines, the requirements are as follows:
 - 1) Welded outlet with screwed connection or threaded coupling or fittings.
 - e. For connections to and between branch line pipes less than 4 inches the requirement is as follows:
 - 1) Threaded pipe couplings and fittings only.
 - f. For connections between drain pipes requirement is as follows:
 - 1) Galvanized threaded or cut-grooved pipe couplings and fittings.
2. Pressure ratings: All fittings and gaskets shall meet or exceed maximum working pressures present within the system.
3. Corrosion protection: All piping and hangers where exposed to the weather or installed in a corrosive atmosphere shall be protected against corrosion.
4. Pipe and Hanger Supports: Provide pipe supports, hangers, and clamps conforming to NFPA 13 and listed by UL or approved by FM. Provide galvanized supports, hangers, and clamps for all galvanized piping.
5. Joint Construction
- a. Plain-end fittings with mechanical couplings, hole-cut mechanical threaded outlet fittings, hole-cut mechanical grooved outlet fittings, and hole-cut saddle clamp outlet fittings are not permitted for new systems. The above fitting types are only permitted on a case-by-case basis when approved by OSHM.
 - b. Procedures for welding outlets shall be in strict conformance with the welding requirements of NFPA 13, including submission of welding certifications. Welding shall not be performed on-site.
 - c. Threaded Joints: Comply with NFPA 13 for pipe thickness and threads. Do not thread pipe with wall thickness less than Schedule 40.

- d. Grooved Joints and Fittings: Assemble joints and fittings with listed coupling and gasket, lubricant, and bolts from the same manufacturer. Fittings and attached couplings shall be from the same manufacturer.
 - e. Steel Pipe: Square-cut or roll-groove piping as indicated. Use grooved-end fittings and rigid, grooved-end-pipe couplings, unless otherwise indicated.
 - f. If the galvanized coating on piping is found to be chipped or cracked upon grooving of joint, two coats of liquid galvanizing material shall be applied to groove. The first coat shall be thoroughly dry prior to applying the second coat.
- G. Use of restriction orifices, reducing flanges, unions, and plain-end fittings is not permitted.
- H. Flanged fittings shall be used in lieu of unions with the exception of drain lines.
- I. The corrosion resistance ratio of pipe and fitting method shall not be less than 0.95.
- J. Pipe Hangers and Supports: Provide in accordance with NFPA 13.
- K. Identification Signs: Attach properly lettered approved metal or polycarbonate signs conforming to NFPA 13 to each valve and alarm device. Polycarbonate signs shall be red with engraved white letters. Signs at valves shall describe the sprinkler zone it controls and state that the valve is to remain open. Permanently affix design data nameplate to the riser of each system.
- L. Inspector's Test Connection: Provide test connections no higher than 6 ft (1.83 m) above the floor for each sprinkler system or portion of each sprinkler system equipped with an alarm device; locate at the control valve/drain assembly. Provide combination drain valve and test connection. Discharge shall be readily visible from the inspector's test connection, either by direct observation of the discharge or through a sight glass. Discharge shall be piped to either an open building drain or to the outside. Drainage and test valves shall be bronze globe, angle, or gate valves.
- M. Drains: Provide drain piping to discharge at safe points outside the building or to sight cones attached to drains of adequate size to readily receive the full flow from each drain under maximum pressure. All drain piping and fittings are to be galvanized. Provide auxiliary drains as required by NFPA 13. Splash guards are to be provided where necessary at discharge outlets.
- N. Pipe Sleeves and Seals. Provide where conduit or piping passes through walls, floors, roofs and partitions. Provide clearance between exterior of piping and interior of sleeve in accordance with NFPA 13. See Section 078413 Penetration Firestopping, for sleeves and seals through fire-rated assemblies. Secure sleeves in proper position and location during construction. Provide sleeves of sufficient length to pass through the entire thickness of walls, floors, roofs, and partitions.
- 1. Sleeves in Masonry and Concrete Walls, Floors, and Roofs: Provide ASTM A53, Schedule 40, zinc-coated steel pipe sleeves. Sleeves in floors shall project 4 inches (101.6 mm) above finished floors to prevent seepage.
 - 2. Sleeves in Partitions and Other than Masonry and Concrete Walls, Floors and Roofs: Sleeves shall be constructed from either zinc-coated schedule 40 steel pipe or zinc-coated 26 gauge steel sheet.
- O. Escutcheon Plates: Provide one piece or split-hinge-type metal plates for piping passing through floors, walls, and ceilings in exposed and concealed areas. Provide chromium-plated or color-

coordinated metal plates where pipe passes through finished ceilings. Securely anchor plates in proper position. Provide sprinkler escutcheon plates to match sprinkler head finish.

2.3 SPRINKLER HEADS

- A. Provide quick response sprinklers in offices and all other areas where their use is listed or approved. Heads located within the air streams of unit heaters or other heat-emitting equipment or skylights shall be selected for proper temperature rating.
- B. Where only a portion of the automatic sprinklers in a space are being replaced as part of a renovation, the response characteristics of the new sprinklers shall match those existing in the space.
- C. Heads shall have a nominal ½ inch (12.7 mm) orifice. Corrosion-resistant sprinkler heads shall be installed where they are exposed to the weather, moisture or corrosive vapors. Heads installed where they might receive mechanical injury or are less than 7 feet (2.13 m) above the floor level, shall be protected with approved guards in accordance with NFPA 13. Provide finish as indicated.

2.4 VALVES:

- A. Provide valves as required by NFPA 13. Valves shall be UL listed or FM approved for fire protection service.
- B. Gate Valves: Gate valves shall be the outside stem & yoke (OS&Y) type, and open by counterclockwise rotation. Gate valves installed higher than 7 feet (2.13 m) above the floor shall be provided with a chain drive or permanently mounted ladder. Provide a gate valve beneath each alarm valve in each riser when more than one alarm valve is supplied from the same water supply pipe. For post indicator valves, provide operating nut located about 3 feet above finish grade. Gate valves for use with indicator posts shall conform to UL 262. Indicator posts shall conform to UL 789. Provide each indicator post with one coat of primer and two coats of red enamel paint.
- C. Check Valves: Check valves shall be clear-opening, swing-check type, with a bronze or stainless steel seat ring and an EPDM rubber clapper facing. Flanged check valves of sizes 4 inch (101.6 mm) and larger shall have flanged inspection and access cover plates.
- D. Butterfly valves: Butterfly valves are permitted only for pipe sizes of 4 inches (100 mm) and smaller.

2.5 ALARMS

- A. Paddle-Type Water Flow Switches: Provide paddle-type water flow switches as indicated on the drawings. Switches shall have sensitivity setting to signal any flow of water that equals or exceeds the discharge from one sprinkler head. Water flow switch mechanisms shall incorporate an instantly recycling, pneumatic, adjustable retard element and shall be set with a 20-second delay. Assembly shall be rated at 175 psi (1206.6 kPa) cold water pressure. Switches shall be compatible with the fire alarm system and shall be connected per Section 283111.

- B. Valve Supervisory Switch: Valve supervisory switches shall be installed on all sprinkler control valves. The switches shall be mounted so as not to interfere with the normal operation of the valve and shall be adjusted to operate within two revolutions of the valve control or when the stem has moved no more than 1/5 of the distance from its normal open position. The mechanism shall be contained in a weather-proof, die-cast, aluminum housing that shall be provided with a ½ inch (12.7 mm) tapped conduit entrance and incorporate the necessary facilities for attachment to the valve. Switch housing shall be finished in red baked enamel. The entire installed assembly shall be tamper-proof and arranged to cause a switch operation if the housing cover is removed or if the unit is removed from its mounting. Devices shall be compatible with and connected to the fire alarm system per Section 283111.

2.6 CABINET

- A. Furnish and install a metal cabinet containing a stock of spare sprinkler heads of all types and ratings installed. The cabinet shall be located near the control valve and where the temperature will not exceed 100°F (37.8°C) The number of spare sprinkler heads shall be as required by NFPA 13.
- B. Provide a sprinkler head wrench in the cabinet for each type of sprinkler installed on the system.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Equipment, materials, installation, workmanship, examination, inspection and testing shall be in accordance with NFPA 13, except as modified herein. Install piping straight and true to bear evenly on hangers and supports. Install piping as close to the ceiling as possible, without interfering with other equipment and construction. Nipples shall be perpendicular to ceilings.
- B. Accurately align sprinkler heads in suspended ceilings symmetrically with diffusers, lights, and ceiling tiles. Install sprinkler heads in the center of the ceiling tiles unless directed otherwise. Concealed head covers shall not be installed until ceiling construction is complete and an inspection to determine height compliance has been conducted by OSHEM.
- C. Keep the interior and ends of all piping affected by Contractor's operations thoroughly clean of water and foreign matter by means of plugs or other approved methods. Inspect piping before placing into position. All pipe, fittings, and gaskets are to be cleaned of oil prior to installation.

3.2 FIELD CHANGES

- A. Do not make field changes in the piping layout, pipe sizes, or type of equipment, without the prior approval of the COTR.

3.3 CONNECTIONS TO EXISTING SPRINKLER SYSTEM

- A. Connection to the existing sprinkler system shall be done only after successfully testing new piping. Connections shall be as shown on the drawings. The COTR shall be notified at least 4 days prior to interruption of sprinkler protection.
- B. A schedule of any interruption of service shall be provided to the COTR and approval received before any service is interrupted. In no case shall the existing sprinkler system be shut off during periods other than normal Contractor construction hours.

3.4 FIELD PAINTING

- A. Painting shall be per Section 099000, Painting, except as modified herein. The above-ground steel piping systems including valves, piping in finished areas, pipe sleeves and hangers shall be cleaned, pre-treated, primed and painted. Coatings shall be applied only to clean, dry surfaces using clean brushes. Surfaces shall be cleaned to remove all dust, dirt, rust and loose mill scale.
- B. Immediately after cleaning, provide the metal surfaces with one coat of primer, applied to a minimum dry film thickness of 1.0 to 1.5 mil. Due care shall be exercised to avoid the painting of sprinkler heads or protective devices or allowing paint to drip or splatter on any equipment, artifacts, building structures, and floors. Materials which are used to protect sprinkler heads while painting is in progress shall be removed upon completion of painting.
- C. The Contractor shall remove all sprinkler heads which are painted and provide new, clean sprinkler heads of the proper type at his own expense. In addition to the primer, surfaces shall receive the following:
 - 1. Above-Ground Piping Systems in Unfinished Areas: Unfinished areas are defined as attic spaces, mechanical equipment spaces, spaces above suspended ceilings, crawl spaces, pipe chases, and spaces where walls or ceilings are not painted or not constructed of a pre-finished material. .
 - 2. Above-Ground Piping Systems in Other Areas: Provide primed surfaces with two coats of paint to match adjacent surfaces, except valves and operating accessories shall receive one coat of red enamel. Provide piping with 2 inch (50.8 mm) wide red enamel bands or self-adhering red plastic bands spaced at a maximum of 10 ft (3.05 m) intervals. In finished areas such as offices, the red bands may be omitted.
 - 3. All other coatings (e.g., zinc for galvanized pipe) shall be in compliance with NFPA 13.

3.5 FIELD TESTING AND FLUSHING

- A. All testing shall be scheduled with the COTR.
- B. At the discretion of the OSHM Fire Protection Engineer, an air pressure test may be required prior to filling the system with water. The test shall be conducted by raising the air pressure in the system to 40 psi (275.8 kPa) and allowing it to stand for 24 hours. There shall be no loss of air pressure greater than 1.5 psi (10.34 kPa) over the 24 hour period. Air pressure during this test shall be tracked via a graph over the 24 hour period by a pressure monitoring device or pressure monitoring gauge.

- C. Hydrostatic tests shall be conducted at the greater of 200 psi (1379 kPa) or the normal system pressure plus 50 psi (345 kPa) for a 2 hour period with no leakage or reduction in gage pressure. Hydrostatic test pressures shall not be maintained on the system overnight. Flush piping with potable water in accordance with NFPA 13.
- D. Preliminary Tests and Procedures: Test the alarms and other devices. Test the water flow alarms by flowing water through the inspector's test connection. Prior to the hydrostatic test, perform an air test on the system. In areas where piping will be concealed by ceilings, walls, or other construction before the system is complete and ready for final testing, the preliminary hydrostatic test shall be conducted prior to piping being concealed. This test shall be witnessed by the COTR and the OSHM Fire Protection Engineer. When all tests and procedures are completed and corrections made, submit a signed and dated certificate, similar to that specified in NFPA 13, with a request for formal inspection and tests.
- E. Formal Inspection and Tests: At this time, all piping, sprinklers, and other system components shall be in-place and all adjustments to the system completed. The OSHM Fire Protection Engineer shall be notified by the COTR, shall witness all tests, and shall approve all systems before they are accepted. Submit a request for a formal inspection at least five working days prior to the date the inspection is to take place. A competent representative of the sprinkler installer shall be present during testing and inspection. As-built drawings shall be on-site for the inspection. At this inspection, the system shall be hydrostatically tested. Any or all of the required tests shall be conducted by the Contractor at his own expense and additional tests made until it has been demonstrated that the systems comply with all contract requirements. The Contractor shall furnish all appliances, equipment, instruments, connecting devices and personnel for the tests. **Any costs incurred by the SI for repeat tests, due to the failure of the Contractor to adequately demonstrate that the system complies with the contract requirements, shall be borne by the Contractor.**
- F. CONNECTIONS TO EXISTING PIPING: During air and hydrostatic testing, the Contractor shall test new piping prior to connecting to the existing system. If the Contractor chooses to test the new piping while it is connected to existing piping or valves, the Contractor shall assume responsibility for all piping and equipment which is pressurized, as well as any damage caused by the failure of existing or new sprinkler systems. **The SI accepts no responsibility for existing valves' performance to withstand hydrostatic pressure testing.**

END OF SECTION 21 13 13

SECTION 22 01 10.62 – PLUMBING PIPING RELINING

PART 1 - GENERAL

1.1 SUMMARY

- A. This section includes furnishing and installing an epoxy repair liner system to seal and reinforce the transition between existing roof scupper outlets and new Schedule 80 PVC piping. The liner shall form a new pipe within the existing deteriorated pipe, typically without excavation, using a cured-in-place pipe (CIPP) method.

1.2 GENERAL REQUIREMENTS

- A. Installation and material tests of CIPP must meet or exceed the following standards:
 - 1. ASTM F1216 – Standard Practice for Rehabilitation of Existing Pipelines and Conduits by Inversion and Curing of a Resin-Impregnated Tube
 - 2. ASTM D638 – Tensile Properties of Plastics (Minimum Tensile Strength: 3,000 psi)
 - 3. ASTM D790 – Flexural Properties of Plastics (Minimum Flexural Strength: 4,500 psi; Flexural Modulus: 250,000 psi)

1.3 SUBMITTALS

- A. Product data and technical data sheets
- B. Installation instructions
- C. Design calculations for liner thickness
- D. Installer qualifications
- E. Warranty documentation

1.4 QUALITY ASSURANCE

- A. - Installer shall have a minimum of 3 years of experience with epoxy liner systems.
- B. - All work shall be performed in accordance with manufacturer's recommendations and ASTM standards.

PART 2 - GENERAL

2.1 EPOXY LINER SYSTEM

- A. Basis of Design: MaxLiner™ System by Applied Felts, Inc. or approved equal.
 - 1. Resin: MaxPox™ Resin and MaxPox™ Hardener, engineered for corrosion resistance and compatibility with cured-in-place pipe (CIPP) applications
 - 2. Liner Tube Options:
 - a. Max LinerTube™ – Standard felt liner with polymer coating
 - b. Max FlexLiner™ – Designed for bends and diameter transitions
 - c. Max WovoLiner™ – For complex geometries and enhanced flexibility
 - 3. Coating: All liners shall have an impermeable polymer coating to form the inner layer of the finished pipe, enhancing corrosion resistance, flow, and abrasion properties.
 - 4. Cure Method: Hot water, ambient, or other approved methods per manufacturer's recommendations.
 - 5. Thickness: Minimum 3 mm cured thickness, designed per project parameters and verified in the field prior to resin impregnation.
 - 6. Design Requirements: Liner shall be monolithic and homogeneous, chemically resistant to domestic sewerage, and structurally capable of supporting dead loads, live loads, and groundwater pressure.
- B. Accessories
 - 1. Surface preparation tools (grinders, brushes)
 - 2. Cleaning agents (non-residue forming)
 - 3. Temporary plugs or bypass piping as needed
 - 4. Pressure inversion apparatus compatible with MaxLiner™ installation methods

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean and dry all surfaces to be lined. Remove rust, scale, debris, and any loose material.
- B. Verify dimensions and alignment of scupper and PVC pipe.
- C. Protect adjacent systems, masonry opening and waterproofing materials.

3.2 INSTALLATION

- A. Cut liner to appropriate length to span from inside scupper throat to a minimum of 2' into the Schedule 80 PVC pipe.

- B. Saturate liner with epoxy resin per manufacturer's instructions.
- C. Insert liner using push or inversion method.
- D. Ensure full contact with host pipe and scupper interior.
- E. Cure per manufacturer's recommended time and temperature.
- F. Trim excess liner and smooth transitions.

3.3 INSPECTION AND TESTING

- A. Visually inspect for full cure, adhesion, and smooth transition.
- B. Perform water test to confirm watertight seal.

3.4 PROTECTION AND CLEANUP

- A. Protect liner from mechanical damage until fully cured.
- B. Remove all debris and restore surrounding areas to original condition.

3.5 WARRANTY

- A. Provide a minimum 1-year warranty on materials and workmanship.

END OF SECTION 22 01 10.62

SECTION 22 05 00 – COMMON MATERIALS AND METHODS FOR PLUMBING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section is intended to describe the common materials and installation methods of the mechanical work and it applies in general to all other Sections under Division 22.
- B. Due to the small scale of the drawings, all work required is not shown on the floor plans and certain work is shown on flow diagrams, riser diagrams and details. Work of Division 22 shall include all required work shown on plans, riser diagrams, flow diagrams and details.
- C. This Section includes the following:
 - 1. Piping materials and installation instructions common to most piping systems
 - 2. Transition fittings
 - 3. Dielectric fittings
 - 4. Mechanical sleeve seals
 - 5. Sleeves
 - 6. Escutcheons
 - 7. Grout
 - 8. Plumbing demolition
 - 9. Equipment installation requirements common to equipment sections
 - 10. Painting and finishing
 - 11. Hangers and supports for plumbing system piping and equipment
 - 12. Identification for plumbing piping and equipment

1.3 DEFINITIONS

- A. Finished Spaces: Spaces other than mechanical and electrical equipment rooms, furred spaces, pipe and duct chases, unheated spaces immediately below roof, spaces above ceilings, unexcavated spaces, crawlspace, and tunnels.
- B. Exposed, Interior Installations: Exposed to view indoors. Examples include finished occupied spaces and mechanical equipment rooms.
- C. Exposed, Exterior Installations: Exposed to view outdoors or subject to outdoor ambient temperatures and weather conditions. Examples include rooftop locations.

- D. Concealed, Interior Installations: Concealed from view and protected from physical contact by building occupants. Examples include above ceilings and chases.
- E. Concealed, Exterior Installations: Concealed from view and protected from weather conditions and physical contact by building occupants but subject to outdoor ambient temperatures. Examples include installations within unheated shelters.

1.4 REFERENCES

- A. Provide work in accordance with all applicable international, state and local, codes, rules, regulations, and standards, including but not limited to, requirements of the following:
 - 1. ASME/ANSI B31: Code for Pressure Piping
 - 2. AWS D1.1: Structural Welding Code-Steel
 - 3. MSS SP58: Pipe Hangers and Supports – Materials, Design, and Manufacturers
 - 4. MSS SP69: Pipe Hangers and Supports – Selection and Application except spacing for hangers
 - 5. ANSI A13.1: Scheme for Identification of Piping Systems
 - 6. NSF/ANSI 61: Drinking Water System Components

1.5 SUBMITTALS

- A. Provide Product List of factory fabricated items, in accordance with Section 01 60 00 “Product Requirements”, including name of proposed manufacturer, for all products specified in various sections of Division 22.
- B. Provide submittals in accordance with Section 01 33 00 “Submittal Procedures” in sufficient detail to verify full compliance with the requirements of the Contract Documents.
- C. Product Data: Provide for each type of factory-fabricated product indicated.
- D. Welding certificates.

1.6 WARRANTY AND CONTRACT CLOSEOUT

- A. Comply with warranty and contract closeout requirements specified in Division 01.
- B. Provide Special Warranties and/or warranty service in accordance with Section 01 60 00 “Product Requirements” where specified in the various sections of Division 22.
- C. Provide manufacturer’s certificates of supervision and startup service as specified in the various sections of Division 22.
- D. Provide testing and cleaning reports. Indicate dates of testing and cleaning operations, procedures used and results obtained for each system. Reports shall be certified as complete.

- E. Provide instructions and demonstration to the Owner's representatives for all equipment and systems installed under Division 22. Instruction and demonstration shall be appropriate for the size and complexity of the installed system.
- F. Include information for all products specified in the operation and maintenance manual.

1.7 QUALITY ASSURANCE

- A. Steel Support Welding: Qualify processes and operators according to AWS D1.1, "Structural Welding Code--Steel."
- B. Steel Pipe Welding: Qualify processes and operators according to ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."
 - 1. Comply with provisions in ASME B31 Series, "Code for Pressure Piping."
 - 2. Certify that each welder has passed AWS qualification tests for welding processes involved and that certification is current.
 - 3. Use welders fully qualified and licensed by the state authorities.
- C. The specifications for certain products and alternative materials may appear in more than one section of Division 22. Work of Division 22 shall be coordinated for all sections of Division 22 to assure that where two or more items of any given product are furnished under Division 22 that they are of the same manufacturer and type and that alternative materials is consistent throughout the work of Division 22.
- D. Except for spacing of hangers, provide hangers and supports in accordance with the latest issue of Manufacturer's Standardization Society (MSS) Specifications SP 58 and 69.
- E. All pipe, valves, fittings, stops, faucets, and domestic water pumps shall comply with the Federal "Reduction of Lead in Drinking Water Act" NSF/ANSI 61 and NSF/ANSI 372 for lead content of 0.25%.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle all material and equipment in accordance with manufacturer's instructions and recommendations. Such instructions and recommendations are hereby made part of these specifications.
- B. Deliver pipes and tubes with factory-applied end caps. Maintain end caps through shipping, storage, and handling to prevent pipe end damage and to prevent entrance of dirt, debris, and moisture.
- C. Deliver products and equipment properly labeled and tagged. Maintain products in original shipping containers and store in a dry area until ready for installation.
- D. Store plastic pipes protected from direct sunlight. Support to prevent sagging and bending.

1.9 COORDINATION

- A. The Mechanical systems are indicated on the Plumbing Drawings. Certain pertinent information and details involving the installation of plumbing work appear on Architectural, Structural, Mechanical and Electrical Drawings. Become familiar with all Drawings and incorporate all pertinent requirements.
- B. Drawings are diagrammatic and indicate general arrangement of systems and requirements of the plumbing work. Do not scale the Drawings to obtain dimensional requirements. Exact locations of equipment must be coordinated and obtained prior to starting the work.
- C. Arrange for pipe spaces, chases, slots, duct shafts and openings in building structure during progress of construction, to allow for plumbing installations.
- D. Coordinate installation of required supporting devices and set sleeves in poured-in-place concrete and other structural components as they are constructed.
- E. Coordinate requirements for access panels and doors for plumbing items requiring access that are concealed behind finished surfaces. Coordinate installation of identification labels with locations of access panels and doors.
- F. Coordinate scheduling, sequencing, movement and positioning of large equipment into the building during construction.
- G. Coordinate installation of identification devices with completion of covering and painting of surfaces where identification devices are to be applied.
- H. Install identification devices prior to installation of ceilings and similar concealment.

1.10 ENERGY PERFORMANCE CRITERIA

- A. All equipment provided under Division 22 shall meet the requirements of the International, or State, Energy Code, ASHRAE Standard 90 or the latest issue of the Standards for Equipment in the National Energy Policy Act (NEPA), whichever is more stringent.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. In other Part 2 articles of various sections of Division 22 where subparagraph titles below introduce lists, the following requirements apply for product selection:
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the manufacturers specified.
 - 2. Unless otherwise noted, substitutions of specified manufacturers shall comply with the requirements of Division 01.

2.2 PIPE, TUBE, AND FITTINGS

- A. Refer to individual Division 22 piping Sections for pipe, tube, and fitting materials and joining methods.
- B. Pipe Threads: ASME B1.20.1 for factory-threaded pipe and pipe fittings.

2.3 JOINING MATERIALS

- A. Refer to individual Division 22 piping Sections for special joining materials not listed below.
- B. Pipe-Flange Gasket Materials: Suitable for chemical and thermal conditions of piping system contents.
 - 1. ASME B16.21, nonmetallic, flat, asbestos-free, 1/8-inch maximum thickness unless thickness or specific material is indicated.
 - a. Full-Face Type: For flat-face, Class 125, cast-iron and cast-bronze flanges.
 - b. Narrow-Face Type: For raised-face, Class 250, cast-iron and steel flanges.
 - 2. AWWA C110, rubber, flat face, 1/8 inch thick, unless otherwise indicated; and full-face or ring type, unless otherwise indicated.
- C. Flange Bolts and Nuts: ASME B18.2.1, carbon steel, unless otherwise indicated.
- D. Plastic, Pipe-Flange Gasket, Bolts, and Nuts: Type and material recommended by piping system manufacturer, unless otherwise indicated.
- E. Solder Filler Metals: ASTM B32, lead-free alloys. Include water-flushable flux according to ASTM B813.
- F. Brazing Filler Metals: AWS A5.8, BCuP Series, copper-phosphorus alloys for general-duty brazing, unless otherwise indicated.
- G. Welding Filler Metals: Comply with AWS D10.12 for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.
- H. Solvent Cements for Joining Plastic Piping:
 - 1. PVC Piping: ASTM D2564. Include primer according to ASTM F656.
- I. Fiberglass Pipe Adhesive: As furnished or recommended by pipe manufacturer.

2.4 TRANSITION FITTINGS

- A. AWWA Transition Couplings: Same size as, and with pressure rating at least equal to and with ends compatible with, piping to be joined.
 - 1. Underground Piping NPS 1-1/2 and Smaller: Manufactured fitting or coupling.

2. Underground Piping NPS 2 and Larger: AWWA C219, metal sleeve type coupling.
 3. Aboveground Pressure Piping: Pipe fitting.
- B. Plastic-to-Metal Transition Fittings: PVC one-piece fitting with manufacturer's Schedule 80 equivalent dimensions; one end with threaded brass insert, and one solvent-cement-joint end.
- C. Plastic-to-Metal Transition Adaptors: One-piece fitting with manufacturer's SDR 11 equivalent dimensions; one end with threaded brass insert, and one solvent-cement-joint end.
- D. Plastic-to-Metal Transition Unions: MSS SP-107, PVC four-part union. Include brass end, solvent-cement-joint end, rubber O-ring, and union nut.
- E. Flexible Transition Couplings for Underground Nonpressure Drainage Piping: ASTM C1173 with elastomeric sleeve, ends same size as piping to be joined, and corrosion-resistant metal band on each end. Fernco, Mission or equal.

2.5 DIELECTRIC FITTINGS

- A. Description: Combination fitting of copper alloy and ferrous materials with threaded, solder-joint, plain, or weld-neck end connections that match piping system materials.
- B. Insulating Material: Suitable for system fluid, pressure, and temperature.
- C. Dielectric Unions: Factory-fabricated, union assembly, for 250-psig minimum working pressure at 180 deg F.
- D. Dielectric Flanges: Factory-fabricated, companion-flange assembly, for 150- or 300-psig minimum working pressure as required to suit system pressures.
- E. Dielectric-Flange Kits: Companion-flange assembly for field assembly. Include flanges, full-face- or ring-type neoprene or phenolic gasket, phenolic or polyethylene bolt sleeves, phenolic washers, and steel backing washers. Separate companion flanges and steel bolts and nuts shall have 150- or 300-psig minimum working pressure where required to suit system pressures.
- F. Dielectric Couplings: Galvanized-steel coupling with inert and noncorrosive, thermoplastic lining; threaded ends; and 300-psig minimum working pressure at 225 deg F.
- G. Dielectric Nipples: Electroplated steel nipple with inert and noncorrosive, thermoplastic lining; plain, threaded, or grooved ends; and 300-psig minimum working pressure at 225 deg F.

2.6 MECHANICAL SLEEVE SEALS

- A. Description: Modular sealing element unit, designed for field assembly, to fill annular space between pipe and sleeve.
1. Sealing Elements: EPDM interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
 2. Pressure Plates: Steel, zinc, dichromate for fire protection reinforced nylon polymer elsewhere. Include two for each sealing element.

3. Connecting Bolts and Nuts: Carbon steel with corrosion-resistant coating of length required to secure pressure plates to sealing elements. Include one for each sealing element.
4. Thunderline link seal or equal.

2.7 SLEEVES

- A. Galvanized-Steel Sheet: 0.0239 inch minimum thickness; round tube closed with welded longitudinal joint.
- B. Steel Pipe: ASTM A53, Type E, Grade B, Schedule 40, galvanized, plain ends.
- C. Cast Iron: Cast or fabricated "wall pipe" equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop, unless otherwise indicated.
- D. Stack Sleeve Fittings; Manufactured, cast-iron sleeve with integral clamping flange. Include clamping ring and bolts and nuts for membrane flashing.
 1. Underdeck Clamp: Clamping ring with set screws.
- E. Molded PE: Reusable, PE, tapered-cup shaped, and smooth outer surface with nailing flange for attaching to wooden forms.

2.8 ESCUTCHEONS

- A. General: Manufactured wall and ceiling escutcheons and floor plates, with an inside diameter to closely fit around pipe, tube, and insulation of insulated piping and an outside diameter that completely covers opening.
- B. One piece construction on exposed piping in finished areas. Elsewhere, split pattern with setscrew. Provide deep pattern type where required to conceal protruding fittings and sleeve.
- C. Provide polished chromium plated escutcheons on pipes passing through walls, floors or ceilings wherever such pipes are exposed to view.

2.9 HANGERS AND SUPPORTS

- A. Acceptable Manufacturers
 1. Other Than Roof Supports
 - a. B-Line Systems, Inc.
 - b. Grinnell Company
 - c. National Pipe Hangers
 - d. Penn Construction Industries
 - e. Other approved United States manufacturer whose products comply with the referenced standards.

B. Reference Standards

1. ASTM A36 - Specification for Structural Steel
2. ASTM A123 - Zinc (Hot-Dip Galvanized Coatings on Products Fabricated from Rolled, Pressed and Forged Steel Shapes, Bars, and Strip)
3. ASTM A653 G90 - Specification for Steel Sheet, Zinc Coated by the Hot-Dip Process
4. ASTM B633 - Specification for Electrodeposited Coatings of Zinc on Iron and Steel
5. AWS D1.1 - Structural Welding Code - Steel
6. MSS SP58 - Manufacturer's Standardization Society: Pipe Hangers and Supports - Materials, Design and Manufacture
7. MSS SP69 - Manufacturer's Standardization Society: Pipe Hangers and Supports - Selection and Application
8. NFPA 13 - Standard for the Installation of sprinkler Systems

C. Quality Assurance

1. Steel pipe hangers and supports shall have the manufacturer's name, part number and applicable size stamped in the part itself for identification.
2. Hangers and supports shall be designed and manufactured in conformance with MSS SP58.

D. General

1. Except for spacing of the hangers, design and fabrication of pipe hangers, supports and welding attachments shall conform to ANSI B31.9 or B31.1 as applicable.
2. Except for spacing of the hangers, hanger types and supports for bare and covered pipe shall conform to MSS SP69 for the temperature range except that only flat wide band hangers shall be used for hangers installed outside of insulation and plastic pipe.
3. Except for spacing of pipe hangers and elsewhere as otherwise indicated, horizontal and vertical piping attachment shall conform to the more stringent of this specification or MSS SP58 or MSS SP69. Continuous inserts and expansion bolts may be used.
4. All ferrous hangers, supports and hardware located outdoors shall be hot dip galvanized after fabrication per ASTM A123.
5. Hangers and clamps for support of bare copper piping shall be coated with copper colored (for identification) baked on epoxy paint. Use additional PVC coating of the epoxy painted hangers where necessary.
6. Provide suitable chromium plated brass supports for chromium plated pipe with exposed heads of bolts and screws chromium plated.
7. Hangers other than described above shall be zinc plated in accordance with ASTM B633 or shall have an electrodeposited epoxy finish.
8. Strut channels shall be pregalvanized in accordance with ASTM A653 G90 or shall have an electrodeposited finish.
9. All hangers and supports shall have some form of adjustment available after installation.

E. Inserts

1. Inserts: Malleable iron case or galvanized steel shell and expander plug for threaded connection with lateral adjustment, top slot for reinforcing rods, lugs for attaching to forms.
2. Size inserts to suit threaded hanger rods.

- F. Pipe Hangers and Supports: For plumbing plastic piping systems provide hangers, supports and support channels as recommended by piping manufacturer, use V-bottom clevis hanger with galvanized 18 gauge continuous support channel where required.
1. Hangers for Pipe Sizes to 1-1/2 Inch: Adjustable carbon steel ring or clevis.
 2. Hangers for Hot or Cold Pipe Sizes 2 Inches to 4 Inches and Cold Pipe Sizes 6 Inches and Over: Adjustable carbon steel clevis.
 3. Multiple or Trapeze Hangers: Factory-enameled steel channels with welded spacers and hanger rods or 12 gauge rolled formed ASTM A570 Grade 33 structural quality steel channels (strut), cast iron roll and stand for hot pipe sizes 6 inches and over. Cross section suitable for span and loading. Suspension by outside hanger rods sized for total load on trapeze.
 4. Wall Support for Pipe Sizes to 3 Inches: Carbon steel hook.
 5. Wall Support for Pipe Sizes 4 Inches and Over: Welded steel bracket and steel clamp, adjustable steel yoke and cast iron roll for hot pipe sizes 6 inches and over.
 6. Vertical Support: Steel riser clamp.
 7. Floor Support for Hot Pipe Sizes to 4 Inches and All Cold Pipe Sizes: Adjustable pipe saddle and pipe nipple attached to steel base stand, and concrete pier or steel support.
 8. Floor Support for Hot Pipe Sizes 6 Inches and Over: Adjustable cast iron roll and stand, steel screws and concrete pier or steel support.
 9. Design hangers to impede disengagement by the movement of supported pipe. Provide spring and neoprene hangers as required.
- G. Beam Clamps: Forged steel C-clamps shall include retaining strap, locking nut or other device for nonslip attachment, except LOCKING NUT NOT ALLOWED for project requiring seismic restraints.
- H. Hanger Rod: Steel hanger rod zinc plated per ASTM B633.

2.10 IDENTIFICATION DEVICES AND LABELS

- A. General
1. Products specified are manufacturer's standard products of categories and types required for each application as referenced in Part 3 of this section and elsewhere on the drawings or in Division 22 specifications. Where more than single type is specified for listed application, selection is Contractor's option, but provide single selection for each product category.
 2. Products shall comply with requirements of ANSI A13.1 and OSHA where applicable.
- B. Snap-On Plastic Pipe Markers: Manufacturer's standard preprinted, semirigid snap-on, color-coded pipe markers conforming to ANSI A13.1.
- C. Pressure-Sensitive Pipe Markers: Manufacturer's standard preprinted, color-coded, pressure-sensitive vinyl pipe markers, with permanent adhesive conforming to ANSI A13.1.
- D. Pipes/Insulation Smaller Than 6 Inches: Full-band pipe markers, extending 360 degrees around pipe/insulation at each location.

- E. Pipes/Insulation 6 Inches And Larger: Either full-band or strip type pipe markers, at least 3 times the letter height and of length required for label.
- F. Arrows: Either integrally with piping system service lettering (to accommodate both directions), or as separate unit, on each pipe marker to indicate direction of flow.
- G. Plastic Tape: Manufacturer's standard color-coded, pressure- sensitive, self-adhesive, vinyl tape, at least 3 mils thick. Width 1-1/2 inches wide on pipes with outside diameters (including insulation) less than 6 inches; 2-1/2 inches wide for larger pipes. Color shall comply with ANSI A13.1 unless otherwise indicated.
- H. Valve Tags: Stamped or engraved brass with 1/4-inch letters for piping system abbreviation and 1/2-inch sequenced numbers. Provide a hole for fastener. Brass wire-link chain, beaded chain, or S-hook fasteners.
- I. Lettering and Graphics: Coordinate names, abbreviations, and other designations used in mechanical identification, with corresponding designations indicated. Use numbers, letters, and terms indicated for proper identification, operation, and maintenance of mechanical systems and equipment.
- J. Multiple Systems: Where multiple systems of same name are indicated, identify individual system number as well as service.

2.11 VALVES

A. General

- 1. Valves are specified by Valve Type and, for some services, several valve types are grouped by "Valve Class" in various sections of Division 22. Application is stated in the "Piping System Requirements". Where more than one Valve Type or Valve Class is listed for a service, use any of the listed Types or Classes, unless otherwise specified or indicated, but selection must be consistent throughout the work.
- 2. It shall be Contractor's responsibility to coordinate the work for all sections of Division 22 to assure that all general service valves throughout the work of Division 22 are of the same manufacture and type and that all valves of the same type number/identification throughout the work of Division 22 are of the same manufacture.
- 3. Valve packing shall not contain asbestos.
- 4. Bronze Valves: Construct body of ASTM B62 for Classes 125 and 150, ASTM B61 for Classes 200 and 300, copper-silicon bronze stem.
- 5. Iron Valves: Construct body of ASTM A126, Class B copper-silicon bronze stem.
- 6. Domestic water system valve construction shall comply with NSF 61, 372 and Federal law 111-380 and shall be certified as lead free compliant.

- B. Types V1 through V4 General Service Valves: Industrial valve construction. Handwheel operator, except where specified to be provided with chain-wheel operator. Figure numbers listed are those of Stockham Valve Co. Valves shall be products of a single manufacturer.

- 1. Type V1: Class 125 bronze, solder-end, for soldered copper piping 2 inch and smaller in size; 150 psig water or oil working pressure at 250° F. with high temperature nonlead solder

joint (95-5 or equivalent as specified in the "Solder" article of this section); 125 psig saturated steam working pressure:

		Swing	
	Ball	Check	Globe
	See Type V9	B309	B14T
*Use B104 where space prevents full extension of rising steam.			

2. Type V2: Class 125 bronze, for screwed piping 2 inch and smaller in size; 125 psig steam working pressure; 200 psig cold water, oil or gas working pressure:

		Swing	Lift		
	Ball	Check	Check	Globe	Angle
	See Type V9	B319	B316	B22T	B222T
*Use B103 where space prevents full extension of rising steam.					

3. Type V3: Class 300 bronze, for screwed piping 2 inch and smaller in size; 300 psig steam working pressure; 600 psig cold water, oil or gas working pressure:

	Swing	Lift		
Ball	Check	Check	Globe	Angle
See Type V9	B375	B367	B74	B266

4. Type V10, General Service: 4 inches and larger ductile iron epoxy coated full port Class 125 ball valve, flanged or grooved ends. FDA epoxy grade coating, PTFE seats and seals, stainless steel ball and stem. CWP rating 200 psig. Apollo 1BV-125-6Q series.

5. Approved Manufacturers:

- a. Apollo
- b. American Valve Co.
- c. Watts
- d. Victaulic
- e. Jomar

- C. Type V11, Nonslam (Silent) Check Valve: Combination Pump Valve Co. figure numbers are indicated.

1. 1/2 Inch Through 2 Inch in Size: Model 36, 300 psi, with union end. Bronze body and trim and stainless steel spring.

- D. Balancing Valves

1. Balancing valves shall be calibrated type with bronze body, Class 125, soldered or screwed ends as required. Valves shall have Venturi flow measuring ports.
2. Approved Manufacturers:

- a. Bell and Gossett Circuit Setter
- b. Taco
- c. Armstrong
- d. Victaulic

E. Drain Valves

- 1. Valves 3/4 Inch and Smaller in Size: Interior hose bibb, brass, compression type, with renewable seat and male threads for attachment of hose and fitted with wheel handle and cap when not piped to drain. Valves exposed in finished spaces chromium plated.
- 2. Valves 1 Inch and Larger in Size: Hose end gate valves fitted with cap except that valves indicated as piped to drain shall be standard shutoff type for system.

F. Refer to other sections of Division 22 for additional valve requirements.

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

A. General

- 1. Furnish, deliver, erect, connect, and finish in every detail all materials, equipment and accessories required for the Work.
- 2. Include in the work and in the bid minor details not usually shown or specified, but manifestly necessary for the proper installation and operation of the various systems, the same as if specified or shown.
- 3. If any departures from the Contract Documents are deemed necessary, submit details of such departure and the reasons therefore to the Architect for approval.
- 4. Be responsible to request clarification from the Architect on any conflicts represented between the drawings and specifications.
- 5. Adequately guard all exposed moving parts of equipment, such that contact by operating personnel will not cause personal damage or injury.

B. Layout and Coordination With Other Trades

- 1. Layout Work from building and property lines and benchmarks provided, verify, and be responsible for the correctness of all measurements in connection with the Work. Any change made in major overall dimensions shown which affect the physical size, shape, or location of any part of the Work, whether due to field check or changes due to use of equipment of a manufacturer other than that used as basis of design shall cause no interference with other Work.
- 2. Examine the Drawings of other trades, cooperate and coordinate with other trades to insure that the Work can be installed properly as designed and planned without interference with other work or delay. Where interferences may occur and departures from the arrangements shown are required, consult with other trade involved. Come to agreement as to changed locations and elevations. Furnish all necessary templates, patterns, measurements, etc., for installation and for the purpose of making adjoining work conform. Furnish setting plans and shop drawings to other trades as required.

3. Investigate the structural and finish conditions affecting the Work. Offsets, bends or other items required may not be shown on the drawings; provide such offsets or bends as required to meet structural or finish conditions.
4. Coordinate layout with architectural ceilings and lighting layouts and similar work.
5. Coordinate and be responsible for the required clearances in shafts, chases, furred partitions and suspended ceilings. Coordinate and cooperate with the trades responsible for constructing such spaces, together with other trades sharing such spaces, and advise other trades of the requirements of the Work. Immediately submit for review large scale composite Drawings showing space requirements that exceed those shown.
6. Install systems so that they do not interfere with any openings, doors or windows, or with other work, and so as to permit proper access.
7. Install material and equipment as high as possible; at minimum, to clear the top of all doors, windows and other structural openings. Maintain maximum headroom and space conditions in every case. Where headroom or space conditions appear inadequate, notify the Architect before proceeding with the installation.
8. Except where greater clearance is specified or required by applicable codes, rules or regulations, install piping, fittings, valves, etc., to provide not less than 1 inch between their finished covering and the structure or adjacent work of any kind. The minimum space between finished hot piping of any kind and adjacent electrical conduit shall be 6 inches.
9. Make reasonable modifications in the layout to provide proper clearances or accessibility, or to prevent conflict with the work of other trades, at no increase in the Contract sum.
10. Prepare large scale composite working drawings, including such section views and details as are necessary to clearly show how the systems are to be installed in relation to the work of other trades. Issue such Drawings to the other trades for coordination of their work. Where such drawings show deviations from the Contract Drawings or conflict with other trades such that reasonable modifications cannot be made, detail and submit such deviation or conflict to the Architect for review.
11. If work is installed before coordinating with other trades so as to cause interference with the work of other trades, or as not to provide proper access for maintenance or repair, make necessary changes to correct the condition at no increase in the Contract sum.
12. For alterations to existing facilities, be fully responsible for coordinating work with all existing conditions. Verify location of existing piping, and equipment in the field. Relocate or offset new piping, and make reasonable modifications to existing piping, as required to fit in available space whether or not such relocation of offset is shown on the Drawings.

C. Manufacturer's Instructions and Recommendations

1. Perform the installation, cleaning, testing, calibration and startup of all material and equipment in accordance with the manufacturer's instructions and recommendations. Such instructions and recommendations are hereby made part of the specifications.
2. Should a conflict exist between specifications and manufacturer's instructions, consult with the Architect.

D. Electrical Rooms

1. Do not install any piping, or equipment in or through an electrical room or similar room containing electrical equipment, other than piping, or equipment exclusively serving the room or equipment in the room.

2. If there is a conflict between the above requirement and the Drawings, the above shall govern. If reasonable modifications cannot be made to accommodate this requirement, obtain instructions from the Architect before proceeding with the work.

E. Painting

1. Except where specified otherwise in Division 22, Work of Division 09 will provide painting of plumbing systems, equipment and components.
2. Protect all equipment from rust, corrosion, and similar damage by either factory applied or field applied protective coatings.
3. Repair marred and damaged factory painted finishes with manufacturer's touch up paint and application procedures to match original factory finish.

F. Wall and Ceiling Access Doors

1. Access Doors shown on Architectural Drawings will be provided under Division 05.
2. Furnish access doors required for access to concealed valves, air vents, traps, cleanouts, unions, expansion joints, and other equipment where no other means of access is available. Access doors shall be of adequate size for the service requirements, minimum clear opening of 14 inches by 16 inches.
3. Coordinate requirements for access panels and doors for plumbing items requiring access that are concealed behind finished surfaces. Access panels and doors shall be specified in Division 08 Section "Access Doors and Frames."

3.2 PENETRATIONS

A. General

1. Coordinate with other trades as to the size and location of openings to be provided in new floors, walls, roofs and ceilings as construction progresses.
2. Do not cut openings in new or existing floors and walls without proper structural reinforcement.
3. Install both piping and seals so as to maintain integrity of seals with expansion and contraction of piping.
4. Sealing Requirements:
 - a. Seal completely all penetrations of piping through any surface.
 - b. Seal completely all penetrations of piping through above ceiling full height partitions.
 - c. Seal completely the perimeter of all access doors or panels in any surface.

B. Sleeves

1. Provide each pipe, passing through a masonry or concrete wall, floor or partition, and elsewhere as indicated, with a sleeve made from standard weight galvanized steel pipe for pipe or conduit.
2. Select sleeves two pipe sizes larger than any pipe to accommodate pipe, insulation, and jacketing without touching the sleeve and shall provide minimum of 3/8 clearance.
3. Be responsible for the proper location and alignment of all sleeves.

4. Extend wall and partition sleeves through and cut flush with each surface unless otherwise indicated or specified.
5. Place sleeves imbedded in concrete floors or walls in the forms before concrete is poured; sleeves shall have integral water stop flanges, where they are to receive either water tight or hydrostatic seals.

C. Fire Rated Penetrations

1. Provide through-penetration fire-stop sealing system for pipe penetrations through fire or smoke rated construction. Refer to Division 07 for through-penetration fire stop sealing system.
2. Coordinate with Division 07 to determine requirements for sleeves and clearances.

D. Interior Non-Rated Wall Pipe Penetrations

1. For acoustically treated partitions, and walls between mechanical equipment rooms and occupied spaces, fill annular void at penetration with acoustical sealant.

E. Resilient Penetration Sleeve/Seal

1. Provide resilient penetration sleeve seal for piping subject to vibration to prevent transmission to the building structure.
2. Maintain an airtight seal around the penetrating element and prevent rigid contact of the penetrating element and the building structure. Fit the sleeve tightly to the building construction and seal airtight on both sides of the construction penetrated with acoustical sealant.

F. Floor Pipe Penetrations

1. Provide water stops for new cored openings for piping where such openings are above grade in toilet rooms and similar wet service areas and in mechanical rooms, penthouses, and pipe chases.
2. Provide water stops for existing pipe floor openings that do not have sleeves extended above the floor.
3. Provide either a sleeve or angle water stop. Sleeve may be used if the fire-stop sealing method selected by Division 07 allows the use of a sleeve, otherwise provide angle water stop.
4. Sleeve Water Stop: Construct of Schedule 40 galvanized steel pipe, length as required for 1 inch above floor and 1 inch below underside of floor. Provide 1 inch by 1 inch by 1/8 inch galvanized angle clips welded to sleeve at 90 degree intervals and securely fastened to underside of floor. Caulk space between floor opening and sleeve water tight with soft setting waterproof silicone sealant.
5. Angle Water Stop: Construct of standard weight pipe 1 inch long welded to a 1/4 inch circular steel base plate ring fastened to floor with expansion anchors. Base plate ring width as required for anchor clearance from edge of cored opening. Seal between base ring and floor and caulk all edges of base ring with waterproof silicone sealant.
6. Floor penetrations through concrete slabs on grade shall be sleeved as required by the plumbing code. For hydrostatic floor slab penetrations provide Link Seal as specified under exterior wall pipe penetrations.

G. Exterior Wall Pipe Penetrations

1. Provide compressible synthetic rubber seal with connecting bolts and pressure plates, equivalent to LINK SEAL, manufactured by the Thunderline Corporation, or THRUWALL manufactured by O.Z. Gedney. Provide sleeve and install seal in accordance with the manufacturer's recommendations to provide air and water tightness above ground and hydrostatic sealing below grade. Caulking or other type mastic is not acceptable.

3.3 ALTERATIONS OF UTILITIES AND SERVICES

- A. Arrange and pay for the relocation, disconnection or removal of existing utilities and services where such utilities and services interfere with new construction. Perform alteration of utilities and services in accordance with rules, regulations and requirements of the involved utility company as well as regulatory agencies having jurisdiction.

3.4 ALTERATIONS AND CONNECTIONS TO EXISTING FACILITIES

- A. Refer to Division 01 Sections "Execution" and "Alteration Project Procedures" and Division 02 Section "Selective Demolition" for general demolition requirements and procedures.
- B. Make all necessary alterations to existing Division 22 systems as required for removing or for connecting or extending these systems to new work and for revisions in existing work as indicated and as required, whether indicated or not. Match new materials in altered systems with existing materials unless otherwise indicated.
- C. Continuity of Existing Services
 1. Perform alterations and connections to existing facilities with a minimum of interruption. Where interruption is necessary, prepare a time schedule for shutdown activities, coordinate with Architect, Owner and other trades, and obtain written approval from Owner prior to proceeding with the work. Include work scheduled for off hours, when Owner requires that shutdown and interruption of facilities occur during unoccupied times.
 2. Prepare and set notices on services and equipment that are temporarily shut off or disconnected.
- D. Provide shutoff valves to isolate new work from existing and temporary or permanent connections to new work as required for proper testing and cleaning of new work.
- E. All relocations of existing work shall be accomplished using new materials and accessories unless specifically noted otherwise.
- F. Where equipment, and/or piping is removed or disconnected under Division 22, perform the work in such a manner that no damage is done to the structure or remaining portions of the existing systems. Do not under any circumstance place stress on existing pipe and fittings that are to be reused. Be fully responsible for and repair, at no additional expense to the Owner, any

leaks developing in existing piping due to failure to take proper precautions when making alterations.

- G. Disconnect, demolish, and remove plumbing systems, equipment, and components indicated to be removed.
 - 1. Piping to Be Removed: Remove portion of piping indicated to be removed and cap or plug remaining piping with same or compatible piping material.
 - 2. Piping to Be Abandoned in Place: Do not abandon any piping in place unless specifically noted to do so. Drain piping and cap or plug piping with same or compatible piping material.
 - 3. Equipment to Be Removed: Disconnect and cap services and remove equipment.
 - 4. Equipment to Be Removed and Reinstalled: Disconnect and cap services and remove, clean, and store equipment; when appropriate, reinstall, reconnect, and make equipment operational.
 - 5. Equipment to Be Removed and Salvaged: Disconnect and cap services and remove equipment and deliver to Owner.
 - 6. All materials and equipment removed or disconnected by Division 22, which is not to be reused or delivered to Owner, shall be removed from the premises.
- H. If pipe, insulation, or equipment to remain is damaged in appearance or is unserviceable, remove damaged or unserviceable portions and replace with new products of equal capacity and quality.
- I. Remove all piping, and equipment hangers and supports.
- J. Cap tight unused connection at mains and risers behind finished surfaces.

3.5 PIPING SYSTEMS - COMMON REQUIREMENTS

- A. Install piping according to the following requirements and Division 22 Sections specifying piping systems.
- B. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated unless deviations to layout are approved on Coordination Drawings.
- C. Install all piping in accordance with best practices of trade and latest code requirements. Locate groups of pipes parallel to each other, spaced to permit valve servicing. Use uniform system materials throughout building.
- D. Install components having pressure rating equal to or greater than system operating pressure.
- E. Keep all piping as high as possible, consistent with proper pitch, to maintain maximum headroom. Cut piping accurately to measurements established at building, work into place without springing, forcing or cutting of the building structure, and install as directly as possible without sags between connecting points parallel with or at right angles to building construction, except as required to obtain pitch.

- F. Pitch all systems for proper venting at high points and to drain at low points where the systems can be completely emptied. Install vents at all high points and drains at all low points, including where offsets and bends in horizontal pipe runs create a low point. Provide drain points with bronze hose end drain valves.
- G. Do not install piping above or through electrical rooms, telecommunication rooms, or similar room having a large collection of electrical equipment.
- H. Keep pipe and fittings clean from cutting burrs, foreign matter and defects in structure and threading. Make all cuts square. Ream after cutting. Bevel plain ends of steel pipe. Clean off scale and dirt inside and outside before assembly. Remove welding slag or other foreign matter inside and outside.
- I. Install piping within building concealed in walls, furred spaces, pipe spaces or above suspended ceilings. Do not build in or bury horizontal piping within partitions. Install exposed piping as closely as possible to walls, ceilings and columns, allowing space for installation of insulation and access for valve operation.
- J. Install piping sections using greatest length possible in all cases. The use of short lengths socketed together will not be allowed, unless approved for tunnel accessibility reasons.
- K. The use of lampwick or other material for packing threads, caulking or wrapping of joints to stop or prevent leaks or correct faults is not permitted. The use of long screws having right and left hand threads or couplings is not permitted.
- L. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.
- M. Install piping to permit valve servicing and application of insulation.
- N. Provide offsets and loops as required for piping crossing a building expansion joint to accommodate building movement, whether or not shown on the drawings. Provide seismic joints for piping crossing building seismic joints to accommodate building movement, whether or not shown on the drawings. Piping seismic joints shall be designed to accommodate maximum building seismic joint movement. Piping seismic joints shall be Metraflex or equal.
- O. Changes in Pipe Size and Direction
 - 1. Make reduction or increase in pipe size with fittings. Use eccentric reducing fittings in horizontal piping. Use reducing tees in pressure piping for side outlet reduction only, not on run. Bushed fittings, notched or straight runs to form tees, or any similar fabrication method will not be permitted.
 - 2. Make changes in direction with standard fittings. Mitering of pipe to form elbows or similar fabrication method will not be allowed. Bending of piping will not be permitted.
- P. Chromium Plated and Stainless Steel Piping: Friction wrenches and clamps must be used exclusively in the installation of chromium plated and stainless steel pipe and fittings. Pipe which is cut, dented or otherwise damaged shall be replaced with new pipe.
- Q. Drain Pans

1. Provide drain pans under entire length of any piping, joints or fittings for soil, waste, rainwater or drain piping system which is installed over food serving facilities, food storage areas, central services areas and electronic data processing areas where indicated on the construction documents and where directed by NMNH facilities personnel.
2. Fabricate each drain pan not less than 2 inches deep and provide 3/4 inch drain pipe to discharge at nearest convenient drain line, floor drain or other approved drain point. Construct of not less than 18 gauge galvanized steel.

R. Electrolysis Control

1. Install copper or brass piping or tubing in such a way as not to touch or come in contact with ferrous metals.
2. Where ferrous piping or equipment is connected to copper or brass piping, make connection with insulating or dielectric union to prevent electrolytic action between the ferrous and nonferrous metals. At branch connections off mains, provide shut off valve upstream of dielectric union in order to isolate downstream union.
3. Where copper or brass piping, tubing or fittings are anchored to, supported by, or come in contact with ferrous metal construction, provide an insulating nonconductor spacer of rubber, plastic or equivalent material to assure prevention of electrolysis.

S. Equipment Piping

1. Verify final locations of equipment for rough in of piping connections.
2. Provide shut off valves in the supply and return to each item of equipment. Suitably locate equipment isolation valves to facilitate removal of equipment.
3. Provide piping from pump glands, drain pans, relief valves or other drainage to spill over open sight drains, floor drains, or other trapped acceptable discharge, terminating with plain end cut at a 45 degree angle.

T. Expansion and Contraction of Piping

1. The piping installation shall be free of stress. Run all piping with full allowance for expansion or contraction. Base expansion calculations on 50 degree F. installation temperature to 140 degree F. for hot water systems.
2. Evaluate the complete piping layout and notify Architect of additional anchors or expansion joints and any deviations required to compensate for expansion.
3. Make connections to equipment in such a manner as to eliminate undue strains in piping and equipment. Install sufficient number of elbow swings to allow for proper expansion and contraction of piping at the point of connection to mains and equipment.
4. Fabricate expansion loops with long radius welded fittings in steel piping and with long radius copper sweat fittings in copper piping.
5. Provide adequate pipe guides as close as possible on each end of the expansion device to preserve alignment and pitch.
6. Install pipe hangers and supports in such a manner as to not cause an anchor condition in any direction.

U. Pipe Anchors

1. Install anchors where required to direct pipe expansion properly into expansion joints, loops or offsets and to prevent transfer of loading and stresses to connected equipment.

2. Pipe anchors may consist of heavy steel clamps bolted or welded to piping and provided with lugs and bolts for clamping and attaching anchor braces. Design anchors to restrict pipe movement and fasten to main members of building structure in most effective manner to secure desired results.
3. Do not attach supports, anchors or stays in places or in such a manner that will damage construction or integrity of the structure, either during installation, by weight of the pipe, or by expansion and contraction of the pipe.

V. Pipe Insulation Inserts and Shields

1. Refer to Section of Division 22, "Plumbing Insulation" to coordinate specific insulation thicknesses and requirements. At hanger locations for insulated piping 1 ½ inches and larger where hanger support is outside the insulation, provide inserts of exploded silica pipe insulation between pipe and hanger. Density and compression strength suitable for pipe size and support spacing as required by MSS SP-58, Paragraph 9 and MSS SP-69, Table 3. Provide inserts as required for smaller piping to prevent deformation of insulation. Inserts of equal thickness to adjoining insulation, provided with vapor retardant seals, and of proper length to fully support pipe at each hanger location. Manufactured by Value Engineered Products Max Span; or equivalent.
2. At all hanger locations for insulated piping where hanger is outside of insulation, provide galvanized sheet steel shield formed to fit insert/insulation, extending up to pipe centerline. Length 12 inch minimum when insert is not required. Where inserts are provided, length of shield 4 inches less than insert length. Provide shields 16 gauge for piping up to 4 inches, 12 gauge for piping 6 inches and larger.
3. Preformed insulated pipe saddles may be used in lieu of insert and shield where appropriate. Thickness of insert or pipe saddle same thickness as pipe insulation.

3.6 VALVES

A. General

1. Provide valves at locations shown, where specified and where required to properly control piping systems. Provide valves recommended or required by equipment manufacturers and codes for proper operation of equipment and shutoff valves to allow isolation of each main and branch service line, whether or not indicated or specified.
2. Examine valve interior for cleanliness, freedom from foreign matter, and corrosion. Remove special packing materials, such as blocks, used to prevent disc movement during shipping and handling.
3. Operate valves from fully open to fully closed positions. Examine guides and seats made accessible by such operation.
4. Examine threads on valve and mating pipe for form and cleanliness.
5. Examine mating flange faces for conditions that might cause leakage. Check bolting for proper size, length, and material. Check gasket material for proper size, material composition suitable for service, and freedom from defects and damage.
6. Do not attempt to repair defective valves; replace with new valves.
7. Install valves in horizontal piping with stem at or above the center of the pipe.
8. Install valves in a position to allow full stem movement.

- B. Check Valves: Provide lift check type in steam, air, gas or vapor service lines and after globe valves, install with stem upright and plumb. Provide nonslam type in vertical piping on discharge side of pumps and elsewhere as indicated or specified. Provide horizontal swing check type elsewhere unless otherwise indicated or required for service intended, install in horizontal position with hinge pin level. Install check valves, including those that are spring loaded, so that force of gravity will operate to close valves.
- C. Provide valve ends to suit character of pipe in which installed. Provide valves designed for working pressure of at least 125% of maximum operating pressure of system in which installed, but not less than 250 psig on high pressure systems, and 125 psig on low pressure systems.
- D. Provide chromium plated valves in chromium plated piping. Provide steam valves in chromium plated piping with composition hand wheels which shall remain reasonably cool in service.

3.7 WELDING, SOLDERING AND BRAZING

- A. Do not employ workers who have not been fully qualified and certified for the specified procedures.
- B. Pipe Welding, Black or Galvanized Steel Pipe: Perform all welding of black or galvanized steel pipe by shielded metallic arc method of fusion welding, in accordance with welding procedures of AWS (American Welding Society) D10.12 recommended procedures for welding low carbon steel pipe, or other approved procedure, conforming to requirements of ASME/ANSI B31.1 for high pressure steam boiler piping and B31.9 elsewhere.
- C. Pipe Welding Stainless Steel Pipe: Refer to other Sections of Division 22 for welding requirements.
- D. Structural Steel Field Welding: Comply with AWS D1.1 procedures for manual shielded metal-arc welding, appearance and quality of welds, methods used in correcting welding work, and the following:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. Finish welds at exposed connections so that no roughness shows after finishing, and so that contours of welded surfaces match adjacent contours.
- E. Soldering: Comply with the requirements of the AWS Soldering Manual.
- F. Brazing: Comply with the requirements of the AWS Brazing Manual and AWS A5.8 specification for filler materials for brazing.

3.8 UNIONS

- A. In Screwed Steel Pipe, 2 inches and smaller: Screwed, Class 250 malleable iron, brass to iron seat, ground joint union with brass seat ring pressed into head piece. Provide galvanized unions in galvanized pipe.
- B. In Welded Steel Pipe, 2 inches and smaller: Class 3000 carbon steel socket welded union, steel to steel seat and ground joint. Provide stainless steel in stainless steel piping.
- C. In Copper tubing, 2 inches and smaller: Class 200 wrought copper, solder type, brass ground joint union.
- D. In Brass Piping, 2 inches and smaller: Class 250 cast bronze, screwed ends, brass ground joint unions. Provide chromium plated unions in chromium plated piping.
- E. Provide companion flanges in piping 2-1/2 inch and larger.

3.9 PIPING CONNECTIONS

- A. Refer to other Sections of Division 22 for additional requirements.
- B. Flanged Connections: Make with nonasbestos gaskets of 1/8 inch thick best quality material as recommended by manufacturer for the service application. For steam piping, factory manufactured for flange/connection size/type as manufactured by Flexitallic. For other piping services either Flexitallic or gaskets factory cut for flange size as manufactured by Garlock Packing Division, Colt Industries, or equal. Align flange surfaces parallel. Install gasket concentrically positioned. Assemble joints by sequencing bolt tightening to make initial contact of flanges and gaskets as flat and parallel as possible. Use suitable lubricants on bolt threads. Tighten bolts gradually and uniformly using torque wrench.
- C. Mechanical Couplings: Prepare pipe and install in accordance with manufacturer's instructions. Standard wall steel pipe either roll or cut grooved at Contractor's option, all sizes, except provide cut grooved as required to accommodate thermal expansion and contraction. Heavy wall steel pipe cut grooved all sizes. Light wall steel pipe roll grooved all sizes. Copper tubing roll grooved.
- D. Soldered Joints: Unless noted otherwise, make with appropriate flux and solder. Clean tubing ends and fittings before assembly. For piping 2 inch and larger tin tubing and fittings before assembly. For tubing 2-1/2 inch and larger use circular flame torch for soldering. The use of lead flux or solder and finishing with 50-50 solder is prohibited.
- E. Threaded Pipe: Make full, clean-cut standard ANSI/ASME B1.20.1 taper pipe threads using sharp dies. Carefully cut, ream or file out to size or bore, removing all chips. Use Schedule 80 pipe for all screwed close and shoulder nipples. Do not use all thread nipples. Provide teflon tape or other approved nontoxic joint compound, applied to male thread only.
- F. Welding Connections: Use only factory made welding fittings, same weight as piping, on welded pipe, except that Bonney Forge WELDOLET or THREADOLET, or Allied Type 1

Branchlet fittings, of same weight as connecting piping, may be used for branch takeoffs two or more commercial pipe sizes smaller than main. All elbows long radius.

- G. Plastic Piping Solvent-Cement Joints: Clean and dry joining surfaces. Join pipe and fittings according to the following:
 - 1. Comply with ASTM F402 for safe-handling practice of cleaners, primers, and solvent cements.
 - 2. PVC Nonpressure Piping: Join according to ASTM D2855.
- H. Plastic Pressure Piping Gasketed Joints: Join according to ASTM D3139.
- I. Plastic Nonpressure Piping Gasketed Joints: Join according to ASTM D3212.

3.10 EQUIPMENT INSTALLATION - COMMON REQUIREMENTS

- A. Install equipment to allow maximum possible headroom unless specific mounting heights are not indicated.
- B. Install equipment level and plumb, parallel and perpendicular to other building systems and components in exposed interior spaces, unless otherwise indicated.
- C. Install plumbing equipment to facilitate service, maintenance, and repair or replacement of components. Connect equipment for ease of disconnecting, with minimum interference to other installations. Extend grease fittings to accessible locations.
- D. Install equipment to allow right of way for piping installed at required slope.

3.11 PROTECTION OF WORK

- A. Clean pipe, tubing, fittings, valves, piping specialties, ductwork and equipment before installation and keep clean while the work is in progress.
- B. Securely close open ends of pipe and tubing and openings in other material and equipment until installed, during installation, and until finally connected or otherwise finished, with caps, plugs or other approved closure devices designed for such service.
- C. Protect factory finished equipment, fixtures and devices with approved temporary covering material where those items are installed so as to be subject to accidental damage or abuse. Contractor shall remove all temporary covering material at the conclusion of the work or as directed.
- D. Protect the work of other trades and property of Owner from damage and assume full responsibility for the cost of repairing or replacing any damage to such work or property caused by the performance of the work under Division 22.

3.12 CLEANING OF SYSTEMS

- A. Refer to other sections of Division 22 for additional cleaning and system flushing requirements.
- B. Following completion of system testing, thoroughly clean all piping systems by flushing with water or other approved method, or as otherwise specified. Completely remove all dirt, scale, oil, grease and other foreign substances that may have accumulated in systems during installation.
- C. Carefully wipe out, wire brush, or if necessary, sand blast sections of pipe lines between temporary or permanent strainers and equipment they are to protect. Replace all permanent strainer screens with temporary screens during cleaning process. Remove temporary screens and reinstall permanent screens after cleaning is completed.
- D. Disconnect automatic devices that can become clogged during cleaning process and do not connect permanently until cleaning process is complete.
- E. Clean all piping and equipment of dirt, scale, plaster, concrete, splattered paint and other foreign matter.
- F. Clean all grease and cuttings from stainless steel piping and trim.
- G. Clean all strainers, dirt pockets, drip legs, traps and other accessories that may collect foreign matter.

3.13 ERECTION OF METAL SUPPORTS AND ANCHORAGES

- A. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor plumbing materials and equipment.
- B. Field Welding: Comply with AWS D1.1.

3.14 GROUTING

- A. Description: ASTM C1107, Grade B, nonshrink and nonmetallic, dry hydraulic-cement grout.
 - 1. Characteristics: Post-hardening, volume-adjusting, nonstaining, noncorrosive, nongaseous, and recommended for interior and exterior applications.
 - 2. Design Mix: 5000-psi, 28-day compressive strength.
- B. Packaging: Premixed and factory packaged. Mix and install grout for equipment base bearing surfaces, pump and other equipment base plates, and anchors.
- C. Clean surfaces that will come into contact with grout. Provide forms as required for placement of grout. Avoid air entrapment during placement of grout.
- D. Place grout, completely filling equipment bases. Place grout on concrete bases and provide smooth bearing surface for equipment. Place grout around anchors. Cure placed grout according to manufacturer's written instructions.

3.15 HANGERS AND SUPPORTS

A. General

1. Support major piping (3 inch and above), tanks and other equipment to the structure above (beams and girders) or by means of struts or brackets to columns. Do not support from terra cotta or brick floor systems or metal roof or floor decks regardless of thickness of concrete above deck. Do not overload structural members to which supports are attached. Hanger spacing not to exceed MSS SP69.
 - a. Provide hangers, rollers, threaded rods, turnbuckles, deflection guides, deflection provisions, inserts, beam clamps and all miscellaneous specialties for attachment of hangers and supports to structure.
 - b. Provide all rods, angles, rails, struts, brace plates, structural steel, platforms and other items required for suspension or support of piping, tanks and equipment.
 - c. Provide supplemental angles, channels, plates or other reinforcement where supports are required between building structural members. Size supports for weight of pipe, pipe contents, equipment fittings and other items, plus a 200 pound live load. Attach supplemental supports in a manner that will not weaken or overload structural members. Weld steel according to AWS D-1.1.
 - d. Attach by welding, clamping, concrete inserts, drilled in mechanical type anchors (Hilti or equal) and other approved means. Adhesive type anchors are not approved.
 - e. Place grout under supports for equipment, and make a smooth bearing surface.
 - f. For seismic restraint, provide double-sided beam clamp loaded perpendicularly to beam for seismic anchor point.
2. No lead shield anchors, powder or power fasteners permitted for attachments.
3. Do not use perforated strap hangers. Do not use steel strap hangers on piping.
4. Wherever possible, support shall be provided directly to main steel or concrete framing beams. If spacing of structure exceeds spacing required to support the mechanical work, supplemental channel or unistrut framing shall be designed and provided by the Contractor.
5. Support all mechanical work independently of other trades. Under no circumstances shall work be supported or suspended from ceiling grids, piping or other supports by other trades.
6. Before drilling concrete for attachments, if attachment locations are not adjustable, scan concrete to determine reinforcing locations and locate drilled holes to miss reinforcing by at least 1 inch. Otherwise, only use hammer drills to make holes, do not cut or damage reinforcing, if reinforcing is encountered move attachment location by approximately 4 inches and reinstall.

B. Pipe Hangers and Supports

1. Unless otherwise required to avoid overloading of structural members or for seismic restraint, support horizontal steel and copper piping as follows:

Nominal Pipe Size (inch)	(a)Maximum Distance Between Support (feet)		Schedule 80 PVC	Hanger Rod Diameter (inch)
	Steel Pipe	Copper Tubing		
up to 3/4	6	5	5	3/8
up to 2	6	6	6	3/8
2-1/2 to 3-1/2	10	8	7	1/2
4 and 5	12	10	9	5/8
6	12	10	10	3/4
8 to 12	14	14	10	7/8
Trapeze Hanger Rod				(b)

- a. Provide additional supports as required to avoid overloading of supporting structure. Reduce distance where so required by applicable codes.
 - b. As required to carry weight of trapeze channel, span of piping with contents, insulation and supports, plus a 200 pound live load.
2. Install hangers to provide minimum 1/2 inch clear space between finished covering and adjacent work.
 3. Install additional attachments at concentrated loads, including valves, flanges, guides, strainers and expansion joints/loops.
 4. Place a hanger within one foot of each horizontal elbow.
 5. Use hangers that are vertically adjustable 1-1/2 inch minimum after piping is erected.
 6. Unless otherwise required to avoid overloading, of structural members or for seismic restraint, support vertical piping with clamps spaced appropriately as to type and weight of piping, minimum spacing at every other floor and below roof. Support vertical soil pipe at each floor at hub. For exposed piping in stairs, walkways and finished areas, locate clamps below floor and secure to structure below floor as required.
 7. Where several pipes can be installed in parallel and at same elevation, provide multiple or trapeze hangers. Space hangers for smallest pipe size or provide intermediate supports for smaller pipe as specified above for individual pipes.
 8. Where practical, support riser piping independently of connected horizontal piping.
 9. Support pipe runs in a manner to minimize stress in the pipe or tubing and on bodies of valves and fittings.
 10. Install hangers and supports to allow controlled movement of piping systems, permit freedom of movement between pipe anchors, and facilitate action of expansion joints, expansion loops, expansion bends, and similar units, and so that piping live and dead loading and stresses from movement will not be transmitted to connected equipment.
 11. Install hangers and supports to provide indicated pipe slopes and so that maximum pipe deflections allowed by ASME B31.9 "Building Services Piping" is not exceeded.
 12. For piping subject to sweating (e.g.: domestic cold water), and for insulated piping requiring roller supports, install hangers outside insulation and provide pipe insulation protection shields as specified in this section. For all other piping, hanger may be attached to the piping before insulation is applied or may be installed outside the insulation with insulation protection shields.
 13. Do not support nonferrous piping with ferrous materials even on a temporary basis.
 14. Do not support piping from other piping or ductwork.
 15. Install hanger rods subjected to tension only. Accomplish lateral and axial movements by proper linkage in rod assembly. Secure hanger to hanger rod with two bottom lock nuts.

3.16 IDENTIFICATION

- A. Identify all new and altered equipment, and new and altered exposed and concealed pipe with legible lettering, applied after finish painting, in a color to contrast with basic color in accordance with ANSI A13.1 and OSHA.
- B. Identify piping by name of pipe content and direction of flow near major equipment items, adjacent to valves or flanges, adjacent to gauges or thermometers, at each tee, at changes in direction, on each side of a penetration of a wall or floor, at each access door or panel and then at maximum 20 foot centers in congested areas and 50 foot intervals elsewhere; indicate flow direction with arrows. Identification shall be by means of plastic markers or tape or painted on the finished pipe surface by using stencils. Lettering shall not be smaller than one third of the pipe diameter and directional arrows not less than 1/2-inch wide and 12 inches long.
- C. Identify equipment and operating devices such as switches, starters and similar equipment, by the equipment numbers shown on Drawings or by the Owner's numbering system, if so directed.
 - 1. Include the type of service or the name of areas served.
 - 2. Lettering minimum 1 inch high.
 - 3. Nameplates shall be two tone plastic, or printed white paper enclosed in a transparent, laminated plastic case with permanently sealed edges.
 - 4. Attach securely to equipment, or where this is not practicable attach by brass link chains.
 - 5. Do not stencil surfaces exposed in public areas.
- D. Furnish for each valve, except those immediately adjacent to apparatus, a 2 inch diameter nonferrous metal tag with figures stamped on the tag.
 - 1. Number Tags for Plumbing: Use Owner's numbering system if so directed.
 - 2. Fasten tags to valves with nonferrous S hooks and nonferrous chains.
 - 3. Where valves are located above removable acoustical tile ceilings, identify the tile section below the valves by an approved color pin system.
 - 4. Furnish duplicate framed schedules showing the location of each valve, system or equipment it serves, manufacturer, and figure number.

3.17 TESTING OF PIPING SYSTEMS – COMMON REQUIREMENTS

- A. Refer to various sections of Division 22 for additional piping system testing requirements.
- B. Provide materials and equipment required for testing. Test and make tight all new piping systems and alterations and connections to existing piping system.
- C. Take precautions during testing to insure safety of personnel and equipment. Provide systems to be pressurized with appropriate gauges and blowouts or relief valve set at a pressure no more than one third higher than test pressure, to protect against damage by expanding liquid or other source of overpressure during test. Make good at no increase in Contract sum damage to work or work of other trades caused by failure to observe proper precautions.
- D. Test piping systems prior to application of insulation. Testing as stipulated herein shall be considered minimum, and where tests stipulated by lawful jurisdictional authorities exceed

these requirements, such more stringent tests shall be performed. Tests shall be witnessed and approved by the authorities having jurisdiction over the work.

- E. Concealed work shall remain uncovered until required tests have been completed. Provide proper sectionalizing devices so that portions of a system may be tested as appropriate.
- F. Isolate and exclude from tests all in line equipment, instruments, gauge glasses, flow meters and all other devices not capable of withstanding test pressure.
- G. Use ambient temperature water as testing medium, except where otherwise specified and except where there is a risk of damage due to freezing.
- H. Apply soap solution to all joints of pneumatically tested systems while system is being subjected to test pressure.
- I. Maintain test pressures sufficient length of time to permit thorough inspection of all joints. Where leaks are observed, replace defective work or material. Caulking of screw joints or holes is not acceptable. Repeat entire test as many times as necessary, until successful completion of test with no leaks.
- J. Prepare written report of testing.

3.18 BALANCING, ADJUSTING AND PERFORMANCE TESTING OF PLUMBING SYSTEMS

- A. Testing, adjusting and balancing of water systems will be provided by the Installing Contractor.
- B. Installing Contractor(s) responsible for the work specified in Division 22 shall perform all work necessary to place systems in full operation prior to start of testing, adjusting and balancing work. In addition, Installing Contractor shall perform certain additional preparatory work required for testing, adjusting and balancing as specified in various Sections of Division 22.
- C. Provide notice upon completion of all preparatory work and all initial operational testing required as part the Work. Perform additional operational testing on equipment, or systems, as directed and to extent and for duration deemed necessary, to demonstrate that systems are performing properly and delivering quantities in accordance with the requirements of the Contract Documents.
- D. Furnish approved manufacturer's technical data and shop drawings for equipment, including pump performance curves.

3.19 INSTRUCTION AND DEMONSTRATION

- A. Upon completion of all work and all tests, and at a time mutually agreed on by Contractor, Architect and Owner, Installing Contractor shall operate systems in all parts and at their expense for sufficient length of time to demonstrate the mode of operation and definitively determine whether the systems as a whole are in first class working condition. Immediately correct, at no cost to Owner, any defects that may develop during this period of operation and place systems in first class working condition before being finally turned over to Owner.

- B. Provide experienced operating personnel to instruct Owner's authorized employees in the operation, adjustment and maintenance of systems and equipment installed under this Contract. Provide instructions for the period of time appropriate for the size and complexity of the system, or as requested by Owner.

3.20 MANUFACTURER'S SUPERVISIONS AND STARTUP SERVICE

- A. Include manufacturer's supervision/startup/certification and special instruction service for equipment as specified in various Sections of Division 22. Be responsible for properly making arrangements for and coordinating with the manufacturer to provide the specified work. Make any corrections/modifications to the installation as required by the manufacturer at no additional cost to Owner.
- B. The manufacturer's engineer or authorized service personnel shall check the equipment for its conformance to the Specifications, for proper installation and run the system in all modes of operation to ascertain that the unit will function properly. Make necessary adjustments to insure optimum efficiency and trouble free service.
- C. After completion of the startup procedures, the manufacturer shall certify, in writing, that the equipment is installed in accordance with their requirements and is operating in accordance with the intent of the Specifications.

3.21 COMMISSIONING

- A. Commissioning will be provided as specified in Division 01 Section "Commissioning". All contractors and subcontractors of the various sections of this specification shall cooperate and participate in the commissioning work in accordance with requirements of Division 01 Section "Commissioning".
- B. Ensure participation of major equipment manufacturers or their representatives.
- C. Equipment and systems/subsystems installed under this section are expected to be in full compliance with the design intent by the commissioning phase. Notify the Commissioning Agent when any specific piece of equipment or specific system/subsystem is ready for commissioning. Be prepared to demonstrate system readiness.
- D. Equipment or systems/subsystems having incomplete work or exhibiting problems related to noncompliance with the design intent shall require commissioning. The contractor for this section shall be fully responsible to make all necessary corrections to incomplete or non-complying work at their own expense and shall pay the Commissioning Agent per diem rate for recommissioning such incomplete or non-complying work.

END OF SECTION 22 05 00

SECTION 22 05 13 - ELECTRICAL REQUIREMENTS FOR PLUMBING EQUIPMENT

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes extent of electrical equipment and electrical wiring that is responsibility of Division 22.
- B. Section includes general requirements for motors installed at equipment manufacturer's factory or shipped separately by equipment manufacturer for field installation.
- C. Related Sections
 - 1. Section 22 05 00, Common Materials and Methods for Plumbing Systems.

1.3 REFERENCE STANDARDS

- A. ANSI/IEEE 112 (C50.20): Test Procedure for single Phase Induction Motors
- B. NFPA 70: National Electric Code (NEC)
- C. UL: Underwriters Laboratories
- D. Smithsonian Institution standards

1.4 SUBMITTALS

- A. Product Data: Include with equipment submittals, data pertinent to electrical characteristics, motor size, type, power requirements, wiring requirements.

1.5 QUALITY ASSURANCE

- A. Provide electrical products, including those factory mounted or factory furnished, which have been tested, listed and labeled with Underwriters' Laboratory (UL) or Electrical Testing Laboratory (ETL).
- B. There shall be no field modifications made to any materials, equipment and systems that would violate the listing and labeling.

- C. Comply with Division 26, NEC and NEMA as applicable to wiring methods, materials and equipment and equipment, construction and installation.

1.6 COORDINATION

A. Wiring Under Division 26 "Electrical"

1. Power wiring under Division 26 will include power feeders from source of building power to wiring terminals on the equipment; unit mounted disconnects, or control panels unless indicated in other sections of Division 22.
2. Where disconnect switches for equipment are provided by Division 26, power wiring under Division 26 will include wiring from disconnect to wiring terminals on the equipment.

- B. Provide all other power and control wiring for Division 22 systems and equipment in accordance with the requirements of Division 26, required for complete operation, including wiring that is specified for factory prewired equipment, but not so provided.

C. Short Circuit Current Rating (SCCR) for Plumbing Equipment

1. Unless otherwise noted, the listed short circuit current rating (SCCR) of all motor controllers, disconnects, contactors, protective devices and associated assemblies that are integral or external to electrically powered mechanical equipment (except for controllers rated less than 2HP at 300V or less and listed exclusively for general purpose branch circuits), shall be equal to or greater than the electrical distribution equipment feeding it. The SCCR value shall be clearly labeled on the equipment. Refer to the electrical drawings, specifically the single line diagrams, panelboard schedules and HPE schedules to obtain this information. Where the minimum SCCR rating is not specifically identified on the documents at the referenced equipment, the SCCR rating of the plumbing equipment shall be equal or greater than the kAIC rating of the electrical distribution equipment feeding the electrically powered mechanical equipment.
2. Plumbing equipment submittals shall include the SCCR rating meeting the above requirements. The contractor may elect to perform short circuit calculations to determine the available short circuit rating at the connection point of the applicable equipment. If the SCCR rating is determined to be less than the values indicated on the contract documents, the submittal shall include the calculations (inclusive of all input and output data), in particular the short circuit reduction on the feeder for each specific piece of equipment, and should show that the equipment rating meets or exceeds this calculated value. The calculations must be signed and sealed by a professional engineer (PE) registered in the project state.
3. All information required to show overall compliance with the above short circuit rating requirements shall be submitted as part of the product submittal. Submittals omitting this required information will be returned "Resubmit" or "Rejected".
4. No change orders or additional costs will be accepted by Owner or Architect to provide upgraded equipment in order to meet the above requirements or to perform any of the calculations described above.

D. Electrical Ratings

1. The motor horsepower and apparatus full load amperage ratings shown or specified are Basis of Design values and the corresponding sizes of feeders and other electrical equipment indicated to serve them are minimum sizes required to meet the Basis of Design requirements. When motors of greater horsepower and apparatus with larger full load amperage ratings are furnished as necessary to meet the design intent of the various sections within the specification, the associated changes to the electrical system (i.e. increase in capacity of the feeders and other electrical equipment serving them) shall be submitted for approval and be completed by the Contractor at no additional cost to the Owner.

PART 2 - PRODUCTS

2.1 ELECTRICAL WIRING

- A. Electrical wiring provided by Division 22 shall be in accordance with the requirements of Division 26.

2.2 MOTORS

A. Single Phase Motors

1. Motors larger than 1/20 hp shall be energy efficient capacitor start type to suit starting torque and requirements of specific motor application:
2. Multispeed Motors: Variable-torque, permanent-split-capacitor type.
3. Bearings: Prelubricated, antifriction ball bearings or sleeve bearings suitable for radial and thrust loading.
4. Motors 1/20 HP and Smaller: Shaded-pole type.
5. Thermal Protection: Internal protection to automatically open power supply circuit to motor when winding temperature exceeds a safe value calibrated to temperature rating of motor insulation. Thermal-protection device shall automatically reset when motor temperature returns to normal range.

PART 3 - EXECUTION

3.1 ELECTRICAL WIRING

- A. Unless otherwise stated herein or on the drawings, power wiring will be provided under Division 26 and control wiring will be provided under Division 22. Provide power and control wiring for Division 22 systems and equipment for interconnecting wiring on apparatus that has not been factory installed.
- B. Coordinate power wiring requirements with Division 26 for locations of electrical panelboards to connect Division 22 system as specified.

END OF SECTION 22 05 13

SECTION 22 07 00 - PLUMBING INSULATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Plumbing Piping Insulation
 - 2. Adhesives, mastics, tapes
 - 3. Recovering

- B. Related Sections

- 1. Warranty, pipe insulation insert and shields are specified in Section 22 05 00, "Common Materials and Methods for Plumbing Systems".

1.3 DEFINITIONS

- A. Cold Surfaces: Normal operating temperatures less than 75° F.
- B. Density: Is expressed in pcf (pounds/cu. ft.).
- C. Dual Temperature Surfaces: Normal operating temperatures that vary from hot to cold.
- D. Hot Surfaces: Normal operating temperatures of 100° F. or higher.
- E. Thermal Conductivity ("K" value): Measure of heat flow through a material at a given temperature difference; conductivity is expressed in units of (Btu x inch)/(h x sq. ft. x ° F.).
- F. Through Resistivity ("R" value): Represents the reciprocal of thermal conductivity ("K" value).

1.4 REFERENCES

- A. ASTM C1136 Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation.
- B. ASTM E84 Surface Burning Characteristics of Building Materials.

- C. MICA Standards: National Commercial & Industrial Insulation Standards published by the Midwest Insulation Contractors Association. Endorsed by National Insulation Contractors Association (NICA) and its regional associations.
- D. NFPA 255 Test of Surface Burning Characteristics of Building Materials.
- E. UL 723 Surface Burning Characteristics of Building Materials.

1.5 SUBMITTALS

- A. Product Data: For each type of product indicated. Include thermal conductivity, thickness, and jackets (both factory and field applied, as applicable).
- B. Product Schedule or List: Prepare a summary of products required and clearly indicate location of their intended use.
- C. Shop Drawings:
 - 1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.

1.6 QUALITY ASSURANCE

- A. Materials, job conditions and installation shall be in compliance with the applicable Building and Plumbing Codes.
- B. Installation shall be done in a workmanlike manner by skilled and experienced workers who are regularly employed in commercial/industrial insulation work, in accordance with manufacturer's recommendations and instructions and best practices of the trade.
- C. Comply with the more stringent of the requirements of this specification or the requirements of the MICA standards.
- D. Insulation materials manufacturing facilities must be certified and registered with an approved registrar for conformance with ISO 9000 quality standard.
- E. Fire Performance Characteristics
 - 1. Insulation, jacketing materials, PVC covers, tapes, adhesives, mastics, cements and finish coatings shall have a composite noncombustible fire and smoke hazard rating and label, as tested in accordance with United States Public Health Service requirements, ASTM E84, NFPA 255 and UL 723 not exceeding Flame Spread 25 and Smoke Developed 50.
 - 2. Indoor Recovering Canvas Jackets: UL listed fabric, 6 ounce per square yard, unless otherwise specified, attached with a lagging fire retardant and waterproof adhesive.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Ship insulation materials in containers marked by manufacturer with appropriate ASTM specification designation, type and grade, and maximum use temperature. Deliver materials to job site in original nonbroken factory packaging, labeled with manufacturer's density and thickness and store in a safe, dry place.
- B. No insulation material shall be installed that has become damaged in any way.
- C. Do not install, and remove from the site, any insulation material that has become wet because of transit or job site exposure to moisture or water. Remove insulation from piping and/or equipment which has become wet. Reinsulate as required.

1.8 COORDINATION

- A. Coordinate size and location of supports, hangers, and insulation shields specified in Division 22 Section "Common Materials and Methods for Plumbing."
- B. Coordinate clearance requirements with installing contractor for insulation application. Before preparing piping Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.

1.9 SCHEDULING

- A. Schedule insulation application after pressure testing of piping systems. Insulation application may begin on segments that have satisfactory test results.
- B. Schedule insulation application on water piping prior to activation of system. Do not install insulation to surfaces where condensation is present.

1.10 ALTERNATIVES

- A. Alternative insulations are subject to Architect's approval. Alternatives shall provide, at normal conditions, thermal resistance within 5% of resistance of materials specified.
- B. Where alternative thermal conductivity ("k") differs from specified thermal conductivity by more than 5%, increase or decrease insulation thickness as follows:

$$\text{New thickness} = \frac{\text{Actual "k"}}{\text{Specified "k"}} \times \text{specified thickness}$$

PART 2 - PRODUCTS

2.1 PIPING INSULATION

- A. Acceptable Manufacturers

1. Johns Manville
2. Armstrong World Industries
3. Owens-Corning Fiberglass
4. Knauf Insulation
5. Certain Teed

B. Insulation Type P1:

1. Piping: Fine fibrous glass insulation, with factory applied vapor barrier jacket, molded to conform to piping, "k" value at 75° F. maximum 0.23. Johns Manville "Micro-Lok AP-T Plus" with jacket of white Kraft reinforced with fiberglass yarn and bonded to aluminum foil, and having a pressure sensitive tape closure system bonded to the longitudinal lap. Wrap all pipe insulation in exposed areas with a canvas covering and paint.
2. Valves and Fittings:
 - a. Glass fiber insert of equal thickness to adjacent pipe insulation and premolded PVC cover, Johns Manville "Zeston" and "Hi-Lo Temp Inserts" for valves and fittings.
 - b. Factory molded fibrous glass fitting covering for fittings of equal thickness to adjacent pipe insulation. Cover with 6 ounce canvas on concealed piping and 8 ounce canvas on exposed piping.
 - c. Mitered sections of pipe covering for valves.

C. Insulation Type P2:

1. Piping: Type P1 with additional PVC jacket Zeston 2000 PVC or equal with minimum thickness of 0.030 inches. Seal with permaweld adhesive.
2. Valves and Fittings: Mitered sections of Type P2 piping insulation with miter seals and snap-straps.

D. Insulation Type P3: Piping, Valves and Fittings: Foamed plastic of closed cell structure, "k" value at 75° F. maximum 0.27. Maximum water vapor transmission rating of 0.1 perms. Insulation shall not drip or melt when exposed to flame. Polyethylene or polyolefin not allowed. Armacell "AP Armaflex", "AP Armaflex SS" and "SSA-2000" (maximum 0.2 perms). Exposed insulation or as indicated on architectural drawings if a custom color is required in finished areas. shall be white. Architect to verify.

E. Insulation Type P4: Piping, Valves and Fittings: Asbestos free hydrous calcium silicate colored gold throughout, molded to conform to piping, "k" value at 200° F. and 500° F. maximum 0.42 and 0.50 Btu/in./sq.ft./° F./hr. respectively. Johns Manville "Thermo-12 Gold" throughout.

F. Insulation Type P5 Mineral-Fiber, Pipe Insulation Wicking System: Preformed pipe insulation complying with ASTM C547, Type I, Grade A, with absorbent cloth factory applied to the entire inside surface of preformed pipe insulation and extended through the longitudinal joint to outside surface of insulation under insulation jacket. Factory apply a white, polymer, vapor-retarder jacket with self-sealing adhesive tape seam and evaporation holes running continuously along the longitudinal seam, exposing the absorbent cloth.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Determine clearances required for installation of work and review such requirements with trades responsible for installing various piping systems, and equipment to be insulated. Where it is determined that working clearances between equipment and material to be insulated and adjacent work will restrict or prohibit proper installation of work, immediately report such conditions to all interested parties and arrange to have affected material relocated or preinsulated before erection, as approved. Failure to so comply will not relieve Contractor of full responsibility for providing specified insulation.
- B. Do not install covering before piping, and equipment has been tested and approved.
- C. Ensure surface is clean and dry prior to installation. Ensure insulation is dry before and during application.

3.2 GENERAL INSTALLATION REQUIREMENTS

- A. Comply with applicable requirements of Division 22 Section, "Common Materials and Methods for Plumbing".
- B. Adhesives and mastics materials shall be compatible with insulation material, jackets and substrates. Apply insulation and adhesives in accordance with manufacturers' instructions.
- C. Do not install any insulation before building is adequately closed in. Where it is necessary to install insulation in any section of building that is not adequately closed in, secure prior permission and, where permission is granted, such insulation shall be in place to form a waterproof covering. Remove and replace all insulation installed that becomes water saturated because of failure to comply with this requirement, at no increase in the Contract sum.
- D. Clean excess adhesive, mastic or cement used in performance of work from all exposed surfaces of insulation jacketing materials. Clean smudges and dirt from all exposed surfaces of insulation jacketing materials at conclusion of this work.
- E. Apply insulation on all cold surfaces with a continuous, unbroken vapor seal. Install hangers outside of insulation for all piping subject to sweating (e.g., domestic water piping, chilled drinking water piping, cold USP water piping) and provide inserts - refer to and coordinate with Section 22 05 00, Part 3, Article titled "PIPING SYSTEMS – COMMON REQUIREMENTS", paragraph titled "Pipe Insulation Inserts and Shields". For all equipment subject to sweating, insulate and vapor seal hangers, supports, anchors, etc., that are secured directly to cold surfaces to prevent condensation.
- F. Extend all surface finishes to protect all surfaces, ends and raw edges of insulation.
- G. Install insulation on and at access doors to allow easy use of access door without damage to insulation.
- H. Finish insulation neatly at hangers, supports and other protrusions.

- I. Install insulation with least number of joints practical. Locate insulation, or cover seams, in least visible locations.
- J. Finish installation with systems at operating conditions. Repair separation of joints or cracking of insulation due to thermal movement or poor workmanship.
- K. Be responsible for proper curing of insulation, etc., in accordance with manufacturers' requirements.
- L. Repair existing insulation damaged through installation of new work to match existing insulation.

3.3 PENETRATIONS

- A. Refer to Division 22 Section "Common Materials and Methods for Plumbing" for additional requirements.
- B. Insulation Installation at Underground Exterior Wall Penetrations: Terminate insulation flush with sleeve seal. Seal terminations with flashing sealant.
- C. Insulation Installation at Non-Rated Interior Wall and Partition Penetrations: Install insulation continuously through walls and partitions.
- D. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Install insulation continuously through penetrations of fire-rated walls and partitions.
 - 1. Comply with requirements in Division 07. Coordinate requirements for sleeves and clearance between finished surface of penetrating item and penetrated construction to achieve proper installation of through-penetration fire-stop system.
- E. Insulation Installation at Floor Penetrations:
 - 1. Pipe: Install insulation continuously through floor penetrations.
 - 2. Seal penetrations through fire-rated assemblies. Comply with requirements in Division 07.

3.4 GENERAL PIPE INSULATION INSTALLATION

- A. Requirements in this article generally apply to all insulation materials except where more specific requirements are specified in various pipe insulation material installation articles.
- B. Carry pipe insulation through sleeves and through hangers which are specified to be installed outside insulation.
- C. Insulation Installation on Fittings, Valves, Strainers, Flanges, and Unions:
 - 1. Install insulation over fittings, valves, strainers, flanges, unions, and other specialties with continuous thermal and vapor-retarder integrity, unless otherwise indicated.

2. Insulate pipe elbows using preformed fitting insulation or mitered fittings made from same material and density as adjacent pipe insulation. Each piece shall be butted tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
 3. Insulate tee fittings with preformed fitting insulation or sectional pipe insulation of same material and thickness as used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
 4. Insulate valves using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
 5. Insulate strainers using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below ambient services, provide a design that maintains vapor barrier.
 6. Insulate flanges and unions using a section of oversized preformed pipe insulation. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker.
 7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below ambient services and a breather mastic for above ambient services. Reinforce the mastic with fabric-reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.
 8. For services not specified to receive a field-applied jacket except for flexible elastomeric and polyolefin, install fitted PVC cover over elbows, tees, strainers, valves, flanges, and unions. Terminate ends with PVC end caps. Tape PVC covers to adjoining insulation facing using PVC tape.
 9. Stencil or label the outside insulation jacket of each union with the word "UNION." Match size and color of pipe labels.
 10. Do not insulate flexible connections and expansion joints. Install insulation on flanges, valves and unions so that it can be removed and replaced without damaging adjacent insulation.
- D. Terminate insulation neatly and finish all exposed ends with plastic material troweled on bevel.
- E. Install removable insulation covers at locations indicated. Installation shall conform to the following:
1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as adjoining pipe insulation.
 2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union long at least two times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless-steel or aluminum bands. Select band material compatible with insulation and jacket.

3. Construct removable valve insulation covers in same manner as for flanges except divide the two-part section on the vertical center line of valve body.
4. When covers are made from block insulation, make two halves, each consisting of mitered blocks wired to stainless-steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.

F. Insulation Type P1

1. Seal longitudinal laps of jackets and wrap butt joints with 3 inch wide strip of jacketing material securely sealed in place.
2. Where premolded PVC fitting covers are used, apply factory precut insulation insert in accordance with manufacturer's instructions and then apply one piece cover. Use two or more layers of inserts on hot piping as required to limit outer surface temperatures of insert(s) to 150° F. maximum. On concealed hot piping, covers may be secured with staples. On exposed hot piping, secure covers by taping ends to adjacent insulation. Seal seam edges of covers on concealed and exposed cold piping with Zeston vapor-barrier adhesive and wrap edges of covers with Zeston vapor-barrier pressure sensitive color matching tape.
3. Where factory-molded fibrous glass fitting covering is used, finish insulation on concealed and exposed hot piping with same jacketing material as adjacent insulation lap-sealed and finished with Foster 30-36, Childers CP-50A or equal. Finish insulation on cold piping same as for hot piping except that sealer shall be Foster 30-35 or Childers CP-30LO vapor sealer.
4. Finish valve covering in hot and cold piping systems same as specified above for fittings.
5. Recover exposed piping in finished areas with 0.02 mil thick PVC jacket or aluminum jacket, except use Alpha Assoc. Fiberglass Scrim Fabric "Luben 58" (20 x 10) white, adhered and then finished with Foster 30-36 or Childers CP-50A and when piping systems are specified to be painted.

G. Insulation Type P2

1. Secure PVC jacket in place by a continuous longitudinal solvent weld type joint.
2. Use mitered sections of PVC jacketed insulation for valves and fittings. Seal joints with solvent weld compound
3. Entire installation of PVC jacketed insulation shall be weatherproof.

H. Insulation Type P3

1. Where possible, slip insulation over tubing as a full cylinder. Where necessary to cut and at all butt joints, tightly butt edges and join by sealing with waterproof vapor barrier adhesive, Armstrong 520. For Self-Seal type, peel adhesive paper from surface and apply firm pressure along entire longitudinal joint.
2. Cover fittings and valves with equivalent thickness of insulation material.

3.5 PIPING INSULATION SCHEDULE

A. Thickness Schedule (TS)

TABLE 1					
Plumbing and Drainage					
Minimum Insulation Thickness for Pipe Sizes* (in inches)					
Schedule No.	Water Temp. Range	< 1 In.	1 In. to < 1-1/2 In.	1-1/2 In. to < 4 In.	4 In. to 8 In.
	Deg F.	In.	In.	In.	In.
TS-11	105-140	1.0	1.5	1.5	1.5
TS-12	40-60	0.5	1.0	1.0	1.0
*Refer to Article titled "Alternatives".					
*Comply with ASHRAE 90.1 - 2016					

B. Type: Unless otherwise specified use Insulation Type P1 on indoor piping and Insulation Type P2 on outdoor piping, except as follows:

1. Contractor may substitute Type P3 of thickness as scheduled for Type P1 or P2 when system operating temperature is less than 210 deg F. and specified thickness is 1/2 inch or less.
2. Contractor may substitute Type P3 of thickness as scheduled for Type P1 or P2 when system operating temperature is less than 210 deg F. and:
 - a. Piping is constructed from rolls of soft copper tubing and space limitations preclude straightening of tubing to allow satisfactory installation of Type P1 or P2.
 - b. Piping is factory fabricated and installed and not sufficiently straight to allow satisfactory installation of Type P1 or P2.
3. Use Type P3 when noted on the following schedules.

C. Plumbing and Drainage Piping System Insulation Requirements

<u>Service</u>	<u>Thickness Schedule</u>
Domestic Cold Water, Above Ground	
1. General Service	1/2 Inch
2. In exterior walls or otherwise subject to freezing	TS-12 Plus 1 inch
3. Chrome Plated Piping	None
Domestic Hot and Recirculating Water	
1. General Service	TS-11
2. In exterior walls or subject to freezing otherwise to freezing otherwise	Listed schedule above plus 1 inch
3. Chromium Plated Piping	None
Drainage and Vent. Insulate Only the following:	
1. Roof drain, vertical connection between	½ inch

<u>Service</u>	<u>Thickness Schedule</u>
drain and horizontal piping, and all above ground horizontal and vertical storm drainage piping. Exposed storm piping insulation shall be colored black and white as required to match exposed painted ceiling spaces	
2. Above ground floor drain service, ice maker, refrigeration and air conditioning system including vertical connection between drain and horizontal piping and connected horizontal piping.	½ inch
3. Electric water cooler and chilled drinking water branch drain, including trap	½ inch
4. Ice maker drain piping	½ inch
5. Vents within 6 lineal feet of roof outlet	TS-12

END OF SECTION 22 07 00

SECTION 22 10 00 - PLUMBING SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section includes the following plumbing materials and equipment for domestic water and drainage systems:
 - 1. Pipe and fittings
 - 2. Valves
 - 3. Water heaters
 - 4. Drains
 - 5. Mixing valves
 - 6. Backflow preventers
 - 7. Interceptors
 - 8. Trap primers
 - 9. Cleanouts
- B. Related Sections include the following:
 - 1. Section 22 11 00: Disinfection of Domestic Water Lines
 - 2. Section 22 05 00: Basic Materials and Methods
 - 3. Section 22 07 00: Plumbing Insulation
 - 4. Section 22 11 23: Plumbing Pumps
 - 5. Section 22 05 13: Electrical Equipment and Wiring for Plumbing Systems
 - 6. Section 22 40 00: Plumbing Fixtures and Trim
 - 7. Division 26: Electrical

1.3 REFERENCES

- A. ASSE 1018 - Trap Seal Primer Valves, Water Supply Fed
- B. NSF 61 - Standard for Potable Domestic Water Piping and Components

1.4 SUBMITTALS

- A. Product data for the following:
 - 1. Pipe, valves and fittings

2. Drains, cleanouts and trap primers
3. Water heating equipment
4. Backflow preventers
5. Mixing valves

B. Welding certificates.

1.5 WARRANTY AND CONTRACT CLOSEOUT

A. Warranty

1. Refer to BIDDING AND CONTRACT REQUIREMENTS and to warranty and contract closeout requirements specified in DIVISION 00.

B. Contract Closeout

1. Comply with requirements of Division 1, Contract Closeout.
2. Include information for all products specified in this section, in the operating and maintenance manual.
3. Provide the following as specified in this section or Section 22 05 00.
 - a. Testing and cleaning reports certified by contractor for each system.
 - b. Balancing report for domestic hot water recirculation system. Include certificate stating degree of accuracy of instruments.
 - c. Portable flow measuring instrument for automatic flow control valves.
 - d. Qualification certificates for plastic piping system installers/solvent welders.

1.6 PRODUCTS FURNISHED BY OTHERS

A. Make piping connections for following, installed under Earthwork Division 31

1. Manholes

1.7 WORK IN OTHER DIVISIONS

A. Underground storm drainage pipe and fittings 5 feet beyond the building wall will be provided under Division 31, EARTHWORK, including all manholes, inlets and headwalls.

B. All footing drainage pipe and fittings will be provided under Division 31, EARTHWORK.

1.8 ENERGY PERFORMANCE CRITERIA

A. Rating temperatures and conditions, Coefficient of Performance (COP) and Energy Efficiency Rate (EER) of all equipment and components provided under this section shall meet the requirements of the State Energy Code or latest issue of ASHRAE Standard 90.1, or of the latest issue of the Standards for Equipment in the National Energy Policy Act, whichever is more stringent.

1.9 QUALIFICATION OF PLASTIC PIPE INSTALLERS AND WELDERS

- A. Obtain and pay for the services of an authorized representative of the plastic pipe manufacturer to qualify and certify plastic pipe installers and plastic pipe welders.
- B. Qualification: Procedures shall be as recommended by the plastic pipe manufacturer and shall include the following as a minimum:
 - 1. Plastic Pipe Installers: Educational material in the expansion and contraction of plastic pipe and comparison to expansion and contraction of metallic pipe materials. Instructions for determining need, location and sizing of expansion loops, offsets and legs to accommodate the expansion and contraction of the plastic pipe.
 - 2. Solvent and Electric Fusion Welders: Instruction on the manufacturer's recommended welding procedures, observation and inspection of a sufficient number of test welds by each welder to be reasonably assured that the welder is capable of producing acceptable welds.
- C. Certificates: Provide a written statement by the manufacturer's authorized representative listing the names of personnel who the manufacturer's authorized representative has qualified as a plastic pipe installer and/or a plastic pipe solvent and/or electric fusion welder.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Subject to compliance with requirements, provide products of the following manufacture:
 - 1. Inline Trap Guard (if approved by AHJ and NMNH facilities)
 - a. Jay R. Smith
 - b. Provent Systems
 - c. Rector Seal - Sure Seal
 - 2. Backflow Preventer
 - a. Hersey Products, Inc.
 - b. ITT Lawler
 - c. Watts Regulator Co.
 - d. Zurn Wilkins Division
 - e. Ames
 - f. Febco
 - 3. Cleanouts, Carriers, Drains, Interceptors
 - a. Watts Drainage
 - b. Enfield
 - c. Jay R. Smith Mfg. Co.
 - d. Josam Mfg. Co.

- e. Precision Plumbing Products, Inc.
 - f. R & G Sloane
 - g. Wade Div. Tyler Pipe Industries, Inc.
 - h. Zurn Industries, Inc.
 - i. Mifab
 - j. Blucher
 - k. Sun Drainage Products
- 4. Domestic Water Heater (Electric Instantaneous Type)
 - a. Eemax
 - b. Chronomite
 - c. Bradford White
 - d. Laars
 - e. Seisco
- 5. Domestic Water Heater (Tank Type)
 - a. State
 - b. A.O. Smith
 - c. Rheem
 - d. Bradford White
- 6. Pipe and Fittings
 - a. General Pipe and Fitting Materials, No Specified Manufacturer
 - 1) Any manufacturer whose products are manufactured in the United States and comply with the reference standards.
 - b. Basic Pipe and Fitting Materials and Accessories not Specified in this Section: Refer to Section 22 05 00.
 - c. PVC Pipe and Fittings
 - 1) Chemtrol
 - 2) Plastic Piping System.
 - 3) R & G Sloan
 - 4) Charlotte
 - d. Ductile Iron Retainer Glands
 - 1) American Cast Iron Pipe Co.
 - 2) Fire Protection Equipment Co.
 - 3) U. S. Pipe.
 - e. PP (Polypropylene) Drainage Pipe and Fittings
 - 1) Lab-Line/Enfield Industrial Corporation
 - 2) R & G Sloane "Fuseal"
 - 3) Sani-Tech, Inc.

- 4) Orion
- f. Copper Pressfit Piping Systems (if approved by NMNH Facilities)
 - 1) Apollo Xpress
 - 2) Nibco
 - 3) Viega
- 7. Valves
 - a. Basic Valve Types not Specified in this Section: Refer to Section 22 05 00.
- 8. Water Hammer Arrestors
 - a. Josam
 - b. J. R. Smith
 - c. Precision Plumbing Products
 - d. Sioux Chief
 - e. Wade
 - f. Zurn
 - g. Watts Drainage

2.2 BACKFLOW PREVENTER

- A. Complete factory assembly including resilient seated shutoff valves before and after device, resilient seated test cocks and protected by strainer having a resilient sealed blowdown valve. Provide pressure differential relief valve located between two positive seating check valves. All bronze body. All internal parts stainless steel. Meet requirements of American Society of Sanitary Engineering Standard 1013 and University of Southern California Foundation for Cross Connection Control and Hydraulic Research, and so certify. Meet local code requirements.

2.3 CLEANOUT AND CLEANOUT ACCESS COVER

- A. Provide cleanouts with brass, screw-in type plugs as listed below. Model numbers shown are for standard of quality.
 - 1. Finished Floors: Duco cast iron body with bronze plug, gasket seal, internally threaded extension with adjustable head, round scoriated nickel bronze top secured with vandalproof screws, Jay R. Smith 4020 Series.
 - 2. Vinyl or Composition Tile Floors: Same as above, except square top with 1/8 inch recess to receive flooring material, Jay R. Smith 4160 Series.
 - 3. Terrazzo or Ceramic Tile Floors: Same as 1. above, except square top with recess of sufficient depth to receive flooring material, Jay R. Smith 4200 Series.
 - 4. Carpeted Floors: Same as 1. above with adjustable carpet clamping frame, Jay R. Smith 4020X Series.
 - 5. Finished Walls: Cast iron cleanout plug or cleanout tee, bronze hex head plug, round nickel-bronze frame and secured cover with vandalproof screws, Jay R. Smith 4720.

6. Exposed Piping in Unfinished Areas: For cast iron pipe, provide a ferrule caulked into the hub of fitting and a heavy rough brass plug screwed into ferrule. For screwed piping, provide heavy rough brass plug screwed into pipe fitting.
7. Driveways and Yards: Duco cast iron spigot body with cutoff grooves and bronze plug. Double flanged housing with heavy-duty scoriated cast iron cover secured with vandalproof screws, Jay R. Smith 4251-U.
8. Plastic Drainage Piping: Duco cast iron housing, less cleanout, with heavy-duty secured scoriated cast iron cover with lifting device. Housing is to be completely free of piping so no load is transmitted to pipe, Jay R. Smith 4880. Provide plastic cleanout and plug.
9. Unfinished Floors: Duco cast iron body with round adjustable scoriated cast iron top secured with vandalproof screws, bronze plug. Jay R. Smith 4220 Series

B. Provide flashing clamp for cleanout installed in waterproof construction.

2.4 DRAINS

- A. Provide floor drains, complete with traps and connections similar to the Jay R. Smith types scheduled.

2.5 UNDERCOUNTER SINGLE FIXTURE TEMPERED WATER MIXING VALVE ASSEMBLY

- A. Provide a thermostatic mixing valve assembly consisting of a Powers Model LFLM-495-1-A lead free mixing valve with union inlet strainer checkstops. Valve capacity shall be 12 gpm at 45 psi pressure drop. Valve shall be CSA 8125 certified and meet the requirements of ASSE 1016 and ASSE 1070. Valve shall have copper silicon alloy body with paraffin-based actuation technology. Valve shall have lockable adjustable outlet temperature range of 80° F. to 120° F. with 105° F. setting. Valve must have union connections to permit servicing of filters.

2.6 PIPE AND FITTINGS

A. General

1. Pipe and fitting materials and joint types are specified in the following "Piping Class" paragraphs. Application material and joint type which will be permitted for specific services are specified hereinafter in the Article titled "Piping System Requirements". Where more than one Piping Class is listed under "Piping System Requirements" for a service, use any of listed classes, unless otherwise specified or indicated, but systems' materials must be consistent throughout the work.
2. Pipe and fittings shall conform to the latest issue of the standards referred to hereinafter. Each length of pipe and each fitting shall be marked with the manufacturer's name brand and specification code designation to which it belongs.

2.7 VALVES

A. General

1. Valves are specified by Valve Type and, for some services, several valve types are grouped by "Valve Class" in various sections of Division 22. Application is stated in the "Piping System Requirements". Where more than one Valve Type or Valve Class is listed for a service, use any of the listed Types or Classes, unless otherwise specified or indicated, but selection must be consistent throughout the work.
2. It shall be Contractor's responsibility to coordinate the work for all sections of Division 22 to assure that all general service valves throughout the work of Division 22 are of the same manufacture and type and that all valves of the same type number/identification throughout the work of Division 22 are of the same manufacture.
3. Valve packing shall not contain asbestos.
4. Bronze Valves: Construct body of ASTM B62 for Classes 125 and 150, ASTM B61 for Classes 200 and 300, copper-silicon bronze stem.
5. Iron Valves: Construct body of ASTM A126, Class B copper-silicon bronze stem.
6. All valves used in domestic water applications shall comply with NSF/ANSI 61 for lead free construction.

B. Refer to Section 22 05 00 for specifications for basic valve types not specified in this section.

C. Valve Classes

1. Valve Class VC-4

SERVICE	LIMITS	ALLOWABLE VALVE TYPES
SHUTOFF	Copper Tubing	V9\V10
THROTTLING (BYPASS)	Copper Tubing	V4 globe, V9, V10
	Steel Pipe	V4 globe, V9, V10
BALANCING		Calibrated type Class 125 bronze body. Bell & Gossett circuit setter or equal. See Section 20 05 00.
CHECK	Copper Tubing	V1 swing check or V14 at storm water or sanitary pump discharge, V1 & V4 elsewhere
	Steel Pipe	V2 swing check or V14 at storm water or sanitary pump discharge, V2 & V4 elsewhere
*Not allowed for pump discharge balancing service.		

2.8 WATER HAMMER ARRESTERS

- A. ASME A112.26.1M, ASSE 1010, or PDI-WH 201, stainless steel bellows with pressurized cushioning chamber. Sizes are based on water-supply fixture units, ASME A112.26.1M sizes A through F and PDI-WH 201 sizes A through F. Piston type are not permitted for applications that exceed 60 psi.

PART 3 - EXECUTION

PLUMBING SYSTEMS

3.1 GENERAL INSTALLATION REQUIREMENTS

- A. Refer to and comply with Section 22 05 00 for basic requirements.
- B. Refer to Section 22 05 48 for vibration isolation and seismic restraints, penetration seals and the like.

3.2 BACKFLOW PREVENTERS

- A. Install backflow preventers on makeup water connections to new coffee machines, ice machines, etc. See detail on Drawings.

3.3 CLEANING OF SYSTEMS

- A. Refer to Section 22 05 00 for basic requirements.
- B. Refer to Section 22 11 00 for Disinfection of Water Lines.

3.4 CLEANOUTS

- A. Provide cleanouts on all drainage piping at 100-foot intervals on piping 4 inches and smaller and at not more than 100 foot intervals on larger pipe sizes, at each change in direction of more than 45°, at the base of drainage stacks and at other locations shown. Cleanouts full size for pipes up to 4 inches and not less than 4 inches for larger pipes except where code requires cleanouts larger than 4 inches. Cleanout sizes shall comply with applicable plumbing code.
- B. Lubricate non-plastic cleanout plugs with mixture of graphite and linseed oil or provide Teflon tape on threads. Sealants for plastic cleanout plugs shall be compatible with plastic piping materials. Prior to building turnover remove cleanout plugs, relubricate and reinstall using only enough force to ensure permanent leakproof joint.

3.5 DOMESTIC WATER PIPING

- A. Install inside domestic water distribution piping with sufficient pitch and in a manner that entire system may be drained at a central point, if possible. Otherwise provide drainage points for each portion of system.
- B. Support risers and stacks by metal brackets attached to building construction or by other approved methods.
- C. Provide a riser shutoff valve at all water supply risers in nearest accessible location.
- D. For each toilet room with two or more plumbing fixtures provide shutoff valves on hot and cold water supply piping within access panel or other accessible location.

- E. For domestic hot water recirculating systems, manually adjust calibrated type balancing valve to provide flow of hot water in each branch with flow rates selected for a maximum temperature drop of 5° F.
- F. Provide isolation valve in new piping where new piping and existing piping interconnect. Provide 3/4 inch tap with hose end valve for pipe disinfection downstream of isolation valve.

3.6 DOMESTIC WATER SYSTEM PIPING INSTALLATION

- A. Refer to Part 1 Article Plastic Pipe Expansion Design Criteria.
- B. Install only by certified installers/welders.
- C. Provide expansion joints and/or expansion loops, offsets and legs located and sized to accommodate expansion and contraction of piping system.
- D. Provide guides at expansion joints and loops, but do not provide any anchors in piping system - allow entire system to float.
- E. Provide a swing joint connection at each branch connection to a main or branch main, consisting of not less than three (3) elbows after a tee connection into the main or branch main. Size length of each section of swing joint connection to accommodate expansion and contraction perpendicular to the specific swing joint section.
- F. Expansion joint shall be telescoping piston joint, material of construction to match piping, of Plastinetics Inc. or equivalent manufacture.
- A. Provide seismic piping joints where piping systems cross building seismic joints.

3.7 DRAINS

- A. Flash drains installed in waterproofed floor or in toilet rooms and mechanical equipment room above grade with 2 foot, 6 inch square, sheet membrane clamped into clamping device of the drain and mopped into waterproofing or cast into concrete. Membrane shall be thermoplastic elastomeric, ASTM D4068, Chloraloy 240, as manufactured by the Noble Company. Installation shall be in strict accordance with the manufacturer's recommendations.

3.8 ELECTRICAL WORK

- A. Refer to Section 22 05 13 for general requirements.
- B. Power wiring will be provided under Division 26, Electrical, to the following:
 - 1. Domestic electric water heater – Point of use, instantaneous.
 - 2. Domestic electric water heater – Tank type.

3.9 PIPING CONNECTIONS

- A. Refer to Section 22 05 00 for basic requirements.
- B. Bell and Spigot, Cast Iron Pipe Joints: Caulk firmly with oakum or hemp and fill with molten lead not less than 1 inch deep and not to extend more than 1/8 inch below rim of hub; no paint, varnish or other coatings permitted on jointing material until after joint has been tested and approved. Neoprene gasketing system or gasket and clamp type mechanical fastener where specified.
- C. Plastic Piping and Fitting Solvent-Cement Joints: Clean and dry joining surfaces by wiping with clean cloth or paper towels. Join pipe and fittings according to the manufacturer's printed instructions.
- D. Plastic Pipe and Fitting Heat-Fusion Joints: Prepare pipe and fittings and join with heat-fusion equipment according to manufacturer's printed instructions.
 - 1. Plain-End Pipe and Fittings: Butt joining
 - 2. Plain-End Pipe and Socket Type Fittings: Socket joining

3.10 SANITARY AND STORM DRAIN PIPING

- A. Set true to line and even slope using grade boards and targets or grade lines. Install cast iron soil pipe and fittings and make joints in accordance with "Cast Iron Soil Pipe & Fittings Handbook".
- B. Slope suspended sanitary and storm piping downward minimum 1/4 inch per foot where possible for pipes 2-1/2 inch and smaller, and 1/8 inch per foot for piping 3-inch diameter and larger, and in all cases, conform to code requirements.
- C. Slope underground sanitary and storm piping downward a minimum of 1/4 inch per foot for 2-1/2 inch pipe and less; a minimum of 1/8 inch per foot for piping larger than 4-inch diameter and larger, and in all cases, conform to code requirements.
- D. Install all piping in accordance with manufacturer's recommendations regarding connections, hangers, hanger spacing, underground installation, etc. Support suspended horizontal soil pipe near each hub, with maximum spacing between hangers of 5 feet for pipe fabricated in 5 foot lengths and 10 feet for pipe fabricated in 10 foot lengths. Provide sway braces and anchorage on horizontal drainage piping sizes 5" and larger at changes of direction, where required by code and/or where recommended by coupling manufacturer and or CISPI standard 4 inch pipe size. Shall be included when required by the plumbing code.
- E. Install buried sewer and water piping in separate trenches.
- F. Provide main trap with fresh air inlet for each main drain line.
- G. For corrosive soil conditions, metallic underground piping shall be provided with loose polyethylene encasement 0.008" thick.

- H. Provide ball joint connections past foundation wall where piping exits building, EBA, Aron flex-tend or equal.

3.11 TESTING OF PIPING SYSTEMS

- A. Refer to Testing of Piping Systems, General in Section 22 05 00.

- B. Field Quality Control

1. Do not enclose, cover, or put into operation water distribution piping system and drainage and vent piping system until each has been inspected and approved by the authority having jurisdiction.
2. Reinspections: When the representative of the authority having jurisdiction finds that piping system will not pass test or inspection, make required corrections and arrange for reinspection by authority having jurisdiction.
3. Reports: Prepare inspection reports signed by the representative of the authority having jurisdiction.
4. Test in accordance with the more stringent of the requirements of the authority having jurisdiction or the following:

- C. Gravity Drainage Systems

1. Combination Water/Air Test

- a. Rough Plumbing Test Procedure: Except for outside leaders and perforated or open-jointed drain tile, test new and/or altered piping of plumbing drainage and venting systems on completion of roughing in piping installation. Tightly close all openings in piping system and fill with water to point of overflow, but not less than 10 feet head of water. Limit test pressure on plastic piping system to 30 feet head of water maximum. Water level shall not drop during the period from 15 minutes before inspection starts through 15 minutes after completion of inspection. Inspect all joints for leaks.
- b. Finished Plumbing Test Procedure: After plumbing fixtures have been set and their traps filled with water, test connections and prove gastight and watertight. Plug stack openings on roof and building drain where it leaves the building and introduce air into the system equal to pressure of 1 inch water column. Use a U tube or manometer inserted in the trap of a water closet to measure this pressure. Air pressure shall remain constant without introducing additional air throughout period beginning 15 minutes before inspection starts and 15 minutes after completion of inspection. Inspect plumbing fixture connections for gas, air and water leaks.

2. Air test only

- a. Rough Plumbing Test Procedure: If tests are made with air, apply a pressure of not less than 5 psig with a force pump and test as specified above. Use mercury-column gauge registering 10 inches in height in air test. Use air tests only when air temperatures around tested system are 32° F. or below and temporary heat is not available.

b. Finished Plumbing Test Procedure: Test as specified above.

- D. Domestic Hot and Cold Water: Test hydrostatically upon completion of the rough-in and before insulating or setting fixtures. Maintain pressure for not less than 4 hours without leakage.
- E. Pumped Systems: Test hydrostatically. Maintain pressure for 4 hours with 2% maximum loss in pressure.
- F. Unless otherwise required by Code, new branch piping connected to existing systems may be tested at normal operating pressure, provided new piping does not exceed 75 feet, or does not include more than 20 fittings.
- G. Gas System: Test in accordance with local codes and/or local gas company requirements.

3.12 VALVES

- A. Refer to Section 22 05 00 for basic requirements.

3.13 WATER HAMMER ARRESTERS

- A. Provide water hammer arresters at ends of all branches in hot and cold supply piping, at each group of flush valve operated fixtures, at equipment having quick closing valves (such as washers) and elsewhere as required and/or indicated on Drawings. Size and install each arrester according to manufacturer's instructions. Where pressure exceeds 60 psi, use next size larger where required by unit manufacturer. Install in an upright position and above ceiling in pipe chase in an accessible location.

END OF SECTION 22 10 00

SECTION 22 11 00 - DISINFECTION OF DOMESTIC WATER LINES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the contract, including General and Supplementary Conditions and Division 01 specification sections, apply to this section.

1.2 SUMMARY

- A. This section includes the disinfection and laboratory testing of domestic water piping systems to certify disinfection of system.
- B. Related Sections: Division 22.

1.3 DEFINITIONS

- A. Disinfectant Residual means the quantity of disinfectant in treated water.
- B. pH Factor means the measure of alkalinity and acidity in water.
- C. Ppm means parts per million.
- D. CFU/mL means colony forming units per milliliter.

1.4 QUALITY ASSURANCE

- A. Water Treatment Contractor: Certified by the Water Quality Association, 477 East Butterfield Road, Lombard, Illinois 60148.
- B. Water Treatment Contractor: At least 5 years' experience performing Work specified herein.
- C. Bacteriological Laboratory: Certified by the District of Columbia and by federal and local authorities having jurisdiction.

1.5 REFERENCE STANDARDS

- A. Comply with the District of Columbia and local Health Department requirements.
- B. AWWA Specification C-651 - Disinfecting Water Mains.
- C. Centers for Disease Control and Prevention (CDC) Standard Methods #9260J.

1.6 SUBMITTALS

- A. Water Treatment Contractor's evidence of certification: Submit four copies.
- B. Water Treatment Contractor's evidence of experience: Submit four copies.
- C. Bacteriological Laboratory's evidence of certification: Submit four copies.
- D. Tests Reports: Submit four copies as follows:
 - 1. Disinfection Report, include:
 - a. Date issued
 - b. Project name and location
 - c. Treatment Contractor's name, address, and phone number and name of person executing disinfection
 - d. Type and form of Disinfectant used
 - e. Time and date of Disinfectant injection start and completion
 - f. Test Locations
 - g. Initial and 24 hour Disinfectant Residuals in ppm for each outlet tested
 - h. Time and date of flushing start and completion
 - i. Disinfectant Residual after flushing in ppm for each outlet tested
 - 2. Bacteriological Report; include:
 - a. Date issued
 - b. Project name and location
 - c. Laboratory's name, certification number, address and phone number
 - d. Time and date of water sample collection
 - e. Name of person collecting samples
 - f. Test locations
 - g. Time and date of laboratory test start
 - h. Coliform bacteria test results for each outlet tested
 - i. Certification that water conforms or fails to conform to bacterial standards or fails to conform to bacterial standards of Federal Safe Drinking Water Act.
 - j. Bacteriologist's signature
 - 3. Legionella Report; include:
 - a. Date issued
 - b. Project name and location
 - c. Laboratory's name, certification number, address and phone number
 - d. Time and date of water sample collection
 - e. Name of person collecting samples
 - f. Test sample point location
 - g. Initial flow temperature of sample
 - h. Stabilized flow temperature of sample
 - i. Time and date of laboratory test start
 - j. Legionella test results for each test sample
 - k. Test method used

1. Bacteriologist's signature

1.7 PRODUCT DELIVERY, STORAGE AND HANDLING

- A. Protect against damage and contamination.
- B. Maintain caution labels on hazardous materials.
- C. Maintain storage room dry and with temperatures as uniform as possible between 60 deg F. and 80 deg F.

1.8 PROTECTION

- A. Provide necessary signs, barricades and notices to prevent any person from accidentally consuming water or disturbing system being treated.

PART 2 - PRODUCTS

2.1 MATERIALS

A. DISINFECTANT

- 1. Sodium Hypochlorite; liquid, powder, tablet, or gas.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Prior to starting Work verify that domestic water system is completed and cleaned.
- B. Notify General Contractor about defects requiring correction.
- C. Do not start Work until conditions are satisfactory.

3.2 SYSTEM TREATMENT

- A. Inject Disinfectant to obtain 200 ppm residual throughout the system.
- B. Starting at outlet closest to water source, bleed water from each outlet until water produces odor of disinfectant. Repeat process at each outlet throughout system.
- C. Maintain disinfectant in system for 3 hours.
- D. At the end of retention period, test for Disinfectant Residual at each of the following locations:

1. Ends of piping runs
2. Remote outlets
3. Tanks
4. At least 15% of outlets on each floor where directed by Architect, but in no case less than 2 outlets

E. If Disinfectant Residual is less than 10 ppm, repeat system treatment.

3.3 FLUSHING

- A. Remove disinfectant by flushing system with clean water until residual rate is reduced to less than 1.0 ppm.

3.4 BACTERIOLOGICAL TEST

- A. Instruct Bacteriological Laboratory to take water samples no sooner than 24 hours after flushing system.
- B. Consult local sewage treatment facility and insure that pH levels of disinfectant to be disposed is satisfactory prior to disposal. Provide all means and methods necessary to meet disposal requirements of authorities having jurisdiction.
- C. Take water samples at each of the following locations:
1. Where water enters system
 2. Ends of piping runs
 3. Remote outlets
 4. Tanks
 5. At least 10% of outlets on each floor other than those used for testing Disinfectant Residual, where directed by Architect, but in no case less than 2 outlets.
- D. Analyze water samples in accordance with AWWA Standard Methods for the Examination of Water & Waste Water, 14th edition.
- E. If bacteriological test proves water quality to be unacceptable, repeat disinfection of system until satisfactory bacteriological results have been obtained.

3.5 LEGIONELLA TEST

- A. Instruct bacteriological laboratory to take water samples no sooner than 24 hours after flushing system.
- B. Take water samples at each of the following locations:
1. Ends of piping runs
 2. Remote outlets
 3. At least 5% of faucets on each floor other than those used for testing disinfectant residual, where directed by Architect, but in no case less than 2 outlets.

- C. Analyze water samples in accordance with CDC Standard Method #9260J, or other internationally recognized test method as recommended by groups such as Special Pathogens Laboratory of the VA Medical Center, Pittsburgh, Pennsylvania.
- D. Samples shall be delivered to the laboratory and analyzed within 48 hours of collection. Samples shall be protected from extreme temperature variations during transport.
- E. Sample collection: Collect samples utilizing the methodology described as follows:
 - 1. Faucets: Moisten the outlet by allowing water to trickle through the opening. A sterile Dacron swab is inserted and rotated four times around the inner circumference and moving up the faucet as far as the swab will reach. Replace the swab into the container. If the swab system does not contain a transport medium, then allow 0.5 ml of water to flow from the faucet into the container to keep the swab moist. The Culturette II System (Becton Dickinson) has a self-contained transport medium, and it is not necessary to add outlet water to the swabs.
 - 2. Shower Head: Moisten the shower head by allowing water to trickle through the opening. Rotate the swab over the entire surface of the shower head 4 times. Place swab into the container. If the swab system does not contain a transport medium, then allow 0.5 ml of water to flow from the shower head into the container.
 - 3. Hot Water Tank: Open the drain valve at the base of the tank. Collect 10 to 50 ml immediately into a sterile specimen container. Then let the water drain out of the pipe for 15-30 seconds to flush out residual water within the drain pipe and then collect another 10 to 50 ml into a second specimen container. This procedure ensures that both residual water in the drain pipe and water from the tank are sampled. Scale and sediment often harbor Legionella bacteria, so it is worthwhile to obtain scale or sediment from tanks or distal sites. Heavy "syrupy" specimens from the bottom of hot water tanks, however, often will not yield the organism.
 - a. Samples should be refrigerated at 2-8° C. until processing. Swab specimens generally give higher yields than water taken from the same site.
 - b. BCYE medium with dyes, glycine, vancomycin and polymyxin will be used (Remel #01338, BBL #99648). Plates overgrown with microflora will undergo acid treatment.
- F. Action Limits: Test results indicating the Legionella CFU/mL shall be acted upon as follows:
 - 1. Individual Sample Test of less than 10 CFU/mL: Report results with recommendation for monthly follow-up testing. After 6 months of follow-up testing with no test above 10 CFU/mL, testing may be suspended. If samples from 30% or more of tested outlets test positive for Legionella, report results with recommendation for prompt cleaning and/or biocide treatment.
 - 2. Individual Sample Test Result Between 10 and 99 CFU/mL: Report results with recommendation for prompt cleaning and/or biocide treatment.
 - 3. Individual Sample Test Result of 100 CFU/mL or Higher: Report results and recommend immediate cleaning and/or biocide treatment with prompt steps to avoid exposure.

3.6 ALTERATIONS TO EXISTING SYSTEMS

- A. Unless required otherwise by Codes or authorities having jurisdiction, new branch piping not exceeding 75 total linear feet that is connected to existing systems need not be disinfected. Thoroughly flush this piping with system water and then test for coliform and Legionella organisms. If tests indicate presence of coliform organisms, or Legionella organisms, treat piping as described above and retest for coliform and Legionella. Provide all necessary valves, drains, piping and accessories required to isolate new piping from existing systems during treatment.

END OF SECTION 22 11 00

SECTION 22 11 23 - PLUMBING PUMPS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following Plumbing Pumps:
 - 1. Centrifugal Pump, Inline, Close Coupled
- B. Related sections include the following:
 - 1. Section 22 05 00 COMMON MATERIALS AND METHODS FOR PLUMBING
 - 2. Section 22 05 13 ELECTRICAL REQUIREMENTS FOR PLUMBING EQUIPMENT

1.3 DEFINITIONS

- A. Buna-N: Nitrile rubber.
- B. EPT: Ethylene propylene terpolymer.

1.4 REFERENCES

- A. ASTM A36 - Specifications for Structural Steel
- B. ASTM A48 - Specification for Gray Iron Castings
- C. ASTM A532 - Specification for Abrasion-Resistant Cast Irons
- D. ASTM B36 - Specification for Brass Plate, Sheet, Strip and Rolled Bar
- E. ASTM B62 - Specification for Composition Bronze or Ounce Metal Castings
- F. ASTM B548 - Specification for Copper Alloy Sand Castings for General Applications

1.5 SUBMITTALS

- A. Product Data: Submit certified pump curves showing pump performance characteristics with pump and system operating point plotted. Include NPSH curve when applicable.

1.6 QUALITY ASSURANCE

- A. Source Limitations: Obtain pumps through one source from a single manufacturer.
- B. Electrical Components, Devices, and Accessories: Shall be listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- C. Standards: Comply with the requirements of the following:
 - 1. "Hydraulic Institute Standards for Centrifugal, Rotary and Reciprocating Pumps" for design, manufacture and installation
 - 2. NEMA MG-1: Motors and Generators
 - 3. NEMA ICS-6: Enclosures for Industrial Control and Systems
 - 4. NEMA 250: Enclosures for Electrical Equipment (1000 volts maximum)
 - 5. NFPA 70: National Electrical Code
- D. Finish: Manufacturer's standard paint, factory applied.
- E. Manufacturer's Preparation for Shipping: Clean flanges and exposed machined metal surface and treat with anticorrosion compound after assembly and testing. Protect flanges, pipe openings and nozzles with wooden flanges or screwed inserts.
- F. Mechanical Pump Seals: Mechanical seal suitable for continuous duty with water/liquid and temperature specified. Fabricate mechanical seals with carbon rotating seal face ring, Type 316 stainless steel springs and, as required and recommended for application.
 - 1. Ni-Resist ceramic or tungsten carbide stationary seat
 - 2. Brass or Type 316 stainless steel retainer
 - 3. Buna N or EPT elastomers
- G. Slingers: Provide for close coupled pumps so that leakage past pump seal cannot enter motor bearings
- H. Motor and Starting Equipment
 - 1. Motor and starting equipment shall comply with requirements of Section 22 05 13 ELECTRICAL REQUIREMENTS FOR PLUMBING EQUIPMENT
 - 2. Motor sizes shown are maximum. Select pump so as not to require larger motor for operation 25% beyond design point on pump curve. Smaller motor acceptable under condition that nominal horsepower of smaller motor exceeds 100% of brake horsepower required for operation of pump provided at 125% of design flow rate on pump curve.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Subject to compliance with requirements, provide products of the following manufacture:

2.2 CENTRIFUGAL PUMP, IN-LINE, CLOSE-COUPLED

A. Acceptable Manufacturers:

1. Armstrong Pumps Inc.
2. Aurora Pump
3. Xylem; Div. of ITT Industries
4. PACO Pumps
5. Patterson Pump Co.
6. Peerless Pump
7. Taco, Inc.

B. Centrifugal, close coupled in-line, supported by piping system in which it is installed, back pullout design for complete servicing without disturbing piping, suitable for horizontal or vertical operation. At Contractor's option, may be substituted for in-line flexible coupled pump as specified hereinafter and in-line, flexible coupled pump may be substituted for this pump when installed in a horizontal position.

C. Cast iron casing Class 30 minimum with inlet and outlet gauge tappings, rated for 175 psig working pressure with air vent plug, drain plug, flanged suction and discharge.

D. Bronze or brass impeller keyed to shaft, removable bronze case wear rings, mechanical seal with 225° F. Ni-Resist Seat and stainless steel metallic parts, internal or external seal flush, either carbon steel shaft with bronze shaft sleeve and O-ring gasket between sleeve and impeller hub or Type 316 stainless steel shaft, grease lubricated ball bearings.

E. Refer to Quality Assurance article in Part 1 of this section.

PART 3 - EXECUTION

3.1 GENERAL

A. General

1. Refer to, and comply with, Section 22 05 00 for basic requirements.
2. Refer to Section 22 05 48 for vibration isolation, penetration seals and the like.

B. Manufacturer's Supervision/Inspection: Contractor shall be fully responsible for properly making arrangements for and coordinating with the manufacturer to provide the specified manufacturer's supervision/inspection services and manufacturer's written certification as specified in Part 1 of this section and shall at his own expense make any corrections/modifications to his installation work as required by the manufacturer.

C. Install all equipment as recommended by manufacturer and in accordance with recommendations of Hydraulic Institute.

D. Align and verify alignment of base mounted pumps.

- E. Where pump connection is smaller than pipe size indicated, valves, strainers and other pipe line accessories shall be line sized.
- F. Support piping adjacent to pump such that no weight is carried on pump casings.
- G. Suspend in-line pump independent from piping.
- H. Where elbows must be used at pump suction, provide long radius reducing type.
- I. Provide line sized shutoff valve and strainer and pressure gauge on suction line and line sized soft-seated check valve, shut-off valve and pressure gauge on discharge line.
- J. Provide supports under elbows on pump suction and discharge line sizes 2 inches and over and on smaller sizes as shown or necessary.

3.2 DOMESTIC HOT WATER CIRCULATING PUMP

- A. Provide in hot water return piping, ahead of circulator, an immersion aquastat, similar to Honeywell L-6006A, for controlling circulator.

END OF SECTION 22 11 23

SECTION 22 40 00 - PLUMBING FIXTURES AND TRIM

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section is intended to describe the plumbing fixtures and devices for the plumbing work.

- B. This section includes the following:

1. Plumbing fixtures and trim
2. Thermostatic mixing valves
3. Electric water coolers
4. Hydrants
5. Miscellaneous plumbing accessories

- C. Related sections include the following:

1. All applicable sections of Division 1
2. Division 7: Sealants
3. Section 22 05 00: Basic Materials and Equipment
4. Section 22 10 00: Plumbing Systems

1.3 REFERENCES AND REQUIREMENTS OF REGULATORY AGENCIES AND THE SMITHSONIAN INSTITUTION STANDARDS

- A. Provide work in accordance with all applicable codes, rules, regulations and reference standards.
- B. NSF/ANSI 61: Drinking Water System Components.

1.4 SUBMITTALS

- A. Provide submittals for all products listed in Article 2.01 of this section in accordance with Section 01 33 00, Submittal procedures in sufficient detail to verify full compliance with the requirements of the contract documents.
- B. For each plumbing fixture category and type specified, include selected fixture, trim, fittings, accessories, appliances, appurtenances, equipment and supports. Indicate materials and finishes, dimensions, construction details and flow-control rates.

- C. Submit wiring diagrams from manufacturer for electrically operated units.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Deliver plumbing fixtures in manufacturer's protective packing, crating and covering.
- B. Store plumbing fixtures on elevated platforms in dry location.

1.6 WARRANTY AND CONTRACT CLOSEOUT

- A. Comply with warranty and contract closeout requirements specified in Division 00.
- B. Warranty
 - 1. Refer to BIDDING AND CONTRACT REQUIREMENTS and to DIVISION 1 - GENERAL REQUIREMENTS. In addition, provide the following extended warranties as specified in this section:
 - a. Five year warranty on electric water cooler refrigeration system.
 - b. Three year nonprorated warranty on flush valves.
- C. Contract Closeout
 - 1. Include information for all products specified in this section, in operating and maintenance manual.

1.7 PROJECT/SITE CONDITIONS

- A. Coordinate roughing in and final fixture locations and verify that plumbing fixtures can be installed in accordance with contract documents.
- B. Check millwork Shop Drawings. Confirm location and size of fixtures and openings before rough-in and installation.

1.8 QUALITY ASSURANCE

- A. Provide fixtures described herein. All trim shall be manufacturer's first grade line, perfect in all respects. "Competition" or "Reserve Grade" material will not be accepted. Manufacturers and model numbers listed are to indicate standard of quality. Equal products of acceptable manufacturers listed in Article 2.01 may be submitted for review.
- B. Faucets, fittings, handles, traps, seat hinges, water and waste connections, and any other exposed parts shall be polished chrome plated brass. Provide chrome plated escutcheons for all exposed plated piping penetrating floors and walls. The metal used in castings shall be red metal, having a copper content of not less than 85% and tubing used at plumbing fixtures, heavy seamless brass tubing, not less than No. 17 gauge.

- C. Fixtures shall be product of one manufacturer.
- D. Fittings of same type shall be product of one manufacturer.

1.9 GENERAL REQUIREMENTS

- A. Plumbing fixtures shall be white, unless indicated otherwise.
- B. Provide supports for all wall mounted fixtures.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Subject to compliance with requirements, provide products of the following manufacture:
 - 1. Plumbing Fixtures (Water Closets, Urinals, Lavatories, Sinks and Mop Receptors)
 - a. Refer to drawing P-501 for Plumbing Fixture Schedule

2.2 GENERAL

- A. All mixing valves, electronic faucets and faucets for service sinks and mop receptors shall have check valves or back checks installed in hot and cold water supply piping, branches or installed integral to faucet inlets.
- B. All supply stops for lavatories, sinks, etc. shall be of lead free construction complying with NSF/ANSI 61. All flex tube risers from stops to fixtures shall be chrome plated copper. Stops shall be all brass, no plastic. Braided flex risers are not permitted.

PART 3 - EXECUTION

3.1 PROTECTION

- A. Protect fixtures against use and damage during construction.

3.2 INSTALLATION

- A. Install each fixture with trap, easily removable for servicing and cleaning.
- B. At completion, thoroughly clean plumbing fixtures and equipment.
- C. Attach floor mounted water closets to floor with wax seal and lag screws. Do not use lead flashing to hold closet in place.

- D. Install faucets with wrist blades requiring "push to close". Wrist blades in the "closed" position should be 45° from the rear wall.
- E. Provide vacuum breakers for hose bibbs and faucets with hose connections.
- F. Securely bolt all fixtures to the building construction and unless special hangers are indicated, provide hangers and/or carriers designed specifically for the fixtures by the fixture manufacturer.
- G. Securely bolt all chair carriers/fixture supports to floor using 4 bolts in each foot of each support. All floor bolt openings in chair carriers to be used.
- H. Caulk all wall hung fixtures between fixture and wall with sealant specified in Section 07 92 00. Provide sealant at all points where mop receptor meets walls and floor.
- I. Install electric water coolers with "P" trap and water shutoff valve within cabinet.
- J. For plumbing fixtures furnished and set by others, rough in piping and provide traps, valved cold and hot water connections and escutcheons. All exposed components shall be chrome plated unless otherwise noted.
- K. For plumbing fixtures installed in millwork, Contractor for this section shall provide template to Contractor for General Construction for required holes. Contractor for this section shall set fixtures and provide rough-in and final connections.
- L. All plumbing fixtures and faucets shall be installed in accordance with manufacturer's recommendations.

3.3 MOUNTING HEIGHTS

- A. Refer to Architectural elevations for all fixture mounting heights.

END OF SECTION 22 40 00

SECTION 22 60 00 - PLUMBING- SPECIAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 RELATED SECTIONS

- A. Section 22 05 00: Common Materials and Methods for Plumbing Systems
- B. Section 22 10 00: Plumbing Systems

1.3 SUBMITTALS

- A. Provide submittals in accordance with Section 01 33 00 "Submittal Procedures" in sufficient detail to verify full compliance with requirements of the contract documents.

1.4 WARRANTY AND CONTRACT CLOSEOUT

- A. Warranty
 - 1. Refer to warranty and closeout requirements in Division 01.
- B. Contract Closeout
 - 1. Comply with the requirements of Division 01.
 - 2. Include information for all products specified in this section in the operating and maintenance manual.
 - 3. Provide the following as specified in this section or Section 22 05 00:
 - a. Testing and cleaning reports, certified by contractor, for each system.
 - b. "Record drawings".
 - c. Instructions and demonstration to the Owner's representatives for the time period specified in Section 22 05 00.
 - d. Qualification certificates for plastic piping system Installers/Solvent Welders.

1.5 QUALIFICATION OF PLASTIC PIPE INSTALLERS AND WELDERS

- A. Obtain and pay for the services of an authorized representative of the plastic pipe manufacturer to qualify and certify plastic pipe installers and plastic pipe solvent and electric fusion welders.

- B. Qualification: Procedures shall be as recommended by the plastic pipe manufacturer and shall include the following as a minimum:
1. Plastic Pipe Installers: Educational material in the expansion and contraction of plastic pipe and comparison to expansion and contraction of metallic pipe materials. Instructions for determining need, location and sizing of expansion loops, offsets and legs to accommodate the expansion and contraction of the plastic pipe.
 2. Solvent and Electric Fusion Welders: Instruction on the manufacturer's recommended welding procedures, observation and inspection of a sufficient number of test welds by each welder to be reasonably assured that the welder is capable of producing acceptable welds.
- C. Certificates: Provide a written statement by the manufacturer's authorized representative listing the names of personnel who the manufacturer's authorized representative has qualified as a plastic pipe installer and/or for a plastic pipe solvent and/or electric fusion welder.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Subject to compliance with requirements, provide products of the following manufacture:
1. Pipe and Fittings
 - a. General Pipe and Fitting Materials
 - 1) Unless otherwise specified, any United States manufacturer whose products comply with the reference standards.
 - b. Basic Pipe and Fitting Materials and Accessories not specified in this section: Refer to Section 22 05 00.
 - c. CPVC and PVC Pipe and Fittings
 - 1) Harvel Plastics, Inc.
 - 2) R & G Sloan
 - 3) Charlotte Pipe Co.
 - 4) Eslon Thermoplastics, Inc.
 - d. PP (Polypropylene) Drainage Pipe and Fittings
 - 1) Lab-Line/Enfield Industrial Corporation.
 - 2) George Fischer "Fuseal"
 - 3) Sani-Tech, Inc.
 - 4) Orion
 - 5) ASAHI Amercian

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

A. General

1. Refer to and comply with Section 220500 for basic requirements.

- #### B. Manufacturers' Supervision/Inspection:
- Contractor shall be fully responsible for properly making arrangements for and coordinating with the manufacturer to provide the specified manufacturer's supervision/inspection services and manufacturer's written certification as specified in Part 1 of this section and shall at his own expense make any corrections/modifications to his installation work as required by the manufacturer.

3.2 ACID NEUTRALIZATION TANK UNDER COUNTER TYPE(ANT-A)

- #### A. Install system in accordance with manufacturer's recommendations.

- #### B. Equipment supplier shall have qualified field service technicians capable of repairing and maintaining the system. Equipment supplier must either inventory or have quick access to any parts or components which may require replacement.

3.3 PIPING CONNECTIONS

- #### A. Refer to Section 22 05 00 for basic requirements.

- #### B. Plastic Pipe and Fitting Solvent-Cement Joints:
- Clean and dry joining surfaces by wiping with clean cloth or paper towels. Join pipe and fittings according to the manufacturer's printed instructions.

- #### C. Plastic Pipe and Fitting Heat-Fusion Joints:
- Prepare pipe and fittings and join with heat-fusion equipment according to manufacturer's printed instructions.

1. Plain-End Pipe and Fittings: Butt joining.
2. Plain-End Pipe and Socket-Type Fittings: Socket joining.

3.4 TESTING OF PIPING SYSTEMS

- #### A. Refer to Testing of Piping Systems, General in Section 22 05 00.

END OF SECTION 22 60 00

SECTION 23 05 00 – COMMON MATERIALS AND METHODS FOR HVAC

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section is intended to describe the common materials and installation methods of the mechanical work and it applies in general to all other Sections under Division 23.
- B. Due to the small scale of the drawings, all work required is not shown on the floor plans and certain work is shown on flow diagrams, riser diagrams and details. Work of Division 23 shall include all required work shown on plans, riser diagrams, flow diagrams and details.
- C. This Section includes the following:
 - 1. Piping materials and installation instructions common to most piping systems
 - 2. Transition fittings
 - 3. Dielectric fittings
 - 4. Mechanical sleeve seals
 - 5. Sleeves
 - 6. Escutcheons
 - 7. Grout
 - 8. HVAC demolition
 - 9. Equipment installation requirements common to equipment sections
 - 10. Painting and finishing
 - 11. Hangers and supports for HVAC system piping and equipment
 - 12. Identification for HVAC Piping and equipment

1.3 DEFINITIONS

- A. Finished Spaces: Spaces other than mechanical and electrical equipment rooms, furred spaces, pipe and duct chases, unheated spaces immediately below roof, spaces above ceilings, unexcavated spaces, crawlspace, and tunnels.
- B. Exposed, Interior Installations: Exposed to view indoors. Examples include finished occupied spaces and mechanical equipment rooms.
- C. Exposed, Exterior Installations: Exposed to view outdoors or subject to outdoor ambient temperatures and weather conditions. Examples include rooftop locations.

- D. Concealed, Interior Installations: Concealed from view and protected from physical contact by building occupants. Examples include above ceilings and chases.
- E. Concealed, Exterior Installations: Concealed from view and protected from weather conditions and physical contact by building occupants but subject to outdoor ambient temperatures. Examples include installations within unheated shelters.

1.4 REFERENCES

- A. Provide work in accordance with all applicable international, state and local, codes, rules, regulations, and standards, including but not limited to, requirements of the following:
 - 1. ASME/ANSI B31: Code for Pressure Piping
 - 2. ASME Boiler and Pressure Vessel Codes
 - 3. AWS D1.1: Structural Welding Code-Steel
 - 4. MSS SP58: Pipe Hangers and Supports – Materials, Design, and Manufacturers
 - 5. MSS SP69: Pipe Hangers and Supports – Selection and Application except spacing for hangers
 - 6. ANSI A13.1: Scheme for Identification of Piping Systems
 - 7. Applicable NFPA Codes and Standards
 - 8. 2021 ICC Codes
 - 9. Smithsonian Facilities Design Standards
 - 10. NMNH Supplemental Design Standards

1.5 SUBMITTALS

- A. Provide Product List of factory fabricated items, in accordance with Section 01 60 00 “Product Requirements”, including name of proposed manufacturer, for all products specified in various sections of Division 23.
- B. Provide submittals in accordance with Section 01 33 00 “Submittal Procedures” in sufficient detail to verify full compliance with the requirements of the Contract Documents.
- C. Product Data: Provide for each type of factory-fabricated product indicated.
- D. Welding certificates.

1.6 WARRANTY AND CONTRACT CLOSEOUT

- A. Comply with warranty and contract closeout requirements specified in Division 00.
- B. Provide Special Warranties and/or warranty service in accordance with Section 01 60 00 “Product Requirements” where specified in the various sections of Division 23.
- C. Provide manufacturer’s certificates of supervision and startup service as specified in the various sections of Division 23.

- D. Provide testing and cleaning reports. Indicate dates of testing and cleaning operations, procedures used and results obtained for each system. Reports shall be certified as complete.
- E. Provide instructions and demonstration to the Owner's representative for all equipment and systems installed under Division 23. Instruction and demonstration shall be appropriate for the size and complexity of the installed system.
- F. Include information for all products specified in the operation and maintenance manual.

1.7 QUALITY ASSURANCE

- A. Steel Support Welding: Qualify processes and operators according to AWS D1.1, "Structural Welding Code--Steel."
- B. Steel Pipe Welding: Qualify processes and operators according to ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."
 - 1. Comply with provisions in ASME B31 Series, "Code for Pressure Piping."
 - 2. Certify that each welder has passed AWS qualification tests for welding processes involved and that certification is current.
 - 3. Use welders fully qualified and licensed by the state authorities.
- C. The specifications for certain products and alternative materials may appear in more than one section of Division 23. Work of Division 23 shall be coordinated for all sections of Division 23 to assure that where two or more items of any given product are furnished under Division 23 that they are of the same manufacturer and type and that alternative materials is consistent throughout the work of Division 23.
- D. Except for spacing of hangers, provide hangers and supports in accordance with the latest issue of Manufacturer's Standardization Society (MSS) Specifications SP 58 and 69.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle all material and equipment in accordance with manufacturer's instructions and recommendations. Such instructions and recommendations are hereby made part of these specifications.
- B. Deliver pipes and tubes with factory-applied end caps. Maintain end caps through shipping, storage, and handling to prevent pipe end damage and to prevent entrance of dirt, debris, and moisture.
- C. Deliver products and equipment properly labeled and tagged. Maintain products in original shipping containers and store in a dry area until ready for installation.
- D. For any equipment that is stored prior to installation, the Contractor shall submit a storage plan.

1.9 COORDINATION

- A. The Mechanical systems are indicated on the Mechanical Drawings. Certain pertinent information and details involving the installation of Mechanical work appear on Architectural, Structural, Plumbing and Electrical Drawings. Become familiar with all Drawings and incorporate all pertinent requirements.
- B. Drawings are diagrammatic and indicate general arrangement of systems and requirements of the Mechanical work. Do not scale the Drawings to obtain dimensional requirements. Exact locations of equipment must be coordinated and obtained prior to starting the work.
- C. Arrange for pipe spaces, chases, slots, duct shafts and openings in building structure during progress of construction, to allow for HVAC installations.
- D. Coordinate installation of required supporting devices and set sleeves in poured-in-place concrete and other structural components as they are constructed.
- E. Coordinate requirements for access panels and doors for HVAC items requiring access that are concealed behind finished surfaces. Coordinate installation of identification labels with locations of access panels and doors.
- F. Coordinate scheduling, sequencing, movement and positioning of large equipment into the building during construction.
- G. Coordinate installation of identification devices with completion of covering and painting of surfaces where identification devices are to be applied.
- H. Install identification devices prior to installation of ceilings and similar concealment.

1.10 ENERGY PERFORMANCE CRITERIA

- A. All equipment provided under Division 23 shall meet the requirements of the International, or State, Energy Code, ASHRAE Standard 90.1 or the latest issue of the Standards for Equipment in the National Energy Policy Act (NEPA), whichever is more stringent.

1.11 ASSET MANAGEMENT

- A. Complete a Smithsonian OFEO-Asset Update Collection form for each piece of mechanical equipment to be removed or relocated. Contractor can obtain form from COTR.
- B. Complete a Smithsonian OFEO – Asset Update/Collection form and assign a barcode for each new or re-installed piece of mechanical equipment. COTR to obtain bar code numbers from the Systems Reliability Branch of SED and provide contractor with form.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. In other Part 2 articles of various sections of Division 23 where subparagraph titles below introduce lists, the following requirements apply for product selection:
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the manufacturers specified.
 - 2. Unless otherwise noted, substitutions of specified manufacturers shall comply with the requirements of Division 01.

2.2 PIPE, TUBE, AND FITTINGS

- A. Refer to individual Division 23 piping Sections for pipe, tube, and fitting materials and joining methods.
- B. Pipe Threads: ASME B1.20.1 for factory-threaded pipe and pipe fittings.

2.3 JOINING MATERIALS

- A. Refer to individual Division 23 piping Sections for special joining materials not listed below.
- B. Pipe-Flange Gasket Materials: Suitable for chemical and thermal conditions of piping system contents.
 - 1. ASME B16.21, nonmetallic, flat, asbestos-free, 1/8-inch (3.2-mm) maximum thickness unless thickness or specific material is indicated.
 - a. Full-Face Type: For flat-face, Class 125, cast-iron and cast-bronze flanges.
 - b. Narrow-Face Type: For raised-face, Class 250, cast-iron and steel flanges.
 - 2. AWWA C110, rubber, flat face, 1/8 inch (3.2 mm) thick, unless otherwise indicated; and full-face or ring type, unless otherwise indicated.
- C. Flange Bolts and Nuts: ASME B18.2.1, carbon steel, unless otherwise indicated.
- D. Solder Filler Metals: ASTM B32, lead-free alloys. Include water-flushable flux according to ASTM B813.
- E. Brazing Filler Metals: AWS A5.8, BCuP Series, copper-phosphorus alloys for general-duty brazing, unless otherwise indicated; and AWS A5.8, BAg1, silver alloy for refrigerant piping, unless otherwise indicated.
- F. Welding Filler Metals: Comply with AWS D10.12 for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.

2.4 DIELECTRIC FITTINGS

- A. Description: Combination fitting of copper alloy and ferrous materials with threaded, solder-joint, plain, or weld-neck end connections that match piping system materials.
- B. Insulating Material: Suitable for system fluid, pressure, and temperature.
- C. Dielectric Unions: Factory-fabricated, union assembly, for 250-psig (1725-kPa) minimum working pressure at 180 deg F (82 deg C).
- D. Dielectric Flanges: Factory-fabricated, companion-flange assembly, for 150- or 300-psig (1035- or 2070-kPa) minimum working pressure as required to suit system pressures.
- E. Dielectric-Flange Kits: Companion-flange assembly for field assembly. Include flanges, full-face- or ring-type neoprene or phenolic gasket, phenolic or polyethylene bolt sleeves, phenolic washers, and steel backing washers. Separate companion flanges and steel bolts and nuts shall have 150- or 300-psig (1035- or 2070-kPa) minimum working pressure where required to suit system pressures.
- F. Dielectric Couplings: Galvanized-steel coupling with inert and noncorrosive, thermoplastic lining; threaded ends; and 300-psig (2070-kPa) minimum working pressure at 225 deg F (107 deg C).
- G. Dielectric Nipples: Electroplated steel nipple with inert and noncorrosive, thermoplastic lining; plain, threaded, or grooved ends; and 300-psig (2070-kPa) minimum working pressure at 225 deg F (107 deg C).

2.5 ESCUTCHEONS

- A. General: Manufactured wall and ceiling escutcheons and floor plates, with an inside diameter to closely fit around pipe, tube, and insulation of insulated piping and an outside diameter that completely covers opening.
- B. One piece construction on exposed piping in finished areas. Elsewhere, split pattern with setscrew. Provide deep pattern type where required to conceal protruding fittings and sleeve.
- C. Provide brushed brass escutcheons, except provide polished chromium plated escutcheons on pipes passing through walls, floors or ceilings wherever such pipes are exposed to view.

2.6 HANGERS AND SUPPORTS

- A. Acceptable Manufacturers
 - 1. Other Than Roof Supports
 - a. B-Line Systems, Inc.
 - b. Grinnell Company
 - c. National Pipe Hangers
 - d. Penn Construction Industries

- e. Other approved United States manufacturer whose products comply with the referenced standards

B. Reference Standards

1. ASTM A36 - Specification for Structural Steel
2. ASTM A123 - Zinc (Hot-Dip Galvanized Coatings on Products Fabricated from Rolled, Pressed and Forged Steel Shapes, Bars, and Strip)
3. ASTM A653 G90 - Specification for Steel Sheet, Zinc Coated by the Hot-Dip Process
4. ASTM B633 - Specification for Electrodeposited Coatings of Zinc on Iron and Steel
5. AWS D1.1 - Structural Welding Code - Steel
6. MSS SP58 - Manufacturer's Standardization Society: Pipe Hangers and Supports - Materials, Design and Manufacture
7. MSS SP69 - Manufacturer's Standardization Society: Pipe Hangers and Supports - Selection and Application
8. NFPA 13 - Standard for the Installation of sprinkler Systems

C. Quality Assurance

1. Steel pipe hangers and supports shall have the manufacturer's name, part number and applicable size stamped in the part itself for identification.
2. Hangers and supports shall be designed and manufactured in conformance with MSS SP58.

D. General

1. Except for spacing of the hangers, design and fabrication of pipe hangers, supports and welding attachments shall conform to ANSI B31.9 or B31.1 as applicable.
2. Except for spacing of the hangers, hanger types and supports for bare and covered pipe shall conform to MSS SP69 for the temperature range except that only flat wide band hangers shall be used for hangers installed outside of insulation and plastic pipe.
3. Except for spacing of pipe hangers and elsewhere as otherwise indicated, horizontal and vertical piping attachment shall conform to the more stringent of this specification or MSS SP58 or MSS SP69. Continuous inserts and expansion bolts may be used.
4. All ferrous hangers, supports and hardware located outdoors shall be hot dip galvanized after fabrication per ASTM A 123.
5. Hangers and clamps for support of bare copper piping shall be coated with copper colored (for identification) baked on epoxy paint. Use additional PVC coating of the epoxy painted hangers where necessary.
6. Provide suitable chromium plated brass supports for chromium plated pipe with exposed heads of bolts and screws chromium plated.
7. Hangers other than described above shall be zinc plated in accordance with ASTM B 633 or shall have an electrodeposited epoxy finish.
8. Strut channels shall be pre-galvanized in accordance with ASTM A 653 G90 or shall have an electrodeposited finish.
9. All hangers and supports shall have some form of adjustment available after installation.

E. Inserts

1. Inserts: Malleable iron case or galvanized steel shell and expander plug for threaded connection with lateral adjustment, top slot for reinforcing rods, lugs for attaching to forms.

2. Size inserts to suit threaded hanger rods.

F. Pipe Hangers and Supports: Provide hangers as follows:

1. Hangers for Pipe Sizes to 1-1/2 Inch (DN 40): Adjustable carbon steel ring or clevis.
2. Hangers for Hot or Cold Pipe Sizes 2 Inches to 4 Inches (DN 50 to DN 100) and Cold Pipe Sizes 6 Inches (DN 150) and Over: Adjustable carbon steel clevis.
3. Hangers for Hot Pipe Sizes 6 Inches (DN 150) and Over: Adjustable steel yoke and cast iron roll.
4. Multiple or Trapeze Hangers: Factory-enameled steel channels with welded spacers and hanger rods or 12 gauge (2.7 mm) rolled formed ASTM A 570 Grade 33 structural quality steel channels (strut), cast iron roll and stand for hot pipe sizes 6 inches (DN 150) and over. Cross section suitable for span and loading. Suspension by outside hanger rods sized for total load on trapeze.
5. Wall Support for Pipe Sizes to 3 Inches (DN 80): Carbon steel hook.
6. Wall Support for Pipe Sizes 4 Inches (DN 100) and Over: Welded steel bracket and steel clamp, adjustable steel yoke and cast iron roll for hot pipe sizes 6 inches (DN 150) and over.
7. Vertical Support: Steel riser clamp.
8. Floor Support for Hot Pipe Sizes to 4 Inches (DN 100) and All Cold Pipe Sizes: Adjustable pipe saddle and pipe nipple attached to steel base stand, and concrete pier or steel support.
9. Floor Support for Hot Pipe Sizes 6 Inches (DN 150) and Over: Adjustable cast iron roll and stand, steel screws and concrete pier or steel support.
10. Design hangers to impede disengagement by the movement of supported pipe. Provide spring and neoprene hangers as required.

G. Beam Clamps: Forged steel C-clamps shall include retaining strap, locking nut or other device for nonslip attachment.

H. Hanger Rod: Steel hanger rod zinc plated per ASTM B 633.

2.7 THERMOMETERS

A. Acceptable Manufacturers

1. Trerice
2. Taylor Instrument Company
3. U.S. Gauge
4. Weksler
5. Weiss

B. Tube Thermometer: ASTM E1, liquid in place thermometer. Cast aluminum case, black baked epoxy enamel finish, 9 inch (225 mm) minimum liquid filled tube, brass stem, adjustable angle type with locking device and with brass union type separable socket. Socket length to suit installation. Mercury filled thermometer not allowed.

C. Select range of thermometer to indicate normal operating temperatures at midpoint of scale. Scale division of 1 degree F (0.5 degree C) for cold service and 2 degree F (1 degree C) for hot service.

- D. Install wells with stem extending to center of pipe. Fill wells with oil or graphite and secure caps.

2.8 PRESSURE GAUGES

A. Acceptable Manufacturers

1. Trerice
2. Taylor Instrument Company
3. U.S. Gauge
4. Weksler
5. Weiss

- B. ASTM B40.1, Grade A phosphor bronze seamless Bourdon spring type with white face, black numerals, 4 ½ inch (125 mm) cast aluminum case, black baked epoxy enamel finish, brass bronze bushed movement and brass socket. Select range of gauge to indicate normal operating pressure of system at midpoint of scale.
- C. Provide ¼ inch (6 mm) brass coil siphon for steam gauge.
- D. For liquid gauge, provide brass snubber of material suitable for system fluid. Provide with needle valve.

2.9 IDENTIFICATION DEVICES AND LABELS

A. General

1. Products specified are manufacturer's standard products of categories and types required for each application as referenced in Part 3 of this section and elsewhere on the drawings or in Division 23 specifications. Where more than single type is specified for listed application, selection is Contractor's option, but provide single selection for each product category.
 2. Products shall comply with requirements of ANSI A13.1 and OSHA where applicable.
- B. Laminated Plastic Pipe Markers: Manufacturer's standard preprinted, wrap-around, color-coded pipe markers conforming to ANSI A13.1.
 - C. Arrows: Either integrally with piping system service lettering (to accommodate both directions), or as separate unit, on each pipe marker to indicate direction of flow.
 - D. Plastic Tape for Ductwork and Piping: Manufacturer's standard color-coded, pressure- sensitive, self-adhesive, vinyl tape, at least 3 mils thick. Width 1-1/2 inches (35 mm) wide on pipes with outside diameters (including insulation) less than 6 inches (DN 150); 2-1/2 inches (65 mm) wide for larger pipes. Color shall comply with ANSI A13.1 unless otherwise indicated.
 - E. Valve Tags: Stamped or engraved brass with 1/4-inch (6 mm) letters for piping system abbreviation and 1/2-inch (12.5 mm) sequenced numbers. Provide a hole for fastener. Brass wire-link chain, beaded chain, or S-hook fasteners.

- F. Access Panel Markers: 1/16 inch (1.5 mm) thick engraved plastic-laminate markers, with abbreviated terms and numbers corresponding to concealed device. Provide center hole for attachment.
- G. Valve Schedule Frames: Glazed display frame, with screws for removable mounting on masonry walls for each page of valve schedule. Polished hardwood or extruded aluminum frame.
- H. Engraved Plastic-Laminate Signs: ASTM D709, Type I, cellulose, paper-base, phenolic-resin-laminate engraving stock; Grade ES-2, black surface, black phenolic core, with white (letter color) melamine subcore, except when other colors are indicated. Fabricate in sizes required for message. Provide holes for mechanical fastening. 1/16 inch (1.5 mm) thick for units up to 20 square inches (0.013 square meters) or 8 inch (200 mm) length, 1/8 inch (3 mm) thick for larger units. Self-tapping stainless steel screws or contact-type permanent adhesive.
- I. Plasticized Tags: Preprinted accident-prevention tags, of plasticized card stock. Size approximately 3-1/4 by 5-5/8 inches (75 mm by 125 mm). Brass grommets and wire fasteners.
- J. Nomenclature: Large-size wording such as "DANGER," "CAUTION," or "DO NOT OPERATE", or as noted on the drawings in the specification.
- K. Lettering and Graphics: Coordinate names, abbreviations, and other designations used in mechanical identification, with corresponding designations indicated. Use numbers, letters, and terms indicated for proper identification, operation, and maintenance of mechanical systems and equipment.
- L. Multiple Systems: Where multiple systems of same name are indicated, identify individual system number as well as service.

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

A. General

1. Furnish, deliver, erect, connect, and finish in every detail all materials, equipment and accessories required for the Work.
2. Include in the work and in the bid minor details not usually shown or specified, but manifestly necessary for the proper installation and operation of the various systems, the same as if specified or shown.
3. If any departures from the Contract Documents are deemed necessary, submit details of such departure and the reasons therefore to the Architect for approval.
4. Be responsible to request clarification from the Architect on any conflicts represented between the drawings and specifications.
5. Adequately guard all exposed moving parts of equipment, such that contact by operating personnel will not cause personal damage or injury.

B. Layout and Coordination With Other Trades

1. Layout Work from building and property lines and benchmarks provided, verify, and be responsible for the correctness of all measurements in connection with the Work. Any change made in major overall dimensions shown which affect the physical size, shape, or location of any part of the Work, whether due to field check or changes due to use of equipment of a manufacturer other than that used as basis of design shall cause no interference with other Work.
2. Examine the Drawings of other trades, cooperate and coordinate with other trades to insure that the Work can be installed properly as designed and planned without interference with other work or delay. Where interferences may occur and departures from the arrangements shown are required, consult with other trade involved. Come to agreement as to changed locations and elevations. Furnish all necessary templates, patterns, measurements, etc., for installation and for the purpose of making adjoining work conform. Furnish setting plans and shop drawings to other trades as required.
3. Investigate the structural and finish conditions affecting the Work. Offsets, bends or other items required may not be shown on the drawings; provide such offsets or bends as required to meet structural or finish conditions.
4. Coordinate layout with architectural ceilings and lighting layouts and similar work.
5. Coordinate and be responsible for the required clearances in shafts, chases, furred partitions and suspended ceilings. Coordinate and cooperate with the trades responsible for constructing such spaces, together with other trades sharing such spaces, and advise other trades of the requirements of the Work. Immediately submit for review large scale composite Drawings showing space requirements that exceed those shown.
6. Install systems so that they do not interfere with any openings, doors or windows, or with other work, and so as to permit proper access.
7. Install material and equipment as high as possible; at minimum, to clear the top of all doors, windows and other structural openings. Maintain maximum headroom and space conditions in every case. Where headroom or space conditions appear inadequate, notify the Architect before proceeding with the installation.
8. Except where greater clearance is specified or required by applicable codes, rules or regulations, install piping, ductwork, fittings, valves, etc., to provide not less than 1 inch between their finished covering and the structure or adjacent work of any kind. The minimum space between finished hot piping of any kind and adjacent electrical conduit shall be 6 inches.
9. Make reasonable modifications in the layout to provide proper clearances or accessibility, or to prevent conflict with the work of other trades, at no increase in the Contract sum.
10. Prepare large scale composite working drawings, including such section views and details as are necessary to clearly show how the systems are to be installed in relation to the work of other trades. Issue such Drawings to the other trades for coordination of their work. Where such drawings show deviations from the Contract Drawings or conflict with other trades such that reasonable modifications cannot be made, detail and submit such deviation or conflict to the Architect for review.
11. If work is installed before coordinating with other trades so as to cause interference with the work of other trades, or as not to provide proper access for maintenance or repair, make necessary changes to correct the condition at no increase in the Contract sum.
12. For alterations to existing facilities, be fully responsible for coordinating work with all existing conditions. Verify location of existing piping, ductwork and equipment in the field. Relocate or offset new piping and ductwork, and make reasonable modifications to existing piping and ductwork, as required to fit in available space whether or not such relocation of offset is shown on the Drawings.

C. Manufacturer's Instructions and Recommendations

1. Perform the installation, cleaning, testing, calibration and start up of all material and equipment in accordance with the manufacturer's instructions and recommendations. Such instructions and recommendations are hereby made part of the specifications.
2. Should a conflict exist between specifications and manufacturer's instructions, consult with the Architect.

D. Electrical Rooms (including IT Rooms, Security Rooms, AV Rooms, etc.)

1. Do not install any piping, ductwork or equipment in or through an electrical room or similar room containing electrical equipment, other than piping, ductwork or equipment exclusively serving the room or equipment in the room.
2. If there is a conflict between the above requirement and the Drawings, the above shall govern. If reasonable modifications cannot be made to accommodate this requirement, obtain instructions from the Architect before proceeding with the work.

E. Painting

1. Except where specified otherwise in Division 23, Work of Division 09 will provide painting of HVAC systems, equipment and components.
2. Protect all equipment from rust, corrosion, and similar damage by either factory applied or field applied protective coatings.
3. Repair marred and damaged factory painted finishes with manufacturer's touch up paint and application procedures to match original factory finish.

F. Wall and Ceiling Access Doors

1. Access Doors shown on Architectural Drawings will be provided under Division 08.
2. Furnish access doors required for access to concealed dampers, valves, air vents, traps, cleanouts, unions, expansion joints, and other equipment where no other means of access is available. Access doors shall be of adequate size for the service requirements, minimum clear opening of 14 inches by 16 inches (350 mm by 400 mm).
3. Access doors shall be as specified in Division 08. Coordinate locations of access doors with all trades.

3.2 PENETRATIONS

A. General

1. Coordinate with other trades as to the size and location of openings to be provided in new floors, walls, roofs and ceilings as construction progresses.
2. Do not cut openings in new or existing floors and walls without proper structural reinforcement.
3. Install both piping and seals so as to maintain integrity of seals with expansion and contraction of piping.
4. Sealing Requirements:

- a. Seal completely all penetrations of piping, ductwork and conduit through any surface.
- b. Seal completely all penetrations of piping, ductwork and conduit through above ceiling full height partitions.
- c. Seal completely the perimeter of all Mechanical fixtures and devices, including diffusers and grilles, applied to any surface.
- d. Seal completely the perimeter of all access doors or panels in any surface.

B. Sleeves

1. Provide each pipe, duct or conduit passing through a masonry or concrete wall, floor or partition, and elsewhere as indicated, with a sleeve made from standard weight galvanized steel pipe for pipe or conduit and 12 gauge (1.3 mm) galvanized sheet steel for ducts, with smooth edges, securely and neatly set in place.
2. Select sleeves two pipe sizes larger than any pipe or conduit to accommodate pipe, insulation, and jacketing without touching the sleeve and shall provide minimum of 3/8 inch (9.5 mm) clearance.
3. Be responsible for the proper location and alignment of all sleeves.
4. Extend wall and partition sleeves through and cut flush with each surface unless otherwise indicated or specified.
5. Place sleeves imbedded in concrete floors or walls in the forms before concrete is poured; sleeves shall have integral water stop flanges, where they are to receive either water tight or hydrostatic seals.

C. Fire Rated Penetrations

1. Provide through-penetration fire-stop sealing system for pipe, duct and conduit penetrations through fire or smoke rated construction. Refer to Division 07 for through-penetration fire stop sealing system.
2. Coordinate with Division 07 to determine requirements for sleeves and clearances.
3. Select duct sleeve sizes to suit requirements of manufacturer of fire and/or smoke dampers.

D. Interior Non-Rated Wall Pipe Penetrations

1. For acoustically treated partitions, and walls between mechanical equipment rooms and occupied spaces, fill annular void at penetration with acoustical sealant.

E. Resilient Penetration Sleeve/Seal

1. Provide resilient penetration sleeve seal for piping and ductwork subject to vibration to prevent transmission to the building structure.
2. Maintain an airtight seal around the penetrating element and prevent rigid contact of the penetrating element and the building structure. Fit the sleeve tightly to the building construction and seal airtight on both sides of the construction penetrated with acoustical sealant.

F. Floor Pipe Penetrations

1. Provide water stops for new cored openings for piping where such openings are above grade, above galleries, in toilet rooms, in similar wet service areas, and in mechanical rooms, penthouses, and pipe chases.
2. Provide water stops for existing pipe floor openings that do not have sleeves extended above the floor.
3. Provide either a sleeve or angle water stop. Sleeve may be used if the fire-stop sealing method selected by Division 07 allows the use of a sleeve, otherwise provide angle water stop.
4. Sleeve Water Stop: Construct of Schedule 40 galvanized steel pipe, length as required for 1 inch (25 mm) above floor and 1 inch (25 mm) below underside of floor. Provide 1 inch by 1 inch by 1/8 inch (25 mm x 25 mm x 3 mm) galvanized angle clips welded to sleeve at 90-degree intervals and securely fastened to underside of floor. Caulk space between floor opening and sleeve watertight with soft setting waterproof silicone sealant.
5. Angle Water Stop: Construct of standard weight pipe 1 inch (25 mm) long welded to a 1/4 inch (6 mm) circular steel base plate ring fastened to floor with expansion anchors. Base plate ring width as required for anchor clearance from edge of cored opening. Seal between base ring and floor and caulk all edges of base ring with waterproof silicone sealant.

3.3 ALTERATIONS AND CONNECTIONS TO EXISTING FACILITIES

- A. Refer to Division 01 Section "Cutting and Patching" and Division 02 Section "Selective Structure Demolition" for general demolition requirements and procedures.
- B. Make all necessary alterations to existing Division 23 systems as required for removing or for connecting or extending these systems to new work and for revisions in existing work as indicated and as required, whether indicated or not. Match new materials in altered systems with existing materials unless otherwise indicated.
- C. Continuity of Existing Services
 1. Perform alterations and connections to existing facilities with a minimum of interruption. Where interruption is necessary, prepare a time schedule for shutdown activities, coordinate with Architect, Owner and other trades, and obtain written approval from Owner prior to proceeding with the work. Include work scheduled for off hours, when Owner requires that shutdown and interruption of facilities occur during unoccupied times.
 2. Prepare and set notices on services and equipment that are temporarily shut off or disconnected.
- D. Provide shutoff valves to isolate new work from existing and temporary or permanent connections to new work as required for proper testing and cleaning of new work.
- E. All relocations of existing work shall be accomplished using new materials and accessories unless specifically noted otherwise.
- F. Where equipment, ductwork and piping is removed or disconnected under Division 23, perform the work in such a manner that no damage is done to the structure or remaining portions of the existing systems. Do not under any circumstance place stress on existing pipe and fittings that are to be reused. Be fully responsible for and repair, at no additional expense to the Owner, any leaks developing in existing piping due to failure to take proper precautions when making alterations.

- G. Disconnect, demolish, and remove HVAC systems, equipment, and components indicated to be removed.
 - 1. Piping to Be Removed: Remove portion of piping indicated to be removed and cap or plug remaining piping with same or compatible piping material.
 - 2. Piping to Be Abandoned in Place: Do not abandon any piping in place unless specifically noted to do so. Drain piping and cap or plug piping with same or compatible piping material.
 - 3. Ducts to Be Removed: Remove portion of ducts indicated to be removed and plug remaining ducts with same or compatible ductwork material.
 - 4. Ducts to Be Abandoned in Place: Do not abandon any ductwork in place unless specifically noted to do so. Cap or plug ducts with same or compatible ductwork material.
 - 5. Equipment to Be Removed: Disconnect and cap services and remove equipment.
 - 6. Equipment to Be Removed and Reinstalled: Disconnect and cap services and remove, clean, and store equipment; when appropriate, reinstall, reconnect, and make equipment operational.
 - 7. Equipment to Be Removed and Salvaged: Disconnect and cap services and remove equipment and deliver to Owner.
 - 8. All materials and equipment removed or disconnected by Division 23, which is not to be reused or delivered to Owner, shall be removed from the premises. Provide owner first right of refusal of equipment prior to removal from site.
- H. If pipe, insulation, or equipment to remain is damaged in appearance or is unserviceable, remove damaged or unserviceable portions and replace with new products of equal capacity and quality.
- I. Remove all piping, ductwork and equipment hangers and supports.
- J. Cap tight unused connection at mains and risers behind finished surfaces.

3.4 PIPING SYSTEMS - COMMON REQUIREMENTS

- A. Install piping according to the following requirements and Division 23 Sections specifying piping systems.
- B. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated unless deviations to layout are approved on Coordination Drawings.
- C. Install all piping in accordance with best practices of trade and latest code requirements. Locate groups of pipes parallel to each other, spaced to permit valve servicing. Use uniform system materials throughout building.
- D. Install components having pressure rating equal to or greater than system operating pressure.
- E. Keep all piping as high as possible, consistent with proper pitch, to maintain maximum headroom. Cut piping accurately to measurements established at building, work into place without springing, forcing or cutting of the building structure, and install as directly as possible without sags between

connecting points parallel with or at right angles to building construction, except as required to obtain pitch.

- F. Pitch all systems for proper venting at high points and to drain at low points where the systems can be completely emptied. Install vents at all high points and drains at all low points, including where offsets and bends in horizontal pipe runs create a low point. Provide drain points with bronze hose end drain valves.
- G. Do not install piping above or through electrical rooms, telecommunication rooms, or similar room having a large collection of electrical equipment.
- H. Keep pipe and fittings clean from cutting burrs, foreign matter and defects in structure and threading. Make all cuts square. Ream after cutting. Bevel plain ends of steel pipe. Clean off scale and dirt inside and outside before assembly. Remove welding slag or other foreign matter inside and outside.
- I. Install piping within building concealed in walls, furred spaces, pipe spaces or above suspended ceilings. Do not build in or bury horizontal piping within partitions. Install exposed piping as closely as possible to walls, ceilings and columns, allowing space for installation of insulation and access for valve operation.
- J. Install piping sections using greatest length possible in all cases. The use of short lengths socketed together will not be allowed.
- K. The use of lampwick or other material for packing threads, caulking or wrapping of joints to stop or prevent leaks or correct faults is not permitted. The use of long screws having right and left hand threads or couplings is not permitted.
- L. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.
- M. Install piping to permit valve servicing and application of insulation.
- N. Provide offsets and loops as required for piping crossing a building expansion joint to accommodate building movement, whether or not shown on the drawings.
- O. Changes in Pipe Size and Direction
 - 1. Make reduction or increase in pipe size with fittings. Use eccentric reducing fittings in horizontal piping. Use reducing tees in pressure piping for side outlet reduction only, not on run. Bushed fittings, notched or straight runs to form tees, or any similar fabrication method will not be permitted.
 - 2. Make changes in direction with standard fittings. Mitering of pipe to form elbows or similar fabrication method will not be allowed. Bending of piping will not be permitted.
- P. Electrolysis Control
 - 1. Install copper or brass piping or tubing in such a way as not to touch or come in contact with ferrous metals.
 - 2. Where ferrous piping or equipment is connected to copper or brass piping, make connection with insulating or dielectric union to prevent electrolytic action between the

ferrous and nonferrous metals. At branch connections off mains, provide shut off valve upstream of dielectric union in order to isolate downstream union.

3. Where copper or brass piping, tubing or fittings are anchored to, supported by, or come in contact with ferrous metal construction, provide an insulating nonconductor spacer of rubber, plastic or equivalent material to assure prevention of electrolysis.

Q. Equipment Piping

1. Verify final locations of equipment for rough in of piping connections.
2. Provide shut off valves in the supply and return to each item of equipment. Suitably locate equipment isolation valves to facilitate removal of equipment.
3. Provide piping from pump glands, cooling coil drain pans, relief valves or other drainage to spill over open sight drains, floor drains, or other trapped acceptable discharge, terminating with plain end cut at a 45-degree angle.

R. Expansion and Contraction of Piping

1. The piping installation shall be free of stress. Run all piping with full allowance for expansion or contraction. Base expansion calculations on 50 degree F (10 degree C) installation temperature to 200 degree F (94 degree C) for hot water systems and temperature of steam pressure for steam systems, plus 30% safety factor.
2. Evaluate the complete piping layout and notify Architect of additional anchors or expansion joints and any deviations required to compensate for expansion.
3. Make connections to equipment in such a manner as to eliminate undue strains in piping and equipment. Install sufficient number of elbow swings to allow for proper expansion and contraction of piping at the point of connection to mains and equipment.
4. Fabricate expansion loops with long radius welded fittings in steel piping and with long radius copper sweat fittings in copper piping.
5. Provide adequate pipe guides on each end of the expansion device to preserve alignment and pitch.
6. Install pipe hangers and supports in such a manner as to not cause an anchor condition in any direction.

S. Pipe Anchors

1. Install anchors where required to direct pipe expansion properly into expansion joints, loops or offsets and to prevent transfer of loading and stresses to connected equipment.
2. Pipe anchors may consist of heavy steel clamps bolted or welded to piping and provided with lugs and bolts for clamping and attaching anchor braces. Design anchors to restrict pipe movement and fasten to main members of building structure in most effective manner to secure desired results.
3. Do not attach supports, anchors or stays in places or in such a manner that will damage construction or integrity of the structure, either during installation, by weight of the pipe, or by expansion and contraction of the pipe.

T. Pipe Insulation Inserts and Shields

1. Refer to Section of Division 23, "HVAC Insulation" to coordinate specific insulation thicknesses and requirements. At hanger locations for insulated piping 1 ½ inches (DN 40) and larger where hanger support is outside the insulation, provide inserts of exploded silica

pipe insulation between pipe and hanger. Density and compression strength suitable for pipe size and support spacing as required by MSS SP-58, Paragraph 9 and MSS SP-69, Table 3. Provide inserts as required for smaller piping to prevent deformation of insulation. Inserts of equal thickness to adjoining insulation, provided with vapor retardant seals, and of proper length to fully support pipe at each hanger location. Manufactured by Value Engineered Products Max Span; or equivalent.

2. At all hanger locations for insulated piping where hanger is outside of insulation, provide galvanized sheet steel shield formed to fit insert/insulation, extending up to pipe centerline. Length 12-inch (300 mm) minimum when insert is not required. Where inserts are provided, length of shield 4 inches (100 mm) less than insert length. Provide shields 16 gauge (1.6 mm) for piping up to 4 inches (DN 100), 12 gauge (1.3 mm) for piping 6 inches (DN 150) and larger. Shields shall have preformed ridges on each side of hanger to prevent hanger from slipping along shield.
3. Preformed insulated pipe saddles may be used in lieu of insert and shield where appropriate. Thickness of insert or pipe saddle same thickness as pipe insulation.

3.5 WELDING, SOLDERING AND BRAZING

- A. Do not employ workers who have not been fully qualified and certified for the specified procedures.
- B. Pipe Welding, Black or Galvanized Steel Pipe: Perform all welding of black or galvanized steel pipe by shielded metallic arc method of fusion welding, in accordance with welding procedures of AWS (American Welding Society) D10.12 recommended procedures for welding low carbon steel pipe, or other approved procedure, conforming to requirements of ASME/ANSI B31.1 for high pressure steam boiler piping and B31.9 elsewhere.
- C. Pipe Welding Stainless Steel Pipe: Refer to other Sections of Division 23 for welding requirements.
- D. Structural Steel Field Welding: Comply with AWS D1.1 procedures for manual shielded metal-arc welding, appearance and quality of welds, methods used in correcting welding work, and the following:
 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 2. Obtain fusion without undercut or overlap.
 3. Remove welding flux immediately.
 4. Finish welds at exposed connections so that no roughness shows after finishing, and so that contours of welded surfaces match adjacent contours.
- E. Soldering: Comply with the requirements of the AWS Soldering Manual.
- F. Brazing: Comply with the requirements of the AWS Brazing Manual and AWS A5.8 specification for filler materials for brazing.

3.6 UNIONS

- A. In Screwed Steel Pipe, 2 inches (DN 50) and smaller: Screwed, Class 250 malleable iron, brass to iron seat, ground joint union with brass seat ring pressed into head piece. Provide galvanized unions in galvanized pipe.
- B. In Welded Steel Pipe, 2 inches (DN 50) and smaller: Class 3000 carbon steel socket welded union, steel to steel seat and ground joint. Provide stainless steel in stainless steel piping.
- C. In Copper tubing, 2 inches (DN 50) and smaller: Class 200 wrought copper, solder type, brass ground joint union.
- D. In Brass Piping, 2 inches (DN 50) and smaller: Class 250 cast bronze, screwed ends, brass ground joint unions. Provide chromium plated unions in chromium plated piping.
- E. Provide companion flanges in piping 2 ½ inch (DN 65) and larger.

3.7 PIPING CONNECTIONS

- A. Refer to other Sections of Division 23 for additional requirements.
- B. Mechanical Connections: Not permitted on any Division 23 piping.
- C. Flanged Connections: Make with nonasbestos gaskets of 1/8 inch (3 mm) thick best quality material as recommended by manufacturer for the service application. For steam piping, factory manufactured for flange/connection size/type as manufactured by Flexitallic. For other piping services either Flexitallic or gaskets factory cut for flange size as manufactured by Garlock Packing Division, Colt Industries, or equal. Align flange surfaces parallel. Install gasket concentrically positioned. Assemble joints by sequencing bolt tightening to make initial contact of flanges and gaskets as flat and parallel as possible. Use suitable lubricants on bolt threads. Tighten bolts gradually and uniformly using torque wrench.
- D. Mechanical Couplings: Prepare pipe and install in accordance with manufacturer's instructions. Standard wall steel pipe either roll or cut grooved at Contractor's option, all sizes, except provide cut grooved as required to accommodate thermal expansion and contraction. Use heavy wall steel piping for cut groove and light wall steel pipe for roll grooves.
- E. Soldered Joints: Unless noted otherwise, make with appropriate flux and solder. Clean tubing ends and fittings before assembly. For piping 2 inch (DN 50) and larger tin tubing and fittings before assembly. For tubing 2-1/2 inch (DN 65) and larger use circular flame torch for soldering. The use of lead flux or solder and finishing with 50-50 solder is prohibited.
- F. Threaded Pipe: Make full, clean-cut standard ANSI/ASME B1.20.1 taper pipe threads using sharp dies. Carefully cut, ream or file out to size or bore, removing all chips. Use Schedule 80 pipe for all screwed close and shoulder nipples. Do not use all thread nipples. Provide teflon tape or other approved nontoxic joint compound, applied to male thread only.
- G. Welding Connections: Use only factory made welding fittings, same weight as piping, on welded pipe, except that Bonney Forge WELDOLET or THREADOLET, or Allied Type 1 Branchlet

fittings, of same weight as connecting piping, may be used for branch takeoffs two or more commercial pipe sizes smaller than main. All elbows long radius.

3.8 EQUIPMENT INSTALLATION - COMMON REQUIREMENTS

- A. Install equipment to allow maximum possible headroom unless specific mounting heights are not indicated.
- B. Install equipment level and plumb, parallel and perpendicular to other building systems and components in exposed interior spaces, unless otherwise indicated.
- C. Install HVAC equipment to facilitate service, maintenance, and repair or replacement of components. Connect equipment for ease of disconnecting, with minimum interference to other installations. Extend grease fittings to accessible locations.
- D. Install equipment to allow right of way for piping installed at required slope.

3.9 PROTECTION OF WORK

- A. Clean pipe, tubing, fittings, valves, piping specialties, ductwork and equipment before installation and keep clean while the work is in progress.
- B. Securely close open ends of pipe and tubing and openings in other material and equipment until installed, during installation, and until finally connected or otherwise finished, with caps, plugs or other approved closure devices designed for such service.
- C. Protect factory finished equipment, fixtures and devices with approved temporary covering material where those items are installed so as to be subject to accidental damage or abuse. Contractor shall remove all temporary covering material at the conclusion of the work or as directed.
- D. Protect the work of other trades and property of Owner from damage and assume full responsibility for the cost of repairing or replacing any damage to such work or property caused by the performance of the work under Division 23.

3.10 CLEANING OF SYSTEMS

- A. Refer to Division 23 "HVAC Water Treatment" for additional cleaning and system flushing requirements.
- B. Following completion of system testing, thoroughly clean all piping systems by flushing with water or other approved method, or as otherwise specified. Completely remove all dirt, scale, oil, grease and other foreign substances that may have accumulated in systems during installation.
- C. Carefully wipe out, wire brush, or if necessary, sand blast sections of pipe lines between temporary or permanent strainers and equipment they are to protect. Replace all permanent strainer screens with temporary screens during cleaning process. Remove temporary screens and reinstall permanent screens after cleaning is completed.

- D. Disconnect automatic devices that can become clogged during cleaning process and do not connect permanently until cleaning process is complete.
- E. Clean all ductwork, piping and equipment of dirt, scale, plaster, concrete, splattered paint and other foreign matter.
- F. Clean all grease and cuttings from stainless steel piping and trim.
- G. Clean all strainers, dirt pockets, drip legs, traps and other accessories that may collect foreign matter.

3.11 HANGERS AND SUPPORTS

A. General

1. Refer to structural drawings for attachment details to existing structure above. Deviations from these details will not be accepted.
2. Support major piping 3 inch (DN 80) and above, ductwork, tanks and other equipment to the structure above (beams and girders) or by means of struts or brackets to columns. Do not support from floor or roof decks. Do not overload structural members to which supports are attached. Hanger spacing not to exceed MSS SP69.
 - a. Provide hangers, rollers, threaded rods, turnbuckles, deflection guides, deflection provisions, inserts, beam clamps and all miscellaneous specialties for attachment of hangers and supports to structure.
 - b. Provide all rods, angles, rails, struts, brace plates, structural steel, platforms and other items required for suspension or support of piping, ductwork, tanks and equipment.
 - c. Provide supplemental angles, channels, plates or other reinforcement where supports are required between building structural members. Size supports for weight of duct, pipe, pipe contents, equipment fittings and other items, plus a 200 pound (90 kg) live load. Attach supplemental supports in a manner that will not weaken or overload structural members. Weld steel according to AWS D-1.1.
 - d. Attach by welding, clamping, concrete inserts, drilled in mechanical type anchors (Hilti or equal) and other approved means. Adhesive type anchors are not approved.
 - e. Place grout under supports for equipment, and make a smooth bearing surface.
 - f. For seismic restraint, provide double-sided beam clamp loaded perpendicularly to beam for seismic anchor point.
3. No lead shield anchors, powder or power fasteners permitted for attachments.
4. Do not use perforated strap hangers. Do not use steel strap hangers on piping.
5. Wherever possible, support shall be provided directly to main steel or concrete framing beams. If spacing of structure exceeds spacing required to support the mechanical work, supplemental channel or unistrut framing shall be designed and provided by the Contractor.
6. Support all mechanical work independently of other trades. Under no circumstances shall work be supported or suspended from ceiling grids, piping or other supports by other trades.
7. Before drilling concrete for attachments, carefully check Drawings and Shop Drawings for such concrete and locate drilled holes to miss reinforcing by at least 1 inch.

B. Inserts

1. Use inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams wherever practicable.
2. Set inserts in position in form work in advance of concrete work. Provide reinforcement rod placed through opening on top and bent over adjacent concrete reinforcement rods.
3. Where concrete slabs form finished ceiling, finish inserts flush with slab surface.
4. Where inserts are omitted, if approved by Architect, drill through concrete slab from below and provide rod with recessed square steel plate and nut above slab or provide drilled in mechanical type anchors, Hilti or equal, after concrete is completely cured.

C. Pipe Hangers and Supports

1. Unless otherwise required to avoid overloading of structural members or for seismic restraint, support horizontal steel and copper piping as follows:

Nominal Pipe Size (inch) (DN)	(a)Maximum Distance Between Support (feet) (m)		Hanger Rod Diameter (inch) (mm)
	Steel Pipe	Copper Tubing	
up to 3/4 (DN 20)	6 (1.8)	5 (1.5)	3/8 (9.5)
1 to 2 (DN 25 to 50)	6 (1.8)	6 (1.8)	3/8 (9.5)
2-1/2 to 3-1/2 (DN 65 to 90)	10 (3.0)	8 (2.4)	1/2 (12.5)
4 & 5 (DN 100 to 125)	12 (3.5)	10 (3.0)	5/8 (15.9)
6 to 8 (DN 150 to 200)	12 (3.5)	10 (3.0)	3/4 (19)
10 to 12 (DN 250 to 300)	12 (3.5)	10 (3.0)	7/8 (22.25)
14 (DN 350) and over	12 (3.5)	-	1 (25)
Trapeze Hanger Rod			(b)

- a. Provide additional supports as required to avoid overloading of supporting structure. Reduce distance where so required by applicable codes.
 - b. As required to carry weight of trapeze channel, span of piping with contents, insulation and supports, plus a 200 pound (90 kg) live load.
2. Install hangers to provide minimum 1/2 inch (12.5 mm) clear space between finished covering and adjacent work.
 3. Install additional attachments at concentrated loads, including valves, flanges, guides, strainers and expansion joints/loops.
 4. Place a hanger within one foot of each horizontal elbow.
 5. Use hangers that are vertically adjustable 1-1/2 inch (35 mm) minimum after piping is erected.
 6. Vertical Piping Support:
 - a. Unless otherwise required to avoid overloading, of structural members or for seismic restraint, support vertical piping with clamps spaced appropriately as to type and weight of piping, minimum spacing at every other floor and below roof. Support vertical soil pipe at each floor at hub. For exposed piping in stairs and finished areas, locate clamps below floor and secure to structure below floor as required.

- b. Support vertical steel pipe at a maximum of 15 feet (4.5 m) spacing.
 - c. Support vertical copper pipe and tubing at a maximum of 10 feet (3.0 m) spacing.
7. Where several pipes can be installed in parallel and at same elevation, provide multiple or trapeze hangers. Space hangers for smallest pipe size or provide intermediate supports for smaller pipe as specified above for individual pipes.
 8. Where practical, support riser piping independently of connected horizontal piping.
 9. Support pipe runs in a manner to minimize stress in the pipe or tubing and on bodies of valves and fittings.
 10. Install hangers and supports to allow controlled movement of piping systems, permit freedom of movement between pipe anchors, and facilitate action of expansion joints, expansion loops, expansion bends, and similar units, and so that piping live and dead loading and stresses from movement will not be transmitted to connected equipment.
 11. Install hangers and supports to provide indicated pipe slopes and so that maximum pipe deflections allowed by ASME B31.9 "Building Services Piping" is not exceeded.
 12. For piping subject to sweating (e.g.: domestic cold water, chilled water, refrigerant suction, drain piping for air conditioning equipment, heat recovery piping, etc.) and for insulated piping requiring roller supports, install hangers outside insulation and provide pipe insulation protection shields as specified in this section. For all other piping, hanger may be attached to the piping before insulation is applied or may be installed outside the insulation with insulation protection shields.
 13. Do not support nonferrous piping with ferrous materials even on a temporary basis.
 14. Do not support piping or ductwork from other piping or ductwork.
 15. Install hanger rods subjected to tension only. Accomplish lateral and axial movements by proper linkage in rod assembly. Secure hanger to hanger rod with two bottom lock nuts.
- D. Duct Hanger and Supports: Refer to Section 23 30 00 "Ductwork and Ductwork Accessories".

3.12 IDENTIFICATION

- A. Identify all new and altered equipment, new and altered exposed and concealed ducts and new and altered exposed and concealed pipe with legible lettering, applied after finish painting, in a color to contrast with basic color in accordance with ANSI A13.1 and OSHA.
- B. Identify piping by name of pipe content and direction of flow near major equipment items, adjacent to valves or flanges, adjacent to gauges or thermometers, at each tee, at changes in direction, on each side of a penetration of a wall or floor, at each access door or panel and then at maximum 20 foot (6 m) centers in congested areas and 50 foot (15 m) intervals elsewhere; indicate flow direction with arrows. Identification shall be by means of laminated plastic markers (affixed with wire ties) or adhesive labels. Lettering shall not be smaller than one third of the pipe diameter and directional arrows not less than 1/2 inch (12.5 mm) wide and 12 inches (300 mm) long. The first color listed is the background color. The second color listed is the text color. "X" denotes an acceptable type of identifier.

Piping System	Colors	Acceptable Marker Type
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	Background	Text	Laminated Plastic	Vinyl Adhesive	Stencils
Low pressure steam:	Yellow	Black	X		
Low pressure steam condensate:	Yellow	Black	X		
Chilled water supply:	Green	White	X	X	
Chilled water return:	Green	White	X	X	
Cooling coil condensate (gravity):	Blue	White	X	X	
Cooling coil condensate (pumped):	Blue	White	X	X	
Heating hot water supply:	Yellow	Black	X	X	
Heating hot water return:	Yellow	Black	X	X	
Domestic water:	Green	White	X	X	
Make-up water:	Green	White	X	X	

- C. Identify equipment and operating devices such as switches, starters and similar equipment, by the equipment numbers shown on Drawings or by the Owner's numbering system, if so directed.
1. Include the type of service or the name of areas served.
 2. Lettering minimum 1 inch (25 mm) high.
 3. Nameplates shall be two tone plastic, or printed white paper enclosed in a transparent, laminated plastic case with permanently sealed edges.
 4. Attach securely to equipment, or where this is not practicable attach by brass link chains.
 5. Do not stencil surfaces exposed in public areas.
- D. Furnish for each valve, except those immediately adjacent to apparatus, a 2 inch (50 mm) diameter nonferrous metal tag with figures stamped on the tag.
1. Number tags for HVAC H-1, H-2, etc.; Use Owner's numbering system if so directed.
 2. Fasten tags to valves with nonferrous S hooks and nonferrous chains.
 3. Where valves are located above removable acoustical tile ceilings, identify the tile section below the valves by an approved color pin system.
 4. Furnish duplicate framed schedules showing the location of each valve, system or equipment it serves, manufacturer, and figure number.
- E. Identify ductwork by name of duct content and direction of flow near major equipment items, at each tee, at changes in direction, on each side of a penetration of a wall or floor, at each access door or panel and then at maximum 20 foot (6 m) centers in congested areas and 50 foot (15 m) intervals elsewhere; indicate flow direction with arrows. Identification shall be by means of adhesive labels. Lettering shall not be smaller than two inches and directional arrows not less than 1/2 inch (12.5 mm) wide and 12 inches (300 mm) long. The first color listed is the background color. The second color listed is the text color.

Ductwork System	Colors		Acceptable Marker Type		
	Background	Text	Laminated Plastic	Vinyl Adhesive	Stencils
Supply Air:	Blue	White		X	*

Return Air:	Blue	White		X	*
Exhaust Air:	Blue	White		X	*
Outside Air:	Blue	White		X	*

* White letter stenciling is required only for exposed ductwork painted black.

- F. Identify access doors to fire dampers with access panel markers or by stencils with the words "Fire Damper Access" except for dynamic fire dampers, use the words "CAUTION Dynamic Fire Damper Access". Provide approved markers to locate fire dampers concealed above ceilings.

3.13 TESTING OF PIPING SYSTEMS – COMMON REQUIREMENTS

- A. Refer to individual piping system specifications elsewhere in Division 23 for additional piping system testing requirements.
- B. Provide materials and equipment required for testing. Test and make tight all new piping systems and alterations and connections to existing piping system.
- C. Take precautions during testing to insure safety of personnel and equipment. Provide systems to be pressurized with appropriate gauges and blowouts or relief valve set at a pressure no more than one third higher than test pressure, to protect against damage by expanding liquid or other source of overpressure during test. Make good at no increase in Contract sum damage to work or work of other trades caused by failure to observe proper precautions.
- D. Test piping systems prior to application of insulation. Testing as stipulated herein shall be considered minimum, and where tests stipulated by lawful jurisdictional authorities exceed these requirements, such more stringent tests shall be performed. Tests shall be witnessed and approved by the authorities having jurisdiction over the work.
- E. Concealed work shall remain uncovered until required tests have been completed. Provide proper sectionalizing devices so that portions of a system may be tested as appropriate.
- F. Isolate and exclude from tests all in line equipment, instruments, gauge glasses, flow meters and all other devices not capable of withstanding test pressure.
- G. Use ambient temperature water as testing medium, except where otherwise specified and except where there is a risk of damage due to freezing.
- H. Apply soap solution to all joints of pneumatically tested systems while system is being subjected to test pressure.
- I. Maintain test pressures sufficient length of time to permit thorough inspection of all joints. Where leaks are observed, replace defective work or material. Caulking of screw joints or holes is not acceptable. Repeat entire test as many times as necessary, until successful completion of test with no leaks.
- J. Prepare written report of testing.

3.14 BALANCING, ADJUSTING AND PERFORMANCE TESTING

- A. Testing, adjusting and balancing of air and water systems will be provided under Division 01 "Testing, Adjusting and Balancing of HVAC Systems".
- B. Installing Contractor(s) responsible for the work specified in Division 23 shall perform all work necessary to place systems in full operation prior to start of testing, adjusting and balancing work. In addition, Installing Contractor shall perform certain additional preparatory work required for testing, adjusting and balancing as specified in various Sections of Division 23.
- C. Provide notice upon completion of all preparatory work and all initial operational testing required as part the Work. Perform additional operational testing on equipment, or systems, as directed and to extent and for duration deemed necessary, to demonstrate that systems are performing properly and delivering quantities in accordance with the requirements of the Contract Documents.
- D. Cooperate with testing, adjusting and balancing Contractor in coordination and scheduling of testing, balancing and adjusting work. Furnish approved manufacturer's technical data and shop drawings for equipment, including fan and pump performance curves.

3.15 INSTRUCTION AND DEMONSTRATION

- A. Upon completion of all work and all tests, and at a time mutually agreed on by Contractor, Architect and Owner, Installing Contractor shall operate systems in all parts and at their expense for sufficient length of time to demonstrate the mode of operation and definitively determine whether the systems as a whole are in first class working condition. Immediately correct, at no cost to Owner, any defects that may develop during this period of operation and place systems in first class working condition before being finally turned over to Owner.
- B. Provide experienced operating personnel to instruct Owner's authorized employees in the operation, adjustment and maintenance of systems and equipment installed under this Contract. Provide instructions for the period of time appropriate for the size and complexity of the system, or as requested by Owner.

3.16 MANUFACTURER'S SUPERVISIONS AND STARTUP SERVICE

- A. Include manufacturer's supervision/startup/certification and special instruction service for equipment as specified in various Sections of Division 23. Be responsible for properly making arrangements for and coordinating with the manufacturer to provide the specified work. Make any corrections/modifications to the installation as required by the manufacturer at no additional cost to Owner.
- B. The manufacturer's engineer or authorized service personnel shall check the equipment for its conformance to the Specifications, for proper installation and run the system in all modes of operation to ascertain that the unit will function properly. Make necessary adjustments to insure optimum efficiency and trouble-free service.

- C. After completion of the startup procedures, the manufacturer shall certify, in writing, that the equipment is installed in accordance with their requirements and is operating in accordance with the intent of the Specifications.

3.17 COMMISSIONING

- A. Commissioning will be provided as specified in Division 01 Section "Commissioning". All contractors and subcontractors of the various sections of this specification shall cooperate and participate in the commissioning work in accordance with requirements of Division 01 Section "Commissioning".
- B. Ensure participation of major equipment manufacturers or their representatives.
- C. Equipment and systems/subsystems installed under this section are expected to be in full compliance with the design intent by the commissioning phase. Notify the Commissioning Agent when any specific piece of equipment or specific system/subsystem is ready for commissioning. Be prepared to demonstrate system readiness.
- D. Equipment or systems/subsystems having incomplete work or exhibiting problems related to noncompliance with the design intent shall require commissioning. The contractor for this section shall be fully responsible to make all necessary corrections to incomplete or non-complying work at their own expense and shall pay the Commissioning Agent per diem rate for recommissioning such incomplete or non-complying work.

END OF SECTION 23 05 00

SECTION 23 07 00 - HVAC INSULATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. HVAC Piping Insulation
 - 2. HVAC Duct Insulation
 - 3. Adhesives, mastics, tapes
 - 4. Recovering

- B. Related Sections

- 1. Pipe insulation insert and shields are specified in Section 23 05 00, "Common Materials and Methods For HVAC".

1.3 DEFINITIONS

- A. Cold Surfaces: Normal operating temperatures less than 75o F (24° C).
- B. Density: Is expressed in pcf (pounds/cu. ft.) (kg/m3).
- C. Dual Temperature Surfaces: Normal operating temperatures that vary from hot to cold.
- D. Hot Surfaces: Normal operating temperatures of 100o F (38° C) or higher.
- E. Thermal Conductivity ("k" value): Measure of heat flow through a material at a given temperature difference; conductivity is expressed in units of (Btu x inch)/(h x sq. ft. x o F) (W/m x °C).
- F. Through Resistivity ("R" value): Represents the reciprocal of thermal conductivity ("k" value).

1.4 REFERENCES

- A. ASTM C1136 Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation.
- B. ASTM E84 Surface Burning Characteristics of Building Materials.

- C. MICA Standards: National Commercial & Industrial Insulation Standards published by the Midwest Insulation Contractors Association. Endorsed by National Insulation Contractors Association (NICA) and its regional associations.
- D. NFPA 90A Installation of Air Conditioning and Ventilating Systems.
- E. NFPA 255 Test of Surface Burning Characteristics of Building Materials.
- F. UL 723 Surface Burning Characteristics of Building Materials.

1.5 SUBMITTALS

- A. Product Data: For each type of product indicated. Include thermal conductivity, thickness, and jackets (both factory and field applied, as applicable).
- B. Product Schedule or List: Prepare a summary of products required and clearly indicate location of their intended use.
- C. Shop Drawings:
 - 1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.

1.6 QUALITY ASSURANCE

- A. Materials, job conditions and installation shall be in compliance with the applicable Building and Mechanical Codes and NFPA 90A.
- B. Installation shall be done in a workmanlike manner by skilled and experienced workers who are regularly employed in commercial/industrial insulation work, in accordance with manufacturer's recommendations and instructions and best practices of the trade.
- C. Comply with the more stringent of the requirements of this specification or the requirements of the MICA standards.
- D. Insulation materials manufacturing facilities must be certified and registered with an approved registrar for conformance with ISO 9000 quality standard.
- E. Fire Performance Characteristics
 - 1. Insulation, jacketing materials, PVC covers, tapes, adhesives, mastics, cements and finish coatings shall have a composite noncombustible fire and smoke hazard rating and label, as tested in accordance with United States Public Health Service requirements, ASTM E 84, NFPA 255 and UL 723 not exceeding Flame Spread 25 and Smoke Developed 50.
 - 2. Indoor Recovering Canvas Jackets: UL listed fabric, 6 ounce per square yard (203 g/m²), unless otherwise specified, attached with a lagging fire retardant and waterproof adhesive.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Ship insulation materials in containers marked by manufacturer with appropriate ASTM specification designation, type and grade, and maximum use temperature. Deliver materials to job site in original nonbroken factory packaging, labeled with manufacturer's density and thickness and store in a safe, dry place.
- B. No insulation material shall be installed that has become damaged in any way.
- C. Do not install, and remove from the site, any insulation material that has become wet because of transit or job site exposure to moisture or water. Remove insulation from ductwork, piping and/or equipment which has become wet. Reinsulate as required.

1.8 COORDINATION

- A. Coordinate size and location of supports, hangers, and insulation shields specified in Division 23 Section "Common Materials and Methods for HVAC."
- B. Coordinate clearance requirements with installing contractor for insulation application. Before preparing piping and ductwork Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.
- C. Coordinate installation and testing of heat tracing.

1.9 SCHEDULING

- A. Schedule insulation application after pressure testing of piping and duct systems. Insulation application may begin on segments that have satisfactory test results.
- B. Schedule insulation application on chilled water piping prior to circulating chilled water and schedule application on supply air ductwork prior to delivering conditioned air. Do not install insulation to surfaces where condensation is present.
- C. Insulation installation shall be reviewed by the project Engineer prior to the installation of the exterior jacketing. If this inspection is not performed prior to installation of the jacketing, installation shall be verified through infrared testing. Provide two weeks' notice to Engineer prior to inspection date.

1.10 ALTERNATIVES

- A. Alternative insulations are subject to Architect's approval. Alternatives shall provide, at normal conditions, thermal resistance within 5% of resistance of materials specified.
- B. Where alternative thermal conductivity ("k") differs from specified thermal conductivity by more than 5%, increase or decrease insulation thickness as follows:

$$\text{New Thickness} = \frac{\text{Actual "k"}}{\text{Specified "k"}} \times \text{Specified Thickness}$$

PART 2 - PRODUCTS

2.1 PIPING INSULATION

A. Acceptable Manufacturers

1. Johns Manville
2. Armacell
3. Owens-Corning Fiberglass
4. Knauf Insulation
5. Certain Teed

B. Insulation Type P1:

1. Piping: Fine fibrous glass insulation, with factory applied vapor barrier jacket, molded to conform to piping, "k" value at 75° F (24° C) maximum 0.23 (0.033). Johns Manville "Micro-Lok AP-T Plus" with jacket of white Kraft reinforced with fiberglass yarn and bonded to aluminum foil and having a pressure sensitive tape closure system bonded to the longitudinal lap. Cover with 6-ounce (170 g) canvas on concealed piping and 8-ounce (227 g) canvas on exposed piping.
2. Valves and Fittings:
 - a. Glass fiber insert of equal thickness to adjacent pipe insulation and premolded PVC cover, Johns Manville "Zeston" and "Hi-Lo Temp Inserts" for valves and fittings.
 - b. Factory molded fibrous glass fitting covering for fittings of equal thickness to adjacent pipe insulation. Cover with 6-ounce (170 g) canvas on concealed piping and 8-ounce (227 g) canvas on exposed piping.
 - c. Mitered sections of pipe covering for valves.

C. Insulation Type P3:

1. Piping: Foamed plastic of closed cell structure, "k" value at 75° F (24° C) maximum 0.25 (0.035). Maximum water vapor transmission rating of 0.05 perms (2.87 ng/s-m²-Pa). Insulation shall not drip or melt when exposed to flame. Polyethylene or polyolefin not allowed. Armacell "AP Armaflex", "AP Armaflex SS" and "AP Armacell W" (maximum 0.2 perms). Exposed or painted insulation shall be white. Cover with 6-ounce (170 g) canvas on concealed piping and 8-ounce (227 g) canvas on exposed piping.
2. Valves and Fitting: Mitered sections of Type P3 insulation. Factory produced or field fabricated. Cover with 6-ounce (170 g) canvas on concealed piping and 8-ounce (227 g) canvas on exposed piping.

2.2 DUCTWORK INSULATION

A. Acceptable Manufacturers

1. Johns Manville
 2. Owens Corning
 3. Knauf
 4. Certain Teed
- B. Insulation Type D1: Semirigid fibrous glass ductwork and casing insulation, density approximately 4 pcf (64 kg/m³) with factory applied reinforced aluminum foil vapor barrier, "k" value at 75° F (24° C) maximum 0.23 (0.033). Johns Manville "815 Spin-Glas" with FSK jacket. Cover all exposed ductwork with a paintable, 8-ounce (227 g) canvas jacket.
- C. Insulation Type D2: Flexible fibrous glass ductwork insulation with factory applied reinforced aluminum foil vapor barrier, density approximately 0.60 to 0.75 pcf (9.6 to 12 kg/m³), "k" value at 75° F (24° C) maximum 0.31 (0.045) Johns Manville "Microlite" with Type FSK jacket. Cover all exposed ductwork with a paintable, 8-ounce (227 g) canvas jacket.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Determine clearances required for installation of work and review such requirements with trades responsible for installing various piping systems, ducts and equipment to be insulated. Where it is determined that working clearances between equipment and material to be insulated and adjacent work will restrict or prohibit proper installation of work, immediately report such conditions to all interested parties and arrange to have affected material relocated or preinsulated before erection, as approved. Failure to so comply will not relieve Contractor of full responsibility for providing specified insulation.
- B. Do not install covering before piping, ductwork and equipment has been tested and approved, or before ductwork has been sealed.
- C. Ensure surface is clean and dry prior to installation. Ensure insulation is dry before and during application.

3.2 GENERAL INSTALLATION REQUIREMENTS

- A. Comply with applicable requirements of Division 23 Section, "Common Materials and Methods for HVAC".
- B. Adhesives and mastics materials shall be compatible with insulation material, jackets and substrates. Apply insulation and adhesives in accordance with manufacturers' instructions.
- C. Clean excess adhesive, mastic or cement used in performance of work from all exposed surfaces of insulation jacketing materials. Clean smudges and dirt from all exposed surfaces of insulation jacketing materials at conclusion of this work.
- D. Apply insulation on all cold surfaces with a continuous, unbroken vapor seal. Install hangers outside of insulation for all piping subject to sweating (e.g., chilled water supply and return, drain piping for air conditioning equipment, etc.) and provide inserts - refer to and coordinate with

Section 23 05 00, Part 3, Article titled "PIPING SYSTEMS – COMMON REQUIREMENTS", paragraph titled "Pipe Insulation Inserts and Shields". For all equipment subject to sweating, insulate and vapor seal hangers, supports, anchors, etc., that are secured directly to cold surfaces to prevent condensation.

- E. Extend all surface finishes to protect all surfaces, ends and raw edges of insulation.
- F. Install insulation on and at access doors to allow easy use of access door without damage to insulation.
- G. Finish insulation neatly at hangers, supports and other protrusions.
- H. Install insulation with least number of joints practical. Locate insulation, or cover seams, in least visible locations.
- I. Finish installation with systems at operating conditions. Repair separation of joints or cracking of insulation due to thermal movement or poor workmanship.
- J. Be responsible for proper curing of insulation, etc., in accordance with manufacturers' requirements.
- K. Repair existing insulation damaged through installation of new work to match existing insulation.

3.3 PENETRATIONS

- A. Refer to Division 23 Section "Common Materials and Methods for HVAC" for additional requirements.
- B. Insulation Installation at Non-Rated Interior Wall and Partition Penetrations: Install insulation continuously through walls and partitions.
- C. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Install insulation continuously through penetrations of fire-rated walls and partitions. Terminate insulation at fire damper sleeves for fire-rated wall and partition penetrations. Externally insulate damper sleeves to match adjacent insulation and overlap duct insulation at least 2 inches (50 mm).
 - 1. Comply with requirements in Division 07. Coordinate requirements for sleeves and clearance between finished surface of penetrating item and penetrated construction to achieve proper installation of through-penetration fire-stop system.
- D. Insulation Installation at Floor Penetrations:
 - 1. Duct: Install insulation continuously through floor penetrations that are not fire rated. For penetrations through fire-rated assemblies, terminate insulation at fire damper sleeves and externally insulate damper sleeve beyond floor to match adjacent duct insulation. Overlap damper sleeve and duct insulation at least 2 inches (50 mm).
 - 2. Pipe: Install insulation continuously through floor penetrations.
 - 3. Seal penetrations through fire-rated assemblies. Comply with requirements in Division 07.

3.4 GENERAL PIPE INSULATION INSTALLATION

- A. Requirements in this article generally apply to all insulation materials except where more specific requirements are specified in various pipe insulation material installation articles.
- B. Carry pipe insulation through sleeves and through hangers which are specified to be installed outside insulation.
- C. Insulation Installation on Fittings, Valves, Strainers, Flanges, and Unions:
 - 1. Install insulation over fittings, valves, strainers, flanges, unions, and other specialties with continuous thermal and vapor-retarder integrity, unless otherwise indicated.
 - 2. Insulate pipe elbows using preformed fitting insulation or mitered fittings made from same material and density as adjacent pipe insulation. Each piece shall be butted tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
 - 3. Insulate tee fittings with preformed fitting insulation or sectional pipe insulation of same material and thickness as used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
 - 4. Insulate valves using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
 - 5. Insulate strainers using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below ambient services, provide a design that maintains vapor barrier.
 - 6. Insulate flanges and unions using a section of oversized preformed pipe insulation. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker.
 - 7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below ambient services and a breather mastic for above ambient services. Reinforce the mastic with fabric-reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.
 - 8. Label the outside insulation jacket of each union with the word "UNION." Match size and color of pipe labels.
 - 9. Insulate flexible connections and expansion joints on cold piping with removable insulation section. Install insulation on flanges, valves and unions so that it can be removed and replaced without damaging adjacent insulation.
- D. Terminate insulation neatly and finish all exposed ends with plastic material troweled on bevel.
- E. Insulate instrument connections for thermometers, pressure gages, pressure temperature taps, test connections, flow meters, sensors, switches, and transmitters on insulated pipes, vessels, and

equipment. Shape insulation at these connections by tapering it to and around the connection with insulating cement and finish with finishing cement, mastic, and flashing sealant.

F. Install removable insulation covers at locations indicated. Installation shall conform to the following:

1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as adjoining pipe insulation.
2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union long at least two times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless-steel or aluminum bands. Select band material compatible with insulation and jacket.
3. Construct removable valve insulation covers in same manner as for flanges except divide the two-part section on the vertical center line of valve body.
4. When covers are made from block insulation, make two halves, each consisting of mitered blocks wired to stainless-steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches (50 mm) over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.

G. Insulation Type P1

1. Seal longitudinal laps of jackets and wrap butt joints with 3 inch (75 mm) wide strip of jacketing material securely sealed in place.
2. Where premolded PVC fitting covers are used, apply factory precut insulation insert in accordance with manufacturer's instructions and then apply one piece cover. Use two or more layers of inserts on hot piping as required to limit outer surface temperatures of insert(s) to 150o F (65.6° C) maximum. Use two layers of inserts on chilled water piping. On concealed hot piping, covers may be secured with staples. On exposed hot piping, secure covers by taping ends to adjacent insulation. Seal seam edges of covers on concealed and exposed cold piping with Zeston vapor-barrier adhesive and wrap edges of covers with Zeston vapor-barrier pressure sensitive color matching tape.
3. Where factory-molded fibrous glass fitting covering is used, finish insulation on concealed and exposed hot piping with same jacketing material as adjacent insulation lap-sealed and finished with Foster 30-36, Childers CP-50A or equal. Finish insulation on cold piping same as for hot piping except that sealer shall be Foster 30-35 or Childers CP-30LO vapor sealer.
4. Finish valve covering in hot and cold piping systems same as specified above for fittings.
5. Recover exposed piping in finished areas with 0.02 mil thick PVC jacket and less than 8 ft (2.5 m) above floor in mechanical room and penthouse with 0.02 mil thick PVC jacket or aluminum jacket, except use Alpha Assoc. Fiberglass Scrim Fabric "Luben 58" (20 x 10) white, adhered and then finished with Foster 30-36 or Childers CP-50A when piping systems are specified to be painted.
6. Cover all exposed piping with a paintable, 8-ounce (227 g) canvas jacket.

H. Insulation Type P3

1. Where possible, slip insulation over tubing as a full cylinder. Push insulation on to pipe. Do not pull insulation on pipe. Where necessary to longitudinally cut, and at all butt joints, tightly butt edges and join by sealing with waterproof vapor barrier adhesive, Armaflex 520. For Self-Seal type, peel adhesive paper from surface and apply firm pressure along entire longitudinal joint.
2. Cover fittings and valves with equivalent thickness of insulation material.

3.5 PIPING INSULATION SCHEDULE

Service Temp	Systems Description	Indoor			Outdoor or Unheated Space, Including Attic			
		Type	Pipe Size	Thickness	Type	Pipe Size	Thickness	Heat Trace
141-200F	Heating and Reheat Hot Water	P1	$\leq 1\ 1/4"$	1 1/2"	P1	All	2 1/2"	Yes
			$\geq 1\ 1/2"$	2"				

Note:

1. Outdoor or Unheated Space insulation thickness shall extend through wall inside the building or heated space minimum of 2 ft.
2. Where systems are required that don't exactly match the systems description listed on the schedule, provide insulation based on the appropriate service temperature range.

3.6 DUCTWORK AND CASINGS INSULATION INSTALLATION

A. Penetrations Through Construction:

1. Except at penetrations through fire or smoke rated construction, ensure that insulation is continuous through floors, walls and partitions. Where duct is enclosed in a fire rated shaft, carry insulation through penetration of intermediate floors in shaft.
2. Omit insulation on portion of duct passing through fire rated construction. Terminate insulation at each face of construction. Provide appropriate vapor seal at termination for ductwork subject to condensation.

B. Insulation Type D1:

1. Apply with edges tightly butted. Impale on pins welded to duct and secure with speed clips. Cut protruding ends of pins flush. Space pins as required to hold insulation firmly in place but not less than one pin per square foot. Seal all joints, exposed ends, speed clips and penetrations of the vapor barrier with minimum 3 inch (75 mm) wide strip of the vapor barrier material applied with Foster 85-75 or Childers CP-82 to both surfaces or with pressure sensitive tape to match facing.
2. Cover all exposed ductwork with a paintable, 8-ounce (227 g) canvas jacket.

C. Insulation Type D2

1. Cut insulation to length longer than duct perimeter to minimize compression and maximize installed "R" value as recommended by manufacturer. Allow maximum fullness at corners of rectangular ductwork and avoid excessive compression.
2. Insulation shall be firmly butted at all joints with a maximum allowable compression of 25%. Secure insulation to underside of ducts 18 inches (450 mm) or greater in width with mechanical fasteners welded to duct and speed clips spaced approximately 18 inches (450 mm) on center. Cut protruding ends of pins flush. Additionally, secure to sides of ducts 18 inches (450 mm) or greater in depth and to the two larger sides of all vertically installed ductwork with mechanical fasteners in the same fashion as for the underside of ducts 18 inches (450 mm) or greater.
3. Overlap all joints at least 2 inches (50 mm) and staple in place. Seal stapled seams, speed clips and breaks in the vapor barrier facing with a minimum 3 inch (75 mm) wide pressure sensitive tape designed for use with the duct insulation. Use pressure sensitive tape and apply additional Foster or Childers sealant in concealed spaces to provide a complete sealed thermal break installation.

3.7 DUCTWORK AND CASING INSULATION SCHEDULE

A. Minimum Thickness

1. Insulation thickness indicated hereinafter in this article and/or on the drawings is minimum thickness. Where duct is constructed with flanged, angle, standing, factory fabricated or similar joints or reinforcing, increase thickness to provide minimum 1/4-inch (6.35 mm) cover on edge of joint or reinforcing.
2. Increased thickness section shall be a minimum of 12-inch (300 mm) inches wide, centered on flange or reinforcing with exposed ends sealed as specified in this article for concealed ductwork. Increase thickness throughout entire length of duct to provide a uniform thickness and appearance for all exposed ductwork.

B. Type

1. Exposed and Concealed Indoor Rectangular Ductwork - Type D1.
2. Apparatus and Equipment Casings – Type D1
3. Indoor Round and Oval Ductwork - Type D2

C. Thickness Schedule

<u>Location</u>	<u>Thickness</u>
1. Equipment Casing	
a. Indoor	2 Inches (50 mm)
2. Ductwork Location	
a. Attic (Unconditioned)	
1) Supply Air	3 Inches (75 mm)
2) Return Air	3 Inches (75 mm)
a. Indoors	1-1/2 Inches (38 mm)

D. Listing of Ductwork and Casings to be Insulated:

1. All air conditioning system supply air, return air, outside air, and relief air ductwork, casings and plenums.
2. Fire damper, smoke damper, combination fire and smoke damper sleeve. Insulate only the portion not within the penetrated construction.
3. Casings of heating and air conditioning systems and equipment which are not factory insulated, including plenums (supply, return, mixing), filter sections, access sections, sound attenuator casings and all other areas subject to condensation, heat gain or heat loss.
4. Flexible ductwork not factory insulated.
5. Outside air ductwork, casings and plenums.
6. Reheat coil casings. Extend insulation minimum of 4 inches upstream and downstream of coil connection.
7. Supply air diffuser plenums/back pans that are not either insulated or acoustically lined at the factory.
8. Ductwork and casings shown on the Drawings to be insulated, whether or not listed above.

3.8 EXISTING INSULATION

- A. Repair existing pipe and ductwork insulation damaged through installation of the new work or alterations or connections to existing work. Match insulation material, thickness and method of application with existing.

3.9 INSULATION OF EXISTING PIPING

- A. Insulate portions of any existing system, within the project scope area, that requires insulation according to this section, and that are not insulated. This includes, but is not limited to, insulation that has been removed from systems because it contained hazardous materials. Provide new insulation in accordance with the requirements of this section.

END OF SECTION 23 07 00

SECTION 23 21 13 - HYDRONIC PIPING SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions, Division 01 Specification Sections and other related sections apply to this Section.

1.2 SUMMARY

- A. This Section includes pipe and fitting materials, joining methods, special-duty valves, and specialties for the following:
 - 1. Pipe, valves and fittings
 - 2. Strainers
 - 3. Hydronic System Specialties and Accessories

1.3 PERFORMANCE REQUIREMENTS

- A. Design working pressure and temperature for all specialties and accessories suitable for system operating temperature and pressure 125 psig (863 kPa) at 225° F (107° C) minimum.
- B. Base expansion calculations on 50° F (10° C) installation temperature to 200° F (100° C) for hot water heating, plus 30% safety factor.

1.4 SUBMITTALS

- A. Product Data: For each type of factory fabricated item indicated, include pressure ratings, construction materials, data sheets, performance characteristics, and furnished accessories.
- B. Field quality-control test reports.
- C. Grooved joint couplings and fittings shall be shown on the drawings and product submittals and shall be specifically identified with the applicable style or series designation.
- D. Shop Drawings: Prepare CAD-generated shop drawings to a scale of not less than 1/4 inch per foot. Show complete piping layout, including piping layout plans clearly and accurately indicating:
 - 1. Pipe size, service and insulation thickness.
 - 2. Elevation and pitch of pipe.
 - 3. Distance of main pipe runs from building grid lines.
 - 4. Piping accessories in pipe over 2", including but not limited to: shut-off valves, strainers, etc.

5. Building structure and beam elevation
 6. Location and service access for equipment including but not limited to: pumps, air handling units and reheat coils.
- E. Prepare pipe layout based on routing indicated on the drawings and make reasonable modifications to layout without increasing system pressure drop in order to coordinate with other trades. Refer to "Layout and Coordination with other Trades" specified in Section 23 05 00. Do not submit pipe shop drawings until multi-discipline coordination drawings specific in Division 1 are completed.
- F. Clearly identify by circle and by note "Deviation" and/or "Interference" in large lettering any and all deviations from Drawings and any and all unresolved interference conditions and assume full responsibility for failure to do so.
- G. Submit all shop drawings for review immediately after coordination drawings have been completed. Modify shop fabrication and coordination drawings as required to incorporate the Architect's review comments.
- H. A complete set of shop drawings shall be maintained at the site and any field modifications recorded on them. At the completion of the installation, the shop drawings shall be modified to reflect as-built conditions. As part of project closeout documentation, final shop drawings labeled "As-Built" and the corresponding CAD files shall be submitted to the Owner for record purposes.

1.5 MANUFACTURER'S SUPERVISION/INSPECTION SERVICE

- A. Flexible Pipe: Provide inspection services by manufacturer's representative for final installing and certify installation is in accordance with manufacturer's recommendations and that connectors are performing satisfactorily.
- B. Grooved Installations: The grooved coupling manufacturer's factory trained representative shall provide on-site training for contractor's field personnel in the use of grooving tools and installation of grooved joint products. The representative shall periodically visit the jobsite and review contractor is following best recommended practices in grooved product installation. (A distributor's representative is not considered qualified to conduct the training or jobsite visit(s).)

1.6 QUALITY ASSURANCE

- A. Steel Support Welding: Qualify processes and operators according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."
- B. Welding: Qualify processes and operators according to ASME Boiler and Pressure Vessel Code: Section IX.
1. Comply with provisions in ASME B31 Series, "Code for Pressure Piping."
 2. Certify that each welder has passed AWS qualification tests for welding processes involved and that certification is current.
- C. Gaskets shall not contain asbestos.

- D. ASME Compliance: Comply with ASME B31.9, "Building Services Piping," for materials, products, and installation. Safety valves and pressure vessels shall bear the appropriate ASME label. Fabricate and stamp air separators and expansion tanks to comply with ASME Boiler and Pressure Vessel Code: Section VIII, Division 01.
- E. All grooved joint couplings, fittings, valves and specialties shall be products of a single manufacturer. Grooving tools shall be of same manufacturer as the grooved component.

1.7 ASSET MANAGEMENT

- A. Complete a Smithsonian OFEO-Asset Update Collection form for each piece of mechanical equipment to be removed. Contractor can obtain form from COTR.
- B. Complete a Smithsonian OFEO - Asset Update/Collection form and assign a barcode for each new piece of mechanical equipment. COTR to obtain bar code numbers from the Systems Reliability Branch of SED and provide contractor with form.

PART 2 - PRODUCTS

2.1 COPPER TUBE AND FITTINGS

- A. Drawn-Temper Copper Tubing: ASTM B88, Type L (ASTM B88M, Type B).
- B. Annealed-Temper Copper Tubing: ASTM B88, Type K (ASTM B88M, Type A).
- C. DWV Copper Tubing: ASTM B306, Type DWV.
- D. Wrought-Copper Fittings: ASME B16.22 for pressure fittings, ASME B16.29 for drainage fittings.
- E. Cast Brass Fittings: ASME B16.18 for pressure fittings, ASME B16.23 for drainage fittings.
- F. Flanges for Copper Tubing: Cast Bronze, ASME B11.24, Classes 150 and 300, solder joint.

2.2 STEEL PIPE AND FITTINGS

- A. Steel Pipe: ASTM A53/A53M, black steel with plain ends; Type E, Grade B (electric resistance welded), wall thickness as indicated in Part 3 "Piping Applications" Article.
- B. Cast-Iron Threaded Fittings: ASME B16.4; Class 125 as indicated in Part 3 "Piping Applications" Article.
- C. Malleable-Iron Threaded Fittings: ASME B16.3, Class 150 as indicated in Part 3 "Piping Applications" Article.
- D. Malleable-Iron Threaded Unions: ASME B16.39; Class 250 as indicated in Part 3 "Piping Applications" Article.

- E. Unions in Welded Steel Pipes 2 Inches (DN 50) and Smaller: Class 3000 carbon steel socket welding union steel to steel seat and ground joint. Stainless steel in stainless steel piping.
- F. Welded Fittings for Black Steel Pipe 2 Inches (DN 50) and Smaller: Forged steel socket welding fittings. ASTM A105 and ASME B16.11, wall thickness to match adjoining pipe. All elbows long radius.
- G. Welding Fittings for Black Steel Pipe 2½ Inches (DN 65) and Larger: ASTM A234/A234M and ASME B16.9, wall thickness to match adjoining pipe. All elbows long radius.
- H. Cast-Iron Pipe Flanges and Flanged Fittings: ASME B16.1, Classes 25, 125, and 250; raised ground face, and bolt holes spot faced as indicated in Part 3 "Piping Applications" Article.
- I. Wrought-Steel Fittings: ASTM A234/A234M, wall thickness to match adjoining pipe.
- J. Forged-Steel Flanges and Flanged Fittings: ASME B16.5, including bolts and nuts in accordance with applicable ASME standards. Flanges to be of slip-on, weld neck, threaded or solder type to suit piping system in which installed.
- K. Flange Gaskets: Flexitallic spiral wound gasket with PTFE filler and stainless steel inner ring rated for 500° F (260° C) service temperature. Gaskets shall be 1/8 inch thick, color coded and selected for service conditions.
- L. Grooved Mechanical-Joint Fittings and Couplings:
 - 1. Acceptable Manufacturers: Subject to compliance with requirements, provide products manufactured by the following manufacturers without substitution:
 - a. Victaulic Company of America.
 - 2. Operating Conditions: -30° F (-34° C) through +250° F (121° C) temperature range according to gasket or valve lining selected and working pressure as shown in manufacturer's current product specification.
 - 3. Couplings
 - a. Cast of ductile iron conforming to ASTM A536. Alkyd enamel coating for black steel piping systems. Galvanized finish for galvanized piping systems. Couplings designed to engage and lock grooved or shouldered piping and fitting ends. Rigid joints only.
 - b. Sizes 2-1/2 inches (DN 65) through 12 inches (DN 300): Coupling housings cast with offsetting, angle pattern bolt pads shall be used to provide system rigidity and support and hanging in accordance with ANSI B31.1 and B31.9. Flange adapter; flat face for direct connection to ANSI Class 125/150 flanged components.
 - c. Gaskets composed of elastomer properties as designated by ASTM D2000. Gaskets for water service Grade "EHP" EPDM, with green color code.
 - d. Coupling Assembly: Housing clamps in two parts, single C-shaped gasket, two or more electroplated steel bolts as required to assemble housing clamps. Couplings with more than 2 parts are not acceptable.
 - e. Equal to Victaulic Styles 107 or W07, for rigid joints.

4. Fittings: Full flow type fittings with grooves designed to accept couplings of the same manufacturer. ASTM A536 cast ductile iron, ASTM A234 forged steel, or ASTM A53 carbon steel, galvanized for galvanized piping systems.
5. Where grooved mechanical joints are used, Contractor may use:
 - a. Grooved design strainers:
 - 1) T-Type 2-1/2 inches through 12 inches (DN 65 - DN 300): 300 psi (2095 kPa), ductile iron body, Type 304 stainless steel frame and removable mesh basket with No. 12 or No. 6 strainer sizes, 57% free open area. Victaulic Style 730.
 - 2) Y-Type: 300 psi (2095 kPa), ductile iron body, Type 304 stainless steel perforated removable metal baskets with 1/16-inch (1.6 mm) or 1/8-inch (3.2 mm) diameter perforations. Victaulic Style 732
 - b. As a substitute for Butterfly Valves, Victaulic Master Seal for sizes 2-1/2 inch (DN 65) through 12 inch (DN 300). Valve stem shall be offset from the disc centerline to provide full 360-degree circumferential seating. Gear operator for valves 6 inches (DN 150) and larger.
 - c. As a substitute for Spring Loaded, Lift Disc Check Valves; Victaulic Series 716 for sizes 2-1/2 inch (DN 65) through 12 inch (DN 300).
- M. Steel Pipe Nipples: ASTM A733, made of same materials as pipe in which they are installed, Schedule 80.

2.3 VALVES, GENERAL REQUIREMENTS

- A. Valves are specified by valve type. Where more than one valve type is listed for a service, use any of the listed types, unless otherwise specified or indicated, but selection must be consistent throughout the work.
- B. Bronze Valves: NPS 2 (DN 50) and smaller with solder or threaded ends, unless otherwise indicated.
- C. Ferrous Valves: NPS 2-1/2 (DN 65) and larger with flanged ends, unless otherwise indicated.
- D. Valve Pressure and Temperature Ratings: Not less than indicated and as required for system pressures and temperatures.
- E. Valve Sizes: Same as upstream pipe, unless otherwise indicated.
- F. Valve Actuators:
 1. Chainwheel: For attachment to valves, of size and mounting height, as indicated in the "Valve Installation" Article in Part 3.
 2. Handwheel: For valves other than quarter-turn types.
 3. Lever Handle: For quarter-turn valves NPS 6 (DN 150) and smaller.
- G. Extended Valve Stems: On insulated valves.

- H. Valve Flanges: ASME B16.1 for cast-iron valves, ASME B16.5 for steel valves, and ASME B16.24 for bronze valves.
- I. Valve Grooved Ends: AWWA C606.
- J. Solder Ends: With sockets according to ASME B16.18.
 - 1. Caution: Use solder with melting point below 840° F (454° C) for angle, check, gate, and globe valves. For ball valves and calibrated orifice balancing valves, manufacturers shall verify that valve construction is satisfactory for use with non-lead solders.
- K. Threaded Ends: With threads according to ASME B1.20.1.
- L. Automatic Temperature-Control Valves, Actuators and Sensors: Comply with requirements specified in Division 25 Section "Instrumentation and Control for HVAC".

2.4 COPPER-ALLOY BALL VALVES

- A. Acceptable Manufacturers
 - 1. Two-Piece, Copper-Alloy Ball Valves
 - a. Conbraco Industries, Inc.; Apollo Div.
 - b. Crane Co.; Crane Valve Group; Crane Valves
 - c. Metso; Jamesbury, Inc.
 - d. Milwaukee Valve Company
 - e. Watts Industries, Inc.; Water Products Div.
 - f. Worcester
- B. Copper-Alloy Ball Valves, General: MSS SP-110.
- C. Two-Piece, Copper-Alloy Ball Valves: Bronze body with full-port, stainless steel ball and trim; PTFE seats and packing; and 600-psig (4140-kPa) minimum CWP rating and blowout-proof stem. Dezincification-resistant bronze body, Bronze ASTM B584 Alloy C84400 (Solder) or Bronze ASTM B62 Alloy C83600 (Threaded) or Bronze ASTM B61, ball and/or trim. Forged or yellow brass bodies will not be accepted.

2.5 HIGH-PRESSURE (HIGH PERFORMANCE) BUTTERFLY VALVES

- A. Acceptable Manufacturers:
 - 1. DeZurik
 - 2. Crane Co.; Crane Valve Group; Flowseal
 - 3. Metso; Jamesbury, Inc.
 - 4. Posi-Seal
 - 5. Tyco International, Ltd.; Tyco Valves & Controls; Keystone Valves
- B. High-Pressure Butterfly Valves:

1. General: MSS SP-68. 2½ inches (DN 65) or larger for Hot or Chilled Water Service.
2. Butterfly valves to have full-lug carbon steel ASTM A216 lug body, one-piece stainless-steel stem with stainless steel bearings and stainless-steel disc manufactured in conformance with MSS SP-68, MSS SP-25, and API-609. Valve shall be capable of bi-directional dead-end service at full pressure rating of the valve with no downstream flanges required. Drive end of shaft to be designed to accept universal ISO 5211 Actuator Mounting top works. The stem journals will be a multiple seal design providing for completely independent seals. Positive stem retention to be provided using 316 stainless steel keys locked in place, to permit removal of handle or actuator while under full operating pressure. Design discs with an offset flat face to reduce dynamic torque, decrease turbulence and maximize flow capacity. Provide disc-to-shaft locking keys of stainless steel and of the tangential or compressive type. The valve seats to consist of replaceable PTFE seating surface with a stainless-steel retaining ring. Shaft and thrust bearings to be of combination permanently lubricated reinforced PTFE with 316 stainless steel. Provide packing of multiple PTFE and graphite V-ring design with adjustable stainless-steel gland follower and gland.
3. Valves 3 to 6 inches (DN 80 to DN 150) to be supplied with multi-position, lever-lock handles. Valve body to be full-lug pattern to comply with MSS-SP-68 and be compatible with ANSI 150 or 300 pattern flanges of appropriate pressure rating.

2.6 BRONZE CHECK VALVES

A. Acceptable Manufacturers:

1. Type 1, Bronze, Lift Check Valves with Metallic Disc:
 - a. Crane Co.; Crane Valve Group; Crane Valves
 - b. Milwaukee Valve Co.
2. Type 3, Bronze, Swing Check Valves with Metallic Disc:
 - a. Crane Co.; Crane Valve Group; Crane Valves
 - b. Milwaukee Valve Company

- B. Bronze Check Valves, General: MSS SP-80. Dezincification-resistant bronze body, Bronze ASTM B584 Alloy C84400 (Solder) or Bronze ASTM B62 Alloy C83600 (Threaded) or Bronze ASTM B61. Forged or yellow brass bodies will not be accepted.
- C. Type 1, Class 125, Bronze, Horizontal or Vertical Lift Check Valves: Bronze body with PTFE disc and bronze seat.
- D. Type 3, Class 125, Bronze, Swing Check Valves: Bronze body with renewable metallic disc and bronze seat.

2.7 GRAY-IRON SWING CHECK VALVES

A. Acceptable Manufacturers:

1. Type II, Gray-Iron Swing Check Valves with Composition to Metal Seats:

- a. Crane Co.; Crane Valve Group; Crane Valves.
 - b. Milwaukee Valve Co.
 - 2. Grooved-End, Ductile-Iron Swing Check Valves:
 - a. Mueller Co.
 - b. Victaulic Co. of America.
 - B. Gray-Iron Swing Check Valves, General: MSS SP-71.
 - C. Type I, Class 125, gray-iron, swing check valves with renewable metallic disc and seats.
 - D. 175-psig (1207-kPa) CWP Rating, Grooved-End, Swing Check Valves: Ductile-iron body with grooved or shouldered ends.
- 2.8 SPRING-LOADED, LIFT-DISC CHECK VALVES
- A. Acceptable Manufacturers:
 - 1. Type III, Globe Lift-Disc Check Valves:
 - a. Metraflex Co.
 - b. Milwaukee Valve Company
 - 2. Type IV, Threaded Lift-Disc Check Valves:
 - a. Metraflex Co.
 - b. Milwaukee Valve Company
 - c. Mueller Steam Specialty
 - B. Lift-Disc Check Valves, General: FCI 74-1 or MSS-SP-125 (Type III), with spring-loaded bronze, PTFE or alloy disc and bronze or alloy seat.
 - C. Type III, Class 125, Globe Lift-Disc Check Valves: Globe style with cast-iron shell and flanged ends.
 - D. Type IV, Class 125, Threaded Lift-Disc Check Valves: Threaded style with bronze shell and threaded ends.
- 2.9 BRONZE GLOBE VALVES
- A. Acceptable Manufacturers:
 - 1. Type 2, Bronze Globe Valves with Nonmetallic Disc:
 - a. Crane Co.; Crane Valve Group; Crane Valves
 - b. Milwaukee Valve Company
 - c. Powell, Wm. Co.

- B. Bronze Globe Valves, General: MSS SP-80, with ferrous-alloy handwheel.
- C. Type 2, Class 125, Bronze Globe Valves: Bronze body with PTFE disc and union-ring bonnet.

2.10 CAST-IRON GLOBE VALVES

- A. Acceptable Manufacturers:
 - 1. Type I, Cast-Iron Globe Valves with Renewable Metal Seats:
 - a. Crane Co.; Crane Valve Group; Crane Valves.
 - b. Milwaukee Valve Company.
- B. Cast-Iron Globe Valves, General: MSS SP-85.
- C. Type I, Class 125, Cast-Iron Globe Valves: Gray-iron body with renewable bronze seats.

2.11 BRONZE, CALIBRATED-ORIFICE, BALANCING VALVES

- A. Acceptable Manufacturers:
 - 1. Armstrong Pumps, Inc.
 - 2. Bell & Gossett Domestic Pump; a division of ITT Industries
 - 3. Taco
 - 4. Tour-Anderson; available through Victaulic Company
- B. Body: Bronze, ball, globe or plug type with adjustable calibrated orifice capable of pre-calculating and pre-setting.
- C. Ball: Brass or stainless steel.
- D. Plug: Resin.
- E. Seat: PTFE or Ametal copper alloy.
- F. End Connections: Threaded or solder.
- G. Pressure Gage Connections: Integral seals for portable differential pressure meter.
- H. Handle Style: Lever, with memory stop to retain set position and marked position settings.
- I. CWP Rating: Minimum 125 psig (860 kPa).
- J. Maximum Operating Temperature: 250° F (121° C).

2.12 CAST-IRON OR STEEL, CALIBRATED-ORIFICE, BALANCING VALVES

- A. Acceptable Manufacturers:

1. Armstrong Pumps, Inc.
 2. Bell & Gossett Domestic Pump; a division of ITT Industries
 3. Taco
 4. Tour & Anderson; available through Victaulic Company
- B. Body: Cast-iron or steel body, ball, plug, or globe pattern with adjustable calibrated orifice venture capable of pre-calculating and pre-setting.
- C. Ball: Brass or stainless steel.
- D. Stem Seals: EPDM O-rings.
- E. Disc: Glass and carbon-filled PTFE.
- F. Seat: PTFE.
- G. End Connections: Flanged or grooved.
- H. Pressure Gage Connections: Integral seals for portable differential pressure meter.
- I. Handle Style: Lever, with memory stop to retain set position and marked position settings.
- J. CWP Rating: Minimum 125 psig (860 kPa).
- K. Maximum Operating Temperature: 250° F (121° C).

2.13 NEEDLE VALVES

- A. Acceptable Manufacturers
1. Stockham
 2. Nibco
 3. Crane
 4. Milwaukee
- B. Bronze body and stem, working pressure to match piping system in which installed.

2.14 DRAIN AND VENT COCKS

- A. Acceptable Manufacturer
1. Ernst Gage Co. or equal.
- B. Bronze needle valve, duplex design, repackable stem, male outlet with cap. Service rating 300 psi steam pressure, 500 psi (3450 kPa) at 100° F (37.8° C) gas liquid pressure.

2.15 AUTOMATIC FLOW-CONTROL VALVES (AFC):

- A. Acceptable Manufacturers:
 - 1. Flow Design Inc.
 - 2. Griswold Controls.
- B. Design working pressure and temperature suitable for system operating temperature and pressure, 125 psig at 225° F minimum
- C. Factory calibrated, direct acting, automatic pressure compensating type valve that limits flow rates to within 5% accuracy, regardless of system pressure fluctuations. Valve control mechanism consists of a tamperproof flow element (or an externally accessible flow cartridge) with open chambers and unobstructed flow passages, with a self-cleaning, spring-loaded moving cup guided at two separate points, utilizing the full available differential pressure to actuate without hysteresis or binding. Flow element 100% 304 passivated stainless steel-plated products not acceptable. Provide valve with a metal tag, chain, showing design flow in gpm, line size, pressure drop at design flow and station identification. Include pressure taps and quick disconnect valves. Factory assembled component figurations of AFC and accessories that meet the features specified will be acceptable.
- D. End connections to suit character of piping system in which installed. Rated at 400 psig, 275° F. Brass body for pipe sizes 2 inches and smaller, steel body for pipe sizes larger than 2 inches. All internal parts of valve either hard, electrodeless nickel plated or stainless steel, control mechanism machined from solid brass.
- E. Provide portable flow measuring instrument, complete with carrying case, pressure gauge, 3-way valve, hoses and connections. Unit to be compatible with automatic flow control valve.

2.16 PRESSURE/TEMPERATURE TEST PLUGS

- A. Acceptable Manufacturers:
 - 1. Peterson Equipment Company, Inc.
 - 2. SISCO
- B. Solid brass test plug with cap and seal, Nordel inner core for up to 300° F, extension to suit insulation thickness. Test plug capable of receiving either a pressure or temperature probe 1/8 inch O.D. Furnish number of 1/8 inch gauges with adapters and 5 inch stem pocket thermometers for appropriate ranges specified in Warranty and Contract Closeout article of Part 1 of this section. Provide one Master Test Kit containing a 2-1/2-inch test gauge and two 5 inch pocket thermometers.

2.17 AIR CONTROL DEVICES

- A. Acceptable Manufacturers:
 - 1. Amtrol, Inc.

2. Armstrong Pumps, Inc.
 3. Bell & Gossett Domestic Pump; a division of ITT Industries
 4. Taco.
- B. Design working pressure and temperature for all specialties and accessories, suitable for system operating pressure and temperature, 125 psig at 225° F minimum.
- C. Air Vent
1. Manual Air Vent: Provide 1/4-inch vent cock.
 2. Automatic Air Vent: Float type with isolating valve, brass or cast iron body, copper float, stainless steel valve and valve seat. Suitable for system operating temperature and pressure, minimum 125 psig. Minimum 3/4-inch inlet for high capacity type, 1/2 inch elsewhere, 1/4 inch outlet.

2.18 STRAINERS GENERAL SERVICE

- A. Acceptable Manufacturers
1. Armstrong Machine Works
 2. Hoffman Specialty, ITT
 3. Illinois
 4. Keckley
 5. Mueller Steam Specialty
 6. Spirax-Sarco
- B. Cast iron (ASTM A126, Grade B), carbon steel (ASTM A216, Grade WCB) or brass (ASTM B62) body as required to meet system pressure and temperature requirements, screwed or flanged ends matching piping in which installed. Provide straight threaded gasket face cap with gasket for screwed end type, cast iron or hot rolled steel cover with gasket for flanged end type and bottom drain connection. Type 304 stainless steel screen unless otherwise required for application with net free area not less than three times the area of the inlet pipe. Maximum 2 psig pressure drop at design flow.
1. Size 4 inch and under, 1/32 inch (20 mesh) perforated screen.
 2. Size 5 inch and larger, 1/8-inch perforated screen.
- C. Y pattern, except basket pattern in low horizontal piping without sufficient clearance for Y pattern screen removal/replacement.

PART 3 - EXECUTION

3.1 PIPING APPLICATIONS

- A. Hot-water heating piping, aboveground, NPS 2 and smaller, shall be the following:
1. Type L (B), drawn-temper copper tubing, wrought-copper fittings, and soldered joints.

- B. Hot-water heating piping, aboveground, NPS 2-1/2 and larger, shall be any of the following:
 - 1. Schedule 40 steel pipe. Wrought-steel fittings and wrought-cast or forged-steel flanges and flange fittings and welded and flanged joints. Nipples Schedule 80.
 - 2. Schedule 40 steel pipe. Grooved, mechanical joint coupling and fittings; and grooved, mechanical joints allowed in accessible locations only. Grooved couplings not allowed within shafts, furred spaces or above inaccessible ceilings.

3.2 PIPING INSTALLATION

- A. Refer to Section 23 05 00, "Common Materials and Methods for HVAC", for general piping installation requirements.
- B. Install drains, consisting of a tee fitting, NPS 3/4 ball valve, and short NPS 3/4 (DN 20) threaded nipple with cap, at low points in piping system mains and elsewhere as required for system drainage.
- C. Reduce pipe sizes using eccentric reducer fitting installed with level side up.
- D. Install branch connections to mains using mechanically formed tee fittings in main pipe, with the branch connected to the bottom of the main pipe. For up-feed risers, connect the branch to the top of the main pipe. Bull head tees are not allowed.
- E. Install strainers on inlet side of each control valve, pressure-reducing valve, solenoid valve, in-line pump, and elsewhere as indicated. Install NPS 3/4 (DN 20) nipple and ball valve in blowdown connection of strainers NPS 2 (DN 50) and larger. Match size of strainer blowoff connection for strainers smaller than NPS 2 (DN 50).

3.3 VALVE APPLICATIONS

- A. Unless otherwise noted, use the following:
 - 1. Shutoff Service: Ball or butterfly valves.
 - 2. Throttling Service: Ball, butterfly, or globe valves.
- B. If valves with specified SWP classes or CWP ratings are not available, the same types of valves with higher SWP class or CWP ratings may be substituted.
- C. Install shutoff-duty valves at each branch connection to supply mains, and at supply connection to each piece of equipment.
- D. Install calibrated-orifice, balancing valves at each branch connection to return main.
- E. Install calibrated-orifice, balancing valves in the return pipe of each heating or cooling terminal.
- F. Heating-Water Piping: Use the following types of valves:
 - 1. Ball Valves, NPS 2 (DN 50) and Smaller: Two piece, 600-psig (4140-kPa) CWP rating, copper alloy.

2. High-Pressure (High Performance) Butterfly Valves, NPS 2-1/2 (DN 65) and Larger: Threaded Lug Type, Class 300.
3. Grooved-End, Ductile-Iron Butterfly Valves, NPS 2-1/2 (DN 65) and Larger: 175-psig (1207-kPa) CWP rating.
4. Lift Check Valves, NPS 2 (DN 50) and Smaller: Type 1, Class 125, horizontal or vertical, bronze.
5. Swing Check Valves, NPS 2 (DN 50) and Smaller: Type 3, Class 125, bronze.
6. Swing Check Valves, NPS 2-1/2 (DN 65) and Larger: Type II, Class 125, gray iron.
7. Grooved-End, Ductile-Iron, Swing Check Valves, NPS 2-1/2 (DN 65) and Larger: 175-psig (1207-kPa) CWP rating.
8. Spring-Loaded, Lift-Disc Check Valves, NPS 2 (DN 50) and Smaller: Type IV, Class 125 minimum.
9. Spring-Loaded, Lift-Disc Check Valves, NPS 2-1/2 (DN 65) and Larger: Type III, Class 125, cast iron.
10. Globe Valves, NPS 2 (DN 50) and Smaller: Type 2, Class 125, bronze.
11. Globe Valves, NPS 2-1/2 (DN 65) and Larger: Type I, Class 125, bronze-mounted cast iron.

3.4 VALVE INSTALLATION

A. General

1. Provide valves at locations shown, where specified and where required to properly control piping systems. Provide valves recommended or required by equipment manufacturers and codes for proper operation of equipment and shutoff valves to allow isolation of each main and branch service line, whether or not indicated or specified.
2. Examine valve interior for cleanliness, freedom from foreign matter, and corrosion. Remove special packing materials, such as blocks, used to prevent disc movement during shipping and handling.
3. Operate valves from fully open to fully closed positions. Examine guides and seats made accessible by such operation.
4. Examine threads on valve and mating pipe for form and cleanliness.
5. Examine mating flange faces for conditions that might cause leakage. Check bolting for proper size, length, and material. Check gasket material for proper size, material composition suitable for service, and freedom from defects and damage.
6. Do not attempt to repair defective valves; replace with new valves.
7. Install valves in horizontal piping with stem at or above the center of the pipe.
8. Install valves in a position to allow full stem movement.

B. Check Valves: Provide lift check type after globe valves, install with stem upright and plumb. Provide nonslam type in vertical piping on discharge side of pumps and elsewhere as indicated or specified. Provide horizontal swing check type elsewhere unless otherwise indicated or required for service intended, install in horizontal position with hinge pin level. Install check valves, including those that are spring loaded, so that force of gravity will operate to close valves.

C. Provide valve ends to suit character of pipe in which installed. Provide valves designed for working pressure of at least 125% of maximum operating pressure of system in which installed, but not less than 250 psig on high pressure systems, and 125 psig on low pressure systems.

3.5 HYDRONIC SPECIALTIES INSTALLATION

A. Air Vents

1. Provide as indicated or required to fully vent air from the system.
2. Provide manual type air vents for small water coils and similar equipment.
3. Provide float type, automatic air vents with valved inlet and discharge piped to floor drain for large capacity water coils, as in air handling units, for large piping mains and other major equipment in equipment rooms, and where indicated on Drawings.
4. Provide manual type air vents at high points of water circulating systems with pipe size air chamber, minimum 2 inches (DN 50). Where high points are not readily accessible, provide 1/4 inch (DN 8) piping and tubing to manual air vent and locate valve in an accessible location.
5. Route all discharges from heating system to a drain cooler prior to discharging to floor drain.

3.6 TERMINAL EQUIPMENT CONNECTIONS

- A. Sizes for supply and return piping connections shall be the same as or larger than equipment connections. Provide full size service to equipment with reducers at equipment connections.
- B. Install control valves in accessible locations close to connected equipment.

3.7 FIELD QUALITY CONTROL

- A. Prepare hydronic piping according to ASME B31.9 and as follows:
 1. Leave joints, including welds, uninsulated and exposed for examination during test.
 2. Provide temporary restraints for expansion joints that cannot sustain reactions due to test pressure. If temporary restraints are impractical, isolate expansion joints from testing.
 3. Flush hydronic piping systems with clean water; then remove and clean or replace strainer screens.
 4. Isolate equipment from piping. If a valve is used to isolate equipment, its closure shall be capable of sealing against test pressure without damage to valve. Install blinds in flanged joints to isolate equipment.
 5. Install safety valve, set at a pressure no more than one-third higher than test pressure, to protect against damage by expanding liquid or other source of overpressure during test.
- B. Grooved joints shall be installed in accordance with the manufacturer's latest published installation instructions. Grooved ends shall be clean and free from indentations, projections, and roll marks in the area from pipe end to groove. Gaskets shall be of an elastomer grade suitable for the intended service, and shall be molded and produced by the coupling manufacturer. The grooved coupling manufacturer's factory trained representative shall provide on-site training for contractor's field personnel in the use of grooving tools and installation of grooved joint products. The representative shall periodically visit the jobsite and review contractor is following best recommended practices in grooved product installation. (A distributor's representative is not considered qualified to conduct the training or jobsite visit(s).)

C. Perform the following tests on hydronic piping:

1. Use ambient temperature water as a testing medium unless there is risk of damage due to freezing. Another liquid that is safe for workers and compatible with piping may be used.
2. While filling system, use vents installed at high points of system to release air. Use drains installed at low points for complete draining of test liquid.
3. Isolate expansion tanks and determine that hydronic system is full of water.
4. Subject piping system to hydrostatic test pressure that is not less than 1.5 times the system's working pressure for a minimum of 4 hours with no drop in pressure. Test pressure shall not exceed maximum pressure for any vessel, pump, valve, or other component in system under test. Verify that stress due to pressure at bottom of vertical runs does not exceed 90 percent of specified minimum yield strength or 1.7 times "SE" value in Appendix A in ASME B31.9, "Building Services Piping."
5. At the beginning of the pressure test, and periodically during the test, examine piping, joints and connections for leakage. Eliminate leaks by tightening, repairing, or replacing components, and repeat hydrostatic test until there are no leaks.
6. Prepare written report of testing.

D. Perform the following before operating the system:

1. Open manual valves fully.
2. Inspect pumps for proper rotation.
3. Set makeup pressure-reducing valves for required system pressure.
4. Inspect air vents at high points of system and determine if all are installed and operating freely (automatic type), or bleed air completely (manual type).
5. Set temperature controls so all coils are calling for full flow.
6. Inspect and set operating temperatures of hydronic equipment, such as boilers, chillers, cooling towers, to specified values.
7. Verify lubrication of motors and bearings.

END OF SECTION 23 21 13

SECTION 23 25 00 - HVAC WATER TREATMENT

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions, Division 01 Specification Sections, and other related sections apply to this Section.

1.2 SUMMARY

- A. This Section includes the following HVAC water-treatment systems:
 - 1. Bypass chemical-feed equipment and controls
 - 2. Chemical treatment test equipment
 - 3. HVAC water-treatment chemicals
 - 4. Chemical cleaning of new systems

1.3 DEFINITIONS

- A. Complete Water Treatment Service: The Owner's personnel has no responsibility for the water treatment program.
- B. Supervised Water Treatment Service: The water treatment is conducted by Owner's personnel.
- C. MSDS: Material Safety Data Sheets.

1.4 PERFORMANCE REQUIREMENTS

- A. Water quality for HVAC systems shall minimize corrosion, scale buildup, and biological growth for optimum efficiency of HVAC equipment without creating a hazard to operating personnel or the environment.
- B. Base HVAC water treatment on quality of water available at Project site, HVAC system equipment material characteristics and functional performance characteristics, operating personnel capabilities, and requirements and guidelines of authorities having jurisdiction.
- C. Contractor shall provide equipment and services necessary and reasonably incidental to proper completion of Water Treatment Systems shown on Drawings and specified herein, except where specifically noted as being furnished under other sections of specifications.
- D. All equipment, once installed, will remain the property of the building Owner.
- E. Provide complete water treatment service for the following systems:

1. Hot water heating

1.5 SUBMITTALS

- A. Product Data: Include for each type of factory fabricated item indicated.
- B. Include name of Water Treatment Subcontractor on Material List.
- C. Informational Submittals:
 1. Water Treatment Service Program: Submit a copy of the service contract for maintenance service and written description of the service program as described in this section.
 2. Water Analysis: Submit a copy of the water analysis to illustrate water quality available at project site.
 3. Pre-Connection Testing: Upon successful completion of new system water treatment (prior to tying in any new systems to existing systems), test water quality of new and existing systems. Submit water analysis to SI COTR and Coastline Water Treatment for approval. Retreat and retest as required to achieve specified water quality as directed by SI COTR.
- D. Field quality control test reports.

1.6 WARRANTY AND CONTRACT CLOSEOUT

- A. Warranty
 1. Products: comply with warranty and contract closeout requirement specified in Division 00.
 2. Performance: Submit a written warranty, stating: conditions to be maintained and chemicals to be used; that the system of treatment prevents corrosion, scale formation and slime and algae growth; that the system of treatment is compatible with the materials in construction of component parts of system, such as pumps, pump seals, valves, piping, equipment; that the system of treatment complies with all applicable codes, rules and regulations.

1.7 SUPERVISION AND STARTUP

- A. The water treatment contractor shall perform the following startup services:
 1. Inspect field-assembled components and equipment installation, including piping and electrical connections. Report results in writing.
 2. Inspect piping and equipment to determine that systems and equipment have been cleaned, flushed and filled with water, and are fully operational before introducing chemicals for water treatment system.
 3. Test water quality and add treatment to return system to proper chemical concentrations.

1.8 QUALITY ASSURANCE

A. Acceptable Water Service Treatment Providers:

1. Coastline Water Treatment (Existing NMNH Provider)

1.9 MAINTENANCE SERVICE

A. Existing Service Contract (Coastline Water Treatment): Once all new systems have been connected to existing systems, SI COTR shall modify the existing water treatment service contract with Coastline Water Treatment to include additional equipment/systems installed under this work.

B. Maintenance service shall include the following:

1. Initial water analysis and HVAC water-treatment recommendations.
2. Startup assistance for Contractor to flush the systems, clean with detergents, and initially fill systems with required chemical treatment prior to operation.
3. Periodic field service and consultation.
4. Customer report charts and log sheets.
5. Laboratory technical analysis.
6. Analyses and reports of all chemical items concerning safety and compliance with government regulations.

1.10 REGULATORY REQUIREMENTS

A. Conform to applicable code for addition of non-potable chemicals to building mechanical systems and for delivery to public sewage systems. Conform to all applicable EPA and local agency's guidelines and regulations.

1.11 ASSET MANAGEMENT

- A. Complete a Smithsonian OFEO-Asset Update Collection form for each piece of mechanical equipment to be removed. Contractor can obtain form from COTR.
- B. Complete a Smithsonian OFEO - Asset Update/Collection form and assign a barcode for each new piece of mechanical equipment. COTR to obtain bar code numbers from the Systems Reliability Branch of SED and provide contractor with form.

PART 2 - PRODUCTS

2.1 CHEMICALS

- A. Chemicals shall be as recommended by water treatment contractor that are compatible with piping system components and connected equipment, and that can attain water quality specified.

2.2 STAINLESS STEEL PIPES AND FITTINGS

- A. Stainless Steel Tubing: Comply with ASTM A269, Type 316.
- B. Stainless Steel Fittings: Complying with ASTM A815 / A815M, Type 316, Grade WP-S.
- C. Three Piece, Full Port, Stainless Steel Ball Valves: ASTM A351, Type 316 stainless steel body; ASTM A276, Type 316 stainless steel stem and vented ball, threaded body design with adjustable stem packing, threaded ends, and 150-psig SWP and 600-psig WP rating.

PART 3 - EXECUTION

3.1 WATER ANALYSIS

- A. Recommended conditions indicated hereinafter are furnished as a guideline only. Coordinate with existing NMNH Water Treatment Provider (Coastline Water Treatment) for site specific requirements. Water Treatment Subcontractor shall obtain water sample, make a chemical analysis of same for pH, total alkalinity, hardness, total solids, chloride and silica and select/change chemicals which will be used to treat system as required for warranty in Part 1 of this section. Chromates shall not be used in Water Treatment System.

3.2 PIPING APPLICATIONS

- A. Chemical Treatment Piping: Stainless steel piping and fittings.

3.3 FIELD QUALITY CONTROL

- A. Perform tests and inspections and prepare test reports.
- B. Tests and Inspections:
 - 1. Inspect field-assembled components and equipment installation, including piping and electrical connections.
 - 2. Inspect piping and equipment to determine that systems and equipment have been cleaned, flushed, and filled with water, and are fully operational before introducing chemicals for water-treatment system.
 - 3. Do not enclose, cover, or put piping into operation until it is tested and satisfactory test results are achieved.
 - 4. Test for leaks and defects. If testing is performed in segments, submit separate report for each test, complete with diagram of portion of piping tested.
 - 5. Leave uncovered and unconcealed new, altered, extended, and replaced water piping until it has been tested and approved. Expose work that has been covered or concealed before it has been tested and approved.
 - 6. Cap and subject piping to static water pressure of 50 psig (345 kPa) above operating pressure, without exceeding pressure rating of piping system materials. Isolate test source

and allow test pressure to stand for four hours. Leaks and loss in test pressure constitute defects.

7. Repair leaks and defects with new materials and retest piping until no leaks exist.

C. Remove and replace malfunctioning units and retest as specified above.

D. Comply with ASTM D3370 and with the following standards:

1. Silica: ASTM D859.
2. Acidity and Alkalinity: ASTM D1067.
3. Iron: ASTM D1068.
4. Water Hardness: ASTM D1126.

3.4 COMPLETE WATER TREATMENT SERVICE

A. For complete water treatment service, Water Treatment Subcontractor shall:

1. Make service visits monthly to closed, recirculating systems to adjust feeding equipment, apply chemicals, obtain and analyze samples and regulate bleed-off, in order to maintain conditions as specified below.
2. Furnish all necessary labor, chemicals and feeding equipment required for the specified treatment. It shall not be necessary for the Owner to handle chemicals, take samples, make tests, regulate or operate the equipment.
3. Obtain a signed service card after each visit and leave a report indicating which systems were serviced.
4. Maintain complete records of treatment program for each system, such records to be made available, upon request, to interested parties.
5. Instruct Contractor in field on installation, piping and wiring of chemical feeding equipment.
6. Maintain chemical feeding equipment in proper working condition for term of contract.
7. Maintain specified conditions in each system.
8. Submit service reports as specified hereinafter.

B. Submit reports to Owner for each service call.

1. Include the following information as a minimum:
 - a. pH before and after treatment.
 - b. Dissolved solids - excluding water treatment compounds.
 - c. Methyl-orange alkalinity - before and after treatment.
 - d. Calcium hardness and total hardness.
 - e. Conductivity - before and after treatment.
 - f. Concentration of each water treatment material - before and after treatment.
 - g. Amount of makeup water used between visits, where water meter is provided in system.
 - h. Legionellae bacteria count per milliter and the results from the previous test.
2. Include items a, e and f on each closed system treatment.

3.5 CLOSED RECIRCULATING SYSTEM (HEATING HOT WATER)

A. Suggested Conditions:

1. For temperatures greater than 140°F.
 - a. Molybdate -100 to 150 ppm (1000 to 1500 ppm Sodium Nitrite) b. pH – 8.0 to 10.5

B. Feeding Equipment: Provide 5-gallon (19 L) bypass feeder with bag filter installed as detailed on drawings. Material and operating pressure shall be consistent with system in which installed.

3.6 CHEMICAL CLEANING OF NEW SYSTEMS

A. Piping Systems

1. The contractor shall provide chemicals and labor for the pre-operational cleaning of all piping systems installed under this Division. This cleaning method is not intended for potable water systems. The cleaning chemicals shall be provided by the Water Treatment Contractor.
2. Contractor shall provide a portable, 20-micron filter with pump for initial system flush. Coordinate procedure and duration with water treatment contractor.
3. Do not use system pumps for cleaning and flushing without prior approval from Owner. In the event system pumps are used, Contractor shall be responsible for extending the warranty period to include duration of cleaning and flushing.
4. The contractor shall flush all systems, including mud from drop legs. Remove, clean and replace all strainers. All systems shall contain the highest quality of water available.
5. Complete circulation must be achieved during the cleaning procedure. A minimum flow rate of 3 ft/sec. needs to be maintained to ensure that the cleaning chemicals will work properly. All manual, electrical, air and thermostatic operated valves must be open. All dead-end runs must be looped together with piping not less than 1/3 the size of the run. This piping is to remain in place until cleaning is complete.
6. A minimum of 1-1/2" ball valve is to be permanently installed in the low point of each system for the purpose of draining each system.
7. The cleaning solution shall be formulated to remove light grease, cutting oils, loose mill scale, organics and extraneous construction debris. The cleaner shall contain corrosion inhibitor, a dispersant and an oil emulsifier. The cleaner shall be as recommended by the water treatment contractor. Enough cleaner should be used to treat all of the piping to remove oil and grease to permit a uniform passivating film to form. This aids in the prevention of flash corrosion when the system is most vulnerable to corrosive attack. The mechanical contractor will provide the water treatment vendor with an estimate of system volume for proper dosage of cleaner. The cleaner shall not require external heat to ensure its effectiveness.
8. Add recommended quantity of passivation chemical directly into the closed loop system before the recirculating pumps to ensure rapid mixing and distribution throughout the system.
9. Recirculate the system for 48 - 72 hours.
10. Open and drain mud legs and low points periodically during the cleaning process.
11. Drain system completely paying particular attention to mud from drop legs and all low points.

12. Refill the system with clean, potable water, check all strainers, re-circulate and drain completely.
 13. Refill the system again. The length of time between the completion of the cleaning procedure and addition of the corrosion inhibitor shall not exceed twenty-four (24) hours.
 14. Add the recommended level of closed loop inhibitor. The system is now ready for operation. The product must also contain an inert tracer that is colorless in the system for use in closed loop leak detection and volume studies.
 15. A service report will be generated on-site by the water treatment representative certifying that the system has been cleaned in accordance with the above procedures. Submit a copy of the service report to the Architect and COTR for record.
 16. Chemical test parameters shall include onsite testing of pH, conductivity, phosphate, total iron, ferrous iron, and copper after the preclean is completed to verify all cleaner is removed and iron and copper levels are low. Total iron must be < 5 ppm, Ferrous Iron <2 ppm, and Copper < 0.5 ppm.
- B. Where new work is an extension of or alteration to the existing system(s), provide shutoff valves to isolate new work from existing, temporary circulating pump and temporary or permanent connections to new work as required for proper circulation and chemical cleaning of the new work. Ensure that new piping is completely flushed, cleaned, and tested prior to opening isolation valves to the main system.

END OF SECTION 23 25 00

SECTION 23 30 00 – DUCTWORK AND DUCTWORK ACCESSORIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Sheet metal ductwork and fittings
 - 2. Sheet metal casings and plenums
 - 3. Duct Accessories; including dampers, flexible connectors, flexible duct, access doors
 - 4. Diffusers, registers and grilles
- B. Related Sections include the following:
 - 1. Duct mounted smoke detectors are specified in Division 28
 - 2. Airflow measuring stations are specified in Division 25
 - 3. Motorized control dampers are specified in Division 25

1.3 DEFINITIONS

- A. Duct Sizes: Inside clear dimensions. For rectangular ductwork in plan, first dimension indicates width and second dimension indicates height.
- B. Pressure Class: The numerical duct construction pressure classification that identifies permissible SMACNA Duct Construction Standards.
- C. Seal Class: Identifies the extent of sealing of duct joints, seams and penetrations per SMACNA Duct Construction Standards.
- D. Leakage Class: Identifies permissible leakage as described in SMACNA HVAC Air Duct Leakage Test Manual.
- E. Low Pressure Ductwork: 3-inch wg (750 Pa) and less.
- F. Medium Pressure Ductwork: 4 inch wg (1000 Pa) and 6 inch wg(1500 Pa).
- G. High Pressure Ductwork: Greater than 6-inch wg (1500 Pa).

1.4 REFERENCES

- A. SMACNA: HVAC Duct Construction Standards – Metal and Flexible
- B. SMACNA: HVAC Air Duct Leakage Manual
- C. ARI 880: Air Volume Terminals
- D. UL: Applicable Standards

1.5 SUBMITTALS

- A. Product Data:
 - 1. Provide booklet of Shop Standards, including the following:
 - a. Duct construction classification and fabrication, assembly, and installation details including SMACNA Tables and Figure numbers clearly marked to identify which are to be used.
 - b. Details of shop fabricated items, fittings, reinforcing details and spacing, seam and joint construction.
 - c. Installation details of duct mounted equipment and accessories, including dampers, coils, access doors, hangers and supports.
 - d. Product data for factory fabricated equipment, dampers, access doors, flexible duct, etc.
 - 2. For each type of product indicated, include performance characteristics, rated capacities, data sheets, and furnished accessories.
 - 3. Shop Standards shall be submitted for review prior to submission of sheet metal shop drawings. Any sheet metal shop drawings submitted prior to the shop standard review will be returned as “Rejected”.
- B. Product Schedule or Lists: Include Diffuser, Register and Grille Schedule, indicating Drawing designation, room location, quantity, model number, size and accessories furnished.
- C. Shop Drawings: Prepare CAD-generated shop fabrication drawings to a scale of not less than 3/8 inch per foot. Show complete ductwork and casing layout, including:
 - 1. Duct layout indicating sizes and pressure classes.
 - 2. Elevations of top and bottom of ducts.
 - 3. Dimensions of main duct runs from building grid lines.
 - 4. Fittings.
 - 5. Clearly show all duct accessories, including access doors, dampers, diffusers and grilles.
 - 6. Acoustical lining and thickness as applicable.
 - 7. Prepare duct layout based on routing indicated on the drawings and make reasonable modifications to layout without increasing duct system pressure drop in order to coordinate with other trades. Refer to “Layout and Coordination with other Trades” specified in Section 23 05 00. Do not submit duct shop drawings until multi-discipline coordination drawings specific in Division 1 are completed.

8. Clearly identify by circle and by note "Deviation" and/or "Interference" in large lettering any and all deviations from Drawings and any and all unresolved interference conditions and assume full responsibility for failure to do so.
9. Submit all shop drawings for review.
10. Modify shop fabrication drawings in accordance with Architect's review comments, if any, and to show any subsequent shop or field changes. At completion of work, submit final shop fabrication drawings labeled "As-Built" to the Owner for record purposes.

D. Field quality-control test reports.

1.6 QUALITY ASSURANCE

- A. Welding: Qualify procedures and personnel according to AWS D1.1, "Structural Welding Code-Steel," for hangers and supports and AWS D9.1, "Sheet Metal Welding Code," for duct joint and seam welding.
- B. NFPA Compliance:
 1. NFPA 90A, "Installation of Air Conditioning and Ventilating Systems."
 2. NFPA 90B, "Installation of Warm Air Heating and Air Conditioning Systems."

1.7 COORDINATION

- A. Coordinate location of duct access doors to allow proper access to dampers, coils, fans, etc. Coordinate with General Contractor proper location of wall and ceiling access panels to permit access to duct access doors.
- B. Coordinate location of duct mounted equipment (coils, humidifiers, filters, smoke detectors) furnished in other Sections for installation under this Section. Provide duct transitions as required.

1.8 ASSET MANAGEMENT

- A. Complete a Smithsonian OFEO-Asset Update Collection form for each piece of mechanical equipment to be removed. Contractor can obtain form from COTR.
- B. Complete a Smithsonian OFEO – Asset Update/Collection form and assign a barcode for each new piece of mechanical equipment. COTR to obtain bar code numbers from the Systems Reliability Branch of SED and provide contractor with form.

PART 2 - PRODUCTS

2.1 SHEET METAL MATERIALS

- A. Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for acceptable materials, material thicknesses, and duct construction methods, unless otherwise

indicated. Sheet metal materials shall be free of pitting, seam marks, roller marks, stains, discolorations, and other imperfections.

- B. Galvanized Sheet Steel: Lock-forming quality; zinc coating each side, complying with ASTM A 653/A 653M and having G90 (Z275) coating designation; ducts shall have mill- phosphatized finish for surfaces exposed to view.
- C. Reinforcements: Structural shapes (channels, angles, and plates) shall be of galvanized steel where installed on galvanized ducts, and Type 304 stainless steel where installed on stainless steel ducts.

2.2 SEALANT MATERIALS

A. Acceptable Manufacturers:

- 1. United McGill Corp.
- 2. H.B. Fuller Co., Foster Products Division
- 3. Precision Adhesives
- 4. Carlisle Hardcast
- 5. General Electric Co.

B. General:

- 1. The term "sealant" is not limited to materials of adhesive or mastic nature but includes tapes and combinations of open-weave fabric strips and mastics.
- 2. Sealants for air distribution systems shall be of liquid and/or mastic type in accordance with SMACNA.

C. All sealants shall be UL Classified and meet NFPA 90A, Class 1 requirements when applied in a manner consistent with its intended use. Ratings shall not exceed a Flame Spread of 25 or a Smoke Developed of 50. All containers and shipping cartons shall bear the UL label indicating flame and smoke ratings and shall include Fire Hazard Classification. Labeling shall also include Hazard Statement required by the Consumers Product Safety Act, CFR Title 16, Chapter II, subchapter C, Federal Hazardous Substances Act Regulations, Part 1500, Section 1-272.

D. Sealants for air distribution systems shall be compatible with the materials, application and operating temperatures of the system. Sealants used for systems handling fumes and chemicals shall be confirmed suitable for the specific application. Sealants exposed to the weather, shall be ultraviolet light and ozone resistant and provide watertight seal.

E. Sealants shall be applied in accordance with manufacturer's instructions. Provide adequate ventilation and follow safety procedures as required. Adequate drying/curing time shall be allowed before operating or testing the systems.

F. Tape Sealing System: Woven-fiber tape impregnated with gypsum mineral compound and modified acrylic/silicone activator to react exothermically with tape to form hard, durable, airtight seal.

- G. Water-Based Joint and Seam Sealant: Flexible, adhesive sealant, resistant to UV light when cured, UL 723 listed, and complying with NFPA requirements for Class 1 ducts.
- H. Flanged Joint Mastic: One-part, acid-curing, silicone, elastomeric joint sealant complying with ASTM C 920, Type S, Grade NS, Class 25, Use O.
- I. Flange Gaskets: Butyl rubber or EPDM polymer with polyisobutylene plasticizer.

2.3 DUCT PRESSURE CLASSIFICATION AND CONSTRUCTION

- A. Construct ducts for pressure class indicated (positive or negative), and seal all seams and joints to achieve Seal Class A, Leakage Class 6 for rectangular duct and Class 3 for round duct, according to the following:
 - 1. Supply Ducts between AHU and Fire/Smoke Damper: 6-inch wg (1500 Pa) positive.
 - 2. Supply Ducts downstream of Fire/Smoke Damper: 4-inch wg (1000 Pa) positive.
 - 3. Temporary Supply Ducts: 2-inch wg (250 Pa) positive.
 - 4. Outside Air Ducts: 2-inch wg (500 Pa) negative.
 - 5. Return Duct between Return Fan and AHU: 4-inch wg (1000 Pa) positive.
 - 6. Return Ducts upstream of Return Fan: 4-inch wg (1000 Pa) negative.
 - 7. Temporary Return Ducts: 2-inch wg (500 Pa) negative.
 - 8. General Exhaust Duct System: 2-inch wg (500 Pa) negative.
- B. Material: All ducts shall be galvanized steel.

2.4 GENERAL DUCT FABRICATION

- A. Detail and fabricate with the fewest possible joints in accordance with SMACNA standards and details, except where more stringent requirements are specified in this section, to keep resistance losses to a minimum.
- B. Size round ducts installed in place of rectangular ducts, and vice versa, from ASHRAE table of equivalent rectangular and round ducts. Aspect ratio of rectangular ducts can be modified for coordination of layout, but in no case exceed 4 to 1, and without reducing free area of duct or increasing pressure drop.
- C. Complete metal ducts within themselves with no single partition between ducts. Open corners are not acceptable.
- D. Lap metal ducts in direction of air flow. Hammer down edges and slips to leave smooth duct interior.
- E. Sleeves:
 - 1. Construct sleeves of galvanized steel minimum 22 gauge unless noted otherwise.
 - 2. Provide sleeves for fire dampers and combination fire/smoke dampers as specified in this Section "Fire and Smoke Dampers".

- a. Breakaway connections are not permitted except for pressure class 2-inch wg (500 Pa) and less.
- F. Square heel and throat elbows, with vanes, are used on Drawings for drafting convenience only. Where space allows, construct tees, bends, and elbows with radius of not less than 1-1/2 times width of duct on centerline. Where not possible on rectangular ductwork, provide smooth radius elbows with full length splitter vanes designed and constructed in accordance with SMACNA Duct Design and Duct Construction Standard to produce a loss coefficient of 0.15 or less, except as follows:
1. Where R/W is such that a vaned radius elbow will not produce the specified loss coefficient and on exposed ductwork in finished areas, provide mitered elbows with square heel and throat and with double thickness turning vanes set into vane runners with a radius of 4-1/2 inches on 3-1/4 inch centers, designed and constructed to produce a loss coefficient of 0.26 or less in accordance with above standards.
 2. Increase/decrease duct sizes gradually, not exceeding 15° divergence/ convergence wherever possible. Maximum divergence upstream of equipment to be 30° and 45° convergence downstream.
- G. Unless otherwise indicated or specified, fabricate branch connections as follows:
1. Rectangular duct branch connection to rectangular ductwork, 45° SMACNA entry/exit butt flange boot with corner filler pieces or proportional splits at Contractor's option. Where size of main duct is not large enough for a boot connection, fabricate ducts with proportional splits.
 2. Round duct branch connections to round ductwork, conical tee.
 3. Round branch connection to rectangular ductwork, bellmouth connection equal to Buckley
 4. BM or BMD or high efficiency boot connection equal to Buckley 3300 or 3300D.
 5. Where round duct is connected to rectangular duct and rectangular duct width shown is not equal to or larger than overall diameter of connecting end of round connector BENDING OF FLANGES OF ROUND CONNECTOR FOR CONNECTION TO RECTANGULAR DUCT IS NOT ALLOWED. Either increase rectangular duct width as required or provide a rectangular 45° SMACNA boot connection to the rectangular duct and a rectangular to round conversion for the round duct connection. For round connection to low pressure ductwork. Contractor may, at his option use a mini bellmouth connection equal to Buckley M-BM or M-BMD or a flat oval bellmouth equal to Buckley FOBM or FOBMD and an oval to round conversion for the round duct connection.
 6. Spin-in, cinch lock or dovetail fittings not allowed except at duct connection to a transfer duct.
 7. Connect branch to top, bottom or side of duct as indicated or as required, whether or not indicated, to suit surrounding conditions and avoid interferences.
 8. Coordinate branch take-off locations with transverse joint spacing. Transverse joints interfering with branch take-offs will not be accepted. Modify joint spacing as required without exceeding maximum spacing.
- H. Rigidly construct ducts with joints mechanically tight, substantially airtight without use of tape, braced and stiffened so as not to breathe, rattle, vibrate, or sag. Caulk duct joints and connections with sealant as ducts are being assembled. Where joints are not accessible for sealing, provide access doors and seal from inside.

- I. Provide easements where low pressure ductwork conflicts with piping and structure. Where easements exceed 10% duct area, split into two ducts maintaining original duct area. No easements or penetrations allowed for medium or high-pressure ductwork.

2.5 RECTANGULAR DUCT FABRICATION

- A. Fabricate ducts, elbows, transitions, offsets, branch connections, and other construction according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" and complying with requirements for metal thickness, reinforcing types and intervals, tie-rod applications, and joint types and intervals.
 1. Construct ducts of minimum 24 gauge.
 2. Do not install tie-rods in areas where system effects apply including upstream and downstream of fans and sound attenuators or within three times the published absorption distance of humidifiers.
 3. Lengths: Fabricate rectangular ducts in lengths appropriate to reinforcement and rigidity class required for pressure class.
 4. Deflection: Duct systems shall not exceed deflection limits according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible."
- A. Transverse Joints: Prefabricated slide-on joints with four bolted flange, fastened to duct section with spot welds (do not use screws), and components constructed using manufacturer's guidelines for material thickness, reinforcement size and spacing, and joint reinforcement. Ductmate Industries, Inc., Nexus Inc., Ward Industries, Inc.
- B. Formed-On Flanges: Construct according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," Figure 1-4, using corner, bolt, cleat, and gasket details. Ductmate Industries, Inc. TDC or Lockformer.
 1. Use joint reinforcing for T-24 or T-25 joints and sheet metal gauges as recommended by SMACNA Standards as a minimum. Do not use lighter gauges shown in joint manufacturer's literature.
- C. Longitudinal Seams: Pittsburgh lock sealed with noncuring polymer sealant. Button punch snaplocks and pocket locks are not permitted.
- D. Cross Breaking or Cross Beading: For duct pressure class 3 inch (750 Pa) or less, cross break or cross bead duct sides 18 inches (440 mm) and larger and 20 gauge thick or less, with more than 10 sq. ft. (0.93 sq. m) of nonbraced panel area unless ducts are lined.

2.6 ROUND DUCT AND FITTING FABRICATION

- A. Acceptable Manufacturer: Factory fabricated or shop fabricated but equal in all respects to factory fabricated items specified herein:
 1. McGill AirFlow Corporation.
 2. SEMCO Incorporated.

- B. Round Duct and Fittings: Lock type spiral seam construction of galvanized steel according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible Stamped elbows and fittings. Gored (segmented) elbows and fitting only for sizes where stamped elbows and fittings not available. Adjustable elbows are not permitted.
- C. Duct Joints:
 - 1. Ducts up to 20 Inches (500 mm) in Diameter: Interior, center-beaded slip coupling, sealed before and after fastening, attached with sheet metal screws.
 - 2. Round Ducts: Prefabricated connection system consisting of double-lipped, EPDM rubber gasket. Manufacture ducts according to connection system manufacturer's tolerances. Ductmate Industries, Inc, Lindab, Inc. or equal.
- D. 90-Degree Tees and Laterals and Conical Tees: Fabricate to comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," with metal thicknesses specified for longitudinal-seam straight ducts.
- E. Diverging-Flow Fittings: Fabricate with reduced entrance to branch taps and with no excess material projecting from fitting onto branch tap entrance.
- F. Mitered-Elbow Radius and Number of Pieces: Welded construction complying with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," unless otherwise indicated.

2.7 FLEXIBLE DUCTWORK

- A. Acceptable Manufacturers: Type numbers indicated are those of Thermaflex.
 - 1. Thermaflex
 - 2. Flexmaster, Inc.
- B. UL listed under UL-181 as Class 1 Air Duct Connector and conforming to NFPA 90A and 90B.
- C. Minimum Rating: 10-inch WG positive all uses, 2-inch WG negative for return or exhaust use, velocity of 4000 feet per minute.
- D. Type "M-KC". Continuous galvanized spring wire helix having a cover of woven fiberglass fabric, vinyl impregnated and coated, or continuous corrugated aluminum for low pressure supply use only, and insulated with 1 inch thick fibrous glass insulation having outer moisture barrier consisting of reinforced metalized Mylar/ neoprene laminate with integral attaching devices. "U" factor at 75° F differential maximum 0.22 btu/sq.ft./° F/hour.
- E. Attachment: Duct clamp stainless steel band with cadmium-plated hex screw to tighten band with a worm-gear action in size to suit duct size.

2.8 HANGERS AND SUPPORTS

- A. Hanger Materials: Galvanized sheet steel or threaded steel rod.

1. Strap and Rod Sizes: Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for steel sheet width and thickness and for steel rod diameters. Perforated strap hangers are not allowed. Wire hanging systems are not permitted.
 2. Galvanized-steel straps attached to aluminum ducts shall have contact surfaces painted with zinc-chromate primer.
- B. Duct Attachments: Sheet metal screws, blind rivets, or self-tapping metal screws; compatible with duct materials. Hanger fasteners shall not pierce medium/high pressure (greater than 3 inches w.g.) ductwork under any circumstance.
- C. Trapeze and Riser Supports: Steel shapes complying with ASTM A 36/A 36M.
1. Supports for Galvanized-Steel Ducts: Galvanized-steel shapes and plates.

2.9 FLEXIBLE CONNECTORS

- A. Acceptable Manufacturers
1. Ductmate Industries, Inc.
 2. Duro Dyne Corp.
 3. Ventfabrics, Inc.
 4. Ward Industries, Inc.
- B. General Description: Flame-retardant or noncombustible fabrics, coatings, and adhesives complying with UL 181, Class 1.
- C. Metal-Edged Connectors: Factory fabricated with a fabric strip 5-3/4 inches (146 mm) wide attached to two strips of 2-3/4-inch- (70-mm-) wide, 0.028-inch- (0.7-mm-) thick, galvanized sheet steel or stainless steel, or 0.032-inch- (0.8-mm-) thick aluminum sheets. Select metal compatible with connected duct system. Comply with SMACNA requirements.
- D. Indoor System, Flexible Connector Fabric: Glass fabric double coated with neoprene.
1. Minimum Weight: 26 oz./sq. yd. (880 g/sq. m).
 2. Tensile Strength: 480 lbf/inch (84 N/mm) in the warp and 360 lbf/inch (63 N/mm) in the filling.
 3. Service Temperature: Minus 40 to plus 200 deg F (Minus 40 to plus 93 deg C).

2.10 VOLUME DAMPERS

- A. General: Factory fabricated, or shop fabricated but equal in all aspects to factory fabricated items, with required hardware and accessories in accordance with SMACNA duct standards except as noted herein.
- B. Damper and damper frames shall be constructed of same material as duct in which they are installed. Fabricate single blade damper of minimum 20 gauge. Fabricate multi-blade damper of minimum 16 gauge.

- C. Damper blades in rectangular ductwork shall be maximum 8 inches wide. Dampers having two or more blades shall be opposed action type with connecting bar and linkage. Multiple blade dampers shall be mounted in frames. Splitter dampers not allowed.
- D. Provide for multiple section damper (larger sizes), appropriately sized jackshaft with bearing assemblies mounted on supports at each mullion and at each end of the multiple damper section. Provide appropriate length and number of mountings to connect linkage of each damper to the jackshaft.
- E. Stiffen blades for stability. Include locking device to hold single-blade dampers in a fixed position without vibration. Device shall include indication of damper position. Provide end bearings and gaskets for damper components to seal duct consistent with pressure class.
- F. Damper Hardware: Dampers shall be provided with all hardware including zinc plated, die-cast core with dial and handle made of 3/32-inch (2.4-mm) thick zinc plated steel, and a 3/4 inch (19-mm) hexagon locking nut. Include center hole to suit damper operating rod size. Include raised quadrant for mounting on insulated duct. Improvised shop or field fabricated hardware will not be accepted.

2.11 FIRE AND SMOKE DAMPERS

- A. Acceptable Manufacturers:
 - 1. Ruskin
 - 2. Greenheck
 - 3. Pottorff.
- B. General: All dampers operable at pressures and velocities required by systems in which installed. Provide single or multiple framed assemblies as required for size of duct in which installed in order to maintain UL listing of entire damper assembly.
- C. Fire Damper (FD):
 - 1. General: Fire dampers shall be UL labeled, constructed in accordance with UL 555.
 - 2. Rating: 1 1/2 hour for use in rated walls 2 hours and less, otherwise 3 hour.
 - 3. Link: Provide replaceable fusible link rated for 165 deg F (74 deg C).
 - 4. Materials: Galvanized steel. Provide rectangular to round adapter when installed in round duct.
 - 5. Sleeve: Factory installed as part of damper assembly furnished by manufacturer, or field fabricated and installed. Construct sleeves for rigid connection to adjoining ductwork in accordance with UL 555 as follows:
 - a. Dampers less than 24 inches, 16 gauge
 - b. Dampers 24 inches and greater, 14 gauge
 - c. Use heavier gauges if required by SMACNA construction standards for pressure class indicated.
 - 6. Type:

- a. Fire dampers for transfer air application shall be shutter type for vertical mounting only.
- b. Fire dampers for other than air transfer, and other than listed below, shall be curtain blade type with closure spring to assure positive closure for either horizontal or vertical mounting with "fans on". Blades retained in a recess out of the air stream, including multiple section applications.
- c. Where UL sizing limitations will not permit the use of a curtain blade damper, or elsewhere as indicated, shall be multi-blade type spring loaded to assure closure for either horizontal or vertical mounting with "fans on". Blades shall be single skin 16 gauge with airfoil profile. Damper linkage rod extended to outside of frame. Bearings shall be stainless steel sleeve in an extruded hole.

D. Combination Fire/Smoke Damper (FD/SD):

1. General: Combination fire and smoke damper shall be UL labeled, for 1 ½ hour fire rating.
2. Materials: Galvanized steel. Factory installed sleeve as part of damper assembly.
3. Provide manually resettable electric heat sensing device rated for 165 deg F (74 deg C).
4. Parallel airfoil blade type with 115-volt, single phase electric motor. Two
5. position motor operation fail closed position upon loss of power.
6. Leakage Class: Class II
7. Include factory-mounted blade position switch. Switch shall contain two individual contacts, one to indicate when damper is open and one to indicate when damper is closed.

2.12 ACCESS DOORS

- A. General: Factory fabricated double wall access doors in accordance with SMACNA standards except as specified herein. Improvised shop or field fabricated access doors will not be permitted.
- B. Fabricate doors airtight and suitable for duct pressure class. Access doors shall be double wall constructed of the same material as the ductwork in which they are installed except that doors located in ducts constructed of plastic, FRP and PVC coated steel shall be constructed of Type 304 stainless steel.
- C. Rectangular Duct: Provide door with closed cell full sealing gaskets and quick turn fastening locking device. Provide hinged doors with butt or piano hinge and 2 cam latches. For medium and high-pressure duct, hingeless with minimum 4 cam latches and factory installed retaining cable.
- D. Round Duct: Provide double wall insulated and gasketed door with minimum 2 compression latches.
- E. For insulated ductwork, fabricate double wall with insulation fill and thickness not less than adjacent duct insulation. Provide raised hinge type.
- F. Access door size 16 inches by 20 inches unless otherwise indicated. Where size of duct will not accommodate this size, provide size as large as possible, minimum 12 inch by 6 inch. Provide view window where indicated.

2.13 INSTRUMENT TEST HOLES

- A. Cast iron or cast aluminum to suit duct material, including screw cap and gasket and a flat mounting gasket. Size to allow insertion of pitot tube and other testing instruments and provide in length to suit duct insulation thickness. Ventlock 699 by Ventfabrics, or approved equal.

PART 3 - EXECUTION

3.1 DUCT INSTALLATION

- A. Construct and install ducts according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," unless otherwise indicated, for the specified duct pressure classification with the fewest possible joints.
- B. Install round ducts in lengths not less than 12 feet (3.7 m) unless interrupted by fittings.
- C. Install fabricated fittings for changes in directions, size, and shape and for connections.
- D. Install couplings tight to duct wall surface with a minimum of projections into duct. Secure couplings with sheet metal screws. Install screws at intervals of 12 inches (300 mm), with a minimum of 3 screws in each coupling.
- E. Install ducts, unless otherwise indicated, vertically and horizontally and parallel and perpendicular to building lines; avoid diagonal runs.
- F. Install ducts close to walls, overhead construction, columns, and other structural and permanent enclosure elements of building.
- G. On ductwork subject to internal condensation, pitch horizontal runs towards equipment, or source of moisture, minimum 1/8 inch per foot slope.
- H. Install ducts with a clearance of 1 inch (25 mm), plus allowance for insulation thickness.
- I. Conceal ducts from view in finished spaces. Do not encase horizontal runs in solid partitions unless specifically indicated.
- J. Coordinate layout with suspended ceiling, fire- and smoke-control dampers, lighting layouts, and similar finished work.
- K. Non-Fire-Rated Partition Penetrations: Where ducts pass through interior partitions and exterior walls and are exposed to view, conceal spaces between construction openings and ducts or duct insulation with sheet metal flanges of same metal thickness as ducts. Overlap openings on 4 sides by at least 1-1/2 inches (38 mm).
- L. Fire-Rated Partition Penetrations: Where ducts pass through interior partitions and exterior walls, install appropriately rated fire dampers, sleeves, and firestopping sealant. Firestopping materials and installation methods are specified in Division 07.

- M. Install ducts with hangers and braces in accordance with Division 23 "Vibration Isolation for HVAC Systems".

3.2 SEAM AND JOINT SEALING

- A. Seal all duct seams and joints according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for duct pressure class indicated, and SMACNA's "HVAC Leakage Test Standards" for leakage and seal class indicated.
 - 1. For pressure classes 2-inch wg (500 Pa) and lower, seal transverse joints.
- B. Seal ducts before external insulation is applied. Apply sealant to male end connectors before insertion, and afterward to cover entire joint and sheet metal screws.

3.3 FLEXIBLE DUCT INSTALLATION

- A. Use only where shown on the drawings, installed in accordance with SMACNA's "HVAC Duct Construction Standards" except as specified herein.
- B. Do not install flexible duct on return or exhaust ductwork unless specifically noted on drawings.
- C. Minimum length 48 inches (1.3 m) and maximum length of 72 inches (2 m). Install runs with minimum centerline radius of bends not less than twice duct diameter and with no more offsets than an equivalent 90° elbow. Provide band strap hangers with saddle supports under flexible duct run to keep supports from sagging and kinking. Stretch duct enough to smooth out internal corrugations.
- D. Connect both ends to collar of rigid ductwork and air delivery device with sheet metal screws in addition to an approved stainless steel worm gear draw band. Plastic tie straps are not permitted.

3.4 FLEXIBLE CONNECTORS INSTALLATION

- A. Provide flexible connections immediately adjacent to fans, and externally isolated air handling units, to isolate and prevent transmission of vibration to ductwork and casings.
- B. Connections shall be made with a 3-inch (75 mm) space between duct and equipment collars, installed in line, and with 1 1/2 inch (38 mm) excess material folded so as not to interfere with airflow through connection. Flexible connectors are not permitted as a means for correcting misalignment.
- C. Do not install flexible connectors on internally isolated air-handling units.

3.5 VOLUME DAMPERS INSTALLATION

- A. Provide volume dampers at all points on low pressure supply, return and exhaust systems where branches are taken from larger ducts as required for air balancing. Do not install volume dampers upstream of air volume control boxes in any circumstance.
- B. Install volume damper as close to main as possible, minimum 2 duct widths from branch takeoff.
- C. Install damper in acoustically lined ducts in such a manner to avoid damage to liner and to avoid erosion of duct liner.
- D. Install orange ribbon tag tied on to volume damper handle for the purpose of visibly identifying volume damper locations.

3.6 FIRE AND SMOKE DAMPER INSTALLATION

- A. Install dampers complete with required perimeter mounting angles and sleeves, in accordance with installation instructions furnished by manufacturer, conforming to installation required of UL as a condition of listing.
- B. Install in such a manner to be accessible for testing and reset. Coordinate size and location access door. Increase duct size to accommodate minimum door size of 12 inch (300 mm) by 12 inch (300 mm) as applicable.
- C. Furnish additional fusible links, one fusible link for every five dampers installed, to Owner.

3.7 ACCESS DOORS INSTALLATION

- A. Install duct access doors to allow inspecting, adjusting and cleaning duct mounted accessories where shown on drawings and as follows:
 - 1. Upstream and downstream of coils, fans, control dampers, airflow measuring devices.
 - 2. Adjacent to fire and fire/smoke dampers, providing adequate access to reset or reinstall fusible link.
 - 3. Adjacent to fire/smoke dampers, providing adequate access to damper actuator/motor.
- B. Where access doors are required within shaft enclosures or above inaccessible ceilings, coordinate with General Contractor and advise proper location of access panel.
- C. Identify all fire damper access doors by stenciling with bright red paint the words "FIRE DAMPER ACCESS". For spring closure dynamic curtain type fire dampers use the words "CAUTION – DYNAMIC FIRE DAMPER ACCESS".

3.8 INSTRUMENT TEST HOLES INSTALLATION

- A. Install instrument test holes where shown on the drawings and in the following locations:

1. Air handling unit supply duct mains; provide multiple test ports properly spaced for traverse airflow measurement. Coordinate location with TAB Contractor.
2. Return fan duct mains; provide multiple test ports properly spaced for traverse airflow measurement. Coordinate with TAB Contractor.
3. Exhaust fan duct mains.
4. Downstream of duct mounted coils.

3.9 DUCT MOUNTED SMOKE DETECTORS INSTALLATION

- A. Install duct mounted smoke detectors furnished by Division 28. Obtain installation instructions and install in accordance with manufacturer's instructions. Mount on ductwork at locations indicated on the drawings.

3.10 DUCT HANGERS AND SUPPORTS INSTALLATION

- A. Suspend and support ductwork, casings, and equipment in accordance with requirements of SMACNA Duct Construction Standards and as specified further in Sections 23 05 00 and 23 05 40, except as specified herein.
- B. No hangers shall be attached to or suspended from any type of metal deck. Attachments made beneath metal deck areas may be made by positive fastening to building structural members (excluding bridging and bracing) in conjunction with the use of miscellaneous auxiliary structural steel. All such steel shall conform to ASTM A36. Attachments relying on friction are not permitted.
- C. Whenever the distance between the top of duct and overhead supporting member is greater than 36 inches, the hangers shall be of structural angles or channels and shall include cross bracing as required to prevent sway. Do not use straps or rods.
- D. Support horizontal ducts within 2 feet (60 mm) of each elbow and within 4 feet (1.3 m) of each branch intersection. Support vertical ducts at a maximum interval of 12 feet (4 m) and at each floor.
- E. Hangers must be arranged to allow the duct insulation to pass through without insulating any part of the hanger.
- F. Unless combined pipe/duct racks are shown on the drawings, provide dedicated hangers for support of ductwork.

3.11 CLEANING AND PROTECTION

- A. Thoroughly clean all air stream surfaces of all equipment, devices and accessories in the air distribution systems and all air stream surfaces of ductwork, casings and plenums and maintain in a clean condition as the work progresses. Clean ductwork piece by piece, section by section as installed.

- B. Protect openings of all equipment, devices and accessories with polyethylene film or another covering to prevent entrance of moisture, dust or debris until final ductwork/casing/plenum connections are made. Similarly protect openings in ductwork, casings and plenums.
- C. After installation, either force air at high velocity through systems or use high power vacuum machines to remove accumulated dust. Protect equipment which may be harmed by dirt with filters, or bypass during cleaning. Provide adequate access into ductwork/casings/plenums for cleaning.
- D. Clean external surfaces of all of the above of foreign substances which might cause corrosive deterioration of metal or, where to be painted, might interfere with painting or cause paint deterioration.
- E. Wipe clean all air terminal units from dust entrained on face during construction.

3.12 FIELD QUALITY CONTROL

- A. Perform duct leakage testing and inspections according to SMACNA's "HVAC Air Duct Leakage Test Manual" and prepare test reports.
- B. Test all high and medium pressure ductwork during installation and before application of any exterior insulation or enclosing of ductwork.
- C. Test all low-pressure ductwork during installation and before application of any exterior insulation or enclosing of ductwork.
- D. Disassemble, reassemble, and seal segments of systems to accommodate leakage testing and for compliance with test requirements.
- E. Conduct tests at static pressures equal to maximum design pressure of system or section being tested. Do not pressurize systems above maximum design operating pressure.
- F. Conduct tests in the presence of TAB Contractor. Give seven days' advance notice for testing.
- G. Maximum Allowable Leakage: Total leakage for pressure class shall not exceed permissible leakage for specified seal and leakage class.
- H. Remake leaking joints and retest until leakage is equal to or less than maximum allowable.

END OF SECTION 23 30 00

SECTION 23 80 00 - HEAT TRANSFER

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes:
 - 1. Radiant ceiling panels
- B. Related Sections:
 - 1. Reheat coils in air volume control boxes are specified with air volume control boxes in Division 23 "Ductwork and Ductwork Accessories".

1.3 SUBMITTALS

- A. Product Data: Include rated capacities, operating characteristics, furnished specialties, and accessories for each type of product indicated.
- B. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 1. Wiring Diagrams: Power, signal, and control wiring.
- C. Color Samples for Initial Selection: For units with factory-applied color finishes.
- D. Operation and Maintenance Data: For equipment include in operation, and maintenance manuals.

1.4 QUALITY ASSURANCE

- A. Product Options: Drawings indicate size, profiles, performance, and dimensional requirements of equipment and are based on the specific equipment indicated. Refer to Division 01 Section "Product Requirements."
- B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

1.5 ASSET MANAGEMENT

- A. Complete a Smithsonian OFEO-Asset Update Collection form for each piece of mechanical equipment to be removed. Contractor can obtain form from COTR.
- B. Complete a Smithsonian OFEO – Asset Update/Collection form and assign a barcode for each new piece of mechanical equipment. COTR to obtain bar code numbers from the Systems Reliability Branch of SED and provide contractor with form.

PART 2 - PRODUCTS

2.1 ELECTRIC RADIANT CEILING PANEL

- A. Acceptable Manufacturers:
 - 1. Qmark
 - 2. Aztec International
 - 3. Airtex Corp
 - 4. TVI Energy Corp
- B. UL listed, voltage as scheduled. White textured surface suitable to stand alone or be field painted.
- C. Design for installation in ceiling system shown on the Architectural Drawings. Include all necessary trim.
- D. Panel installed in toilet rooms and shower rooms shall be constructed as recommended by manufacturer for high moisture area installation.
- E. Provide line voltage wall mounted room thermostat.

PART 3 - EXECUTION

3.1 FIELD QUALITY CONTROL

- A. Perform the following field tests and inspections and prepare test reports:
 - 1. Operational Test: After electrical circuitry has been energized, operate electric coils to confirm proper unit operation.
 - 2. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

END OF SECTION 23 80 00

SECTION 25 09 00 - INSTRUMENTATION AND CONTROLS FOR HVAC

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.
- B. Refer to Division 23 for general provisions, installation requirements and additional HVAC equipment control information.
- C. All electrical work shall be in accordance with Division 26 Specification Sections.
- D. Related Sections include the following:
 - 1. Refer to Division 1 Section "TESTING, ADJUSTING, AND BALANCING FOR HVAC" for additional work related to system testing and balancing.
 - 2. Refer to Division 1 Section "COMMISSIONING" for work associated with HVAC system commissioning.
 - 3. Refer to Division 23 Section "Hydronic Piping Systems" for water flow measuring devices that relates to this Section.

1.2 SUMMARY

- A. This Section includes all labor, materials, equipment, and service necessary for a complete and operating control system for all HVAC equipment including control of units not supplied with factory-wired controls and installation and wiring of loose controls shipped with equipment.
- B. All new HVAC equipment to be provided with Direct Digital Controls (DDC) monitored/controlled thru the new Building Automation System (BAS).
- C. Furnish all labor, materials, equipment, and service necessary for a complete and operating Building Automation System (BAS), utilizing a high speed peer to peer network of interoperable Direct Digital Controls (DDC), Graphical User Interface (GUI) with color graphic displays with at least 64 licenses, and electronic interfaces and actuation devices, as shown on the drawings and as described herein.
- D. The new BAS system shall be fully compatible with the existing campus control system. All new controls shall be fully accessible through the existing operator's terminals. Division 25 contractor is responsible for determining compatibility prior to submitting bid
- E. Owner presently has an existing Siemens Building Automation System as part of past projects. The intent of this specification is to extend and interoperate with this system and to provide a peer-to-peer, networked control system for the control work that is part of this project. All components, software and operation shall be interoperable with the existing building automation system. The installed system will interface directly with the existing system. The existing

software and database will be modified to accept the new equipment being installed under this project to maintain integrity for centralized scheduling, trending, programming and alarming. PC Desktop icons that “link” to a separate system are not acceptable. Any costs associated with connecting to the existing energy management system, including licensed software, programming, training etc., shall be part of the controls contractor’s bid. The contractor must demonstrate their ability to perform the integration to the existing systems prior to submittal acceptance. All systems as described in the sequence of operation will be shown via dynamic graphics with all pertinent system alarms for proper operation and maintenance. The use of separate PC workstations, gateways, metalinks, replacement of existing controllers and control devices and additional software graphic packages to accomplish this integration will not be accepted.

- F. Provide surge and over-voltage protection of all electronic controllers serving HVAC equipment. This shall include protection of all controllers provided with equipment where this protection is not factory installed.

1.3 DEFINITIONS

- A. BAS: Building Automation System
- B. Control Contractor: Contractor for this section
- C. DDC: Direct digital control
- D. I/O: Input/output
- E. MS/TP: Master slave/token passing
- F. NIST: National Institute of Standards and Technology
- G. PC: Personal computer
- H. PID: Proportional plus integral plus derivative
- I. OWS: Operator Work Station
- J. RTD: Resistance temperature detector

1.4 SYSTEM PERFORMANCE

- A. Comply with the following performance requirements:
 - 1. Graphic Display: Display graphic with minimum 20 dynamic points with current data within 10 seconds.
 - 2. Graphic Refresh: Update graphic with minimum 20 dynamic points with current data within 8 seconds.
 - 3. Object Command: Reaction time of less than two seconds between operator command of a binary object and device reaction.

4. Object Scan: Transmit change of state and change of analog values to control units or work station within six seconds.
5. Alarm Response Time: Annunciate alarm at work station within 45 seconds. Multiple work stations must receive alarms within five seconds of each other.
6. Program Execution Frequency: Run capability of applications as often as five seconds, but selected consistent with mechanical process under control.
7. Performance: Programmable controllers shall execute DDC PID control loops, and scan and update process values or update changes and outputs at least once per second.
8. Reporting Accuracy and Stability of Control: Report values and maintain measured variables within minimum tolerances as follows:
 - a. Water Temperature: Plus or minus 1 deg F (0.5 deg C)
 - b. Water Flow: Plus or minus 5 percent of full scale
 - c. Water Pressure: Plus or minus 2 percent of full scale
 - d. Space Temperature: Plus or minus 0.5 deg F (0.3 deg C)
 - e. Ducted Air Temperature: Plus or minus 0.5 deg F (0.3 deg C)
 - f. Outside Air Temperature: Plus or minus 0.5 deg F (0.3 deg C)
 - g. Dew Point Temperature: Plus or minus 3 deg F (1.5 deg C)
 - h. Temperature Differential: Plus or minus 0.25 deg F (0.15 deg C)
 - i. Space Relative Humidity: Plus or minus 1.7 percent (up to 90% RH), Plus or minus 2.5% (above 90% RH)
 - j. Ducted Relative Humidity: Plus or minus 1.0 percent (up to 90% RH), Plus or minus 1.7% (above 90% RH)
 - k. Airflow (Pressurized Spaces): Plus or minus 3 percent of full scale
 - l. Airflow (Measuring Stations): Plus or minus 2 percent of full scale
 - m. Airflow (Terminal): Plus or minus 10 percent of full scale
 - n. Air Pressure (Space): Plus or minus 0.01-inch wg (2.5 Pa)
 - o. Air Pressure (Ducts): Plus or minus 0.1-inch wg (25 Pa)
 - p. Carbon Dioxide: Plus or minus 50 ppm
 - q. Electrical: Plus or minus 5 percent of reading

1.5 SUBMITTALS

- A. The Controls contractor shall provide a complete set of DDC Drawings as part of shop drawings; these shop drawings shall show all logic, set points and control schedules. A complete points list with range and scale for electrical and engineering units shall be provided as part of the submittal as well.
- B. During the submittal process, full point names must be submitted. All point names must adhere to SI point naming conventions for both points and panels. Naming convention list will be provided by SI upon request.
- C. Product Data: Include manufacturer's technical literature for each control device. Indicate dimensions, capacities, performance characteristics, electrical characteristics, finishes for materials, and installation and startup instructions for each type of product indicated.
 1. DDC System Hardware: Bill of materials of equipment indicating quantity, manufacturer, and model number. Include technical data for interface equipment, control units,

- transducers/transmitters, sensors, actuators, valves, relays/switches, control panels, and operator interface equipment.
 - 2. Control System Software: Include technical data for operating system software, operator interface, color graphics, and other third-party applications.
 - 3. Controlled Systems: Instrumentation list with element name, type of device, manufacturer, model number, and product data. Include written description of sequence of operation including schematic diagram.
- D. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
- 1. Bill of materials of equipment indicating quantity, manufacturer, and model number.
 - 2. Schematic flow diagrams showing fans, pumps, coils, dampers, valves, and control devices. Contractor's detailed installation drawings will not be accepted in lieu of schematic flow diagrams.
 - 3. Wiring Diagrams: Power, signal, and control wiring. All wiring diagrams shall reference new BAS point names.
 - 4. Written description of sequence of operation.
 - 5. Schedule of dampers including size, leakage, and flow characteristics.
 - 6. Schedule of valves including flow characteristics.
 - 7. All data sheets shall indicate accessories and options included.
 - 8. DDC System Hardware:
 - a. Wiring diagrams for control units with termination numbers. All wiring diagrams shall reference new BAS point names.
 - b. Schematic diagrams and floor plans for field sensors and control hardware.
 - c. Schematic diagrams for control, communication, and power wiring, showing trunk data conductors and wiring between operator work station and control unit locations.
 - 9. Control System Software: List of color graphics indicating monitored systems, data (connected and calculated) point addresses, output schedule, and operator notations.
 - 10. Controlled Systems:
 - a. Schematic diagrams of each controlled system with control points labeled and control elements graphically shown, with wiring.
 - b. Written description of sequence of operation including schematic diagram.
 - c. Points list.
- E. Data Communications Protocol Certificates: Certify that each proposed DDC controller complies with ASHRAE 135.
- F. Software and Firmware Operational Documentation: Include the following:
- 1. Software operating and upgrade manuals.
 - 2. Program Software Backup: On a magnetic media or compact disc, complete with data files.
 - 3. Device address list.
 - 4. Printout of software application and graphic screens.
 - 5. Software license required by and installed for DDC workstations and control systems.

- G. Software Upgrade Kit: For Owner to use in modifying software to suit future systems revisions or monitoring and control revisions.
- H. Qualification Data: For Installer and manufacturer.
- I. Field quality-control test reports.
- J. Graphics: Submit graphics package for approval by SI COTR.
- K. At project closeout, the DDC contractor shall submit as-built drawings with operations and maintenance manuals in .pdf format. The contractor shall also import the as-built drawings into the FMCS database for accessibility to end users.
- L. Operation and Maintenance Data: For HVAC instrumentation and control system to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Operation and Maintenance Data," include the following:
 - 1. Maintenance instructions and lists of spare parts for each type of control device.
 - 2. Index sheet, listing contents in alphabetical order.
 - 3. Manufacturer's equipment parts list of all functional components of the system, Auto-CAD disk of system schematics, including wiring diagrams.
 - 4. Description of Sequence of Operations.
 - 5. As-built interconnection wiring diagrams.
 - 6. Operator's manuals.
 - 7. Trunk cable schematic showing all remote electronic panel locations, and all trunk data wiring runs.
 - 8. All commissioning documentation specified herein.
 - 9. Copies of all graphic screens.
 - 10. Keyboard illustrations and step-by-step procedures indexed for each operator function.
 - 11. Inspection period, cleaning methods, cleaning materials recommended, and calibration tolerances.
 - 12. Calibration records and list of set points.
 - 13. All new graphics shall match the existing screens and be reviewed with the SI COTR.

1.6 WARRANTY

- A. Provide all services, materials and equipment necessary for the successful operation of the entire system for a period of one year after acceptance. The adjustment, required testing, and repair of the system includes all computer equipment, transmission equipment and all sensors and control devices. Provide cost for second year.
- B. The on-line support services shall allow the system supplier to dial out over telephone lines to monitor and control the facility's building automation system. This remote connection to the facility shall be within 2 hours of the time that the problem is reported. This coverage shall be extended to include normal business hours, after business hours, weekends and holidays.
- C. If the problem cannot be resolved on-line by the local office, the national office of the building automation system manufacturer shall have the same capabilities for remote connection to the facility. If the problem cannot be resolved with on-line support services, the system supplier shall

dispatch the appropriate personnel to the job site to resolve the problem within 3 hours of the time that the problem is reported.

1.7 QUALITY ASSURANCE

A. Installer Qualifications

1. Building automation system manufacturer's authorized representative who is trained and approved for installation of system components required for this Project.
2. The system shall be installed, commissioned, and serviced by manufacturer employed, factory trained personnel.
3. Installer of control system shall have a branch office within 50 miles of the site with technical staff, spare parts inventory and necessary test and diagnostic equipment.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

C. Comply with ASHRAE 135 for DDC system controllers.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Factory-Mounted Components: Where control devices specified in this Section are indicated to be factory mounted on equipment, arrange for shipping of control devices to equipment manufacturer.

B. System Software: Update to latest version of software at Project completion.

1.9 COORDINATION

A. The location of all thermostats, humidistats, exposed control (and monitoring) sensors, and control panels shall be reviewed with, and approved by, SI COTR prior to rough-in.

B. Coordinate location of control panels, dampers, valves, and devices such that clearance can be maintained for proper access to all components.

C. Coordinate equipment and wiring with Division 26 requirements to achieve compatibility of communication interfaces, drives, motor starters and annunciation devices.

D. Coordinate equipment with Division 28 to achieve compatibility with equipment that interfaces with Fire Alarm system.

E. Coordinate and assist Testing, Adjusting and Balancing (TAB) Contractor with proper set up and operation of HVAC Systems.

F. The minimum quantity of DDC/ATC panels are located on the contract documents. Provide additional panels as required. All panel locations must be approved by the SI COTR and Architect

and coordinated with all trades prior to installation. If approval and/or coordination are not completed, then panels shall be relocated at no cost to owner.

- G. Do not locate DDC panels above ceilings. Panels shall be located in mechanical rooms or in equipment systems rooms.
- H. Automatic temperature control valves and thermowells furnished by Control Contractor shall be installed by Division 23 Contractor under the supervision of Control Contractor.
- I. All automatic control dampers shall be furnished by Control Contractor and shall be installed by Contractor for Section 23 30 00 under supervision of Control Contractor except where dampers are specified to be provided by unit manufacturer.
- J. Combination fire/smoke and smoke dampers in ducts with electric motors will be provided by Contractor for Section 23 30 00. Division 26 Contractor shall provide power wiring and Control Contractor shall wire to the Building Automation System.
- K. Smoke detectors in ducts and at air handling units shall be wired into the Fire Alarm System by Division 28. Required power for those smoke detectors shall also be provided by Division 28. Provide wiring from smoke detectors/interface modules to respective air handling unit(s) and fan(s) for shutdown in the event of smoke conditions. Contractor for Section 23 30 00 will install detectors in ductwork where shown or where required.
- L. Furnish Contractors for Sections 01 91 00 and 01 91 13 approved temperature control technical data and shop drawings, information relating to changes or revisions in work, and all other information required for proper balancing, adjusting and commissioning of systems.
- M. Furnish duct mounted airflow measuring stations and static probes to Contractor for Section 23 30 00 for installation.
- N. Through-penetration fire-stop systems at penetrations through floors or fire or smoke rated walls and partitions will be provided by Division 07. Control Contractor shall be responsible to coordinate quantity and locations of all penetrations.
- O. Provide all power wiring and devices required for electric/electronic motors.
- P. Provide air volume (constant or variable) box controls. Furnish control valve for field installation and provide all additional wiring and tubing as required for a complete installation.
- Q. All deviations from specifications shall be documented separately. Obtain approval for deviations prior to fabrication or installation. Include all costs, including delays to other trades, to remedy deviations in fabrication, installation or other issues. All issues shall be reviewed.
- R. All mechanical equipment sent with loose controls shall be mounted and wired by Division 25.
- S. Control Contractor shall provide all conduit and accessories, etc. required for power and control wiring to his devices.
- T. Control Contractor shall include time in his bid required to meet with specialty equipment manufacturers (air-handlers, etc.) and program their equipment to fully interface with the BAS.

- U. Control Contractor shall interlock fans or pumps through hard wiring where indicated on contract documents; software interlocks shall not be acceptable.
- V. Provide communication cards or ports to interface with equipment such as drives, etc. as required by the contract documents.
- W. Electronic communication with drives shall be used for monitoring. System functions such as start/stop and status shall be hard wiring for reliability.
- X. All control panels require 120V Emergency Power and an integral UPS.

1.10 ASSET MANAGEMENT

- A. Complete a Smithsonian OFEO-Asset Update Collection form for each piece of mechanical equipment to be removed. Contractor can obtain form from COTR.
- B. Complete a Smithsonian OFEO – Asset Update/Collection form and assign a barcode for each new piece of mechanical equipment. COTR to obtain bar code numbers from the Systems Reliability Branch of SED and provide contractor with form.

PART 2 - PRODUCTS

2.1 CONTROL SYSTEM

- A. Acceptable Manufacturers: Subject to compliance with requirements, provide system by one of the following:
 - 1. Basis of Design Installer/Manufacturer: Siemens, Beltsville, MD Branch Office
- B. Provide a direct digital control (DDC) system consisting of sensors, indicators, actuators, final control elements, interface equipment, other apparatus, and accessories to control mechanical systems and to perform functions as specified.
- C. Control system shall consist of sensors, indicators, actuators, final control elements, interface equipment, other apparatus, accessories, and software connected to distributed controllers operating in multiuser, multitasking environment on token-passing network and programmed to control mechanical systems. An operator workstation permits interface with the network via dynamic color graphics with each mechanical system, building floor plan, and control device depicted by point-and-click graphics.
- D. Provide extension of existing control system consisting of sensors, controllers, indicating devices, interface equipment, other apparatus, and accessories to operate mechanical equipment and to perform functions as specified. Update all existing data bases and incorporate new graphics.
- E. Provide all materials and field work necessary for a complete system.
- F. Provide an electric motor for each damper and valve to be controlled, unless one is specified elsewhere.

- G. Unless specified otherwise, provide fully modulating components.
- H. Unless specified otherwise, provide proportional/integral/inverse derivative components for variable air volume controls, proportional/integral components for air handling unit discharge control, and fully proportional/integral components elsewhere.
- I. Motors that respond to incremental "pulse" signals or do not fail to the specified position shall not be acceptable.
- J. Provide all electrical wiring, communication cabling, relays or other devices for interlocking of equipment as described in Sequence of Operations or as shown on drawings.
- K. DDC system shall be capable of operating in environmental conditions of 30 deg F to 120 F and 10% RH to 90% RH noncondensing. Sensors and final control elements shall be capable of operating in environment in which they are installed.
- L. Graphics: New global graphics to provide a user-friendly interface to the new and existing detail graphics. Provide an overall riser diagram page which will allow instant access to new floor plan graphic pages, individual air handling units and central plants. An individual floor plan graphic will be provided for each floor of the building. The floor plan will show air handling zone layout and provide a link to the associated air handler graphic within each zone. Each space temperature available on the DDC system shall be interactively displayed on the floor plan. Provide sub-area graphics as required to fit all temperatures.
- M. All DDC panels shall include a minimum 10% spare capacity of each point type (AI, AO, DI, DO).

2.2 DDC CONTROLLER INTEGRATION TO EXISTING BAS NETWORK

- A. It shall be the responsibility of the Controls contractor to coordinate all work associated with the FMCS connection. The new DDC controls shall be connected/integrated to the FMCS over the SI.EDU WAN via Ethernet connections.
- B. Prior to physically connecting the new DDC Controllers to the existing system, the BAS contractor shall print a predefined report listing any points that are failed, in alarm, or overridden in the system. Once the integration is complete, verify through the same reports that no additional existing points are failed, in alarm, or overridden.
- C. Once the new DDC controllers are commissioned, the BAS contractor shall assist the Owner's rep to make the physical connection to the existing network. The DDC system shall be connected to the FMCS database server within the HDC Herndon Data center for full database management via the Siemens software. All new controls shall be fully accessible through the existing site-wide operator's terminals and FMCS Database Server.
- D. It shall be possible to access any new DDC panel through the network. Such access shall include full read-write capability from an operational as well as programming standpoint. Total system information shall be available to the FMCS Database Server at all new or existing operators' workstations.

- E. The control system shall be set up and configured in such a way that the communications between BLN/ALN is eliminated. Cross trunk service will not be utilized.
- F. The control system shall be set up and configured in such a way that the communications between panels is eliminated or kept to an absolute minimum.
- G. Once the tie-in is complete, the BAS contractor shall confirm communication with the server.
- H. Upload all data to the server.
- I. Verify there are no new failed existing points on the system. If so, take corrective action to resolve discrepancies.
- J. Create graphics that represent the new systems, including but not limited to AHU layouts, navigation, screens, and room graphics.
- K. Map all alarmable points into the existing remote notification software installed on the server.

2.3 ETHERNET COMMUNICATIONS AND EQUIPMENT (SI.EDU Network)

- A. The DDC Controllers shall communicate via TCP/IP over Ethernet. SI shall furnish and install cables, switches, and signal repeaters to connect the new project to the existing IP network.
- B. The BAS contractor shall coordinate node names, IP addresses, access privileges and system configuration with COTR prior to startup.
- C. SI shall provide a modular, 8-pin, Category 6 information outlet at all DDC controllers and shall terminate the cable inside the field panel at the information outlet. A patch cable shall be provided by the BAS contractor to connect the field panel to the information outlet.

2.4 DDC EQUIPMENT

- A. Control Units: Modular, comprising processor board with programmable, nonvolatile, random-access memory; local operator access and display panel; integral interface equipment; and backup power source.
 - 1. Units monitor or control each I/O point; process information; execute commands from other control units, devices, and operator stations; and download from or upload to operator workstation or diagnostic terminal unit.
 - 2. Stand-alone mode control functions operate regardless of network status. Functions include the following:
 - a. Global communications.
 - b. Discrete/digital, analog, and pulse I/O.
 - c. Monitoring, controlling, or addressing data points.
 - d. Software applications, scheduling, and alarm processing.
 - e. Testing and developing control algorithms without disrupting field hardware and controlled environment.

3. Application Programs:
 - a. Include control programs capable of performing functions as described in Sequence of Operations.
 - b. Programming Application Features: Include trend point; alarm processing and messaging; weekly, monthly, and annual scheduling; energy calculations; run-time totalization; and security access.
 - c. Remote communications.
 - d. Maintenance management.
 - e. Units of Measure: Inch-pound and SI (metric).
 4. Local operator interface provides for download from or upload to operator workstation or diagnostic terminal unit.
 5. ASHRAE 135 Compliance: Control units shall use ASHRAE 135 protocol and communicate using ISO 8802-3 (Ethernet) datalink/physical layer protocol.
- B. Local Control Units: Modular, comprising processor board with electronically programmable, nonvolatile, read-only memory; and backup power source.
1. Units monitor or control each I/O point, process information, and download from or upload to operator workstation or diagnostic terminal unit.
 2. Stand-alone mode control functions operate regardless of network status. Functions include the following:
 - a. Global communications
 - b. Discrete/digital, analog, and pulse I/O
 - c. Monitoring, controlling, or addressing data points
 3. Local operator interface provides for download from or upload to operator workstation or diagnostic terminal unit.
 4. ASHRAE 135 Compliance: Control units shall use ASHRAE 135 protocol and communicate using ISO 8802-3 (Ethernet) datalink/physical layer protocol.
- C. I/O Interface: Hardwired inputs and outputs may tie into system through controllers. Protect points so that shorting will cause no damage to controllers.
1. Binary Inputs: Allow monitoring of on-off signals without external power.
 2. Pulse Accumulation Inputs: Accept up to 10 pulses per second.
 3. Analog Inputs: Allow monitoring of low-voltage (0-10VDC), current (4 to 20 mA), or resistance signals.
 4. Binary Outputs: Provide on-off or pulsed low-voltage signal, selectable for normally open or normally closed operation with three-position (on-off-auto) override switches and status lights.
 5. Analog Outputs: Provide modulating signal, low-voltage (0-10VDC) or current (4 to 20 mA) with status lights, two-position (auto-manual) switch, and manually adjustable potentiometer.
 6. Universal I/Os: Provide software selectable binary or analog outputs.

- D. Power Supplies: Transformers with Class 2 current-limiting type or overcurrent protection; limit connected loads to 80 percent of rated capacity. DC power supply shall match output current and voltage requirements and be full-wave rectifier type with the following:
1. Output ripple of 5.0 mV maximum peak to peak.
 2. Combined 1 percent line and load regulation with 100-mic.sec. response time for 50 percent load changes.
 3. Built-in overvoltage and overcurrent protection and be able to withstand 150 percent overload for at least 3 seconds without failure.
- E. Power Line Filtering: Internal or external transient voltage and surge suppression for work stations or controllers with the following:
1. Minimum dielectric strength of 1000 V
 2. Maximum response time of 10 nanoseconds
 3. Minimum transverse-mode noise attenuation of 65 dB
 4. Minimum common-mode noise attenuation of 150 dB at 40 to 100 Hz

2.5 DDC CONTROLLERS

A. General

1. DDC controllers shall be capable of stand-alone operation with sufficient memory to support its operating system, database, and programming requirements, and with sufficient I/O capacity for the application.
2. Configuration: Local keypad and display; diagnostic LEDs for power, communication, and processor; wiring termination to terminal strip or card connected with ribbon cable; memory with bios; and 72-hour battery backup.
3. Operating System: Manage I/O communication to allow distributed controllers to share real and virtual object information and allow central monitoring and alarms. Perform scheduling with real-time clock. Perform automatic system diagnostics; monitor system and report failures.
4. ASHRAE 135 Compliance: Communicate using read (execute and initiate) and write (execute and initiate) property services defined in ASHRAE 135. Controller shall have service communication port for connection to diagnostic terminal unit.

- B. Each DDC controller shall be provided for control of each of the following types of equipment. DDC controllers shall reside on a peer-to-peer network.

1. DDC controllers shall be a minimum of 16-bit stand-alone, multitasking, multiuser, real-time digital control processors.
2. Each primary networked DDC controller shall house a minimum of 32 MB RAM and 8 MB Flash ROM to support its own operating system, databases, and stand-alone software functions including:
 - a. Control Processes
 - b. Energy Management Applications
 - c. Mathematical Modeling, Equipment Learning, Part Load Curve Updating Functions to support advanced algorithms for energy reduction.

- 1) Provide capabilities for mathematical modeling of VAV air delivery systems to optimize static pressure setpoints and reduce fan energy while maintaining cooling and ventilation constraints in each of the areas served by the VAV system.
 - 2) Provide capabilities for equipment learning such that load and part load curves are built for equipment controlled to optimize equipment selection at given loads.
 - d. It shall be possible for the controller to determine equipment degradation from internal plotting of load curves to schedule maintenance.
 - e. Alarm Management Applications including custom alarm messages for each level alarm for each point in the system.
 - f. Historical/Trend data for points specified.
 - g. Maintenance support applications.
 - h. Custom Processes.
 - i. Operator I/O
 - j. Remote Communications
3. DDC controllers shall provide a communication port for operation of operator I/O devices such as industry standard printers, operator terminals, modems and portable laptop operator's terminals.
 4. DDC controllers shall be provided with digital input and output LED status indication for visual confirmation of point conditions.
 5. The operator shall have the ability to manually override automatic or centrally executed commands at the Networked DDC Controller via local, point discrete, on-board hand/off/auto operator override switches for digital control type points and gradual switches for analog control type points.
 6. DDC Controllers shall be provided with communication ports for the control and monitoring of application specific controllers to coordinate control of major mechanical equipment with downstream terminal equipment.
 7. Each DDC Controller shall continuously perform self-diagnostics, communication diagnosis and diagnosis of all components. The DDC Controller shall provide both local and remote annunciation of any detected component failures, low battery conditions or repeated failure to establish communication to alert facility personnel of failure.
 8. Surge and Transient Protection: Provide isolation at all network terminations, as well as all field point terminations to suppress induced voltage transients consistent with IEEE Standard 587.
 9. In the event of the loss of normal power, there shall be an orderly shutdown of all DDC controllers to prevent the loss of database or operating system software. Nonvolatile memory shall be incorporated for all critical controller configuration data and battery backup shall be provided to support the real-time clock and all volatile memory for a minimum of 72 hours.
 - a. Upon restoration of normal power, the DDC controller shall automatically resume full operation, incorporating time delays to prevent surges, without manual intervention.
 - b. Should DDC controller memory be lost for any reason, the user shall have the capability of reloading the DDC controller via the local communication port, via telephone line dial-in or from a network workstation PC.

10. Controllers shall be provided with the capability to communicate TCP/IP directly over Ethernet, without the use of an external network interface card. Devices must:

- a. Auto-sense 10/100 Mbps networks.
- b. Receive an IP Address from a Dynamic Host configuration Protocol (DHCP) Server or be configured with a Fixed IP Address. (Owner shall provide IP addresses and relevant network information for each DDC controller provided under this specification.)
- c. Resolve Name to IP Address for devices using a Domain Name Service (DNS) Server on the Ethernet network.
- d. Allow access using Telnet.

C. Each Application Specific Controller (ASC) shall operate as a stand-alone controller capable of performing its specified control responsibilities independently of other controllers in the network. Each ASC shall be a microprocessor-based, multitasking, real-time digital control processor. Each ASC shall be capable of control of the terminal device independent of the manufacturer of the terminal device.

2.6 ELECTRONIC SENSORS

A. Description: Vibration and corrosion resistant; for wall, immersion, or duct mounting as required.

B. Thermistor Temperature Sensors and Transmitters:

1. Wire: Twisted, shielded-pair cable.
2. Insertion Elements in Ducts: Single point, 18 inches (460 mm) long; use where not affected by temperature stratification or where ducts are smaller than 9 sq. ft. (0.84 sq. m).
3. Averaging Elements in AHU/Ducts: Minimum 72 inches (1830 mm) long, flexible; use where prone to temperature stratification or where ducts are larger than 10 sq. ft. (1 sq. m).
4. Insertion Elements for Liquids: Brass or stainless-steel socket with minimum insertion length of 2-1/2 inches (64 mm).
5. Room Sensor Cover Construction: Manufacturer's standard locking covers matching NMNH standard.

C. RTDs and Transmitters:

1. Wire: Twisted, shielded-pair cable.
2. Insertion Elements in Ducts: Single point, 18 inches (460 mm) long; use where not affected by temperature stratification or where ducts are smaller than 9 sq. ft. (0.84 sq. m).
3. Averaging Elements in Ducts: Minimum 8 feet (2.4 m) in length; use where prone to temperature stratification or where ducts are larger than 9 sq. ft. (0.84 sq. m); length as required.
4. Insertion Elements for Liquids: Brass socket with minimum insertion length of 2-1/2 inches (64 mm).
5. Room Sensor Cover Construction: Manufacturer's standard locking covers matching NMNH standard.

D. Humidity Sensors:

1. Acceptable Manufacturers:
 - a. Vaisala
 - 1) Duct Mounted Sensor shall be: Vaisala Model HMT331
 2. Duct Sensor: Provide element guard and mounting plate.
- E. Pressure Transmitters/Transducers:
1. Acceptable Manufacturers:
 - a. Veltron (High Accuracy)
 - b. Fisher-Rosemount (High Accuracy)
 - c. AutoTran (Mid Range)
 - d. Setra (Low Range)
 2. Differential-Pressure Transmitter: Nondirectional sensor with suitable range for expected input, and temperature compensated.
 - a. High Accuracy Differential Pressure Transmitter: Shall be microprocessor based pressure and flow smart transmitters with automatic zeroing circuit.
 - 1) Range: 0 to 0.1 inch wg (0 to 25 Pa).
 - 2) Accuracy: 0.17 of natural span, including nonlinearity, hysteresis, deadband, nonrepeatability.
 - 3) Span Range: Calibrated span can be down ranged to 40 percent of natural span.
 - 4) Transducer Response Time: 0.5 seconds to reach 98 percent of a step change.
 - 5) Temperature Affect: Zero corrected by auto zero.
 - 6) Mounting Position Affect: None.
 - 7) Stability: Auto zeroing accuracies within 0.17 of calibrated span frequency every 1 to 24 hours on 1 hour intervals.
 - 8) Temperature Limit: -20 to 120 deg F (-20 to 48 deg C).
 - 9) Humidity Limit: 0-90 percent RH.
 - 10) Display: LCD data display and LED indicating lights.
 - 11) Outputs: 4-20 mA.
 - 12) Power Supply: 24 VAC/120 VAC.
 - b. Midrange Differential Pressure Transmitter: Shall be smart pressure transmitter.
 - 1) Range: 0.1 to 3.0 inch wg (25 to 750 Pa).
 - 2) Temperature Limits: -40 to 185 deg F (-40 to 85 deg C).
 - 3) Humidity Limits: 0-100 percent RH.
 - 4) Stability: 5 year stability.
 - 5) Outputs: 4-20 mA.
 - c. Low Range Differential Pressure Transmitter: Shall be zero maintenance pressure transducer.

- 1) Range: 0.1 to 5 inch wg (25 to 1250 Pa).
 - 2) Accuracy: Plus or minus 1 percent at less than 1 inch wg.
 - 3) Stability: Auto zero plus or minus 0.5 percent FSO over life of device.
 - 4) Temperature Limits: 0 to 65 deg F (32 to 150 deg C).
 - 5) Humidity Limits: 0-90 percent RH.
 - 6) Outputs: 4-20 mA.
3. Water Pressure Transducers: Stainless-steel diaphragm construction, suitable for service; minimum 150-psig (1034-kPa) operating pressure; linear output 4 to 20 mA.
 4. Water Differential-Pressure Transducers: Stainless-steel diaphragm construction, suitable for service; minimum 150-psig (1034-kPa) operating pressure and tested to 300-psig (2070-kPa); linear output 4 to 20 mA.
 5. Differential-Pressure Switch (Air or Water): Snap acting, with pilot-duty rating and with suitable scale range and differential.
 6. Pressure Transmitters: Direct acting for gas, liquid, or steam service; range suitable for system; linear output 4 to 20 mA.

2.7 STATUS SENSORS

A. Current Switches

1. The current switch shall be a Veris Industries Hawkeye Current Sensor.
2. The current sensor shall be induce powered from the monitored load.
3. The current sensor shall provide on/off status indication of electrical loads from 1.5 to 200 amperes.
4. The selected switch shall match current VFD System output requirements.
5. The current sensor shall be capable of providing accurate status at temperatures from -15° to 60° C.
6. The current sensor shall be isolated to 600 VAC rms.
7. The current sensor output shall be N.O. solid-state 1.0 ampere at 30 VAC/DC.
8. The current sensor shall be a self-gripping split-core type.
9. The current sensor shall detect drive belts slipping, breaking, or pump couplings shearing.

B. Electronic Valve/Damper Position Indicator: Visual scale indicating percent of travel and 2- to 10-V dc, feedback signal.

C. Water-Flow Switches: Bellows-actuated mercury or snap-acting type with pilot-duty rating, stainless-steel or bronze paddle, with appropriate range and differential adjustment, in NEMA 250, Type 1 enclosure.

2.8 THERMOSTATS

A. Electric, Low-Limit Duct Thermostat (Low Temperature Limit Thermostat): Two sets of contacts, snap-acting, single-pole, single-throw, manual reset switch that trips if temperature sensed across any incremental segment of capillary length is equal to or below set point.

1. Capillary Length: Minimum 20 feet (6 m)
2. Quantity: One thermostat for every 20 sq. ft. (2 sq. m) of coil surface

2.9 ACTUATORS

- A. Electric Motors: Size to operate with sufficient reserve power to provide smooth modulating action or two-position action.
1. Comply with requirements in Division 23 Section "Electrical Requirements for HVAC Equipment."
 2. Permanent Split-Capacitor or Shaded-Pole Type: Gear trains completely oil immersed and sealed. Equip spring-return motors with integral spiral-spring mechanism in housings designed for easy removal for service or adjustment of limit switches, auxiliary switches, or feedback potentiometer.
 3. Nonspring-Return Motors for Valves Larger Than NPS 2-1/2 (DN 65): Size for running torque of 150 in. x lbf (16.9 N x m) and breakaway torque of 300 in. x lbf (33.9 N x m).
 4. Spring-Return Motors for Valves Larger Than NPS 2-1/2 (DN 65): Size for running and breakaway torque of 150 in. x lbf (16.9 N x m).
 5. Nonspring-Return Motors for Dampers Larger Than 25 Sq. Ft. (2.3 sq. m): Size for running torque of 150 in. x lbf (16.9 N x m) and breakaway torque of 300 in. x lbf (33.9 N x m).
 6. Spring-Return Motors for Dampers Larger Than 25 Sq. Ft. (2.3 sq. m): Size for running and breakaway torque of 150 in. x lbf (16.9 N x m).
- B. Electronic Actuators: Direct-coupled type designed for minimum 60,000 full-stroke cycles at rated torque.
1. Acceptable Manufacturers:
 - a. Belimo Aircontrols (USA), Inc.
 2. Valves: Size for torque required for valve close off at maximum pump differential pressure.
 3. Dampers: Size for running torque calculated as follows:
 - a. Parallel-Blade Damper with Edge Seals: 7 inch-lb/sq. ft. (86.8 kg-cm/sq. m) of damper.
 - b. Opposed-Blade Damper with Edge Seals: 5 inch-lb/sq. ft. (62 kg-cm/sq. m) of damper.
 - c. Dampers with 2- to 3-Inch wg (500 to 750 Pa) of Pressure Drop or Face Velocities of 1000 to 2500 fpm (5 to 13 m/s): Increase running torque by 1.5.
 - d. Dampers with 3- to 4-Inch wg (750 to 1000 Pa) of Pressure Drop or Face Velocities of 2500 to 3000 fpm (13 to 15 m/s): Increase running torque by 2.0.
 4. Coupling: V-bolt and V-shaped, toothed cradle.
 5. Overload Protection: Electronic overload or digital rotation-sensing circuitry.
 6. Fail-Safe Operation: Mechanical, spring-return mechanism. Provide external, manual gear release on nonspring-return actuators.
 7. Power Requirements (Two-Position Spring Return): 24-V ac.
 8. Power Requirements (Modulating): Maximum 10 VA at 24-V ac or 8 W at 24-V dc.
 9. Proportional Signal: 2- to 10-V dc or 4 to 20 mA, and 2- to 10-V dc position feedback signal.
 10. Temperature Rating: 40 to 104 deg F (5 to 40 deg C).

11. Run Time: 30 seconds.

2.10 CONTROL VALVES

- A. Control Valves: Factory fabricated, of type, body material, and pressure class based on maximum pressure and temperature rating of piping system, unless otherwise indicated.
- B. Hydronic system valves shall have the following characteristics:
 1. NPS 2 (DN 50) and Smaller:
 - a. Globe Valve: Class 125 bronze body, bronze trim, stainless steel rising stem, renewable composition disc, and screwed ends with backseating capacity repackable under pressure. Replaceable plugs and stainless-steel or brass seats. Cage trim provides seating and guiding surfaces for plug on top and bottom of guided plugs.
 - b. Ball Valve: Class 125 bronze body with full-port, stainless steel ball and stem; TFE seats and packing; and 600-psig (4140-kPa) minimum CWP rating and blowout-proof stem. Threaded ends.
 2. NPS 2-1/2 (DN 65) and Larger:
 - a. Globe Valve: Class 125 iron body, bronze trim, stainless steel rising stem, plug-type disc, flanged ends, and renewable seat and disc. Replaceable plugs and stainless-steel or brass seats. Cage trim provides seating and guiding surfaces for plug on top and bottom of guided plugs.
 - b. Butterfly Valve: 150 psig (1035 kPa) WOG, bubbletight shutoff, 250°F (121°C) continuous: Butterfly type with one piece semisteel body (split body design not acceptable), threaded lugs (same number of lugs as connecting flange), extended neck to suit insulation thickness, bronze disc, bronze bearings, stainless steel shaft and continuous retained EPDM resilient seat to provide end or isolation service without use of downstream flanges.
 3. Sizing:
 - a. Two Position: Line size; ball or butterfly valve.
 - b. Two-Way Modulating: 3-psig (21-kPa) maximum pressure drop at design flow rate; globe or ball valve.
 - c. Three-Way Modulating: Line size; globe, ball or butterfly valve.
 - d. Differential Pressure: Typically 50% of pump head with full pump flow.
 - e. Valves which are sized 6 inch and larger shall be high performance butterfly type.
 4. Flow Characteristics: Two-way valves shall have equal percentage characteristics; three-way valves shall have linear characteristics.
 5. Close-Off (Differential) Pressure Rating: Combination of actuator and trim shall provide minimum close-off pressure rating equal to pump dead head (zero flow) pressure for two-way valves and 100 percent of pressure differential across valve for three-way valves.
 6. One third and two thirds of total capacity is design and actual split will be the ratio of each valve's Cv to the total Cv.

2.11 DAMPERS

A. Acceptable Manufacturers:

1. Ruskin
2. Air Balance Inc.
3. Greenheck

B. Dampers: AMCA-rated, airfoil design; 0.108-inch- (2.8-mm-) minimum thick, galvanized-steel or 0.125-inch- (3.2-mm-) minimum thick, extruded-aluminum frames with holes for duct mounting; damper blades shall not be less than 0.064-inch- (1.6-mm-) thick galvanized steel or extruded aluminum with maximum blade width of 6 inches (150 mm) and length of 48 inches (1220 mm). Dampers installed in stainless steel ductwork shall be completely constructed of stainless steel.

1. Secure blades to 1/2-inch- (13-mm-) diameter, zinc-plated axles using zinc-plated hardware, with oil-impregnated sintered bronze blade bearings, blade-linkage hardware of zinc-plated steel and brass, ends sealed against spring-stainless-steel blade bearings, and thrust bearings at each end of every blade.
2. Operating Temperature Range: From minus 40 to plus 200 deg F (minus 40 to plus 93 deg C).
3. Edge Seals: Use inflatable blade edging or replaceable rubber blade seals and spring-loaded stainless-steel side seals, rated for leakage at less than 10 cfm per sq. ft. (50 L/s per sq. m) of damper area, at differential pressure of 4-inch wg (1000 Pa) when damper is held by torque of 50 in. x lbf (5.6 N x m); when tested according to AMCA 500D.

C. Provide parallel type blades for mixing applications and opposed blade dampers for all other reasons.

D. Damper End Switches: Whisker type end switches.

2.12 DUCT STATIC PROBE

A. Furnish duct static pressure probe and place for static pressure sensing 2/3 of way downstream in system ductwork or as indicated on the mechanical drawings.

B. Construct with multiple static pressure sensors located along exterior surface of cylindrical probe, of extruded aluminum and (except for 3/4 inch dia. probes with lengths of 24 inches or less) complete with threaded end support rod, sealing washer and nut, and mounting plate with gasket and static pressure signal fitting. Produce steady, nonpulsating signal of standard static pressure, without need for correction factors, with instrument accuracy of 0.5%.

2.13 LIMIT SWITCHES

A. Separate independent position switches shall be installed on dampers including smoke isolation, as needed. The DPST switches shall change state in response to a damper that has modulated beyond 95% full open or closed as dictated by the application.

- B. Limit switches shall be the flexible spring-return rod type, with a mechanical endurance of 30×10^6 cycles.
- C. The contacts shall be rated for 10A at 120VAC.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that conditioned power supply is available to control units.
- B. Verify that duct-, pipe-, and equipment-mounted devices are installed before proceeding with installation.

3.2 INSTALLATION

A. General

1. Install software in control units and operator workstation(s). Implement all features of programs to specified requirements and as appropriate to sequence of operation.
2. Connect and configure equipment and software to achieve sequence of operation specified.
3. Furnish automatic control dampers to Division 23 Section "Duct and Duct Accessories" for installation.
4. Install damper motors on outside of duct in tempered areas, not in locations exposed directly to outdoor temperatures.
5. Install labels and nameplates to identify control components according to Division 23 Section "Identification for HVAC Piping and Equipment."
6. Coordinate location of hydronic instrument wells, valves, and other accessories installed by Division 23 Section "Hydronic Piping Systems."
7. Coordinate location of steam and condensate instrument wells, valves, and other accessories installed by Division 23 Section "Steam and Condensate Piping Systems."

B. Thermostats and Temperature Sensors

1. Verify location of thermostats and/or temperature sensors where shown on drawings and room interior elevations. Coordinate location with other wall mounted devices.
2. Install space thermostats and/or temperature sensors 48 inches (1200 mm) above finished floor unless noted otherwise on the drawings.
3. Provide insulation pads for thermostats and/or temperature sensors mounted on exterior walls and columns.
4. Install averaging elements in ducts and plenums in serpentine, crossing or zigzag pattern across the area of duct or plenum in order to sense true average temperature. Secure averaging elements in such a manner as to prevent vibration from causing element fatigue.
5. Secure duct mounted sensors to ductwork in a vibration free area.
6. Furnish thermal wells for sensors to be installed in piping. Furnish extension necks where installed in insulated piping. Material for wells shall be compatible with material of piping where installed.

C. Humidity Sensors

1. Verify location of humidistats and/or humidity sensors where shown on drawings and room interior elevations. Coordinate location with other wall mounted devices.
2. Secure duct mounted sensors to ductwork in a vibration free area.

D. Control Valves

1. Tag each valve with brass or aluminum tag with corresponding number on control drawings. Tag shall identify valve number and be attached to valve with nonferrous metal chain.

E. Control Dampers

1. Verify size and locations of control dampers with Division 23 Contractor prior to fabrication. Locations of dampers shall be reviewed to insure that maximum velocity rating is not exceeded.

F. Current Switches

1. Shall be installed such that core is securely in place.
2. Shall be adjusted such that calibration trip point will detect drive belts slipping, breaking, or pump coupling shear.

3.3 ELECTRICAL WIRING AND CONNECTION INSTALLATION

- A. Provide signal and power wiring to all panels and devices furnished under the contract and signal and safety device wiring to all equipment controlled under this contract.
- B. Provide all interlock wiring between equipment being sequenced as required to accomplish the sequence of operations, which shall include supply and return air fans, exhaust fans, coil circulating pumps, chilled and condenser water pumps, cooling tower fans and chiller control panels, flow switches, etc.
- C. Mount and wire all loose control components provided with packaged equipment.
- D. Provide all required power wiring and conduit for all panels furnished by the contractor for the project. All panels shall be circuited to the nearest emergency essential equipment panel and utilize 20-amp single pole spare circuit breaker in panelboard. Refer to electrical documents to ascertain exact location of nearest emergency essential equipment panelboard. Multiple panels may use same circuit within the electrical limitations. Indicate panelboard name and circuit number for each panel on shop drawings.
- E. Install raceways, boxes, and cabinets according to Division 26 Section "Raceway and Boxes for Electrical Systems."
- F. Install building wire and cable according to Division 26 Section "Low-Voltage Electrical Power Conductors and Cables."

- G. Install signal and communication cable according to Division 26 Section "Communications Horizontal Cabling." All control cabling must be installed in blue conduit, minimum ¾" (DN20) and shall not exceed 50% fill capacity.
1. Conceal cable, except in mechanical rooms and areas where other conduit and piping are exposed.
 2. Install exposed cable in conduit.
 3. Install concealed cable in conduit.
 4. Bundle and harness multiconductor instrument cable in place of single cables where several cables follow a common path.
 5. Fasten flexible conductors, bridging cabinets and doors, along hinge side; protect against abrasion. Tie and support conductors.
 6. Number-code or color-code conductors for future identification and service of control system, except local individual room control cables.
 7. Install wire and cable with sufficient slack and flexible connections to allow for vibration of piping and equipment.
- H. Connect manual-reset limit controls independent of manual-control switch positions. Automatic duct heater resets may be connected in interlock circuit of power controllers.
- I. Connect hand-off-auto selector switches to override automatic interlock controls when switch is in hand position.

3.4 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust field-assembled components and equipment installation, including connections, and to assist in field testing. Report results in writing.
- B. Perform the following field tests and inspections and prepare test reports:
1. Operational Test: After electrical circuitry has been energized, start units to confirm proper unit operation. Remove and replace malfunctioning units and retest.
 2. Test and adjust controls and safeties.
 3. Test calibration of electronic controllers by disconnecting input sensors and stimulating operation with compatible signal generator.
 4. Test each point through its full operating range to verify that safety and operating control set points are as required.
 5. Test each control loop to verify stable mode of operation and compliance with sequence of operation. Adjust PID actions.
 6. Test each system for compliance with sequence of operation.
 7. Test software and hardware interlocks.
- C. DDC Verification:
1. Verify that instruments are installed before calibration, testing, and loop or leak checks.
 2. Check instruments for proper location and accessibility.
 3. Check instrument installation for direction of flow, elevation, orientation, insertion depth, and other applicable considerations.

4. Check instrument tubing for proper fittings, slope, material, and support.
5. Check flow instruments. Inspect tag number and line and bore size and verify that inlet side is identified and that meters are installed correctly.
6. Check temperature instruments and material and length of sensing elements.
7. Check control valves. Verify that they are in correct direction.
8. Check air-operated dampers. Verify that pressure gages are provided and that proper blade alignment, either parallel or opposed, has been provided.
9. Check DDC system as follows:
 - a. Verify that DDC controller power supply is from emergency power supply, if applicable.
 - b. Verify that wires at control panels are tagged with their service designation and approved tagging system.
 - c. Verify that spare I/O capacity has been provided.
 - d. Verify that DDC controllers are protected from power supply surges.

D. Replace damaged or malfunctioning controls and equipment and repeat testing procedures.

3.5 ADJUSTING

A. Calibrating and Adjusting:

1. Calibrate instruments.
2. Make three-point calibration test for both linearity and accuracy for each analog instrument.
3. Calibrate equipment and procedures using manufacturer's written recommendations and instruction manuals. Use test equipment with accuracy at least double that of instrument being calibrated.
4. Control System Inputs and Outputs:
 - a. Check analog inputs at 0, 50, and 100 percent of span.
 - b. Check analog outputs using milliamper meter at 0, 50, and 100 percent output.
 - c. Check digital inputs using jumper wire.
 - d. Check digital outputs using ohmmeter to test for contact making or breaking.
 - e. Check resistance temperature inputs at 0, 50, and 100 percent of span using a precision-resistant source.
5. Flow:
 - a. Set differential pressure flow transmitters for 0 and 100 percent values with 3-point calibration accomplished at 50, 90, and 100 percent of span.
 - b. Manually operate flow switches to verify that they make or break contact.
6. Pressure:
 - a. Calibrate pressure transmitters at 0, 50, and 100 percent of span.
 - b. Calibrate pressure switches to make or break contacts, with adjustable differential set at minimum.

7. Temperature:
 - a. Calibrate resistance temperature transmitters at 0, 50, and 100 percent of span using a precision-resistance source.
 - b. Calibrate temperature switches to make or break contacts.
 8. Stroke and adjust control valves and dampers without positioners, following the manufacturer's recommended procedure, so that valve or damper is 100 percent open and closed.
 9. Stroke and adjust control valves and dampers with positioners, following manufacturer's recommended procedure, so that valve and damper is 0, 50, and 100 percent closed.
 10. Provide diagnostic and test instruments for calibration and adjustment of system.
 11. Provide written description of procedures and equipment for calibrating each type of instrument. Submit procedures review and approval before initiating startup procedures.
- B. Adjust initial temperature and humidity set points.
- C. Occupancy Adjustments: When requested within 12 months of date of Substantial Completion, provide on-site assistance in adjusting system to suit actual occupied conditions. Provide up to three visits to Project during other than normal occupancy hours for this purpose.

3.6 TESTING, ADJUSTING AND BALANCING

- A. Testing, adjusting and balancing of air and water systems will be provided under Division 01 "Testing, Adjusting and Balancing of HVAC Systems".
- B. Cooperate with testing, adjusting and balancing Contractor in coordination and scheduling of testing, balancing and adjusting work, as well as determining appropriate set point adjustments required for proper system operation.
- C. Provide notice upon completion of all preparatory work and all initial operational testing required as part the Work. Perform additional operational testing on equipment, or systems, as directed and to extent and for duration deemed necessary, to demonstrate that systems are performing properly and delivering quantities in accordance with the requirements of the Contract Documents.
- D. BAS Contractor shall set up and calibrate the mass flow control devices to the design contract values. BAS Contractor shall adjust the AVCB control so that final setup does not deviate more than plus or minus 5 percent from the design value.
- E. BAS Contractor shall index the system configuration as requested by the TAB Contractor.
- F. BAS Contractor shall obtain static pressure readings from TAB Contractor at the various points in the system for programming and tuning final set point conditions.

3.7 COMMISSIONING

- A. Commissioning will be provided as specified in Division 01 Section "Commissioning". All contractors and subcontractors of the various sections of this specification shall cooperate and participate in the commissioning work in accordance with requirements of Division 01 Section "Commissioning".
- B. Ensure participation of major equipment manufacturers or their representatives.
- C. Equipment and systems/subsystems installed under this section are expected to be in full compliance with the design intent by the commissioning phase. Notify the Commissioning Agent when any specific piece of equipment or specific system/subsystem is ready for commissioning. Be prepared to demonstrate system readiness.
- D. Equipment or systems/subsystems having incomplete work or exhibiting problems related to noncompliance with the design intent shall require commissioning. The contractor for this section shall be fully responsible to make all necessary corrections to incomplete or non-complying work at their own expense and shall pay the Commissioning Agent per diem rate for recommissioning such incomplete or non-complying work.

3.8 DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain HVAC instrumentation and controls. Refer to Division 01 Section "Demonstration and Training."
- B. Upon completion of all work and tests, operate systems for a sufficient length of time to demonstrate to Owner, mode of operation and definitively determine whether the system as a whole is in first class working condition. Before systems are turned over to Owner, a final demonstration test of 48 continuous hours, during which systems shall operate without adjustment, shall be performed.
- C. Before installation is accepted, provide certification to Owner and Architect that control system and equipment have been inspected and found to be properly installed and functioning satisfactorily.

END OF SECTION 25 09 00

SECTION 26 05 00 - COMMON MATERIALS AND METHODS FOR ELECTRICAL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section is intended to describe the basic materials and installation methods for electrical work; it applies in general to all Sections under DIVISION 26. All materials and equipment specified and/or shown on Drawings are new unless noted otherwise.
- B. All new materials, equipment and systems shall be listed and labeled by a licensed nationally recognized testing laboratory as defined by OSHA and used for the specific purpose, environment or application for which it was tested and approved. No field modifications and/or noncompliant installation whatsoever shall be made to any materials, equipment and systems that would violate the listing and labeling.
- C. This section includes the following:
 - 1. Electrical equipment coordination and installation
 - 2. Sleeves for raceways and cables
 - 3. Sleeve seals
 - 4. Grout
 - 5. Common electrical installation requirements

1.3 DEFINITIONS

- A. Finished Spaces: Spaces other than mechanical and electrical equipment rooms, furred spaces, pipe and duct chases, unheated spaces immediately below roof, spaces above ceilings, unexcavated spaces, crawlspace, and tunnels.
- B. Exposed, Interior Installations: Exposed to view indoors. Examples include finished occupied spaces and mechanical equipment rooms.
- C. Concealed, Interior Installations: Concealed from view and protected from physical contact by building occupants. Examples include above ceilings and chases.
- D. EPDM: Ethylene-propylene-diene terpolymer rubber.
- E. NBR: Acrylonitrile-butadiene rubber.

1.4 REFERENCES

- A. Provide work in accordance with all applicable international, state and local, codes, rules, regulations, and standards, including but not limited to, requirements of the following:
 - 1. National Electrical Code, NFPA 70 (NEC)
 - 2. Underwriters' Laboratories, Inc. (UL)
 - 3. National Electrical Manufacturer's Association (NEMA)
 - 4. National Electrical Contractors Association (NECA)
 - 5. The Institute of Electrical and Electronics Engineers, Inc. (IEEE)
 - 6. Applicable NFPA Codes and Standards
 - 7. See Drawing LS001 for a complete list of applicable codes
- B. Conflicts
 - 1. Nothing stated or shown in Specifications or on Drawings is intended to conflict with the above standards and regulations. Should Contractor find any apparent conflict, it shall be his responsibility to notify Architect before any of the work in question is performed or material purchased.

1.5 SUBMITTALS

- A. Provide Product List of factory fabricated items, in accordance with Section 016000 "Product Requirements", including name of proposed manufacturer, for all products specified in various sections of Division 26.
- B. Provide submittals in accordance with Section 013300 "Submittal Procedures" in sufficient detail to verify full compliance with the requirements of the Contract Documents.
- C. Product Data: Provide for each type of factory-fabricated product indicated.
- D. Submit testing reports.

1.6 WARRANTY AND CONTRACT CLOSEOUT

- A. Comply with warranty and contract closeout requirements specified in Division 01, GENERAL REQUIREMENTS.
- B. Provide Special Warranties and/or warranty service in accordance with Division 00 and 01 requirements where specified in the various sections of Division 26.
- C. Provide manufacturer's certificates of supervision and startup service as specified in the various sections of Division 26.
- D. Upon completion of work and tests, and at a time mutually agreed to by Contractor, Architect and Owner, operate all systems installed, in all parts, at Contractor's expense for sufficient length of time to demonstrate the mode of operation and definitely determine whether systems as a whole are in first class working condition. Defects and malfunctions that may develop during this period

of operation shall be immediately corrected by Contractor at his own expense, and systems placed in first class working condition before being finally turned over to Owner.

- E. Include information for all products specified in the operation and maintenance manual.
- F. Provide electrical certificate(s) from electrical inspection agency - see Article titled "Inspections".
- G. Provide manufacturer's certification and warranty of system operation - see Article titled "Tests".

1.7 QUALITY ASSURANCE

- A. The specifications for certain products and alternative materials may appear in more than one section of Division 26. Work of Division 26 shall be coordinated for all sections of Division 26 to assure that where two or more items of any given product are furnished under Division 26 that they are of the same manufacturer and type and that alternative materials is consistent throughout the work of Division 26.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle all material and equipment in accordance with manufacturer's instructions and recommendations. Such instructions and recommendations are hereby made part of these specifications.
- B. Deliver products and equipment properly labeled and tagged. Maintain products in original shipping containers and store in a dry area until ready for installation.

1.9 PHASE ARRANGEMENT

- A. Exercise great care in maintaining a uniform and consistent arrangement of phase conductors on all systems. Throughout the entire wiring systems, each phase conductor must always be in the same physical position with respect to the other phase wires at equipment terminals.
- B. Identify phase wires by color coded conductors. Refer to Section 26 05 53.
- C. Determine the existing established phase arrangement and use it throughout the renovation. Use special care in determination and execution.

1.10 INSPECTIONS

- A. Before starting any Work under this Contract, file for inspection with the agency approved by the AHJ. Upon completion of the work, furnish electrical certificates from said agency for all electrical equipment and systems installed or furnished and installed as part of the work.
- B. Electrical equipment or systems that are modified in the field shall be reinspected. Furnish a new electrical certificate covering such modifications.

1.11 GENERAL COORDINATION.

- A. The electrical systems are indicated on the Electrical Drawings. Certain pertinent information and details required by the electrical work appear on the Architectural, Structural, Mechanical and Fire Protection Drawings. Become familiar with all drawings and incorporate all pertinent requirements
- B. Drawings are diagrammatic and indicate general arrangement of systems and requirements of the Electrical work. Do not scale the drawings to obtain dimensional requirements. Exact locations of equipment must be coordinated and obtained prior to starting the work.
- C. Coordinate scheduling, sequencing, movement and positioning of large equipment into the building during construction.
- D. Coordinate installation of identification devices with completion of covering and painting of surfaces where identification devices are to be applied.
- E. Coordinate arrangement, mounting, and support of electrical equipment:
 - 1. To allow maximum possible headroom unless specific mounting heights that reduce headroom are indicated.
 - 2. To provide for ease of disconnecting the equipment with minimum interference to other installations.
 - 3. To allow right of way for piping and conduit installed at required slope.
 - 4. So connecting raceways, cables, wireways, cable trays, and busways will be clear of obstructions and of the working and access space of other equipment.
- F. Coordinate installation of required supporting devices and set sleeves in cast-in-place concrete, masonry walls, and other structural components as they are constructed.
- G. Coordinate location of access panels and doors for electrical items that are behind finished surfaces or otherwise concealed. Access doors and panels are specified in Division 08 Section "Access Doors and Frames."
- H. Coordinate sleeve selection and application with selection and application of firestopping specified in Division 07 Section "Penetration Firestopping."

1.12 ELECTRICAL WORK ASSOCIATED WITH OTHER DIVISIONS

- A. WORK NOT REQUIRED BY DIVISION 26
 - 1. Unless otherwise noted or specified, setting of mechanical equipment and associated safety and control devices will be done by Division 22 and 23 Contractors.
 - 2. Control devices provided under Division 22 or 23 will include, but are not necessarily limited to the following:
 - a. Limit Switches: Aquastats, thermostats, air, gas and water pressure
 - b. Sensing Devices: Air, gas and water flow and their presence in atmosphere
 - c. Level Switches: Liquidometer, float switches or valves

- d. Flow Switches: Air, medical gas, water
 - e. Proximity Switches: O.S.&Y. valve tamper
 - f. Valve Operators: Electric or air motors or pistons
- 3. Magnetic motor starters and their associated auxiliary contacts (other than Motor Control Center), pushbutton stations, selector switches, pilot lights and control switches that are integral with factory prewired packaged equipment.
 - 4. Unless otherwise noted or specified, setting of mechanical equipment and associated safety and control devices will be done by Division 22 and 23 Contractors.
 - 5. Automatic Temperature Control System including all necessary power wiring.
 - 6. Motors for mechanical equipment.
 - 7. Field wiring, other than power wiring, for electrical accessories furnished as an integral part of equipment furnished under Division 22 and 23.
 - 8. Integral fused disconnect switch, circuit breaker or motor circuit breaker in control panels for Division 22 equipment, unless otherwise indicated.
 - 9. Furnishing, installing and wiring of pilot switches, relays and other control devices required for the proper sequence, electrical or pneumatic control of the apparatus furnished under Division 22 and 23, and not specified or shown on Drawings as being furnished under Automatic Temperature Control System or under Division 26 of the Specifications.
 - 10. Variable frequency drives (VFD) furnished by Division 23 that are either integral with factory prewired packaged equipment or installed separately.
 - 11. Passive communication cabling, terminations and devices are provided by Division 27.
 - 12. Furnishing and installing heat trace cable and electric heaters in mechanical equipment, including control wiring.
 - 13. Furnishing and installing humidifiers including control wiring.

B. WORK REQUIRED BY DIVISION 26

- 1. Provide all labor, materials, equipment, components and tools for required wiring and connections to control devices and instruments, as specified herein.
- 2. Provide all recording, indicating devices, contactors, time switches, photoelectric devices and lighting dimming equipment, as specified.
- 3. Install, completely wire and connect all systems specified herein in accordance with details on drawings and manufacturer's instructions.
- 4. Install and adjust all mechanical and electrical interlocks. Repair or replace faulty equipment.
- 5. Install, wire and connect electric heater non-integral thermostats.
- 6. Provide magnetic and manual motor starters for equipment furnished by others except where starters and controllers are furnished as an integral part of prewired packaged equipment.
- 7. Provide power wiring to starters and from starters to motor including all connections. Wire capacitors, VFD and associated filter when they are provided by Division 23. Exception: When motors, starters, controllers, etc., are furnished as an integral part of any equipment furnished under Division 23, only power wiring to power terminals on said equipment shall be included herein.
- 8. Provide Motor Control Centers and/or Motor Starters including integral pushbuttons, pilot light, control switches, auxiliary contacts and the like, required for proper sequence and control function specified under Division 23.
- 9. Provide and wire all manual motor switches as required and specified.

10. Provide motor and circuit disconnect safety switches, both fusible and nonfusible types, as shown or as required by the NEC. Integral disconnect switches for power roof ventilators will be included under Division 23 and in equipment control panels under Division 22.
11. Provide branch circuits for pipe tracing electric heating cable, strip type space heaters, humidifiers and snow melting systems furnished and installed by Divisions 22 and 23.
12. Installation of electric unit heaters, cabinet unit heaters, baseboard heaters, and duct heaters including their power and control wiring.
13. Provide all interlock and control wiring shown or referred to on the Electrical and Mechanical Drawings.
14. Provide grounded convenience outlet, or grounded junction box for solid connection when required, for each electric water cooler. Locate properly so connection is not visible.
15. Installation of variable frequency drives (VFD). Exception: When furnished as an integral part of prewired packaged equipment. Provide power wiring to VFD and from VFD to motor including all connections. Obtain manufacturer's installation instructions and wiring diagrams from Division 22 and 23.

C. Automatic Temperature Controls (ATC)

1. All temperature control equipment and devices will be provided under Division 25, with the exception of starters contained in Motor Control Centers (M.C.C.'s) and any devices indicated or specified otherwise under Division 26.
2. The following controls will be furnished and installed by Division 23, and as noted herein, wired and connected by Division Contractor indicated.
 - a. Line Voltage Thermostats – Division 25
 - b. Aquastats, Freezestats, Firestats – Division 25
 - c. Air Flow Switches, Float and Pressure Switches – Division 25
 - d. Pneumatic Electric (P.E.) and Electric Pneumatic (E.P.) Switches – Division 25
 - e. Duct Mounted Smoke Detectors - Furnished and wired to Fire Alarm System - Division 28. Installed and wired to dampers, fan controls and other mechanical devices – Division 25
 - f. Initiating Alarm Contacts for Supervisory (Surveillance) Alarm Wiring - Division 28
 - g. Control Transformers for HVAC Control Equipment - Division 25; exception control transformers in Motor Control Centers provided under Division 26.
 - h. Selector Switches for indexing spare chilled and condenser water pumps to primary pump service - provided by Division 25, wired by Division 26
3. Coordinate all temperature control equipment device locations and test with the respective Division 23 Contractor.

D. Air Volume Control Box Wiring

1. Unless otherwise noted in the Panelboard Schedule, connect no more than ten (10) air volume control boxes on a 20A, 120V branch circuit.
2. Connect all air volume control boxes to the nearest normal 208/120V power panelboard within the respective electrical circuit zone, unless the air handler that serves the air volume control box is on emergency power and/or indicated in the air volume control box schedule shown on the mechanical drawings. Air volume control boxes requiring emergency power shall be connected to the nearest 208/120V emergency panel that is fed by the same

automatic transfer switch (emergency branch) that supplies power to the respective air handler.

3. Refer to Mechanical Drawings for quantity and locations of air volume control boxes.

1.13 EQUIPMENT LOCATIONS

- A. Locations are subject to changes in order to avoid obstacles in building construction. Verify all dimensions and conditions at site. Check layout for sizes and clearances, so that the apparatus and material may be installed and operated satisfactorily in space provided. Install equipment and raceways to preserve headroom and to keep openings and passageways clear.
- B. Install equipment, boxes and outlets in accessible locations. Obtain final locations of all outlets and equipment from details on drawings and from Architect. Examine drawings of other trades and avoid interferences with their work.
- C. In case of conflict in location of flush outlets, architectural details shall take precedence.
- D. Install conduit to avoid mechanical and/or structural obstructions, minimizing crossovers.
- E. Install all exposed conduits parallel or perpendicular to building lines.
- F. Provide minimum of 6 inches clearance between electrical work and flues, steam pipes and other heat sources.
- G. Mounting heights of outlets and equipment shall be as indicated on "Mounting Height" Schedule, or as specified herein.
- H. Verify all door swings before installing switch boxes. In case of conflict between drawings, Architectural details shall take precedence.
- I. Architect reserves the right to change, without additional cost, location of any luminaire, wall switch, receptacle or power outlet, provided such changed location is not more than 10 feet, and is ordered changed before said work is completely "roughed in".
- J. Locations of electrical equipment and connections to all other equipment are approximately correct and are subject to such modifications as are required at time of installation, in order to meet field conditions or the dimensions of equipment actually being supplied.
- K. No changes are to be made in the original design without written approval by Architect.

1.14 SEISMIC REQUIREMENTS

- A. Seismic Design
 1. Installation of all new electrical systems (equipment and raceways) to be in accordance with seismic requirements of the applicable building code and site specific seismic design parameters, considering exemptions where applicable.
 2. Refer to Division 01, "Summary" for seismic criteria to be used on this project.

PART 2 - PRODUCTS

2.1 GROUT

- A. Nonmetallic, Shrinkage-Resistant Grout: ASTM C1107, factory-packaged, nonmetallic aggregate grout, noncorrosive, nonstaining, mixed with water to consistency suitable for application and a 30-minute working time.

PART 3 - EXECUTION

3.1 COMMON REQUIREMENTS FOR ELECTRICAL INSTALLATION

A. General

1. Comply with NECA 1.
2. Measure indicated mounting heights to bottom of unit for suspended items and to center of unit for wall-mounting items.
3. Headroom Maintenance: If mounting heights or other location criteria are not indicated, arrange and install components and equipment to provide maximum possible headroom consistent with these requirements.
4. Equipment: Install to facilitate service, maintenance, and repair or replacement of components of both electrical equipment and other nearby installations. Connect in such a way as to facilitate future disconnecting with minimum interference with other items in the vicinity.
5. Right of Way: Give to piping systems installed at a required slope.
6. Furnish, deliver, erect, connect and finish in every detail, all materials, equipment and accessories required for the Work. Select and arrange to fit properly into the building spaces.
7. Perform all work in accordance with the drawings, specifications, including manufacturer's installation instructions, all applicable codes and NECA's Standard of Installation guidelines.
8. Include in the Work and in the bid proposal minor details not shown or specified, but manifestly necessary for the proper installation and operation of the various systems, as if specified or shown.
9. Position and install all material and equipment to permit proper access and in such a manner that maintenance, adjustment, calibration, inspection, repair and replacement of the material and equipment can be accomplished with minimum effort and cost.
10. Perform the installation, wiring, cleaning, testing, calibration and startup of all material and equipment in accordance with the manufacturers' instructions and recommendations. Such instructions and recommendations are hereby made a part of these Specifications.
11. If any departures from Contract Documents are deemed necessary, submit details of such departures and the reasons there for to Architect for approval.
12. Pull and junction boxes shall be located and sized by the electrical contractor in accordance with NEC, EIA/TIA, utility company requirements and/or owner standards, unless otherwise noted on the drawings.

B. Layout and Coordination

1. Lay out all work from approved building and property lines and benchmarks. Verify and be responsible for the correctness of all measurements in connection with work. Any change made in major overall dimensions as shown which affect the physical size, shape, or location of any part of the Work, whether due to field check or changes due to the use of equipment of a manufacturer other than that used as the basis of design shall not cause any interference with other work.
2. Examine the drawings of other trades and initiate cooperation and coordination of the Work with the work of other trades to ensure that the Work can be installed properly as designed and planned without interference with other work or delay. Furnish all necessary templates, patterns, measurements, etc., for installing work and for the purpose of making adjoining work conform; furnish setting plans and shop details to other trades as required.
3. Investigate the structural and finish conditions affecting the Work. Offsets, bends or other items required by the Work may not be shown due to the small scale of the drawings; provide such offsets, bends or other items as required to meet structural or finish conditions.
4. Coordinate and be responsible for the required clearances of the Work in shafts, chases, double partitions and suspended ceilings. Coordinate and cooperate with the trades responsible for constructing such spaces, together with other trades sharing such spaces, and advise other trades of the requirements of the Work. Immediately submit for review space requirements that exceed those shown.
5. Install material and equipment as high as possible; at minimum, to clear the top of all doors, windows and other structural openings. Maintain maximum headroom and space conditions in every case. Where headroom or space conditions appear inadequate, notify the Architect before proceeding with the installation.
6. Install conduit, fittings, etc., to provide not less than 1/2 inch between their finished covering and the structure or adjacent work of any kind.
7. Electrical equipment shall not interfere in any way with other material or equipment and shall be provided with adequate working space; see the National Electrical Code working space requirements.
8. Make reasonable modifications in the layout of the Work, as directed, to provide proper clearances or accessibility, or to prevent conflict with the work of other trades, at no increase in the Contract sum.
9. Cooperate fully with the Contractor for General Construction in regards to location of electrical equipment and work progress schedules. Notify him of all flush panelboard locations so that wall of proper thickness is provided.
10. Prepare large scale composite working drawings, including such section views and details as are necessary to clearly show how the Work is to be installed in relation to the work of other trades. Issue such drawings to the other trades for coordination of their work. Where such drawings show deviations from the Contract Drawings or conflict with other trades, detail and submit such deviation or conflicts to the Architect for review.
11. Locate wall switches at strike side of doors and at height indicated on "Mounting Height" schedule. Review all door swings with Contractor for General Construction prior to rough-in.
12. Locate receptacles at heights indicated in "Mounting Height" schedule. Mount receptacles vertically, ground pole at top. In special areas such as kitchens, laboratories, utility areas, coordinate locations with counters, benches and casework.
13. All devices and wiring are to be concealed where possible. Where specifically shown or where approved by the Architect, install exposed outlet boxes and branch circuit wiring in

finished areas in formed metallic surface raceway systems using suitable factory fabricated fittings and devices as specified herein.

14. If work is installed before coordinating with all other trades and Owner's work, or so as to cause interference with the work of other trades, or so as not to provide proper access for maintenance or repair, make necessary changes in work to correct the condition at no cost to the Owner.

C. Cutting and Patching

1. Except where specified otherwise in Division 26, provide cutting, patching and refinishing work in accord with the requirements of Division 01, GENERAL REQUIREMENTS.
2. Horizontal chases shall not be cut into existing walls or partitions without approval of Architect.

D. Painting

1. Except where specified otherwise in Division 26, general painting will be provided under Division 09, FINISHES.
2. Touch up or paint out damage done to items having a factory applied finish, and which are installed under Division 26, utilizing materials and methods specified in Division 09. FINISHES.

3.2 PROTECTION OF WORK

- A. Protect all conduit, fittings, panelboards, switchgear, transformers and other equipment before and during installation and keep clean.
- B. Protect factory finished equipment, luminaires, panels, switchgear and devices with approved temporary protective material where these items are subject to accidental damage or abuse. Electrical equipment and switchgear shall be stored indoors or otherwise securely protected and kept free of condensation by adequate electric heat. Contractor shall remove all temporary protective material at the conclusion of the Work or as directed.
- C. The Contractor shall assume full responsibility for the cost of repairing or replacing any damaged Work or material caused by employees working under this Division.

3.3 TESTS

- A. Test equipment, including switchgear, motor starters, motors, panelboards and all other equipment to verify that items are free from unintended grounds, short circuits, and open circuits and that equipment will operate as specified. Test feeders for insulation resistance; for load balance of the final installation, and for overall operation of systems. Furnish labor and material required for making such tests and make corrections necessary to balance the load and to obtain proper operation.
- B. Measure secondary voltages after energizing transformers and adjust transformer taps to provide rated voltages listed on drawings and/or specifications. After building is in use, measure these voltages during a period of "normal load" and "light load". Report results to Architect, who will

advise Contractor whether or not further adjustment of voltages is required. Make changes as directed.

- C. Test all wiring systems up to 600 volts, for insulation resistance with a megger in accordance with the NEC. Determine values with switchboards, panelboards, fuseholders, switches, receptacles and overcurrent devices in place.
- D. Determine ampere rating of thermal overload relays in motor starters and compare with equipment nameplate and actual current measurement of the motor. If overload relays are found to be incorrect, provide proper size. Provide proper motor rotation.
- E. Arrange for each system to be fully tested and adjusted by manufacturer or his authorized representative. Each element of each system shall be individually operated to insure that it will function as intended. Furnish all labor and material required to correct all defects.
- F. Submit to the Architect a letter from manufacturer (or authorized representative) of each system, attesting to the fact that all necessary tests and adjustments have been made and that the entire system is functioning properly in every respect.
- G. This article shall not be construed as deleting other tests specifically outlined in other sections of this Specification.

3.4 WORKMANSHIP

- A. Electrical equipment shall be installed in a neat and workmanlike manner in accordance with latest and best practices of the trade.
- B. Only mechanics skilled in this type of Work shall be employed and utilized by Contractor for this Division in the execution of this Work.

3.5 REFINISHING

- A. All surfaces of boxes, cabinets and equipment shall have suitable lacquer, enamel or plated finishes. Touch up any finishes marred during construction. Supports and other metal work not furnished with a protective coating, shall be given two coats of approved paint after completion of the work.

3.6 FIRESTOPPING

- A. Apply firestopping to penetrations of fire-rated floor and wall assemblies for electrical installations to restore original fire-resistance rating of assembly. Firestopping materials and installation requirements are specified in Division 07, THERMAL AND MOISTURE PROTECTION.
- B. Install all fire-rated cable pathways in accordance with manufacturer's recommendations.

3.7 CONTINUITY OF EXISTING SERVICES

- A. Perform alterations and connections to existing facilities with a minimum of interruption. Where interruption is necessary, prepare a time schedule for same, coordinate with Architect, Owner and other sections, and obtain prior written clearance from Owner. Provide and place notices in affected areas, and on luminaires or equipment, etc., which will be temporarily out of use. Remove notices when interruption has been completed.
- B. When Owner requires that work involving shutdown and interruption of existing facilities be performed on an overtime basis, Contractor shall pay all overtime costs.

3.8 DEMOLITION

- A. Disconnect and cap existing DIVISION 26 services to demolished areas of existing building(s) as required. Cap services at points where indicated by Contractor for Division 2, EXISTING CONDITIONS. Contractor for Division 2 will remove abandoned DIVISION 26 services between capped services and demolished areas.

3.9 ALTERATIONS AND CONNECTIONS TO EXISTING FACILITIES

- A. Make all necessary alterations to existing DIVISION 26 systems to permit connecting or extending these systems to new work and to permit existing systems to remain in use whether indicated or not. New materials used to alter existing systems shall match existing materials unless otherwise indicated. Record modifications for Owner's future use.
- B. Make all necessary alterations to existing DIVISION 26 systems to permit the installation of new DIVISION 23 equipment, i.e. ductwork and piping, to permit connecting or extending these systems to new work and to permit existing systems to remain in use whether indicated or not. New materials used to alter existing systems shall match existing materials unless otherwise indicated. Record modifications for Owner's future use.
- C. Where equipment, ductwork and piping is removed or disconnected under DIVISION 26, perform the work in such a manner that no damage is done to the structure or remaining portions of the systems.
- D. Remove exposed conduit, hangers and supports made obsolete due to this modification.
- E. Where existing concealed conduit is not to be reused, abandon same in place unless otherwise indicated or specified.
- F. Unless otherwise specified, all materials and equipment removed or disconnected by Contractor which are not to be reused shall become the property of Contractor and be removed from premises.

3.10 COMMISSIONING

- A. Commissioning will be provided as specified in Division 01 Section "Commissioning". All contractors and subcontractors of the various sections of this specification shall cooperate and

participate in the commissioning work in accordance with requirements of Division 01 Section "Commissioning".

- B. Ensure participation of major equipment manufacturers or their representatives.
- C. Equipment and systems/subsystems installed under this section are expected to be in full compliance with the design intent by the commissioning phase. Notify the Commissioning Agent when any specific piece of equipment or specific system/subsystem is ready for commissioning. Be prepared to demonstrate system readiness.
- D. Equipment or systems/subsystems having incomplete work or exhibiting problems related to noncompliance with the design intent shall require commissioning. The contractor for this section shall be fully responsible to make all necessary corrections to incomplete or non-complying work at their own expense and shall pay the Commissioning Agent per diem rate for recommissioning such incomplete or non-complying work.

END OF SECTION 26 05 00

SECTION 26 05 19 - LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Building wires and cables rated 600 V and less.
 - 2. Connectors, splices, and terminations rated 600 V and less.
- B. Related sections include the following:
 - 1. Division 26 Section "Common Material and Methods for Electrical"
 - 2. Division 26 Section "Identification for Electrical Systems"

1.3 DEFINITIONS

- A. EPDM: Ethylene-propylene-diene terpolymer rubber.
- B. NBR: Acrylonitrile-butadiene rubber.
- C. NETA: International Electrical Testing Association.
- D. NRTL: Nationally Recognized Testing Laboratory.
- E. Homerun: The circuit conductors between the branch circuit outlet and/or device and the panel board over current protective device.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

1.5 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For testing agency.
- B. Field quality control reports.

1.6 QUALITY ASSURANCE

- A. Testing Agency Qualifications: Member company of NETA or an NRTL.
 - 1. Testing Agency's Field Supervisor: Certified by NETA to supervise on-site testing.
- B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended location and application.
- C. Comply with NFPA 70.

PART 2 - PRODUCTS

2.1 CONDUCTORS AND CABLES

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. 600V Insulated Building Wire
 - a. Alpha Wire Corp.
 - b. General Cable Technologies Corporation
 - c. Anixter
 - d. Southwire Company
 - e. Encore Wire Corporation
 - f. The Okonite Company
 - 2. 600V Insulated Metallic Clad Cable: Metal-Clad, Type MC.
 - a. AFC Cable Systems Inc.
 - b. Aetna Insulated Wire
 - c. Southwire Company
 - d. Kaf Tech Inc.
- B. Copper Conductors: Comply with NEMA WC 70/ICEA S-95-658.
- C. Conductor Insulation: Comply with NEMA WC 70/ICEA S-95-658 for Types THW-2, THHN-2, THWN-2, XHHW-2, UF, USE and SO.
- D. Multiconductor Cable: Comply with NEMA WC 70/ICEA S-95-658 for; metal-clad cable, Type MC.
- E. Multiconductor Cable Sheathing: Aluminum or steel.

2.2 CONNECTORS AND SPLICES

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. AFC Cable Systems, Inc.
 2. Hubbell Power Systems, Inc.
 3. O-Z/Gedney; EGS Electrical Group LLC
 4. 3M Company
 5. Ideal Industries, Inc.
 6. Tyco Electronics Corp.
- B. Description: Factory-fabricated connectors and splices of size, ampacity rating, material, type, and class for application and service indicated. Die cast type connectors and fittings are not acceptable.

PART 3 - EXECUTION

3.1 CONDUCTOR MATERIAL APPLICATIONS

- A. Feeders: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.
- B. Branch Circuits: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger, unless otherwise indicated.

3.2 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

- A. Exposed Feeders: Type THHN-2-THWN-2, single conductors in metal raceway.
- B. Feeders Concealed in Ceilings, Walls, Partitions, and Crawlspace: Type THHN-2-THWN-2, single conductors in metal raceway.
- C. Exposed Branch Circuits, Including in Crawlspace: Type THHN-2-THWN-2, single conductors in metal raceway.
- D. Normal Power and Lighting Branch Circuits Concealed in Ceilings, Walls, and Partitions: Type THHN-2-THWN-2, single conductors in metal raceway.
- E. Emergency Power and Lighting Branch Circuits Concealed in Ceilings, Walls, and Partitions: Type THHN-2-THHW-2, single conductors in metal raceway.
- F. Class 1 Control Circuits: Type THHN-2-THWN-2, in raceway.
- G. Class 2 Control Circuits: Type THHN-2-THWN-2, single conductors in metal raceway.
- H. VFD Output Circuits: Type XHHW-2, single conductors in metal raceway.

3.3 INSTALLATION OF CONDUCTORS AND CABLES

- M. Conceal cables in finished walls, ceilings, and floors, unless otherwise indicated.
- N. Complete raceway installation between conductor and cable termination points according to Division 26, Section "Raceways and Boxes for Electrical Systems" prior to pulling conductors and cables through raceways.
- O. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.
- P. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips, that will not damage cables or raceway.
- Q. Install exposed cables parallel and perpendicular to surfaces of exposed structural members and follow surface contours where possible.
- R. Support cables according to Division 26 Section "Hangers and Supports for Electrical Systems."
- S. Identify and color-code conductors and cables according to Division 26 Section "Identification for Electrical Systems."
- T. Where specifically shown or where approved by the Architect, install exposed branch circuit wiring in finished areas in formed metallic surface raceway systems using suitable factory fabricated fittings and devices as specified herein.
- U. Multi-wire branch circuits will not be allowed. A separate neutral conductor shall be provided for each branch circuit.
- V. The continuity of a neutral conductor shall not be dependent upon device and luminaire connections, where removal of such device would interrupt the continuity of circuit.
- W. All conductors of a parallel feeder shall be of the same length.
- X. Lace or clip groups of conductors at lighting and distribution panels, pull boxes and wireways.
- Y. The use of MC Cable is strictly prohibited except as follows: connections from nearest branch circuit junction box to a ceiling mounted lighting control device or luminaire may use metal-clad type MC Cable in lengths not to exceed 6 feet. Install per requirements of NEC and support at each junction box.
- Z. Insulated conductors and cables installed in underground enclosures or raceways shall be listed for use in wet locations and shall comply with NEC Article 310.10(C). Any connections or splices in an underground installation shall be approved for wet locations.
- AA. Wireways and junction boxes shall not be used for routing and/or splicing of branch circuit, power and control wiring in excess of thirty current carrying conductors.
- BB. Do not install conduits and cables in the cell of the metal deck above bar joists or other structural member.

CC. For branch circuits a maximum of six current carrying conductors shall be permitted to be installed in a raceway.

DD. The smallest size conductors permitted are:

1. Lighting and Power Branch Circuits - No. 12 AWG
2. Control Circuits - No. 14 AWG
3. Luminaires (Pig Tails) - No. 16 AWG
4. Fire Alarm System - No. 14 AWG
5. Branch circuits rated 20 amperes, 120 volts and longer than 100 circuit feet from panelboard shall be #10 AWG.
6. Branch circuits rated 20 amperes, 277 volts and longer than 150 circuit feet from panelboard shall be #10 AWG.

3.4 CONNECTIONS

- A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A and UL 486B.
- B. Make splices and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.
- C. Wiring at Outlets: Install conductor at each outlet, with at least 6 inches of slack.

3.5 IDENTIFICATION

- A. Identify and color code conductors and cables according to Division 26, Section "Identification for Electrical Systems."
- B. Identify each spare conductor at each end with identity number and location of other end of conductor and identify as spare conductor.

3.6 FIRESTOPPING

- A. Apply firestopping to electrical penetrations of fire-rated floor and wall assemblies to restore original fire-resistance rating of assembly according to Division 26, Section "Common Materials and Methods for Electrical" and Division 7, Section "Firestopping."

3.7 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.
- B. Tests and Inspections:
 1. Test all conductors and cables to verify that no short circuits or accidental grounds exist.

2. After installing conductors and cables and before electrical circuitry has been energized, test service entrance and feeder conductors, per inspection and test procedures Section 7.3.2 as stated in the NETA Acceptance Testing Specification. Certify compliance with test parameters.
- C. Test and Inspection Reports: Prepare a written report to record the following:
1. Procedures used.
 2. Results that comply with requirements.
 3. Results that do not comply with requirements and corrective action taken to achieve compliance with requirements.
- D. Cables will be considered defective if they do not pass tests and inspections. Perform corrective action, including replacement and retest.

END OF SECTION 26 05 19

SECTION 26 05 26 - GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes methods and materials for grounding systems and equipment.
 - 1. Common ground bonding.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated.

1.4 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- B. Comply with UL 467 for grounding and bonding materials and equipment.

PART 2 - PRODUCTS

2.1 CONDUCTORS

- A. Insulated Conductors: Copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction.
- B. Equipment grounding conductor insulation color shall be continuous green, except for wire sizes No. 4 AWG and larger which shall be identified per NEC.
- C. Bare Copper Conductors:
 - 1. Solid Conductors: ASTM B3
 - 2. Stranded Conductors: ASTM B8

2.2 CONNECTORS

- A. Listed and labeled by a nationally recognized testing laboratory acceptable to authorities having jurisdiction for applications in which used, and for specific types, sizes, and combinations of conductors and other items connected.
- B. Bolted Connectors for Conductors and Pipes: Copper or copper alloy, bolted pressure-type, with at least two bolts.
 - 1. Pipe Connectors: Clamp type, sized for pipe.
- C. Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.

PART 3 - EXECUTION

3.1 APPLICATIONS

- A. Conductors: Install solid conductor for No. 10 AWG and smaller, and stranded conductors for No. 8 AWG and larger, unless otherwise indicated.
- B. Conductor Terminations and Connections:
 - 1. Pipe and Equipment Grounding Conductor Terminations: Bolted connectors.
 - 2. Connections to Ground Rods at Test Wells: Compression or welded connectors.
 - 3. Connections to Structural Steel: Exothermic-welding connectors.

3.2 EQUIPMENT GROUNDING

- A. Install insulated equipment grounding conductors with all feeders and branch circuits.
- B. Air-Duct Equipment Circuits: Install insulated equipment grounding conductor to duct-mounted electrical devices operating at 120 V and more, including air cleaners, heaters, dampers, humidifiers, and other duct electrical equipment. Bond conductor to each unit and to air duct and connected metallic piping.

3.3 INSTALLATION

- A. Grounding Conductors: Route along shortest and straightest paths possible, unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.

3.4 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing and inspecting agency to perform the following field tests and inspections and prepare test reports:

1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements.
 2. Test completed grounding system at each location where a maximum ground-resistance level is specified, at service disconnect enclosure grounding terminal. Make tests at ground rods before any conductors are connected.
 - a. Measure ground resistance not less than two full days after last trace of precipitation and without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance.
 - b. Perform tests by fall-of-potential method according to IEEE 81.
- B. Report measured ground resistances that exceed the following values:
1. Power and Lighting Equipment or System with Capacity 500 kVA and Less: 10 ohms
 2. Power and Lighting Equipment or System with Capacity 500 to 1000 kVA: 5 ohms
 3. Power and Lighting Equipment or System with Capacity More Than 1000 kVA: 3 ohms
 4. Power Distribution Units or Panelboards Serving Electronic Equipment: 3 ohm(s)
- C. Excessive Ground Resistance: If resistance to ground exceeds specified values, notify Architect promptly and include recommendations to reduce ground resistance.

END OF SECTION 26 05 26

SECTION 26 05 29 - HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Hangers and supports for electrical equipment and systems

1.3 DEFINITIONS

- A. EMT: Electrical metallic tubing
- B. IMC: Intermediate metal conduit
- C. RMC: Rigid metal conduit

1.4 PERFORMANCE REQUIREMENTS

- A. Design supports for multiple raceways capable of supporting combined weight of supported systems and its contents.
- B. Design equipment supports capable of supporting combined operating weight of supported equipment and connected systems and components.

1.5 SUBMITTALS

- A. Product Data: For the following:
 - 1. Steel slotted support systems
 - 2. Nonmetallic slotted support systems
- B. Shop Drawings: Show fabrication and installation details and include calculations for the following:
 - 1. Trapeze hangers. Include Product Data for components.
 - 2. Steel slotted channel systems. Include Product Data for components.
 - 3. Nonmetallic slotted channel systems. Include Product Data for components.

- 4. Equipment supports.
- C. Welding certificates.

1.6 QUALITY ASSURANCE

- A. Welding: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."
- B. Comply with NFPA 70.

PART 2 - PRODUCTS

2.1 SUPPORT, ANCHORAGE, AND ATTACHMENT COMPONENTS

- A. Steel Slotted Support Systems: Comply with MFMA-4, factory-fabricated components for field assembly.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Allied Tube & Conduit
 - b. Cooper B-Line, Inc.; a division of Cooper Industries
 - c. ERICO International Corporation
 - d. GS Metals Corp.
 - e. Thomas & Betts Corporation
 - f. Unistrut; Tyco International, Ltd.
 - g. Wesanco, Inc.
 - 2. Metallic Coatings: Hot-dip galvanized after fabrication and applied according to MFMA-4.
 - 3. Channel Dimensions: Selected for applicable load criteria.
- B. Raceway and Cable Supports: As described in NECA 1 and NECA 101.
- C. Conduit and Cable Support Devices: Steel hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.
- D. Support for Conductors in Vertical Conduit: Factory-fabricated assembly consisting of threaded body and insulating wedging plug or plugs for non-armored electrical conductors or cables in riser conduits. Plugs shall have number, size, and shape of conductor gripping pieces as required to suit individual conductors or cables supported. Body shall be malleable iron.
- E. Structural Steel for Fabricated Supports and Restraints: ASTM A36/A36M, steel plates, shapes, and bars; black and galvanized.
- F. Mounting, Anchoring, and Attachment Components: Items for fastening electrical items or their supports to building surfaces include the following:

1. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete, steel, or wood, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
 - a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1) Hilti Inc.
 - 2) ITW Ramset/Red Head; a division of Illinois Tool Works, Inc.
 - 3) MKT Fastening, LLC
 - 4) Simpson Strong-Tie Co., Inc.; Masterset Fastening Systems Unit
2. Mechanical-Expansion Anchors: Insert-wedge-type, stainless steel, for use in hardened portland cement concrete with tension, shear, and pullout capacities appropriate for supported loads and building materials in which used.
 - a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1) Cooper B-Line, Inc.; a division of Cooper Industries
 - 2) Empire Tool and Manufacturing Co., Inc.
 - 3) Hilti, Inc.
 - 4) ITW Ramset/Red Head; a division of Illinois Tool Works, Inc.
 - 5) MKT Fastening, LLC
3. Concrete Inserts: Steel or malleable-iron, slotted support system units similar to MSS Type 18; complying with MFMA-4 or MSS SP-58.
4. Clamps for Attachment to Steel Structural Elements: MSS SP-58, type suitable for attached structural element.
5. Through Bolts: Structural type, hex head, and high strength. Comply with ASTM A325.
6. Toggle Bolts: All-steel springhead type.
7. Hanger Rods: Threaded steel.
8. Plastic Nylon Ties: Not acceptable.

PART 3 - EXECUTION

3.1 APPLICATION

- A. Comply with NECA 1 and NECA 101 for application of hangers and supports for electrical equipment and systems except if requirements in this Section are stricter.
- B. Maximum Support Spacing and Minimum Hanger Rod Size for Raceway: Space supports for EMT, IMC, and RMC as required by NFPA 70. Minimum rod size shall be 1/4 inch in diameter.
- C. Multiple Raceways or Cables: For horizontal pendent installation, install trapeze-type supports fabricated with steel slotted support system, sized so capacity can be increased by at least 25 percent in future without exceeding specified design load limits. For vertical surface

installation, install steel slotted support system anchored to structural wall system for the entire length of run.

1. Secure raceways and cables to these supports with two-bolt conduit clamps.

- D. Spring-steel clamps designed for supporting single conduits without bolts may be used for 1-1/2-inch and smaller raceways serving branch circuits and communication systems above suspended ceilings and for fastening raceways to trapeze supports.

3.2 SUPPORT REQUIREMENTS

A. General

1. Provide materials, equipment, supplies and labor necessary as required to adequately support, brace and strengthen all equipment and materials furnished as part of this work.
2. Materials, equipment, apparatus supports and mounting hardware shall be approved for use in the location installed. For example, use electroplated galvanized, hot dipped galvanized, epoxy coated, PVC coated, stainless steel or aluminum in wet/outdoor locations.

B. Attachment to Building Structure

1. Light fixtures, raceways, boxes, cable trays, wireways, and other utilities shall not be supported by roof deck.
2. Light fixtures, raceways, boxes, cable trays, wireways may be suspended from floor deck provided the attachment is designed and supplied by the Contractor.
3. Transformers, switches, starters, apparatus, or any group of conduits in excess of 100 pounds shall not be supported by floor deck, regardless of concrete thickness above steel deck.
4. Transformers, switches, starters, apparatus, or any group of conduits in excess of 100 pounds shall not be suspended by concrete roof or floor slabs unless specifically shown on the contract drawings.
5. Whenever possible, support shall be provided directly to the main steel or concrete framing beams. If spacing of structure exceeds the spacing required to sufficiently support the electrical work, supplemental steel member or steel channel shall be designed and provided by the Contractor.
6. Support all Electrical work independently of any other trades. Under no circumstances shall work be supported or suspended from ceiling grids, ductwork, piping, or other supports for other trades.
7. Before drilling concrete for attachments, carefully check Drawings and Shop Drawings for such concrete and locate drilled holes to miss reinforcing by at least 1 inch.
8. Inserts in precast concrete to support Work of Division 26 will be furnished and installed by precast concrete supplier. Prepare drawings locating such inserts for review by Architect before distribution.

3.3 SUPPORT INSTALLATION

- A. Comply with NECA 1 and NECA 101 for installation requirements except as specified in this Article.

- B. Raceway Support Methods: In addition to methods described in NECA 1, EMT may be supported by openings through structure members, as permitted in NFPA 70.
- C. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb.
- D. Mounting and Anchorage of Surface-Mounted Equipment and Components: Install steel slotted channel-racks to finish wall anchored and fastened to structural floor and ceiling, and wall were crossing structural members. Anchor and fasten electrical items to the steel slotted support system. The steel slotted channel-racks shall be anchored and fastened by the following methods unless otherwise indicated by code:
 - 1. To Wood: Fasten with lag screws or through bolts.
 - 2. To New Concrete: Bolt to concrete inserts.
 - 3. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
 - 4. To Existing Concrete: Expansion anchor fasteners.
 - 5. Structural Steel: Beam clamps, angle clips welded or powder-driven studs.
 - 6. Exterior: Anchors, screws, bolts, etc., exposed to weather and corrosion shall be rustproof finished or nonferrous material.
 - 7. Adhesive Anchors: The use of adhesive anchors is prohibited.
 - 8. Instead of expansion anchors, powder-actuated driven threaded studs provided with lock washers and nuts may be used in existing standard-weight concrete 4 inches thick or greater. Do not use for anchorage to lightweight-aggregate concrete or for slabs less than 4 inches thick.
 - 9. To Steel: Beam clamps (MSS Type 19, 21, 23, 25, or 27) complying with MSS SP-69.
 - 10. To Light Steel: Sheet metal screws.
- E. Drill holes for expansion anchors in concrete at locations and to depths that avoid reinforcing bars.

3.4 INSTALLATION OF FABRICATED METAL SUPPORTS

- A. Comply with installation requirements in Division 05 Section "Metal Fabrications" for site-fabricated metal supports.
- B. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor electrical materials and equipment.
- C. Field Welding: Comply with AWS D1.1/D1.1M.

3.5 PAINTING

- A. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.

1. Apply paint by brush or spray to provide minimum dry film thickness of 2.0 mils.
- B. Touchup: Comply with requirements in Division 09 painting Sections for cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal.
- C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A780.

END OF SECTION 26 05 29

SECTION 26 05 33 - RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes raceways, fittings, boxes, enclosures, and cabinets for electrical wiring.

1.3 DEFINITIONS

- A. EMT: Electrical metallic tubing
- B. ENT: Electrical nonmetallic tubing
- C. EPDM: Ethylene-propylene-diene terpolymer rubber
- D. FMC: Flexible metal conduit
- E. IMC: Intermediate metal conduit
- F. LFMC: Liquidtight flexible metal conduit
- G. LFNC: Liquidtight flexible nonmetallic conduit
- H. NBR: Acrylonitrile-butadiene rubber
- I. RNC: Rigid nonmetallic conduit

1.4 SUBMITTALS

- A. Product Data: For surface raceways, conduits, wireways and fittings, boxes, enclosures, and cabinets.
- B. Qualification Data: For professional engineer and testing agency.
- C. Source quality-control test reports.

1.5 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- B. Comply with NFPA 70.

PART 2 - PRODUCTS

2.1 METAL CONDUIT AND TUBING

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. AFC Cable Systems, Inc.
 - 2. Alfex Inc.
 - 3. Allied Tube & Conduit; a Tyco International Ltd. Co.
 - 4. Anamet Electrical, Inc.; Anaconda Metal Hose
 - 5. Electri-Flex Co.
 - 6. Manhattan/CDT/Cole-Flex
 - 7. Maverick Tube Corporation
 - 8. O-Z Gedney; a unit of General Signal
 - 9. Wheatland Tube Company
- B. Rigid Steel Conduit: ANSI C80.1.
- C. Aluminum Rigid Conduit: ANSI C80.5.
- D. IMC: ANSI C80.6.
- E. EMT: ANSI C80.3.
- F. FMC: Zinc-coated steel or aluminum.
- G. LFMC: Flexible steel conduit with PVC jacket.
- H. Fittings for Conduit (Including all Types and Flexible and Liquidtight), EMT, and Cable: NEMA FB 1; listed for type and size raceway with which used, and for application and environment in which installed.
 - 1. Conduit Fittings for Hazardous (Classified) Locations: Comply with UL 886.
 - 2. Fittings for EMT: Set-screw or compression type.
 - 3. Coating for Fittings for PVC-Coated Conduit: Minimum thickness, 0.040 inch, with overlapping sleeves protecting threaded joints.
 - 4. Fittings, couplings and connectors in general shall be of the same material as the conduit to which applied. Die cast fittings are not acceptable.
 - 5. Fittings and couplings for use with rigid and intermediate metal conduit shall be threaded type.

- I. Joint Compound for Rigid Steel Conduit or IMC: Listed for use in cable connector assemblies, and compounded for use to lubricate and protect threaded raceway joints from corrosion and enhance their conductivity.

2.2 BOXES, ENCLOSURES, AND CABINETS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 1. Cooper Crouse-Hinds; Div. of Cooper Industries, Inc.
 2. EGS/Appleton Electric
 3. Erickson Electrical Equipment Company
 4. Hoffman
 5. Hubbell Incorporated; Killark Electric Manufacturing Co. Division
 6. O-Z/Gedney; a unit of General Signal
 7. RACO; a Hubbell Company
 8. Robroy Industries, Inc.; Enclosure Division
 9. Scott Fetzer Co.; Adalet Division
 10. Spring City Electrical Manufacturing Company
 11. Thomas & Betts Corporation
 12. Walker Systems, Inc.; Wiremold Company (The)
 13. Woodhead, Daniel Company; Woodhead Industries, Inc. Subsidiary
- B. Interior Sheet Metal Outlet and Device Boxes: NEMA OS1.
 1. General: Provide pressed galvanized sheet steel boxes with knockouts to suit raceway system to be used. Supports shall also be galvanized steel.
 2. Ceiling: Provide flush, 4 inch, octagonal boxes not less than 1-1/2 inches deep with 4 knockouts. Provide 3/8 inch luminaire studs where required. Surface, 4 inch, square boxes not less than 1-1/2 inches deep.
 3. Walls: Provide flush 4 inch boxes not less than 2 inches deep with matching plaster ring for single or 2 gang outlets. For larger boxes use 1-piece construction to suit application. In masonry use deep boxes with square corners so masonry units will fit closely to box.
 4. Do not use sectional or handy boxes unless specifically noted
- C. Nonmetallic Outlet and Device Boxes: Not allowed.
- D. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.
- E. Hinged-Cover Enclosures: NEMA 250, Type 1, with continuous-hinge cover with flush latch, unless otherwise indicated.
 1. Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
- F. Cabinets:
 1. NEMA 250, Type 1, galvanized-steel box with removable interior panel and removable front, finished inside and out with manufacturer's standard enamel.
 2. Hinged door in front cover with flush latch and concealed hinge.

3. Key latch to match panelboards.
4. Metal barriers to separate wiring of different systems and voltage.
5. Accessory feet where required for freestanding equipment.

PART 3 - EXECUTION

3.1 RACEWAY APPLICATION

A. Comply with the following indoor applications, unless otherwise indicated:

1. Exposed in Unfinished Areas, Not Subject to Physical Damage: EMT
2. Exposed in Unfinished Areas and Subject to Physical Damage: Rigid steel conduit or IMC. Includes raceways installed below 10 feet, in the following locations:
 - a. Loading dock
 - b. Corridors used for traffic of mechanized carts, forklifts, and pallet-handling units
 - c. Mechanical rooms
 - d. Boiler rooms
 - e. Garages
 - f. Tunnel
 - g. Shipping and receiving areas
 - h. Fire pump rooms
3. Exposed in finished areas; Formed metallic surface raceway systems using suitable factory fabricated fittings and devices.
4. Concealed in Ceilings and Interior Walls and Partitions: EMT
5. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): FMC, except use LFMC in damp or wet locations.
6. Damp or Wet Locations: Rigid steel conduit or IMC.
7. Boxes and Enclosures: NEMA 250, Type 1, except use NEMA 250, Type 4, stainless steel in damp or wet locations.

B. Minimum Raceway Size: 3/4-inch trade size for power,

C. Raceway Fittings: Compatible with raceways and suitable for use and location.

1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings, unless otherwise indicated.
2. Use UL approved expansion fittings complete with grounding jumpers where conduits cross building expansion joints. Install bends or offsets in conduit adjacent to building expansion joints where conduit is installed above suspended ceilings.

3.2 RACEWAY INSTALLATION

A. Comply with NECA 1 for installation requirements applicable to products specified in Part 2 except where requirements on Drawings or in this Article are stricter.

- B. Keep raceways at least 6 inches away from parallel runs of flues and steam or hot-water pipes. Install horizontal raceway runs above water and steam piping.
- C. Complete raceway installation before starting conductor installation.
- D. Support raceways as specified in Division 26 Section "Hangers and Supports for Electrical Systems."
- E. Install no more than the equivalent of three 90-degree bends in any conduit run except for communications conduits, for which fewer bends are allowed.
- F. Conceal conduit and EMT within finished walls, ceilings, and floors, including walls of unfinished spaces, unless otherwise indicated.
- G. Install pull wires in empty raceways. Use polypropylene or monofilament plastic line with not less than 200-lb tensile strength. Leave at least 12 inches of slack at each end of pull wire.
- H. Install raceway sealing fittings at suitable, approved, and accessible locations and fill them with listed sealing compound. For concealed raceways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces. Install raceway sealing fittings at the following points:
 - 1. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
 - 2. Where otherwise required by NFPA 70.
 - 3. Install fitting(s) that provide expansion and contraction for at least 0.00041 inch per foot of length of straight run per deg F of temperature change.
 - 4. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at the time of installation.
- I. Flexible Conduit Connections: Use maximum of 72 inches of flexible conduit for recessed and semirecessed lighting fixtures, equipment subject to vibration, noise transmission, or movement; and for transformers and motors.
 - 1. Use LFMC in damp or wet locations subject to severe physical damage.
 - 2. Use LFMC or LFNC in damp or wet locations not subject to severe physical damage.
- J. Recessed Boxes in Masonry Walls: Saw-cut opening for box in center of cell of masonry block and install box flush with surface of wall.

3.3 BOX INSTALLATION

- A. Mount outlet boxes flush in areas other than mechanical rooms, electrical rooms, and above removable ceilings. Secure firmly in place and set true and square with finished surfaces. Provide raised covers for boxes as required to suit the wall or ceiling, construction, and finish.
- B. Do not fasten boxes to ceiling support wires or other piping systems.

- C. Install pull and junction boxes above accessible ceilings and in unfinished areas only.
- D. In Accessible Ceiling Areas: Install outlet and junction boxes no more than 6 inches from ceiling access panel or from removable recessed luminaire. Note: Removable luminaires can only be considered an access point for pull and splice boxes (housing no active components nor devices requiring routine maintenance or repair).
- E. Locate flush mounted boxes in masonry walls so that only a corner need be cut from masonry units. Coordinate masonry cutting to achieve neat opening.
- F. Do not install flush mounted boxes back to back within the same wall; Install with minimum 6 inch separation. For acoustic rated walls, boxes shall be separated by at least one stud and a minimum 12 inch separation.
- G. Do not use two-gang and multi-gang boxes in such a manner that one section contains a switch or receptacle circuit or combination of such circuits operating at more than 277V.
- H. For boxes mounted in exterior walls install insulation behind outlet boxes to prevent condensation in boxes and air movement. Install boxes without damaging wall insulation or reducing its effectiveness.
- I. All boxes shall be provided with a far side support bracket to prevent movement.
- J. Remove only knockouts as required and plug unused openings. Use threaded plugs for cast metal boxes and snap in metal covers for sheet metal boxes.
- K. Support boxes independently of conduit.
- L. Install stamped steel bridges to fasten flush mounted outlet boxes between studs.
- M. Install adjustable steel channel fasteners for hung ceiling outlet boxes.
- N. Fire-resistance-rated walls and ceilings; Device outlet boxes shall only penetrate one face of a fire-resistance-rated assembly. Steel electrical boxes shall not exceed 16 square inches (i.e. a typical 4" x 4" outlet box), and the sum of such penetrations shall not exceed 100 square inches within 100 square feet of wall or ceiling space. Additionally, electrical boxes on opposite sides of the walls must either be separated by a distance of 24 inches, by a distance not less than the depth of a wall cavity when filled with insulation, by solid fireblocking, by listed putty pads, or by other approved listed materials and methods. Contractor can utilize a shallow (4" x 4" x 1 1/2") depth box to meet the above criteria if; volume of the box meets the minimum conductor fill requirements of the NEC and box is of sufficient depth to accommodate the device and terminations.

3.4 PROTECTION

- A. Provide final protection and maintain conditions that ensure coatings, finishes, and cabinets are without damage or deterioration at time of Substantial Completion.

1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.

END OF SECTION 26 05 33

SECTION 26 05 53 - IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Identification for raceway and metal-clad cable.
 - 2. Identification for conductors and communication and control cable.
 - 3. Warning labels and signs.
 - 4. Instruction signs.
 - 5. Equipment identification labels.
 - 6. Miscellaneous identification products.

1.3 SUBMITTALS

- A. Product Data: For each electrical identification product indicated.
- B. Identification Schedule: An index of nomenclature of electrical equipment and system components used in identification signs and labels.

1.4 QUALITY ASSURANCE

- A. Comply with ANSI A13.1 and ANSI C2.
- B. Comply with NFPA 70.
- C. Comply with 29 CFR 1910.145.

1.5 COORDINATION

- A. Coordinate identification names, abbreviations, colors, and other features with requirements in the Contract Documents, Shop Drawings, manufacturer's wiring diagrams, and the Operation and Maintenance Manual, and with those required by codes, standards, and 29 CFR 1910.145. Use consistent designations throughout Project.
- B. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.

- C. Coordinate installation of identifying devices with location of access panels and doors.
- D. Install identifying devices before installing acoustical ceilings and similar concealment.

PART 2 - PRODUCTS

2.1 RACEWAY AND METALLIC-CLAD CABLE IDENTIFICATION MATERIALS

- A. Comply with ANSI A13.1 for minimum size of letters for legend and for minimum length of color field for each raceway and cable size.
- B. Color for Printed Legend:
 - 1. Power Circuits: Black letters on an orange field.
 - 2. Legend: Indicate system or service and voltage, if applicable.
- C. Self-Adhesive Vinyl Labels: Preprinted, flexible label laminated with a clear, weather- and chemical-resistant coating and matching wraparound adhesive tape for securing ends of legend label.
- D. Snap-Around Labels: Slit, pretensioned, flexible, preprinted, color-coded acrylic sleeves, with diameter sized to suit diameter of raceway or cable it identifies and to stay in place by gripping action.
- E. Self-Adhesive Vinyl Tape: Colored, heavy duty, waterproof, fade resistant; 2 inches wide; compounded for outdoor use.

2.2 CONDUCTOR AND COMMUNICATION- AND CONTROL-CABLE IDENTIFICATION MATERIALS

- A. Color-Coding Conductor Tape: Colored, self-adhesive vinyl tape not less than 3 mils thick by 1 to 2 inches wide.
- B. Marker Tapes: Vinyl or vinyl-cloth, self-adhesive wraparound type, with circuit identification legend machine printed by thermal transfer or equivalent process.
- C. Aluminum Wraparound Marker Labels: Cut from 0.014-inch-thick aluminum sheet, with stamped, embossed, or scribed legend, and fitted with tabs and matching slots for permanently securing around wire or cable jacket or around groups of conductors.
- D. Metal Tags: Brass or aluminum, 2 by 2 by 0.05 inch, with stamped legend, punched for use with self-locking nylon tie fastener.
- E. Write-On Tags: Polyester tag, 0.015 inch thick, with corrosion-resistant grommet and polyester or nylon tie for attachment to conductor or cable.
 - 1. Marker for Tags: Permanent, waterproof, black ink marker recommended by tag manufacturer.

2.3 WARNING LABELS AND SIGNS

- A. Comply with NFPA 70 and 29 CFR 1910.145.
- B. Self-Adhesive Warning Labels: Factory printed, multicolor, pressure-sensitive adhesive labels, configured for display on front cover, door, or other access to equipment, unless otherwise indicated.
- C. Baked-Enamel Warning Signs: Preprinted aluminum signs, punched or drilled for fasteners, with colors, legend, and size required for application. 1/4-inch grommets in corners for mounting. Nominal size, 7 by 10 inches.
- D. Metal-Backed, Butyrate Warning Signs: Weather-resistant, nonfading, preprinted, cellulose-acetate butyrate signs with 0.0396-inch galvanized-steel backing; and with colors, legend, and size required for application. 1/4-inch grommets in corners for mounting. Nominal size, 10 by 14 inches.
- E. Warning label and sign shall include, but are not limited to, the following legends:
 - 1. Multiple Power Source Warning: "DANGER - ELECTRICAL SHOCK HAZARD - EQUIPMENT HAS MULTIPLE POWER SOURCES."
 - 2. Workspace Clearance Warning: "WARNING - OSHA REGULATION - AREA IN FRONT OF ELECTRICAL EQUIPMENT MUST BE KEPT CLEAR FOR 36 INCHES."

2.4 INSTRUCTION SIGNS

- A. Engraved, laminated acrylic or melamine plastic, minimum 1/16 inch thick for signs up to 20 sq. in. and 1/8 inch thick for larger sizes.
 - 1. Engraved legend with black letters on white face.
 - 2. Punched or drilled for mechanical fasteners. Attach with self-tapping screws.
 - 3. Framed with mitered acrylic molding and arranged for attachment at applicable equipment.

2.5 EQUIPMENT AND DEVICE IDENTIFICATION LABELS

- A. Self-Adhesive, Engraved, Laminated Acrylic or Melamine Label: Adhesive backed, with white letters on a dark-gray background. Minimum letter height shall be 3/8 inch (10 mm).
- B. Engraved, Laminated Acrylic or Melamine Label: Punched or drilled for screw mounting. White letters on a dark-gray background. Minimum letter height shall be 3/8 inch (10 mm).

2.6 MISCELLANEOUS IDENTIFICATION PRODUCTS

- A. Cable Ties: Fungus-inert, self-extinguishing, 1-piece, self-locking, Type 6/6 nylon cable ties.
 - 1. Minimum Width: 3/16 inch.
 - 2. Tensile Strength: 50 lb, minimum.
 - 3. Temperature Range: Minus 40 to plus 185 deg F.

- 4. Color: Black, except where used for color-coding.
- B. Paint: Paint materials and application requirements are specified in Division 09 painting Sections.
- C. Fasteners for Labels and Signs: Self-tapping, stainless-steel screws or stainless-steel machine screws with nuts and flat and lock washers.

PART 3 - EXECUTION

3.1 APPLICATION

- A. Accessible Raceways and Metallic-Clad Cables, 600 V or Less, for Service, Feeder, and Branch Circuits More Than 30 A: Identify with snap-around label.
- B. Accessible Raceways and Cables of Auxiliary Systems: Identify the following systems with color-coded, self-adhesive vinyl tape applied in bands:
 - 1. Fire Alarm System: Red.
 - 2. Fire-Suppression Supervisory and Control System: Red and yellow.
 - 3. Combined Fire Alarm and Security System: Red and blue.
 - 4. Security System: Blue and yellow.
 - 5. Mechanical and Electrical Supervisory System: Green and blue.
 - 6. Telecommunication System: Green and yellow.
 - 7. Control Wiring: Green and red.
- C. Emergency Systems: Each junction and pull box shall be painted orange. Use black indelible liquid marker to label "EMERG" in 3/8 inch letters minimum.
- D. Fire Alarm System: Each junction and pull box shall be painted Red. Use Black indelible liquid marker to label "FA" in 3/8 inch letters minimum.
- E. Feeders Shown on Single Line Diagram: Each junction and pull box shall be marked with black indelible liquid marker with the assigned feeder number (example: "FDR #3B) in 3/8 inch letters minimum.
- F. Branch-Circuit Conductor Identification: Where there are conductors for more than three branch circuits in same junction or pull box, use aluminum wraparound marker labels. Identify each ungrounded conductor according to source and circuit number.
- G. Conductors to Be Extended in the Future: Attach write-on tags to conductors and list source and circuit number.
- H. Auxiliary Electrical Systems Conductor Identification: Identify field-installed alarm, control, signal, sound, intercommunications, voice, and data connections.
 - 1. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, and pull points. Identify by system and circuit designation.

2. Use system of marker tape designations that is uniform and consistent with system used by manufacturer for factory-installed connections.
 3. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and Operation and Maintenance Manual.
- I. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting: Comply with 29 CFR 1910.145 and apply baked-enamel warning signs. Identify system voltage with black letters on an orange background. Apply to exterior of door, cover, or other access.
1. Equipment with Multiple Power or Control Sources: Apply to door or cover of equipment including, but not limited to, the following:
 - a. Power transfer switches.
 - b. Controls with external control power connections.
 2. Equipment Requiring Workspace Clearance According to NFPA 70: Unless otherwise indicated, apply to door or cover of equipment but not on flush panelboards and similar equipment in finished spaces.
- J. Instruction Signs:
1. Operating Instructions: Install instruction signs to facilitate proper operation and maintenance of electrical systems and items to which they connect. Install instruction signs with approved legend where instructions are needed for system or equipment operation.
- K. Equipment Identification Labels: On each unit of equipment, install unique designation label that is consistent with wiring diagrams, schedules, and Operation and Maintenance Manual. Apply labels to disconnect switches and protection equipment, central or master units, control panels, control stations, terminal cabinets, and racks of each system. Systems include power, lighting, control, communication, signal, monitoring, and alarm systems unless equipment is provided with its own identification.
1. Labeling Instructions:
 - a. Indoor Equipment: Self-adhesive, engraved, laminated acrylic or melamine label. Unless otherwise indicated, provide a single line of text with 1/2-inch-high letters on 1-1/2-inch-high label; where 2 lines of text are required, use labels 2 inches high.
 - b. Elevated Components: Increase sizes of labels and letters to those appropriate for viewing from the floor.
 2. Equipment to Be Labeled:
 - a. Panelboards, electrical cabinets, and enclosures.
 - b. Access doors and panels for concealed electrical items.
 - c. Emergency system boxes and enclosures.
 - d. Disconnect switches.
 - e. Enclosed circuit breakers.
 - f. Motor starters.
 - g. Push-button stations.
 - h. Power transfer equipment.

- i. Contactors.
 - j. Remote-controlled switches, dimmer modules, and control devices.
 - k. Battery inverter units.
 - l. Battery racks.
 - m. Power-generating units.
 - n. Fire-alarm control panel and annunciators.
 - o. Security and intrusion-detection control stations, control panels, terminal cabinets, and racks.
 - p. Monitoring and control equipment.
3. Devices to be Labeled:
- a. Receptacles
 - 1) Normal Power Receptacles: Provide self-adhesive 3/8-inch-high clear label with 3/16 inch high black lettering mounted at top of device cover plate.
 - 2) Emergency Power Receptacles: Provide self-adhesive 3/8 inch high clear label with 3/16 inch high red lettering mounted at top of device cover plate.
 - 3) Wrap edges of self-adhesive label around edges of cover plate such that label is pressed between cover plate and wall.
 - 4) Lettering may be printed either on the back or front of label. Lettering shall be permanently affixed to label and resist damage from normal building operations.
 - 5) Label shall indicate branch circuit number and panelboard from which device is fed. Example: "A-1C-1 #2". Review nomenclature with the Owner.
4. Panel Schedule: For all new and/or altered existing panelboards, provide neatly typed Excel format panel schedules for each panel. Provide Owner with an electronic copy of Excel files for all panel schedules and post copy of each printed schedule on inside of panelboard door.

3.2 INSTALLATION

- A. Verify identity of each item before installing identification products.
- B. Location: Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment.
- C. Apply identification devices to surfaces that require finish after completing finish work.
- D. Self-Adhesive Identification Products: Clean surfaces before application, using materials and methods recommended by manufacturer of identification device.
- E. Attach nonadhesive signs and plastic labels with screws and auxiliary hardware appropriate to the location and substrate.
- F. System Identification Color Banding for Raceways and Cables: Each color band shall completely encircle cable or conduit. Place adjacent bands of two-color markings in contact, side by side.

Locate bands at changes in direction, at penetrations of walls and floors, at 50-foot maximum intervals in straight runs, and at 25-foot maximum intervals in congested areas.

- G. Color-Coding for Phase and Voltage Level Identification, 600 V and Less: Use the colors listed below for ungrounded service, feeder, and branch-circuit conductors.
1. Color shall be factory applied.
 2. Colors for 208/120-V Circuits:
 - a. Phase A: Black.
 - b. Phase B: Red.
 - c. Phase C: Blue.
 - d. Neutral: White.
 3. Colors for 480/277-V Circuits:
 - a. Phase A: Brown.
 - b. Phase B: Orange.
 - c. Phase C: Yellow.
 - d. Neutral: Gray.
 4. Field-Applied, Color-Coding Conductor Tape: Apply in half-lapped turns for a minimum distance of 6 inches from terminal points and in boxes where splices or taps are made. Apply last two turns of tape with no tension to prevent possible unwinding. Locate bands to avoid obscuring factory cable markings.
- H. Aluminum Wraparound Marker Labels and Metal Tags: Secure tight to surface of conductor or cable at a location with high visibility and accessibility.
- I. Painted Identification: Prepare surface and apply paint according to Division 09 painting Sections.

END OF SECTION 26 05 53

SECTION 26 09 36 - MODULAR DIMMING CONTROLS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Integrated, multi-preset modular dimming controls.
 - a. Low voltage wall stations
 - b. Control interfaces and modules
 - c. Occupancy, Vacancy, and Photocell Sensors
 - d. Emergency Lighting Control

1.3 REFERENCE STANDARDS

- A. ANSI C12.20 - American National Standards for Electricity Meters – 0.2 and 0.5 Accuracy Classes; 2010.
- B. ANSI C82.11 – American National Standard for Lamp Ballasts – High Frequency Fluorescent Lamp Ballasts – Supplements; 2011.
- C. ANSI/ESD S20.20 – Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices); 2007.
- D. ASTM D4674 – Standard Practice for Accelerated Testing for Color Stability of Plastics Exposed to Indoor Office Environments; 2002a (Reapproved 2010).
- E. IEC 60669-2-1 – Switches for Household and Similar Fixed Electrical Installations – Part 2-1: Particular Requirements – Electronic Switches; 2009.
- F. IEC 60929 – AC and/or DC-Supplied Electronic Control Gear for Tubular Fluorescent Lamps – Performance Requirements; 2011.
- G. IEC 61000-4-2 – Electromagnetic Compatibility (EMC) – Part 4-2: Testing and Measurement Techniques – Electrostatic Discharge Immunity Test; 2008.
- H. IEC 61000-4-5 - IEC 61000-4-5 - Electromagnetic Compatibility (EMC) - Part 4-5: Testing and Measurement Techniques - Surge Immunity Test; 2008.

- I. IEC 61347-2-3 - Lamp Control Gear - Part 2-3: Particular Requirements for A.C. and/or D.C. Supplied Electronic Control Gear for Fluorescent Lamps; 2011.
- J. IEC 61347-2-13 - Lamp Control Gear - Part 2-13: Particular Requirements for D.C. or A.C. Supplied Electronic Control Gear for LED Modules; 2006.
- K. IEEE C62.41.2 - Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and less) AC Power Circuits; 2002 (R2008).
- L. ISO 9001 - Quality Management Systems-Requirements; 2008.
- M. NECA 1 - Standard for Good Workmanship in Electrical Construction; 2010.
- N. NECA 130 - Standard for Installing and Maintaining Wiring Devices; National Electrical Contractors Association; 2010.
- O. NEMA WD 1 - General Color Requirements for Wiring Devices; National Electrical Manufacturers Association; 1999 (R 2005).
- P. NFPA 70 - National Electrical Code; National Fire Protection Association; Most Recent Edition Adopted by Authority Having Jurisdiction, Including All Applicable Amendments and Supplements.
- Q. UL 94 - Tests for Flammability of Plastic Materials for Parts in Devices and Appliances; Current Edition, Including All Revisions.
- R. UL 508 - Industrial Control Equipment; Underwriters Laboratories Inc.; Current Edition, Including All Revisions.
- S. UL 508A - Industrial Control Panels; Underwriters Laboratories Inc.; Current Edition, Including All Revisions.
- T. UL 924 - Emergency Lighting and Power Equipment; Current Edition, Including All Revisions.
- U. UL 935 - Fluorescent-Lamp Ballasts; Current Edition, Including All Revisions.
- V. UL 1472 - Solid-State Dimming Controls; Current Edition, Including All Revisions.
- W. UL 2043 - Fire Test for Heat and Visible Smoke Release for Discrete Products and Their Accessories Installed in Air-Handling Spaces; Current Edition, Including All Revisions.
- X. UL 8750 - Light Emitting Diode (LED) Equipment for Use in Lighting Products; Current Edition, Including All Revisions.

1.4 DEFINITIONS

- A. Fade Rate: The time it takes each zone to arrive at the next scene, dependent on the degree of change in lighting level.

- B. Low Voltage: As defined in NFPA 70 for circuits and equipment operating at less than 50 V or for remote-control, signaling and power-limited circuits.
- C. Zone: A fixture or group of fixtures controlled simultaneously as a single entity.
- D. Scene: The lighting effect created by adjusting several zones of lighting to the desired intensity.
- E. SCR: Silicon-controlled rectifier.

1.5 SUBMITTALS

- A. Product Data: For all products, including, but not limited to; multi preset modular dimming control; combined modular control systems; interfaces and devices; low voltage communications cabling; control devices, including, but not limited to; switches, sensors, dimming interface, wall plates, and conductors.
 - 1. Catalog cut sheets with performance specifications demonstrating compliance with specified requirements.
 - 2. Project specific elevation, dimension, features, characteristics, ratings, and labels
 - 3. Device plates and plate color, material and finish.
 - 4. Required engraved switch labels and manufacturer engraving template.
 - 5. Required ballasts and lamp combinations compatible with dimmers.
- B. Shop Drawings: Detailed assemblies of components, custom assembled for specific application on this project.
 - 1. Floor Plans: Location, orientation, and coverage area of each sensor; group designations; and other specific design symbols and designations as required defining the installation, location, and configuration of all control devices. Indicate digital control loops on drawings.
 - 2. Provide room/area details including products and sequence of operation for each room, area, and/or networked system.
 - 3. Wiring Diagrams: Power, signal, and control wiring. Include complete system wiring diagram illustrating all interconnections between components specified in this Section and devices furnished with power distribution system components required for building control. Include network cable specification and end-of-line termination details. Indicate data communication paths and identify networks, data buses, data gateways, concentrators, and other devices to be used. Describe characteristics of network and other data communication lines. Coordinate nomenclature and presentation with room wiring diagram. Coordinate integration with mechanical and/or other trades.
 - 4. Wire Termination Diagrams and Schedules: Coordinate nomenclature and presentation with Drawings and block diagram. Differentiate between manufacturer-installed and field-installed wiring.
 - 5. Outline Drawings: Indicate dimensions, weights, arrangement of components, and clearance and access requirements.
 - 6. Sequence of Operations for each space.

- C. Coordination Drawings: Submit evidence that lighting controls are compatible with connected monitoring and control devices and systems specified in other Sections.
 - 1. Show interconnecting signal and control wiring and interfacing devices that prove compatibility of inputs and outputs.
 - 2. For networked controls, list network protocols and provide statements from manufacturers that input and output devices meet interoperability requirements of the network protocol.
 - 3. Interfaces with other control systems including protocol and cabling requirements.
- D. Software and Firmware Operational Documentation:
 - 1. Software operating and upgrade manuals.
 - 2. Program Software Backup: On a magnetic media or compact disc, complete with data files. (2) copies provided to Owner.
 - 3. Device address list.
 - 4. Printout of software application and graphic screens.
- E. Field quality-control test reports.
- F. Software licenses and upgrades required by and installed for operation and programming of digital and analog devices.
- G. Operation and Maintenance Data: For lighting controls to include in emergency, operation, and maintenance manuals.
 - 1. Describe system features, operation and architecture in electronic and printed documentation. Include user account information, network access information and technical support contact information.
- H. Lighting Control System As-Built Report
 - 1. Provide a written report of all final lighting control settings, device settings, presets and lighting uses on a room by room basis.
- I. Warranty: Special warranty specified in this Section.
- J. Samples: Samples shall be provided for master and remote-control stations, faceplates with factory-applied color finishes, technical features and any other required system component as required by the architect/engineer.
- K. Packaging: Modular dimming controls shall not be submitted as part of a luminaire submittal package and will be rejected as such without review.
- L. Substitutions: Not permitted.

1.6 QUALITY ASSURANCE

- A. Manufacturer to have a minimum of 10 years' experience in manufacturing of architectural lighting controls.

- B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- C. Comply with NFPA 70.
- D. Source Limitations: Obtain lighting control module and power distribution components through one source from a single manufacturer.
- E. Installer & Commissioner Qualifications
 - 1. The system shall include manufacturer start up services by a trained and approved technician. References shall be provided upon request.

1.7 COORDINATION

- A. Coordinate lighting control components to form an integrated interconnection of compatible components.
 - 1. Match components and interconnections for optimum performance of lighting control functions.
 - 2. Coordinate lighting controls with ballast, drivers and dimmer type for all luminaires for a fully compatible system.
 - 3. Coordinate lighting controls with BAS either through IP based intercommunication of system or hardwired auxiliary relay outputs.
- B. Coordinate lighting control components specified in this Section with components specified in Division 26 Section "Panelboards."

1.8 PROJECT CONDITIONS

- A. Do not install equipment until following conditions can be maintained in spaces to receive equipment:
 - 1. Ambient temperature: 0 degrees to 40 degrees C (32 degrees to 104 degrees F).
 - 2. Relative humidity: Maximum 90 percent, non-condensing.
 - 3. Lighting control system must be protected from dust during installation.

1.9 WARRANTY

- A. Provide a five year manufacturer's warranty on all hardware and software
- B. The hardware warranty shall cover repair or replacement any defective products within the warranty period.

1.10 MAINTENANCE MATERIAL SUBMITTALS

- A. Make ordering of new equipment for expansions, replacements, and spare parts available to end-user.
- B. Make new replacement parts available for minimum of 10 years from date of manufacture.

PART 2 - PRODUCTS

2.1 GENERAL DIMMING DEVICE REQUIREMENTS

- A. Compatibility: Dimming control components shall be compatible with other elements of lighting fixtures, ballasts, transformers, and lighting controls. Dimming control components shall interface with all additional subordinate equipment.
- B. Dimmers and Dimmer Modules: Comply with UL 508.
 - 1. Audible Noise and Radio-Frequency Interference Suppression: Solid-state dimmers shall operate smoothly over their operating ranges without audible lamp or dimmer noise or radio-frequency interference. Modules shall include integral or external filters to suppress audible noise and radio-frequency interference.
 - 2. Dimmer or Dimmer-Module Rating: Not less than 125 percent of connected load unless otherwise indicated.
 - 3. Control all light sources in a smooth continuous manner. Dimmers with visual stepping are not acceptable.
 - 4. Withstand surges without impairment of performance when subjected to surges of 6,000 volts, 3,000 amps per ANSI/IEEE C62.41B.

2.2 MULTIPRESET MODULAR DIGITAL LIGHTING CONTROLS

- A. Basis-of-Design Product: Wattstopper DLM. Subject to compliance with requirements, provide product indicated on Drawings or a comparable product by one of the following:
 - 1. Lutron Electronics, Inc.
 - 2. Cooper
 - 3. Crestron Electronics
 - 4. Acuity
- B. Description: Modular digital lighting control system providing manual dimming control consisting of power (relay) packs controlled by wall stations, occupancy and / or vacancy sensors, daylight photosensors, and plug-load controllers. Wall stations shall be adjustable to indicated number of scenes.
- C. Operation: System shall provide standalone lighting control zones for individual spaces. System shall be capable of utilizing room or circuit controllers, digital occupancy sensors, switches, daylight sensors, and accessories required by the lighting and or electrical system parameters.

System shall be capable of incorporating a network backbone for connection of multiple spaces and to allow for remote or time-based operation.

2.3 POWER PACKS AND ROOM CONTROLLERS

A. Description: Power packs and room controllers are individual control units or relays that can control single or multiple zones of luminaires. The control units can be connected to devices in the space wirelessly or via low voltage cable. The control units shall include the following features:

1. Power packs shall incorporate one or more Class 1 relays and contribute low voltage power to the rest of the system. Secondary Packs shall incorporate the relay(s), shall have an optional 2nd relay, 0-10 VDC dimming output, or line voltage dimming output, but shall not be required to contribute system power. Power Supplies shall provide system power only but are not required to switch line voltage circuit. Auxiliary Relay Packs shall switch low voltage circuits only.
2. Power Packs shall accept 120 or 277 VAC (or optionally 347 VAC), be plenum rated, and provide Class 2 power to the system.
3. Power Pack shall securely mount to junction location through a threaded ½ inch chase nipple or be capable of being secured within a luminaire ballast channel

B. SWITCHING POWER PACKS AND ROOM CONTROLLERS

1. On/off control of lighting circuits through single or multiple relays.
 - a. Provide power packs and/or room controllers that are UL924 listed for switching and dimming of Emergency Power circuits. The unit shall provide normal ON/OFF control of emergency lighting along with the normal lighting. Upon normal power failure the emergency lighting circuit will close, forcing the emergency lighting ON until normal power is restored.
2. Every Power Pack parameter shall be available and configurable remotely from the software and locally via the device push-button
3. When required by local code, Power Pack must install inside standard electrical enclosure and provide UL recognized support to junction box. All Class 1 wiring is to pass through chase nipple into adjacent junction box without any exposure of wire leads

C. DIMMING POWER PACKS AND ROOM CONTROLLERS

1. On/off and dimming control of lighting circuits through single or multiple relays.
 - a. Provide power packs and/or room controllers that are UL924 listed for switching and dimming of Emergency Power circuits. The unit shall provide normal ON/OFF control of emergency lighting along with the normal lighting. Upon normal power failure the emergency lighting circuit will close, forcing the emergency lighting ON until normal power is restored.
2. Every Power Pack parameter shall be available and configurable remotely from the software and locally via the device push-button

3. When required by local code, Power Pack must install inside standard electrical enclosure and provide UL recognized support to junction box. All Class 1 wiring is to pass through chase nipple into adjacent junction box without any exposure of wire leads
4. One dimming output per relay
 - a. 0-10V Dimming - Where indicated, one 0-10 volt analog output per relay for control of compatible ballasts and LED drivers. The 0-10 volt output shall automatically open upon loss of power to the Room Controller to assure full light output from the controlled lighting.
 - b. Line Voltage, Forward Phase Dimming - Where indicated, one forward phase control line voltage dimming output per relay for control of compatible two-wire or three-wire ballasts, LED drivers, MLV, forward phase compatible ELV, neon/cold cathode and incandescent loads
 - c. Each dimming output channel shall have an independently configurable minimum and maximum calibration trim level to set the dimming range to match the true dynamic range of the connected ballast or driver
 - d. Each dimming output channel shall have an independently configurable minimum and maximum trim level to set the dynamic range of the output within the new 0-100% dimming range defined by the minimum and maximum calibration trim

2.4 WALL OR CEILING MOUNTED OCCUPANCY SENSORS

A. Quality Assurance

1. Sensors and related relays shall be compatible with the specific lighting types controlled.
2. All sensors shall be of the same manufacturer, mixing brands of sensors is not acceptable.
3. All sensors and related equipment shall have a five-year warranty.
4. All sensors and control modules connected to more than 50VAC shall be listed by Underwriters Laboratories.
5. All sensors and related equipment shall be manufactured in the United States of America.

B. General Description: Wall- or ceiling-mounting, solid-state units with a separate relay unit.

1. Operation: Unless otherwise indicated, turn lights on when covered area is occupied and off when unoccupied; with a time delay for turning lights off, adjustable over a minimum range of 1 to 30 minutes.
2. Sensor Output: Contacts rated to operate the connected relay, complying with UL 773A. Sensor shall be powered from the relay unit.
3. Relay Unit: Dry contacts rated for 20-A ballast load at 120- and 277-V ac, for 13-A tungsten at 120-V ac, and for 1 hp at 120-V ac. Power supply to sensor shall be 24-V dc, 150-mA, Class 2 power source as defined by NFPA 70.
4. Mounting:
 - a. Sensor: Suitable for mounting in any position on a standard outlet box.
 - b. Relay: Externally mounted through a 1/2-inch knockout in a standard electrical enclosure.
 - c. Time-Delay and Sensitivity Adjustments: Recessed and concealed behind hinged door.

5. Indicator: LED, to show when motion is being detected during testing and normal operation of the sensor.
 6. Bypass Switch: Override the on function in case of sensor failure.
 7. Automatic Light-Level Sensor: Adjustable from 2 to 200 fc; keep lighting off when selected lighting level is present.
 8. Communication and Class 2 low voltage power shall be delivered to each device via standard CAT-5 low voltage cabling with RJ-45 connectors.
 9. Every sensor parameter shall be available and configurable remotely from the software and locally via the device push-button.
 10. Sensors shall be able to function together with other sensors in order to provide expanded coverage areas by simply daisy-chain wiring together the units with CAT-5 cabling.
- C. Dual-Technology Type: Detect occupancy by using a combination of PIR and ultrasonic detection methods in area of coverage. Particular technology or combination of technologies that controls on-off functions shall be selectable in the field by operating controls on unit. All sensors shall have selectable auto-on/auto-off and manual-on/auto-off operations.
1. Sensitivity Adjustment: Separate for each sensing technology.
 2. Detector Sensitivity: Detect occurrences of 6-inch-minimum movement of any portion of a human body that presents a target of not less than 36 sq. in., and detect a person of average size and weight moving not less than 12 inches in either a horizontal or a vertical manner at an approximate speed of 12 inches/s.
 3. Detection Coverage (Standard Room): Detect occupancy anywhere within a circular area of 300 sq. ft. when mounted on a 96-inch-high ceiling.
 4. Occupancy sensors shown on floor plans are based on the following Wattstopper catalogue numbers. Approved equal manufacturers must demonstrate equal coverage patterns and equal characteristics to the specified sensors.

Ceiling Mounted Dual Technology

Sensor nLight Cat. No

LMUC-100-2

Application Area

300 to 1000 sq.ft. area

5. Wall-switch sensors – Dual/Adaptive Technology Occupancy Sensor
 - a. Single-relay: Wattstopper Cat. No. DW-100.
 - b. Description: Adaptive-technology type, 120/277 V, adjustable time delay up to 20 minutes, 180-degree field of view, with a minimum coverage area of 250 sq. ft. Auto on.
6. Wall-Switch Sensors – Dual/Adaptive Technology Vacancy Sensor
 - a. Single-relay: Wattstopper Cat No. DW-100
 - b. Description: Adaptive-technology type, 120/277 V, adjustable time delay up to 20 minutes, 180-degree field of view, with a minimum coverage area of 250 sq. ft. Manual on, auto off.
7. Wall-Switch Sensors – Dual/Adaptive Technology Dimming Occupancy
 - a. Single-relay: Wattstopper Cat No. DW-311

- b. Description: Adaptive-technology type, 0-10 V dimming, adjustable time delay up to 20 minutes, 180-degree field of view, with a minimum coverage area of 250 sq. ft. Auto on, auto off.
- 8. Wall -Switch Sensors – Dual/Adaptive Technology Dimming Vacancy
 - a. Single-relay: Wattstopper Cat No. DW-311
 - b. Description: Adaptive-technology type, 0-10 V dimming, adjustable time delay up to 20 minutes, 180-degree field of view, with a minimum coverage area of 250 sq. ft. Manual on, auto off.
- 9. All ceiling and wall mounted occupancy sensors shall be powered via Cat 5e or low voltage cables. When daisy chained using CAT-5e or connected wirelessly to other sensors, power packs, or switches, a lighting control zone is created.
- 10. Where ceiling mounted occupancy sensors are provided, provide local wall switch for additional user control, to turn "off" lights when room is occupied.
- 11. Wall-mounted occupancy sensors shall have manual "on/off" override.

2.5 DIGITAL WALL SWITCHES

- A. Description: Low voltage momentary pushbutton switches in 1,2,3,4,5, and 8 button configuration.
 - 1. Removable buttons for field replacement with customizable engraved buttons and/or alternate color buttons. Button replacement may be completed without removing the switch from the wall.
 - 2. Configuration LED on each switch that blinks to indicate data transmission.
 - 3. Two RJ-45 ports for connection to additional controllers, switches, and sensor
- B. Operation: Automatically change variable dimmer settings of indicated number of zones simultaneously from one preset scene to another when a push button is operated. Provide raise / lower and or on / off control of indicated zones.
- C. When multiple single rooms solutions are connected to a network, device shall be capable of selecting a lighting profile be run by the system's upstream gateway so as to implement selected lighting profile across multiple zones.
 - 1. LED luminaires.
 - 2. Fluorescent lamps with electronic ballasts.
- D. Memory: Provided with nonvolatile type to retain preset scenes through power failures without the use of battery backup for up to one year. In the event of a power interruption, upon return of power, system shall return to the same lighting levels prior to power interruption.
- E. Device Plates: Style, material, and color shall comply with Division 26 Section "Wiring Devices."

F. Dimmers

1. Dimming Curve: Modified "square law" as specified in IESNA's "IESNA Lighting Handbook".
2. Dimming Range: 0 to 100 percent, full output voltage not less than 98 percent of line voltage.
3. Voltage Regulation: Dimmer shall maintain a constant light level, with no visible flicker, when the source voltage varies plus or minus 2 percent in RMS voltage.

G. Low Voltage Control Interfaces

1. Contact Closure Interface: For momentary and maintained dry contact closure input and output to remote devices including, but not necessarily limited to, Automatic Transfer Switches, Occupancy Sensors and motorized shade control.
2. Partition Control: Automatically combine and separate lighting and accessory traction controls as spaces are configured with movable partitions.
3. Short-Circuit Rating: 10 kA for 120 V, 14 kA for 277 V.

2.6 WALL BOX ACCESSORIES

A. Wall Plates

1. Listed to UL 514C, CSA C22.2 #42.1-00
2. Provide an adapter plate for proper device alignment and wall plate attachment.

2.7 SOURCE QUALITY CONTROL

- A. Perform full-function testing on completed assemblies at end of line. Statistical sampling is not acceptable.

2.8 CONDUCTORS AND CABLES

- A. Wiring to Supply Side of Remote-Control Power Sources: Not smaller than No. 12 AWG. Comply with requirements in Division 26 Section "Low-Voltage Electrical Power Conductors and Cables."
- B. Class 2 Control Cable: Multiconductor cable with stranded-copper conductors shall be as required by the equipment manufacturer. Comply with requirements in Division 26 Section "Low-Voltage Electrical Power Conductors and Cables."
- C. CAT 5e Cable: Communication and Class 2 low voltage power shall be delivered to each device via standard CAT-5 low voltage cabling with RJ-45 connectors. Pre-terminated Cat 5e cables with RJ-45 connectors. Systems that utilize RJ-45 patch cords but do not provide serial communication data from individual end devices are not acceptable. If manufacturer's pre-terminated Cat 5e cables are not used for the installation, the contractor is responsible for testing each cable following installation and supplying manufacturer with test results.

PART 3 - EXECUTION

3.1 WIRING INSTALLATION

- A. Comply with NECA 1.
- B. Wiring Method: Comply with requirements in Division 26 Section "Low-Voltage Electrical Power Conductors and Cables." Minimum conduit size shall be 3/4 inch.
- C. Wiring within Enclosures: Bundle, lace, and train conductors to terminal points. Separate power-limited and nonpower-limited conductors according to conductor manufacturer's written instructions.
- D. Size conductors according to lighting control device manufacturer's written instructions unless otherwise indicated.
- E. Splices, Taps, and Terminations: Make connections only on numbered terminal strips in junction, pull, and outlet boxes; terminal cabinets; and equipment enclosures.

3.2 INSTALLATION

- A. Install equipment in accordance with manufacturer's installation instructions.
- B. Provide complete installation of system in accordance with Contract Documents.
- C. Provide equipment at locations and in quantities indicated on Drawings. Provide any additional equipment required to provide control intent.
- D. Define each dimmer's/relay's load type, assign each load to a zone, and set control functions.
- E. Mount exterior daylight sensors to point due north with constant view of daylight.
- F. Ensure that daylight sensor placement minimizes sensors view of electric light sources; ceiling mounted and fixture-mounted daylight sensors shall not have direct view of luminaries.

3.3 STARTUP SERVICE AND SUPPORT

- A. Install and program software with initial settings of adjustable values. Make backup copies of software and user-supplied values. Provide current licenses for software.
- B. Engage a factory-authorized service representative to perform startup service. Owner's Representative, Architect and Commissioning Agent shall be notified when startup service visits are scheduled and completed.
 - 1. Complete initial review meeting with construction team and commissioning agent to review installation techniques, project specific wiring requirements, and product information prior to installation of equipment.

2. Complete pre-wire installation and startup checks according to manufacturer's written instructions.
 3. Activate light fixtures and verify that all lamps are operating at 100 percent and dimming is functioning.
 4. Confirm correct communications wiring, initiate communications between devices and controller/gateways, and program the lighting control system according to approved configuration schedules, time-of-day schedules, and input override assignments.
 5. Establish lighting levels for all rooms per sequence of operations.
 6. Calibrate all photo cells to appropriate lighting levels.
 7. Program all low voltage end devices and test to meet sequence of operation and owner requirements.
- C. Startup C: Startup service shall include the following on site activities as part of deployment requirement.
1. The commissioning agent shall visit the site on a scheduled basis as per contract provisions, and shall confirm all manufacturer start-up service visits and tasks have been completed throughout the construction process.
 2. Wiring and Hardware Review: All wiring connections and electrical equipment included in the scope of the lighting control system shall be assessed.
 3. Field Testing: All communication connections, sensor connections and Ethernet connections shall be verified in accordance to a specified testing procedure.
 4. Final Inspection: All of the connections relating to the lighting control system shall be tested and verified.

3.4 STARTUP AND PROGRAMMING

- A. Provide factory certified field service engineer to complete all onsite tasks and start up requirements as indicated for all luminaires and components. Quantity of days on site shall be provided as required for a complete fully functioning start up. Coordinate testing and demonstration of products specified in this section with testing and demonstration requirements for low voltage, programmable lighting control system specified in Division 26 Section "Network Lighting Controls". Not to be less than 2 weeks (14 days) to ensure proper system installation and operation under following parameters:
1. Qualifications for factory certified service engineer:
 - a. Minimum experience of 2 years training in the electrical/electronic field.
 - b. Minimum of 5 installations with the specific lighting control system.
 - c. Certified by the equipment manufacturer on the system installed.
 - d. Visit duration shall be suitable to accomplish required tasks.
- B. First Visit Task: Make first visit prior to installation of wiring.
1. Review:
 - a. Low voltage wiring requirements.
 - b. Separation of power and low voltage/data wiring.
 - c. Wire labeling.

- d. Lighting Management Panel locations and installations.
 - e. Control locations.
 - f. Computer jack locations.
 - g. Load circuit wiring.
 - h. Network wiring requirements.
 - i. Connections to other equipment.
 - j. Installer responsibilities.
 - k. Power Panel locations.
- 2. System Integration: Lighting Control System Manufacturer shall provide a factory certified Field Service Engineer on-site to meet with Owners Representatives, Architect, Commissioning Agent and other related equipment manufacturers to discuss equipment and integration procedures. Each site visit shall be scheduled and shall include a summary report completed by the technician, including site conditions, action items, and follow-up items for review by the owner's representative, commissioning agent and architect.
- C. Second Visit Task: make visits to the site at periodic timeframe during installation of the network lighting control system to review ongoing installation. Include a minimum of 4 visits to the site during product installation.
- D. Third Visit Task: Make visits upon completion of installation of Network Lighting Control System.
 - 1. Review:
 - a. Verify connection of power wiring and load circuits.
 - b. Verify connection and location of controls.
 - c. Energize Lighting Management Panels and download system data program.
 - d. Address devices.
 - e. Verify proper connection of panel links (low voltage/data) and address panel.
 - f. Download system panel data to dimming/switching panels.
 - g. Check dimming panel load types and currents and supervise removal of by-pass jumpers.
 - h. Verify system operation control by control.
 - i. Verify proper operation of manufacturers interfacing equipment.
 - j. Verify proper operation of manufacturers supplied PC and installed programs.
 - k. Configure initial groupings of led drivers for wall controls, daylight sensors and occupant sensors.
 - l. Initial calibration of sensors and setting of photocell to maintain indicated illuminance levels in all spaces.
 - m. Obtain sign-off on system functions.
 - 2. Tuning: Lighting Control System Manufacturer shall coordinate an on-site meeting with Commissioning Agent, Owner and Architect to make required lighting adjustments to the system for conformance with the original design intent.
- E. Fourth Visit Task: Adjustment/Training

1. Engage a factory-authorized service representative on site to train Owner's maintenance personnel to adjust, operate, and maintain the control unit and operator interface. Training shall be digitally recorded and provided to the Owner.
2. The end customer shall be trained in the usage of the system within 1 month of completion of the startup process. A second training session shall be scheduled within 6 months of final start up.
3. On-site Walkthrough: Lighting Control System Manufacturer shall provide a factory certified Field Service Engineer on-site to demonstrate system functionality to the Commissioning Agent and Owner. Lighting control manufacturer shall provide final report of as-built programming and control for systems upon completion of start up services.

3.5 SOFTWARE SERVICE AGREEMENT

- A. Technical Support: Beginning at Substantial Completion, service agreement shall include software support for five years.
- B. Upgrade Service: At Substantial Completion, update software to latest version. Upgrading software shall include operating system and new or revised licenses for using software.
 1. Upgrade Notice: At least 30 days to allow Owner to schedule and access the system and to upgrade computer equipment if necessary
 2. Startup
 - a. System configuration
 - 1) Naming and association of areas and lighting zones.
- C. Training of customer representatives for lighting control system.
 1. Configuration Software used to make system programming and configuration changes
 2. Control and Monitor
- D. Tech Support
 1. Provide factory direct technical support hotline 24 hours per day, 7 days per week.

3.6 FIELD QUALITY CONTROL

- A. Manufacturer Services
 1. Aim and Focus Visit
 - a. Facility Representative to coordinate on-site meeting with Lighting Control System Manufacturer and Lighting Design Consultant to make required lighting adjustments to the system for conformance with the Lighting Design Consultant's original design intent.

3.7 CLOSEOUT ACTIVITIES

- A. Training Visits: Lighting Control System Manufacturer shall provide (2) 2 day additional on-site system training to Owners personnel. Time frame shall be determined by owner and shall be post building occupancy.
- B. System Optimization Visit: Lighting Control System Manufacturer shall provide a factory certified Field Service Engineer on-site within 6 months after completion of final construction phase to evaluate system usage and discuss opportunities to make efficiency improvements that will fit with the current use of the facility.

3.8 MAINTENANCE

- A. Capable of providing on-site service support within 24 hours anywhere in continental United States and within 72 hours worldwide except where special visas are required.
- B. Offer renewable service contract on yearly basis, to include parts, factory labor, and annual training visits. Make service contracts available up to ten years after date of system startup.
- C. System Optimization Visit
 - 1. Lighting Control System Manufacturer to visit site 6 months after system start-up to evaluate system usage and discuss opportunities to make efficiency improvements that will fit with the current use of the facility.

END OF SECTION 26 09 36

SECTION 26 27 26 - WIRING DEVICES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Receptacles, receptacles with integral GFCI, and associated device plates.
 - 2. Snap switches and wall-box dimmers.
- B. Related Sections include the following:
 - 1. Division 26 Section "Lighting Control Devices" for indoor and outdoor occupancy sensors, and wall box dimmers.

1.3 DEFINITIONS

- A. EMI: Electromagnetic interference.
- B. GFCI: Ground-fault circuit interrupter.
- C. Pigtail: Short lead used to connect a device to a branch-circuit conductor.
- D. RFI: Radio-frequency interference.
- E. TVSS: Transient voltage surge suppressor.
- F. UTP: Unshielded twisted pair.

1.4 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Shop Drawings: List of legends and description of materials and process used for premarking wall plates.
- C. Samples: One for each type of device and wall plate specified, in each color specified.
- D. Field quality-control test reports.

- E. Operation and Maintenance Data: For wiring devices to include in all manufacturers' packing label warnings and instruction manuals that include labeling conditions.

1.5 QUALITY ASSURANCE

- A. Source Limitations: Obtain each type of wiring device and associated wall plate through one source from a single manufacturer. Insofar as they are available, obtain all wiring devices and associated wall plates from a single manufacturer and one source.
- B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- C. Comply with NFPA 70.

1.6 COORDINATION

- A. Receptacles for Owner-Furnished Equipment: Match plug configurations.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers' Names: Shortened versions (shown in parentheses) of the following manufacturers' names are used in other Part 2 articles:
 - 1. Eaton Wiring Devices/Arrow Hart.
 - 2. Hubbell Incorporated; Wiring Device-Kellems (Hubbell).
 - 3. Leviton Mfg. Company Inc. (Leviton).
 - 4. Pass & Seymour/Legrand; Wiring Devices & Accessories (Pass & Seymour).

2.2 STRAIGHT BLADE RECEPTACLES

- A. Convenience Receptacles, 125 V, 20 A: Comply with NEMA WD 1, NEMA WD 6 configuration 5-20R, and UL 498.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Eaton/Arrow Hart; 1877 (single), 5362 (duplex).
 - b. Hubbell; HBL5351 (single), CR5362 (duplex).
 - c. Leviton; 5891 (single), PS5362 (duplex).
 - d. Pass & Seymour; 5381 (single), 5362 (duplex).

2.3 GFCI RECEPTACLES

- A. General Description: Straight blade, non-feed-through type. Comply with NEMA WD 1, NEMA WD 6, UL 498, and UL 943, Class A, and include indicator light that is lighted when device is tripped.
- B. Duplex GFCI (Self Test) Convenience Receptacles, 125 V, 20 A:
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Eaton/Arrow Hart; SGF20.
 - b. Pass & Seymour; 2084.
 - c. Hubbell
 - d. Leviton
- C. Receptacles located in damp or wet locations shall be listed weather-resistant (WR) type.

2.4 SNAP SWITCHES

- A. Comply with NEMA WD 1 and UL 20.
- B. Switches, 120/277 V, 20 A:
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Eaton/Arrow Hart; 1221 (single pole), 1222 (two pole), 1223 (three way), 1224 (four way).
 - b. Hubbell; CS1221 (single pole), CS1222 (two pole), CS1223 (three way), CS1224 (four way).
 - c. Leviton; 1221-2 (single pole), 1222-2 (two pole), 1223-2 (three way), 1224-2 (four way).
 - d. Pass & Seymour; 20AC1 (single pole), 20AC2 (two pole), 20AC3 (three way), 20AC4 (four way).

2.5 WALL PLATES

- A. Single and combination types to match corresponding wiring devices.
 - 1. Plate-Securing Screws: Metal with head color to match plate finish.
 - 2. Material for Finished Spaces: 430 satin-finished stainless steel.
 - 3. Material for Unfinished Spaces: Galvanized steel.
 - 4. Material for Damp Locations: Cast aluminum with spring-loaded lift cover, and listed and labeled for use in "wet locations."

2.6 FINISHES

- A. Color: Wiring device catalog numbers in Section Text do not designate device color.

1. Wiring Devices Connected to Normal Power System: As selected by Architect, unless otherwise indicated or required by NFPA 70 or device listing.
2. Wiring Devices Connected to Emergency Power System: Red.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Comply with NECA 1, including the mounting heights listed in that standard, unless otherwise noted.
- B. Coordination with Other Trades:
 1. Take steps to ensure that devices and their boxes are protected. Do not place wall finish materials over device boxes and do not cut holes for boxes with routers that are guided by riding against outside of the boxes.
 2. Keep outlet boxes free of plaster, drywall joint compound, mortar, cement, concrete, dust, paint, and other material that may contaminate the raceway system, conductors, and cables.
 3. Install device boxes in brick or block walls so that the cover plate does not cross a joint unless the joint is troweled flush with the face of the wall.
 4. Install wiring devices after all wall preparation, including painting, is complete.
- C. Conductors:
 1. Do not strip insulation from conductors until just before they are spliced or terminated on devices.
 2. Strip insulation evenly around the conductor using tools designed for the purpose. Avoid scoring or nicking of solid wire or cutting strands from stranded wire.
 3. The length of free conductors at outlets for devices shall meet provisions of NFPA 70, Article 300, without pigtails.
 4. Existing Conductors:
 - a. Cut back and pigtail or replace all damaged conductors.
 - b. Straighten conductors that remain and remove corrosion and foreign matter.
 - c. Pigtailling existing conductors is permitted provided the outlet box is large enough.
- D. Device Installation:
 1. Replace all devices that have been in temporary use during construction or that show signs that they were installed before building finishing operations were complete.
 2. Keep each wiring device in its package or otherwise protected until it is time to connect conductors.
 3. Do not remove surface protection, such as plastic film and smudge covers, until the last possible moment.
 4. Connect devices to branch circuits using pigtails that are not less than 6 inches (152 mm) in length.
 5. When there is a choice, use side wiring with binding-head screw terminals. Wrap solid conductor tightly clockwise, 2/3 to 3/4 of the way around terminal screw.

6. Use a torque screwdriver when a torque is recommended or required by the manufacturer.
7. When conductors larger than No. 12 AWG are installed on 15- or 20-A circuits, splice No. 12 AWG pigtails for device connections.
8. Tighten unused terminal screws on the device.
9. When mounting into metal boxes, remove the fiber or plastic washers used to hold device mounting screws in yokes, allowing metal-to-metal contact.

E. Light Switch Installation

1. Neon or LED lighted handle type light switches - Illuminated when load is "OFF" (meeting the device type, style and color requirements described in other specification sections and drawings) shall be provided for the following applications:
 - a. All "E" designated switches shown on the drawings.
 - b. Manual light switches serving normal and emergency lighting within the following spaces:
 - 1) Electrical Rooms
 - 2) Communications Rooms
 - 3) Mechanical Equipment Rooms
 - 4) Elevator Pits

F. Receptacle Orientation:

1. Install ground pin of vertically mounted receptacles up, and on horizontally mounted receptacles to the right.

G. Device Plates: Do not use oversized or extra-deep plates. Repair wall finishes and remount outlet boxes when standard device plates do not fit flush or do not cover rough wall opening.

H. Arrangement of Devices: Unless otherwise indicated, mount flush to finished wall surface, with long dimension vertical. Group adjacent switches under single, multigang wall plates.

I. Adjust locations of floor service outlets and service poles to suit arrangement of partitions and furnishings.

3.2 IDENTIFICATION

- A. Comply with Division 26 Section "Identification for Electrical Systems."

3.3 FIELD QUALITY CONTROL

- A. Perform tests and inspections and prepare test reports.

1. Test Instruments: Use instruments that comply with UL 1436.
2. Test Instrument for Convenience Receptacles: Digital wiring analyzer with digital readout or illuminated LED indicators of measurement.

- B. Tests for Convenience Receptacles:

1. Line Voltage: Acceptable range is 105 to 132 V.
2. Percent Voltage Drop under 15-A Load: A value of 6 percent or higher is not acceptable.
3. Ground Impedance: Values of up to 2 ohms are acceptable.
4. GFCI Trip: Test for tripping values specified in UL 1436 and UL 943.
5. Using the test plug, verify that the device and its outlet box are securely mounted.
6. The tests shall be diagnostic, indicating damaged conductors, high resistance at the circuit breaker, poor connections, inadequate fault current path, defective devices, or similar problems. Correct circuit conditions, remove malfunctioning units and replace with new ones, and retest as specified above.

END OF SECTION 26 27 26

SECTION 26 51 00 - INTERIOR LIGHTING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Interior lighting fixtures, lamps, and ballasts
 - 2. Exit signs
 - 3. Lighting fixture supports
- B. Related Sections include the following:
 - 1. Division 26 Section "Lighting Control Devices" for automatic control of lighting, including time switches, photoelectric relays, occupancy sensors, wall box dimmers, and multipole lighting relays and contactors.

1.3 DEFINITIONS

- A. BF: Ballast factor
- B. CRI: Color-rendering index
- C. HID: High-intensity discharge
- D. LER: Luminaire efficacy rating
- E. Luminaire: Complete lighting fixture, including ballast housing if provided

1.4 SUBMITTALS

- A. Product Data: For each type of lighting fixture, arranged in order of fixture designation. Include data on features, accessories, finishes, and the following:
 - 1. Physical description of lighting fixture including dimensions.
 - 2. Emergency lighting units including battery and charger.
 - 3. Ballast, including manufacturer.
 - 4. Energy-efficiency data, including lamp type, manufacturer.
 - 5. Lamp and ballast warranty information.

6. Life, output, and energy-efficiency data for lamps.
7. Photometric data, in IESNA format, based on laboratory tests of each lighting fixture type, outfitted with lamps, ballasts, and accessories identical to those indicated for the lighting fixture as applied in this Project.
 - a. Photometric data shall be certified by a manufacturer's laboratory with a current accreditation under the National Voluntary Laboratory Accreditation Program (NVLAP) for Energy Efficient Lighting Products.
- B. Luminaire submittals not including lamp and ballast types and manufacturer's name will be returned marked "not reviewed". Lamp and ballast types and manufacturer's name must be submitted with the luminaire submittals in order to review.
- C. Shop Drawings: Show details of nonstandard or custom lighting fixtures. Indicate dimensions, weights, methods of field assembly, components, features, and accessories.
 1. Wiring Diagrams: Power and control wiring.
- D. Coordination Drawings: Reflected ceiling plan(s) and other details, drawn to scale, on which the following items are shown and coordinated with each other, based on input from installers of the items involved:
 1. Lighting fixtures.
 2. Suspended ceiling components.
 3. Structural members to which suspension systems for lighting fixtures will be attached.
 4. Other items in finished ceiling including the following:
 - a. Air outlets and inlets
 - b. Speakers
 - c. Sprinklers
 - d. Smoke and fire detectors
 - e. Occupancy sensors
 - f. Access panels
 - g. HVAC ductwork and piping
5. Perimeter moldings.
- E. Product Certificates: For each type of ballast for bi-level and dimmer-controlled fixtures, signed by product manufacturer.
- F. Qualification Data: For agencies providing photometric data for lighting fixtures.
- G. Field quality-control test reports.
- H. Operation and Maintenance Data: For lighting equipment and fixtures to include in emergency, operation, and maintenance manuals.
- I. Warranties: Special warranties specified in this Section.

1.5 QUALITY ASSURANCE

- A. Luminaire Photometric Data Testing Laboratory Qualifications: Provided by manufacturers' laboratories that are accredited under the National Volunteer Laboratory Accreditation Program for Energy Efficient Lighting Products.
- B. Luminaire Photometric Data Testing Laboratory Qualifications: Provided by an independent agency, with the experience and capability to conduct the testing indicated, that is an NRTL as defined by OSHA in 29 CFR 1910.7.
- C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- D. Comply with NFPA 70
- E. Comply with the recommended standards of the Illuminating Engineering Society of North America.
- F. Comply with the requirements of ASHRAE 90.1, Energy Standard for Buildings, Except Low-Rise Residential Buildings.
- G. LED Compliance: LED components, lamps, drivers and fixtures shall comply with: PCC 47 CFR Part 15; UL 8750; ANSI/NEMA Standards C78.377, NEMA SSL-1, ANSI C82.77, IESNA Standards TM-16-05, RP-16, LM-79, LM-80 and TM-21.

1.6 COORDINATION

- A. Coordinate layout and installation of lighting fixtures and suspension system with other construction that penetrates ceilings or is supported by them, including HVAC equipment, fire-suppression system, and partition assemblies.

1.7 WARRANTY

- A. Special Warranty for Drivers: Manufacturer's standard form in which ballast manufacturer agrees to repair or replace ballasts that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period for Electronic Drivers: Five years from date of Substantial Completion.
 - 2. Should a driver fail during the warranty period, a new driver shall be supplied to the Owner at no charge, and a labor allowance of \$15.00 for each defective driver shall be supplied to the Owner. The replacement driver shall be identical to, or an improvement upon, the original design of the malfunctioning driver.
- B. Special Warranty for LED luminaires: Provide a five year manufacturer's warranty on LED luminaires including full replacement of any failed components and \$50 labor allowance for each luminaire. Warranty shall be from the date of Substantial Completion.

1.8 EXTRA MATERIALS

- A. Furnish extra materials described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
1. Lamps and LED Modules: 10 for every 100 of each type and rating installed. Furnish at least one of each type.
 2. Plastic Diffusers and Lenses: 1 for every 100 of each type and rating installed. Furnish at least one of each type.
 3. Drivers: 1 for every 100 of each type and rating installed. Furnish at least one of each type.
 4. Globes and Guards: 1 for every 20 of each type and rating installed. Furnish at least one of each type.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Acceptable Manufacturers
1. Luminaires
 - a. Refer to "Luminaire Schedule", herein or on the Drawings.
 2. LED Lamps and Modules:
 - a. Philips
 - b. General Electric
 - c. Osram-Sylvania
 - d. Cree
 - e. Nichia
 - f. Xicato
 - g. Bridgelux
 3. LED Power Supplies
 - a. Osram-Sylvania
 - b. General Electric
 - c. Philips
 4. LED Dimming Drivers
 - a. Lutron
 5. Substitutions - Not Permitted

2.2 LIGHTING FIXTURES AND COMPONENTS, GENERAL REQUIREMENTS

- A. Recessed Fixtures: Comply with NEMA LE 4 for ceiling compatibility for recessed fixtures.
- B. Luminaires shall bear UL inspection label, suitable to application and installed environment.
- C. Furnish on request I.T.L. and/or E.T.L. test reports for luminaires furnished.
- D. Metal Parts: Free of burrs and sharp corners and edges.
- E. Sheet Metal Components: Steel, unless otherwise indicated. Form and support to prevent warping and sagging.
- F. Doors, Frames, and Other Internal Access: Smooth operating, free of light leakage under operating conditions, and designed to permit relamping without use of tools. Designed to prevent doors, frames, lenses, diffusers, and other components from falling accidentally during relamping and when secured in operating position.
- G. Reflecting surfaces shall have minimum reflectance as follows, unless otherwise indicated:
 - 1. White Surfaces: 85 percent
 - 2. Specular Surfaces: 83 percent
 - 3. Diffusing Specular Surfaces: 75 percent
 - 4. Laminated Silver Metalized Film: 90 percent
- H. Plastic Diffusers, Covers, and Globes:
 - 1. Acrylic Lighting Diffusers: 100 percent virgin acrylic plastic. High resistance to yellowing and other changes due to aging, exposure to heat, and UV radiation.
 - a. Lens Thickness: At least 0.125 inch minimum unless different thickness is indicated.
 - b. UV stabilized.
 - 2. Glass: Annealed crystal glass, unless otherwise indicated.
- I. Luminaire housing and door frame shall be fully sealed against light leakage. Light leaks between ceiling trims of recessed luminaires and ceiling will not be acceptable.
- J. Alzak parabolic cones shall be guaranteed against fading for a minimum of 2 years. In the event of premature fading, luminaire manufacturer shall replace cones and pay for both labor and material costs.
- K. Luminaires with triphosphor lamps (T8, compact fluorescent, etc.) shall have reduced iridescent reflectors, louvers, etc.
- L. Adjustable luminaires shall be capable of being locked into position. Aim and adjust all adjustable luminaires to the satisfaction of the Owner and the Architect.

- M. Maximum depth of recessed tubular-fluorescent luminaires shall not exceed 6 inches including mounting yokes. Reflector and lamp positions shall provide highest efficiency and even brightness to eliminate lamp lines on diffusers.
- N. Recessed luminaires shall be removable from below to provide access to outlet or prewired luminaire box.
- O. A snap type local disconnect shall be provided for all luminaires. The disconnect shall disconnect all supply conductors simultaneously, including the grounded conductor. The line side terminals of the disconnecting means shall be guarded. The disconnecting means shall be located inside or outside the luminaire housing, but must be accessible to qualified persons before servicing or maintaining the ballast or lamps.

2.3 EXIT SIGNS

- A. Description: Comply with UL 924; for sign colors, visibility, luminance, and lettering size, comply with authorities having jurisdiction.
- B. Internally Lighted Signs:
 - 1. Lamps for AC Operation: LEDs, 70,000 hours minimum rated lamp life.
 - 2. Self-Powered Exit Signs (Battery Type): Integral automatic charger in a self-contained power pack.
 - a. Battery: Sealed, maintenance-free, nickel-cadmium type.
 - b. Charger: Fully automatic, solid-state type with sealed transfer relay.
 - c. Operation: Relay automatically energizes lamp from battery when circuit voltage drops to 80 percent of nominal voltage or below. When normal voltage is restored, relay disconnects lamps from battery, and battery is automatically recharged and floated on charger.
 - d. Test Push Button: Push-to-test type, in unit housing, simulates loss of normal power and demonstrates unit operability.
 - e. LED Indicator Light: Indicates normal power on. Normal glow indicates trickle charge; bright glow indicates charging at end of discharge cycle.
 - f. Integral Self-Test: Factory-installed electronic device automatically initiates code-required test of unit emergency operation at required intervals. Test failure is annunciated by an integral audible alarm and flashing red LED.
 - 3. Type "X" Directional Exit Signs shall have directional indicating chevrons pointing in direction of travel.

2.4 LED SYSTEMS

- A. The LED module itself and all its components must not be subjected to mechanical stress.
- B. Assembly must not damage or destroy conducting paths on the circuit board.

- C. Installation of LED modules (with power supplies) shall adhere to all applicable electrical and safety standards.
- D. Correct electrical polarity shall be clearly identified.
- E. LED module must be protected from unbalanced voltage drop, and/or overload.
- F. Ensure that the power supply is of adequate power to operate the total load.
- G. Utilize standard ESD precautions when installing the module.
- H. Install system according to the manufacturers heat sinking parameters.
- I. For applications involving exposure to humidity and dust, the module shall be protected by a fixture or housing with a suitable protection glass. The module shall be protected against condensation water by treatment with an appropriate circuit board conformal coating. The conformal coating should have the following features.
 - 1. Optical transparency
 - 2. UV resistance
 - 3. Thermal expansion properties matching those of the module (15-30 x 10-6cm/cm/K)
 - 4. Low permeability of steam for all climate conditions
 - 5. Resistance against corrosive environments
- J. The LED Module shall be operated with an electronically stabilized power supply offering protection against short circuits, overload and overheating.
- K. All LED products shall respect Intellectual Property Rights and shall have a Solid State License for the product. Unlicensed products are not acceptable.
- L. LED Driver shall be installed inside an electrical enclosure suitable for use in dry, damp, or wet location as required by fixture listing.
- M. Wiring inside electrical enclosure shall comply with 600V/105°C rating or higher.
- N. LED Driver shall meet the following:
 - 1. Class A sound rating.
 - 2. An operating ambient temperature range of -20°C to 55°C.
 - 3. A maximum life expectancy of 50,000 hours at Tcase of $\leq 70^{\circ}\text{C}$.
 - 4. A maximum life expectancy of 100,000 hours at Tcase of $\leq 62^{\circ}\text{C}$.
 - 5. Function to reduce output power to LEDs if its maximum allowable case temperature is exceeded.
 - 6. A failure rate of $\leq 0.01\%$ per 1,000 hours at Tcase of $\leq 70^{\circ}\text{C}$.
 - 7. A failure rate of 0.01% - 0.02% per 1,000 hours at Tcase of 70°C - 80°C .
 - 8. Tolerance for sustained open circuit and short circuit output conditions without damage.
 - 9. Power Factor $> 90\%$.
 - 10. Total harmonic Distortion $< 20\%$.
 - 11. Compliance with FCC rules and regulations, as per Title 47 CFR Part 15 Non-Consumer (Class A).

- O. Where addressable dimming LED driver is specified, provide Lutron A series driver.
- P. Where 0-10V, ELV or 3-wire dimming LED driver is specified, ensure compatibility of driver and dimming control system of controller.

2.5 LIGHTING FIXTURE SUPPORT COMPONENTS

- A. Comply with Division 26 Section "Hangers and Supports for Electrical Systems" for channel- and angle-iron supports and nonmetallic channel and angle supports.
- B. Single-Stem Hangers: 1/2-inch steel tubing with swivel ball fittings and ceiling canopy. Finish same as fixture.
- C. Twin-Stem Hangers: Two, 1/2-inch steel tubes with single canopy designed to mount a single fixture. Finish same as fixture.
- D. Wires: ASTM A641/A641M, Class 3, soft temper, zinc-coated steel, 12 gage.
- E. Wires for Humid Spaces: ASTM A580/A580M, Composition 302 or 304, annealed stainless steel, 12 gage.
- F. Rod Hangers: 3/16-inch minimum diameter, cadmium-plated, threaded steel rod.
- G. Hook Hangers: Integrated assembly matched to fixture and line voltage and equipped with threaded attachment, cord, and locking-type plug.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Lighting fixtures: Set level, plumb, and square with ceilings and walls. Install lamps in each fixture.
- B. Support for Lighting Fixtures in or on Grid-Type Suspended Ceilings: Use grid as a support element. Coordinate with ceiling installer (Division 9).
 - 1. Install a minimum of four ceiling support system rods or wires for each fixture. Attach to grid and locate not more than 6 inches from lighting fixture corners.
 - 2. Support Clips: Fasten to lighting fixtures and to ceiling grid members at or near each fixture corner with clips that are UL listed for the application.
 - 3. Fixtures of Sizes Less Than Ceiling Grid: Install as indicated on reflected ceiling plans or center in acoustical panel, and support fixtures independently with at least two 3/4-inch metal channels spanning and secured to ceiling tees.
- C. Suspended Lighting Fixture Support:
 - 1. Pendants and Rods: Where longer than 48 inches, brace to limit swinging.
 - 2. Stem-Mounted, Single-Unit Fixtures: Suspend with twin-stem hangers.

3. Continuous Rows: Use tubing or stem for wiring at one point and tubing or rod for suspension for each unit length of fixture chassis, including one at each end.
- D. Adjust aimable lighting fixtures to provide required light intensities. Adjust to satisfaction of Architect and Owner.
- E. Connect wiring according to Division 26 Section "Low-Voltage Electrical Power Conductors and Cables."
- F. Do not install reflector cones and visible trim of luminaires until completion of plastering, ceiling tile work, painting, and general cleanup. Installation of reflector cones and visible trim of luminaires shall be carefully coordinated with ceiling openings to prevent light leaks at the ceiling plane. Handle cones and trim carefully to avoid scratching or finger printing. Luminaires shall be completely clean at time of acceptance by Owner.
- G. Plaster frames (frame-in-kits) for all recessed fixtures shall be installed per manufacturer's instructions and in a rigid manner so as not to allow fixture frame or housing to move or shift when trim is removed or fixture is relamped. If plaster frame shifts when relamped or trim is not tight to ceiling, or the fixture has not been installed per the manufacturer's instructions, the Contractor shall resecure frames and have ceiling patched and refinished at their expense. General Contractor to spackle gap between rough g.w.b. opening and the plaster frame of light fixture to avoid light-leak and allow for narrow or flangeless trims to be used.
- H. Drywall suspension ceiling systems will be provided with supplemental steel stud channels at luminaire locations for supporting luminaires to ceiling system under Section 09 26 00, Gypsum Wallboard Systems. Securely fasten recessed and/or surface mounted luminaires to steel channels with approved earthquake clips or clamping devices. Where required by local codes, where ceiling systems are not seismically braced or where the weight of the luminaire may cause deformation of the suspended ceiling, provide independent supports for luminaires connected to the structure above. The amount of deformation allowed is specified in Section 09 26 00.
- I. Where MC and/or AC cable is not allowed for branch circuiting as described in Specification Section 260519, extend conduit to an accessible junction box within six feet of luminaires and provide a maximum six-foot length of flexible conduit (Greenfield) and wire to each luminaire. Where MC and/or AC cable is allowed for branch circuiting as described in Specification Section 260519, extend AC and/or MC cable to an accessible junction box within six feet of luminaires and provide a maximum six-foot length of AC and/or MC cable to each luminaire. Locate box so that luminaires can be readily moved into adjacent modules and to provide access to space above ceiling. Support flexible conduit or cable from structure; do not lay on ceiling tiles or attach to any support systems.
- J. Properly align all surface type luminaires. Bolt together so that alignment will be permanent.
- K. Mount under cabinet luminaires as indicated on architectural details. If cabling to luminaires is not concealed, provide wire mold cover to conceal wiring, color as directed by Architect.
- L. Provide aiming of adjustable fixtures to the satisfaction of Engineer/Architect/Owner.

3.2 FIELD QUALITY CONTROL

- A. Test for Emergency Lighting: Interrupt power supply to demonstrate proper operation. Verify transfer from normal power to battery and retransfer to normal.
- B. Prepare a written report of tests, inspections, observations, and verifications indicating and interpreting results. If adjustments are made to lighting system, retest to demonstrate compliance with standards.

END OF SECTION 26 51 00

SECTION 28 31 11 – ADDRESSABLE FIRE ALARM SYTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and General Provision of Contract, including General and Special Conditions and Division I Specification Section, apply to work of this Section.
- B. Related Sections:
 - 1. General and Special conditions
 - 2. Division 1: Submittals
 - 3. Division 7: Firestopping
 - 4. Division 8: Door Release and unlocking
 - 5. Division 9: Painting
 - 6. Division 14: Elevators
 - 7. Division 21: Sprinkler System, Fire Pumps
 - 8. Division 23: Air Handling Systems
 - 9. Division 26: Basic Electrical Requirements

1.2 SUMMARY

- A. Scope: The contract drawings are conceptual. This work includes designing and modifying the existing addressable fire alarm system as described herein and on the contract drawings for The Smithsonian National Museum of Natural History. The existing system is anticipated to be a fully operational Siemens XLS. The system shall include all wiring, raceways, pull boxes, terminal cabinets, outlet and mounting boxes, control equipment, dual channel, alarm and supervisory signals, initiating devices, alarm notification appliances, interfaced equipment, and all other accessories and miscellaneous items required for a complete operating system even though each item is not specifically mentioned or described.
- B. Existing Equipment: Existing or temporary fire alarm equipment shall be maintained operational until the new equipment has been tested and accepted by the Smithsonian Institution Authority Having Jurisdiction. As new equipment is installed, it shall be labeled "NOT IN SERVICE" until the new equipment is accepted. Once the new system is accepted by the Smithsonian, the labels on the new equipment shall be removed and the existing equipment shall be labeled "NOT IN SERVICE" until removed from the building. Just prior to demolition and construction activities (dust producing activities), dust covers shall be installed over smoke detectors and the duct detector(s) for the air handling unit serving the area on a daily/per shift basis. Duct detector sampling tubes and interior housing shall be cleaned prior to removal of detector dust cover. The dust covers are to be removed at the end of every workday/shift. An in-line filter shall be installed for each air-sampling unit to prevent dust infiltration if it is not possible to shut down the unit in question during construction. This filter shall be maintained and replaced by the Contractor for the duration of construction. Air sampling piping shall be cleaned at the end of the construction contract.

- C. Equipment Removal: After acceptance of the new system by the Smithsonian, all existing equipment not connected to the new system shall be removed and all damaged surfaces shall be restored to finishes similar to surrounding walls/ceiling/floor. Detectors shall remain covered during removal and while in storage. Operational equipment which was removed shall be carefully packaged, labeled, and turned over to the SI at the discretion of the COTR.
 - 1. Other equipment, such as conduit and electrical boxes, shall be removed from the site and disposed of by the Contractor.
- D. Repair/Service Replacement Parts: Repair services and replacement parts for the system shall be furnished under this contract after the date of final acceptance of work by the Smithsonian Institution. On-site service during the warranty period shall be provided within 24 hours after notification. All repairs shall be completed within 48 hours after notification.

1.3 RELATED SECTIONS

- A. General and Special conditions
- B. See Division 01 Section SUBMITTALS
- C. See Division 07 Section FIRESTOPPING
- D. See Division 09 Section PAINTING
- E. See Division 26 BASIC ELECTRICAL REQUIREMENTS

1.4 DEFINITIONS

- A. COTR: Contracting Officer Technical Representative
- B. Designer: Person or persons responsible for the development of fire alarm or signaling system plans and specifications.
- C. EoR: Engineer of Record
- D. FACU: Fire Alarm Control Unit
- E. Fire Alarm Contractor: Any company with the qualified experience and personnel as laid out in Section 1.9 to complete fire alarm system installation work.
- F. FM: FM Global (Factory Mutual)
- G. FPE: Fire Protection Engineer
- H. Furnish: To supply the stated equipment or materials
- I. Install: To set in position and connect or adjust for use

- J. Installer: The person or persons working on the wiring, conduit, devices, and panels associated with the fire alarm system.
- K. NAC: Notification Appliance Circuit
- L. NFPA: National Fire Protection Association
- M. NICET: National Institute for Certification in Engineering Technologies
- N. OSHM: Office of Safety Health and Environmental Management
- O. Provide: To furnish and install the stated equipment or materials
- P. Quality Control Personnel: The person or persons responsible inspection for and quality control activities associated with the wiring, conduit, devices, and panels of the fire alarm system.
- Q. SLC: Signaling Line Circuit
- R. Technician: The qualified person in charge of fire alarm system programming.
- S. UL: Underwriters Laboratories

1.5 SYSTEM DESCRIPTION

- A. The System shall be a complete, supervised, noncoded, addressable multiplex fire alarm system with voice/strobe evacuation, dual channel intelligent analog alarm initiation, and complying with all aspects of the applicable documents listed herein.

1.6 PERFORMANCE REQUIREMENTS

- A. Comply with NFPA 72, all contract documents, and the requirements of the Smithsonian Design Standards.
- B. The system shall be classified as a proprietary protective signaling system.
- C. Control features:
 - 1. Any intelligent analog smoke detector or conventional smoke detector zone shall include a selectable alarm confirmation capability. Alarm conditions on these devices are processed through a confirmation period of 45 seconds. Over the next 300 seconds, a signal justification period is initiated where any subsequent alarms are reported immediately.
 - 2. A subprogram shall be provided to allow environmental compensating for smoke detector sensitivity. Each smoke detector shall be programmed with this capability.
 - 3. The system shall provide a field-test function where one person can test the complete system or a specified area at the fire alarm control unit while maintaining full operational function of other areas not under test. Alarms, troubles, device types, and the initiation device addresses shall be logged to historical memory.

D. Supervision

1. Where connecting to existing system, match existing wiring scheme (e.g., Class A, B)
2. Class B signaling line circuits for each floor.
3. Class X signaling line circuits for panel-to-panel communication and audio network(s).
4. Class B notification appliance circuits.
5. Provide electrical supervision of the circuits leading to interfacing modules for the monitoring of contact type initiation devices, the control of electrical devices, fire pump controllers, load control relays (controlling elevators and HVAC equipment), and each independent smoke detection, kitchen, and gaseous fire suppression systems.
6. Connections to Public Address/Emergency Communications Systems shall meet NFPA 72 criteria for circuit monitoring.

E. Spare capacity:

1. All installed signaling line circuits and notification appliance circuits shall have 20 percent spare capacity.
2. All amplifiers shall have 20 percent spare capacity.
3. Battery size shall be a minimum of 125 percent of the calculated requirement.

F. Each independent fire detection, kitchen, and fire suppression system shall be monitored for supervisory conditions. Each monitored condition shall be provided with a separate address.

G. Fire Alarm Signal Initiation Shall Be By One Or More Of The Following Devices:

1. Manual pull station
2. Heat detector
3. Smoke detector
4. Projected beam detector
5. Aspiration smoke detector Alarm
6. Automatic sprinkler system water flow switch or pressure switch.
7. Combustible gas detection system
8. Conventional initiation device zone (for legacy systems)
9. Flame detector
10. Releasing control panel Alarm
11. Fire pump running

H. Supervisory signal initiation shall be by one or more of the following devices or actions:

1. Operation of a fire-protection system valve tamper switch.
2. Operation of the building fire pump or pump running signal and/or any associated fire pump system troubles/ supervisory signals.
3. Duct smoke detector associated with air handling unit or smoke damper.
4. Laser scanner aspiration smoke detector alert and action
5. Operation of any non-fire system as designated on the Matrix.
6. Generator running
7. Generator fault
8. Fire pump fault
9. Bypass Functions
10. Strobe power supply panel trouble conditions

11. Carbon monoxide detector
12. Any sub-system such as releasing panel trouble contacts.
13. Opening, tampering, or removal of alarm-initiating and supervisory signal-initiating devices.
14. Critical failure of emergency communications system hardware
15. General failure of emergency communications system hardware
16. Abnormal position of any switch at the FACU or annunciator.

I. System trouble signal initiation shall be by one or more of the following devices or actions:

1. Loss of primary power at the FACU or remote panel(s).
2. Ground or a single break in FACU internal circuits.
3. Abnormal A/C voltage at the FACU.
4. A break in standby battery circuitry.
5. Open circuits, shorts, and grounds of wiring for initiating device, signaling line, and notification-appliance circuits.
6. Failure of battery charging.
7. Amplifier failure

1.7 SUBMITTALS

- A. General: Refer to Division 01 Section "SUBMITTALS" for basic information relating to submittal requirements. Partial submittals will not be acceptable and will be returned without review. Complete submittals shall include drawings, product data, calculations, and qualifications. Before any work is commenced, the submittal shall be approved by the Office of Safety, Health and Environmental Management (OSHEM). Any work performed by the contractor prior to their approval will be at the contractor's own risk. If such work is contrary to applicable codes and contract documents, the contractor shall bear all costs including, but not limited to, demolition, reconstruction, and all costs and expenses associated with revising the fire alarm system to meet all applicable codes and contract document requirements. All working drawings shall be a minimum 24 in by 36 in. Electronic submittals in PDF based format are also acceptable.
- B. System Description: Submit a detailed description of the control unit as it shall operate for this specific installation. General system descriptions from the catalog cuts and copies of the Systems Design Operation portion of this specification will not be acceptable. All equipment shall be compatible and listed/approved for operation with one another. Any accessory panel shall be compatible with the fire alarm unit(s).
- C. Comply and provide documents associated with the recommendations and requirements in the "Documentation" section of the "Fundamentals" chapter in NFPA 72.
- D. Equipment: Where altered or providing new, include annotated catalog data showing manufacturer's name, model, voltage, and catalog numbers for all equipment and components of the following:
 1. Strobe power extender panels
 2. Storage batteries
 3. Battery charger

4. Cabinet
 5. Manual pull station
 6. Addressable interface devices
 7. Terminal cabinets/assemblies
 8. Addressable relays and interface modules
 9. Annunciation devices (e.g., speakers, strobes, bells)
 10. Fire detector (e.g., smoke, heat, flame, video)
 11. Air aspirating smoke detection system
 12. Strobe, speaker, and speaker/strobe notification appliances
 13. Amplifiers (primary and back-up)
 14. Tone generators
 15. Firefighter telephones
 16. Waterflow switch
 17. Tamper switch
 18. Fire alarm cabling
 19. Boxes
 20. Terminal strips
 21. Wire nuts
 22. Relays
 23. Conduit
 24. Support
- E. Shop Drawings: Provide three sets of working drawings not smaller than 24 inches by 36 inches (609 mm by 914 mm) and an electronic file. Shop drawings shall be prepared on a Computer Aided Drafting (CAD) System. As a minimum, the shop drawing submittal shall include all items required by NFPA 72 Section 7.4, and the following:
1. Point to point wiring diagrams on floor plans at a scale of not less than $1/8" = 1' - 0"$ (1:100), showing all field devices (indicating and initiating devices, relays, switches, etc.), field interconnections, the routing of conduit and circuits between devices, electrical boxes, terminal cabinets, risers, and the FACU. All device circuit numbers and addresses shall be indicated. The device circuit number and addresses on the point-to-point wiring diagram shall be in logical/sequential order (e.g., L1D1, L1D2).
 2. Field wiring color code scheme: The following wire jackets shall be used for fire alarm cabling.
 - a. XNET/Class X Network Circuit: White
 - b. DACNET/Audio Risers: Blue
 - c. Phone Riser/Circuit: Yellow
 - d. Master Sync: Red with White Stripe
 - e. VESDAnet: Black
 - f. 24VDC Power, Door Holds, etc.: White with Red Stripe
 - g. Strobe: Red with Blue Stripe
 - h. Speaker: Blue with Red Stripe
 - i. SLC: Red
 - j. IDC: Green
 - k. Control Circuit: Orange
 - l. Additions to existing systems and circuitry shall match existing cabling.

3. Wire nut color code scheme shall be provided in the wiring schedule for each wire size/type.
 4. Locations for all ceiling mounted equipment shall be coordinated with lighting fixtures, air outlets, ductwork, and other fixtures. Shop drawings shall include reflective ceiling plans that show at a minimum lights and HVAC diffusers. All detectors shall be centered and aligned with ceiling tiles and/or other ceiling mounted devices.
 5. Complete riser diagrams indicating the wiring sequence of all devices and their connections to the control equipment. Provide a color code schedule for the wiring.
 6. Wiring diagrams of all load control relays. Each device shall be provided with a diagram demonstrating wiring terminations of that device, specific to the project. Installation responsibilities of specific parts of the detail (if applicable), shall be shown.
 7. Detailed sequence of operations and matrix.
 8. Utilize NFPA 170 standard symbols.
- F. As-Built (Record) Working Drawings: On a daily basis the contractor's superintendent shall record as-built conditions on a set of Shop Drawings maintained at the job site. Two sets of Shop Drawings reflecting as-built conditions shall be available prior to and for use during the final acceptance test. Two weeks after the acceptance test and before final acceptance of the work, furnish four complete sets of as-built drawings. The drawings shall be prepared on uniform sized sheets not less than 24 inches by 36 inches (609 mm by 914 mm). The drawings shall include:
1. As-built locations of all devices and equipment. Point to point wiring diagrams on floor plans at a scale of not less than 1/8" = 1'-0" (1:100), showing all field devices (indicating and initiating devices, relays, switches, etc.), field interconnections, the routing of conduit and circuits between devices, electrical boxes, terminal cabinets, risers, and the FACU. All device circuit numbers and addresses shall be October 2024 Volume 2- Specifications 283111 - 17 indicated. Complete wiring diagrams showing connections between all devices and equipment. Each conductor shall be numbered at every junction point with indication of origination and termination points. Device addresses shall be listed next to each device.
 2. Conduit pathways shall be shown. T-taps and other circuit deviations shall be shown.
 3. Riser diagram
 4. All deviations from the project drawings and approved shop drawings. Deviations shall be noted by clouds or other means.
 5. Record Drawing Software: Provide three (3) USB drives containing CAD based drawings in DWG and PDF format of all as-built drawings and schematics. All asbuilt submittals shall also be digitally transmitted to the COTR.
 6. Record Programming: Provide one (1) USB drive containing the final programming for the FACU.
 7. Device Addresses: Prior to fire system installation, provide for approval a complete list of device addresses with corresponding commands, controls, and sequence of operation.
 8. LCD Display Messages: The fire alarm control unit shall display messages in the following format: Device type, Floor, Room/Location. All messages for tamper and waterflow switches shall also display the zone and valve location. System programming and device descriptions shall be provided to OSHM with as-built drawings prior to final acceptance. Any changes required to the programming/descriptions shall be completed before final acceptance.
 9. LCD messaging shall be standardized using the following rules and abbreviations:
 - a. Abbreviations:

- 1) AMPS-24 – Power Supply
- 2) DD – Duct Detector
- 3) FFT – Firefighter Telephone
- 4) FS – Flow Switch
- 5) FSD – Fire/Smoke Damper (usually followed by a number) October 2024
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- 6) HD – Heat Detector
- 7) MON – Monitor Module
- 8) MS – Manual Station
- 9) NAP – Notification Appliance Power Supply
- 10) RLY – Relay Module
- 11) SD – Smoke Detector
- 12) TS – Tamper Switch

b. Examples:

- 1) Standard Device: (Device Type) (Floor Level) (Location Description including room number)
Example: SD L1 GALLERY 203 BY STAIR 2
- 2) Air Handler: (Device Type) (Floor Level) (Air Handler Number) (Room Number where AHU is present)
Example: DD LL AHU10 SUPPLY PM10
- 3) Fire/Smoke Dampers: (Device Type) (Floor Level) (Location) (Damper Number) (Air Handler number, if possible) (Room Number where AHU is present)
Example: DD LL 3103 FSD-OA-1-8 AHU10 PM10
- 4) Fire Sprinkler Valve: (Device Type) (Floor Level) (System Number) (Location/Area covered by sprinkler system) (Location of valve if possible)
Example: FS L2 2-1 G203 VLV IN STAIR 1
- 5) Relays and Monitors: (Device Type) (Floor Level) (Location Description) (Function of Relay/Monitor)
Example: RLY L1 EL10 EMR 110A SHUNT TRIP

- G. Qualification Data: For Designer, Installer, and Technician as outlined in Section 1.6.
- H. Strobe and Voice Evacuation Circuit Labels: Prior to fire system installation, provide for approval a complete list of area descriptions for strobe and voice evacuation circuits to be labeled on the fire alarm control unit(s).
- I. Service Manuals and Equipment Descriptions: Thirty days prior to the final acceptance test and after the preliminary testing has been completed submit the following:
1. Furnish four (4) bound copies of complete service manuals to include: device and board specifications, operation, installation, and maintenance manual; manufacturers installation instructions for all aspects of the installation; Test Mode Operating Instructions; manufacturer's wiring specifications for the system; training manual.
 2. Maintenance checklists for equipment.
 3. As-built circuit diagrams, complete with color-code scheme, and device descriptions.
 4. Complete parts list by make model number and manufacturer.
 5. List of smoke detector addresses and corresponding sensitivity readings.

6. Copies of approved submittal materials.
7. Operating instructions shall be framed and mounted at the FACU.

J. Calculations:

1. Battery capacity calculations: The battery calculations shall be detailed to show device quantity, standby current, alarm current and total current to justify proposed battery size. Ampere-hour requirements for each system component and each panel component shall be submitted with the calculations.
2. Supervisory power requirements for all equipment.
3. Alarm power requirements for all equipment.
4. Power supply rating justification showing power requirements for each of the system power supplies. October 2024 Volume 2- Specifications 283111 - 20
5. Justification showing power requirements of the system amplifiers. Amplifiers shall be sized for a minimum of 1 watt per connected speaker in common areas and 0.5 watt per speaker in all other locations.
6. Voltage-drop calculations for NAC wiring runs demonstrating worst-case condition. Show capability of 25 or 70.7 vrms circuits for wire runs.
7. Provide complete battery calculations for both the alarm and supervisory power requirements. Ampere-hour requirements for each system component shall be submitted with the calculations.
8. Updated battery calculations shall be provided prior to Acceptance Testing to account for as-built conditions.

K. Terminal Cabinet Wire Chart: Prepare a wire chart of the wires in each terminal cabinet. Chart every wire showing the wire number, color, size, type of circuit, designation, origination point and termination point. The chart shall be typewritten with minimum 12-point lettering. The format of the wire chart shall be as shown on the contract drawings. The chart shall be protected with a clear laminate and mounted in each cabinet so that it does not interfere with the wiring or terminals.

L. Work Schedule: All work shall be coordinated with facility operations. Museum operations may require limited access to areas, arranging for museum security personnel to accompany contractors in non-public areas, and working during off-hours. Prior to initial work, meet with museum staff to establish a work schedule. A work schedule shall be submitted for approval prior to initial work.

M. Certificate of Compliance: Within two weeks after passing the acceptance test, submit a certificate of code and contract compliance to the COTR in accordance with NFPA 72.

1.8 QUALITY ASSURANCE

A. General Qualifications:

1. A submittal shall be provided, prior to installation taking place, specifying personnel for each of the roles outlined below.

B. Manufacturer Qualifications

1. Testing Services or Laboratories: Construct all fire alarm and fire detection equipment in accordance with the latest edition of the following publications from Underwriters Laboratories (UL) and Factory Mutual Engineering Corporation (FM):
 - a. UL Fire Protection Equipment Directory
 - b. UL Electrical Construction Materials Directory
 - c. UL 38 – Standard for Manual Signaling Boxes for Fire Alarm Systems
 - d. UL 228 – Door Closers-Holders, With or Without Integral Smoke Detectors
 - e. UL 268 - Smoke Detectors for Fire Alarm Systems
 - f. UL 268A - Smoke Detectors for Duct Application
 - g. UL 464 - Audible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories
 - h. UL 497A – Standard for Secondary Protectors for Communications Circuit
 - i. UL 521 - Heat Detectors for Fire Protective Signaling Systems
 - j. UL 864 - Standard for Control Units and Accessories for Fire Alarm Systems
 - k. UL 1283 – Standard for Electromagnetic Interference Filters
 - l. UL 1449 - Standard for Surge Protective Devices
 - m. UL 1480 - Speakers for Fire Alarm and Signaling Systems, Including Accessories
 - n. UL 1971 – Standard for Signaling Devices for the Hearing Impaired
 - o. FM Approval Guide
 - p. Codes and Standards, Latest Edition
 - q. International Building Code
 - r. NFPA 70, National Electrical Code
 - s. NFPA 72, National Fire Alarm and Signaling Code
 - t. NFPA 101, Life Safety Code
 - u. Americans with Disabilities Act Accessibility Guidelines for Buildings and Facilities (ADAAG).
 - v. ASME/ANSI A 17.1, Safety Code for Elevators and Escalators

C. Qualifications of Installer:

1. Installation shall be accomplished by a fire alarm contractor with a minimum of five years' experience in the installation of fire alarm system of comparable size and capacity.
2. Contractor shall have an office, which has been in existence for at least 3 years, within a 121-kilometer (75 mile) radius of the project site.
3. The Installer foremen or team leaders shall be at minimum NICET Level II in fire alarm systems and shall be onsite when any work on the fire alarm system is performed. Work shall not be performed on the fire alarm system if one of these individuals is not present and overseeing work. Key personnel shall be present on the Contractor's daily log.
4. Quality Control Personnel shall be at minimum NICET Level III in fire alarm systems and shall visit the site weekly to inspect the work installed, including conduit, wiring, devices, and panels. This person or persons shall not be the Installer or the Technician.
5. The Technician shall be at minimum NICET Level III in fire alarm systems and shall supervise installation, adjustments, and tests of the system. If a Technician is on October 2024 Volume 2- Specifications 283111 - 23 site daily whenever work is performed, they can qualify as the Installer as well. The Technician shall be an employee of the fire alarm contractor.
6. The certifications of the Installer, Technician, and Quality Control Personnel shall be part of the submittals and shall be approved by OSHM prior to the start of their work. If the Installer or Technician is changed at any point during the project, the credentials of the

replacement shall be submitted and approved prior to the start of their work. Certifications for corporate personnel or personnel who will not be on-site for installation of the project will not be accepted.

D. Qualifications of Distributor/ Service Organization/ Designer:

1. Design Personnel certified by NICET as Fire Alarm Level III or IV or a registered fire protection engineer. The manufacturer's equipment distributor shall show evidence of certification by the manufacturer in the technical support of the system installed under this contract.
2. The distributor shall show evidence of certification of at least one employee by the NICET at Level III or IV in the Fire Alarm Systems subfield of Fire Protection Engineering Technology. If such a certified individual is not employed, adequate documentation shall be provided to show comparable training and experience of an existing employee. At a minimum, comparable training and experience shall consist of ten years of progressive experience in the installation and design of fire alarm systems of comparable size and complexity to that specified herein.
3. In lieu of an employee with NICET Level III or IV certification, the distributor shall show evidence of at least one employee with a minimum of ten years of progressive experience in the design of fire alarm systems. In addition, the distributor shall show evidence of technical support in the design, installation, and testing of the systems from a manufacturer-affiliated company, which shall show evidence of certification of at least one employee by the NICET at level III or IV in the Fire Alarm Systems subfield of Fire Protection Engineering Technology.
4. The contractor shall furnish evidence that the fire alarm equipment supplier has an experienced and effective service organization, which carries a stock of repair parts for the system being furnished. Should the Contractor fail to comply with the service requirements of this section, the Smithsonian will then have the option to make the necessary repairs and back-charge contractor without any loss of warranty as provided by the contract documents.
5. The control equipment manufacturer shall provide an employee certified in the inspection, testing and maintenance of the panel and/or equipment being October 2024 Volume 2- Specifications 283111 - 24 installed, to assist in the final connections to the panel. This individual shall also assist with pre-testing and acceptance testing.

E. Qualifications of Instructor:

1. Instructor: Include in the project the services of an instructor, who shall have received specific training from the manufacturer for the training of other persons regarding the inspection, testing and maintenance of the system provided. The instructor shall train the Smithsonian employees designated by the Contracting Officer, in the care, adjustment, maintenance, and operation of the fire alarm system. Qualifications shall be provided to OSHM for review prior to any training taking place.

F. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

1.9 DELIVERY STORAGE AND HANDLING

- A. Deliver products to project site in original, unopened packages with intact and legible manufacturers' labels identifying product and manufacturer, date of manufacture, and shelf life if applicable.
- B. Store materials inside, under cover, above ground, and kept dry and protected from physical damage until ready for use. Remove from site and discard wet or damaged materials.

1.10 PROJECT CONDITIONS

- A. Interruption of Existing Fire Alarm Service: Do not interrupt fire alarm service to facilities occupied by The Smithsonian Institution or others unless permitted under the following conditions and then only after arranging to provide temporary guard service according to requirements indicated:
 - 1. Notify COTR no fewer than 48 hours in advance of proposed interruption of fire alarm service.
 - 2. Do not proceed with interruption of fire alarm service without The Smithsonian Institution's written permission.
 - 3. No system shall be out of service during non-business hours unless a fire watch is established.
 - 4. At no time shall the work area not be protected by either the temporary fire alarm system or the building fire alarm system.
 - 5. When working on the fire alarm system, the Contractor shall be present in the control room or at the main fire alarm control unit for the duration of the work and be aware of any abnormal conditions resulting from their work. If their work causes any abnormal conditions (alarm, supervisory, or trouble signals), the Contractor shall be present to acknowledge them and clear them by the end of the work shift. Issues that would incapacitate the majority or all of the system shall be brought to the attention of the COTR and OSHM and rectified immediately. If other abnormal conditions are received (not the result of the Contractor's work), the Contractor will discontinue work as needed by SI/OPS to resolve the condition.
 - 6. COTR shall be notified in advance if disruption is required in areas open to the public. The affected area shall be closed to the public during service disruption. Work shall be done during closed-hours, or Smithsonian Institution shall give permission to close the public areas during open hours.
 - 7. All contractor(s) working on the fire alarm system (or subsystem) shall check the system(s) status before and after each shift to ensure no faults are present in the area of work. Any existing alarm, trouble, or supervisory signals shall be recorded at the beginning and end of each shift. Photographs of the panel troubles shall be provided to the COTR and OSHM before and after each shift. Any troubles in either the area of work or the remainder of the facility determined to be at the fault of the Contractor shall be rectified at no additional cost to the SI within 24 hours of the issue being discovered. Issues determined to be critical by the COTR or OSHM shall be rectified immediately.
 - 8. If a temporary fire alarm panel is present, photographs of the panel troubles shall be provided to the COTR and OSHM daily. Any troubles shall be rectified within 24 hours of the issue being discovered.

1.11 TEMPORARY FIRE PROTECTION

- A. MC cable may be utilized for temporary fire alarm systems. All temporary MC cable shall be removed prior to project close-out.
- B. A Listed and Approved temporary wireless fire alarm system shall be provided. The system shall be monitored by the building fire alarm system.

1.12 COORDINATION

- A. Coordinate device layout with reflected ceiling plan and all ceiling-mounted equipment, including diffusers, lights, security cameras, fire alarm devices, exit signs, and other devices.
- B. Coordinate major equipment and aspiration smoke detection piping layouts with other trades to avoid obstructions and excessive changes in direction for piping.

1.13 WARRANTY

- A. The contractor shall guarantee labor, materials, and equipment provided under this contract against defects for a period of one year after the final acceptance of this work by the Smithsonian and after the receipt of as-built drawings and schematics of all equipment.
- B. A full system warranty fire alarm test shall be provided by the installer one month prior to warranty expiration. All interfaced systems (e.g., smoke management systems, lighting systems), shall be tested. All deficiencies found shall be corrected for the warranty to be considered complete. If the warranty is phased, then the testing shall also be phased. The warranty period shall not start when the panel is powered on but rather when the project or phase has been completed.

1.14 EXTRA MATERIALS

- A. Spare parts shall be directly interchangeable with the corresponding components of the installed system. Spare parts shall be suitably packaged and identified by nameplate, stamping, or tagging.
- B. Furnish the following spare parts. Quantity shall be two percent of the installed number of devices, but not less than the quantities listed:
 - 1. Smoke Detectors of each type installed: 5
 - 2. Audio/visual devices of each type installed: 5
 - 3. Addressable monitor modules: 5
 - 4. Addressable relay modules: 10
 - 5. Duct detector housing: 1
 - 6. Keys for all fire alarm cabinets: 5
 - 7. Wrenches or special tools required to gain access to all lockable equipment: 5 of each type required

PART 2 - PRODUCTS

2.1 GENERAL

- A. All components provided shall be listed for use with the selected system.
- B. With the exception of the air aspirating smoke detection control panels, all fire alarm equipment (including releasing panels) shall be from a single manufacturer.
- C. Only appliances that comply with the manufacturer's installation instructions shall be permitted.

2.2 MANUFACTURERS

- A. Available Manufacturers: Subject to compliance with requirements and compatibility to Smithsonian monitoring system, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. FACU, cabinets, and peripheral devices:
 - a. Notifier
 - b. FireLite
 - c. Siemens
 - d. Approved equal. (Shall be approved by OSHEM prior to submitting shop drawings)
 - 2. Wire and Cable:
 - a. As per Division 26 this specification.
 - 3. Conduit:
 - a. As per Division 26 this specification
 - 4. Boxes, supports, terminal blocks, and appurtenances:
 - a. As per Division 26 this specification.

2.3 FIRE DETECTORS

- A. Photoelectric light scattering type smoke detectors shall be provided as follows:
 - 1. The detector shall be suitable for two-wire 24VDC operation and two-way communications on the intelligent analog signaling circuit. Smoke detectors shall be UL listed for use with the FACU and environmental conditions.
 - 2. Detectors shall be self-compensating for ambient temperature and humidity.
 - 3. Detector bases shall be installed on an industry standard, 4 inch (101 mm) square or octagonal electrical outlet box. Bases shall be universal for ionization, heat, and photoelectric (light scattering type) detectors.
 - 4. Detectors shall be twist-lock type on to the base with self-wiping contacts.

5. Screw clamp terminals shall be provided for all conductor terminations.
6. The detector shall be addressed, tested, and programmed prior to installation. The detector readout shall yield a discreet electrical value for status tracking and logging for determining maintenance and cleaning requirements. The control unit shall provide a sensitivity readout from the detector without removal from the base.
7. Provide self-restoring type detectors that do not require any readjustment after actuation to restore them to normal operation.
8. All components shall be rust and corrosion resistant. Vibration shall have no effect on the detector's operation. Protect the detection chamber with a fine mesh metallic screen that prevents the entrance of insects or air born materials. The screen shall not inhibit the movement of smoke particles into the chamber.
9. The detector shall display a differentiating indicating LED when in the alarm state when the system is operating from normal or standby power.

B. Photoelectric - Beam Type:

1. Detectors shall be UL listed for use with the fire alarm control unit and environmental conditions.
2. Detectors with active receivers are required.
3. Field programmable sensitivity settings of 20%, 35%, and 60% beam obscuration.
4. Latching alarm LED is built into the cover of the receiver or transmitter.
5. Compensation circuit for environmental variations.

C. Duct Detectors:

1. Duct smoke detectors shall meet the requirements for photoelectric light scattering type detectors. With the addition that duct detectors are UL 268A listed for installation in air duct sampling housings for the detection of smoke in HVAC system ducts.
2. Provide remote indication where the detector is obstructed or not visible from the floor level. Location of remote indicators shall be approved by OSHM.
3. Provide a remote test station where the detector is more than six feet above floor level. An operating key switch shall initiate an alarm test. Location of key switches shall be approved by OSHM and shall be accessible from the floor level. Key switches shall be provided no more than 8 feet above finished floor.
4. Weatherproof Enclosure: NEMA 250, Type 4X; NRTL Listed for use with the supplied detector for smoke detection in HVAC system ducts.
5. Sampling Tubes: Design and dimensions as recommended by the manufacturer for the specific duct size, air velocity, and installation conditions where installed.
6. Relay Fan Shutdown: Rated to interrupt fan motor-control circuit.

D. Heat Detectors:

1. Detectors shall be Listed or Approved for use with the FACU and environment conditions.
2. Detectors shall be addressed, tested and programmed prior to installation.
3. Detectors shall display a steady LED when in the alarm state when the system is operating from normal or standby power.
4. Detectors shall be equipped with screw terminals for each conductor.
5. Detectors shall be hermetically sealed and of the automatically resetting type which will operate when ambient air temperature reaches detector setting regardless of rate of temperature rise.

6. Detector operation shall not be subject to thermal lag.
7. Heat detectors in hoistways and elevator machine rooms shall be intelligent, low temperature (135°F-140°F), rate-compensating detectors.

2.4 NOTIFICATION APPLIANCES

- A. Speakers and strobes shall be 24VDC (strobes) or 70 VAC (speakers) and shall be equipped with terminal strip and pressure-style screw terminals for the connection of field wiring. Devices shall be mounted to 4 inch (101 mm) square outlet boxes.
- B. Speakers: Provide fire alarm speakers conforming to UL 464.
 1. Speakers shall be provided with at least the following four tap settings: 2-watt, 1-watt, 1/2-watt, and 1/4-watt.
 2. The audible signal shall have two alternate and distinct audible patterns; horn tone and recorded voice message; each field switchable for peak output levels of 100db or 106db measured at 3048mm (10 feet). When multiple tones are activated simultaneously, the tones shall be prioritized to avoid interference.
 3. System shall meet intelligibility requirements of NFPA 72 and this specification document.
 4. Where the design has determined that high-challenge acoustic areas are present, high-fidelity speakers are to be provided to achieve audibility and intelligibility in the subject areas.
 5. If determined to be needed, design team to specify specific make/model options in this section.
- C. Strobe lights
 1. Provide with white finish plate and with the word "ALERT" horizontally printed for ceiling mounting.
 2. Xenon strobe with a minimum repetition rate of 1 Hz, not exceeding 3 Hz and a maximum duty cycle of 40% with pulse duration of 0.2 seconds.
 3. Visual alarm signals shall be furnished with minimum light intensity of 75 candelas and meet the requirements of ADA and UL 1971. In large rooms with ceilings over 10 feet, 110 candela models shall be provided. Any device up to 110 cd shall have adjustable candela ratings.
 4. Provide strobe light visual alarm signals that operate on 24 VDC.
 5. Synchronization shall be provided as required in NFPA 72. All strobe lights within view of one another shall be synchronized.
- D. Audio-Visual Combination Assemblies shall be provided where strobes and speakers are identified at the same locations.

2.5 AMPLIFIERS, PREAMPLIFIERS, TONE GENERATORS

- A. General: Amplifiers, preamplifiers, tone generators, digitalized voice drives and all other hardware necessary for a complete, operational dual channel voice/alarm signaling service conforming to NFPA 72 shall be housed in a remote fire alarm control unit, terminal cabinet, or

in the fire alarm control unit. Each amplifier shall have two channels: one to broadcast a message and the other for paging.

- B. Construction: Amplifiers shall utilize solid-state components and shall be provided with output protection devices sufficient to protect the amplifier against any transient voltage up to ten times the highest rated voltage in the system.
- C. Inputs: Each system shall be equipped with separate inputs from the tone generator, digitalized voice driver and panel mounted microphone. Microphone inputs shall be of the low impedance, balanced line type. Both microphone and tone generator input shall be operational on any amplifier.
- D. Tone Generator: The tone generator shall be of the modular, plug-in type with securely attached labels to identify the component as a tone generator and to identify the specific tone it produces. The tone generator shall be capable of producing multiple tones and be constantly repeated until interrupted by the digitalized voice message, the microphone input, or the alarm silence mode as specified. Tones shall be approved by OSHM before panel programming. The tone generator shall be single channel with an automatic backup generator per channel such that failure of the primary tone generator causes the backup generator to automatically take over the functions of the failed unit and causes transfer of the common trouble relay.
- E. Protection Circuits: Each amplifier shall be constantly supervised for any condition that could render the amplifier inoperable at its maximum output. Failure of any component shall cause automatic transfer to a designated backup amplifier, illumination of a visual "amplifier trouble" indicator on the control unit, appropriate logging of the condition and other actions for trouble conditions as specified.

2.6 POWER SUPPLIES

- A. Primary power for the FACU shall be 120VAC service obtained from the emergency power panel board. Red colored breaker locks shall be provided for all fire alarm circuit breakers.
- B. Secondary power for the FACU shall be provided by sealed gelled electrolyte batteries. Batteries shall be housed in the control cabinet or a separate cabinet with adequate cell separation to prevent accidental discharge.
- C. Battery Capacity: Battery supply shall be calculated to operate its load in a supervisory mode for 24 hours with no primary power applied, and after that time, operate its alarm mode for 15 minutes. (In addition, an alarm reserve correction of 1.3 shall be included.)
- D. Battery Charger: Secondary power battery chargers shall be obtained from the emergency power panel board. Provide battery-charging circuitry for each standby battery bank in the system low voltage power supply or as a separate circuit. The charger shall be automatic in design, adjusting the charge rate to the condition of the batteries. Battery charge rate and terminal voltage shall be read using the fire alarm control unit LCD display in the service mode, indicating directly in volts and amps. Charger shall be housed in the main fire alarm control unit or the battery cabinet.
- E. Fast tab battery terminals are not permitted for batteries larger than 12 ampere-hours.

2.7 EQUIPMENT LOAD RELAYS

- A. Relays for the control of air handler contactors and elevator recall circuits shall be rated for use with circuits up to 240VAC at 7A inductive. Relays shall be of the sealed pluggable type, and terminations shall be made to pressure-type screw terminals.
- B. Wiring powering equipment load relays shall be supervised.

2.8 INTERFACE/CONTROL MODULES

- A. Furnish intelligent analog signaling circuit interface modules for the monitoring of contact type initiation devices, the control of electrical devices, fire pump controllers, load control relays (controlling elevators and HVAC equipment), and each independent smoke detection, kitchen, and gaseous fire suppression systems. The modules shall be capable of monitoring three separate functions: alarm, trouble, and supervisory conditions.
- B. The module shall be addressed, tested, and programmed prior to installation using a UL listed programmer/tester.
- C. The module shall display a steady or contrasting intermittent LED for each circuit, in both normal and standby power conditions, when in the alarm state or during control circuit is activation. The LED shall flash during system address polling.

2.9 CABLE FASTENERS

- A. Wire nuts shall be rated to the same temperature as the cabling in question.
- B. Wire nuts shall be provided at the size appropriate for the cabling in question.
- C. Terminal blocks shall be provided at the size appropriate for the cabling in question.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Equipment, materials, installation, workmanship, examination, inspection, and testing shall be in accordance with Division 26 of the project specifications and NFPA 72, except as modified herein.

3.2 SYSTEM FIELD WIRING AND CONDUIT

- A. Wiring within Cabinets and Junction Boxes: Provide wiring installed in a neat and workmanlike manner and installed parallel with or at right angles to the sides and back of any box or cabinet.
- B. Conductor Type and Size: Wire size shall be sufficient to prevent dropping voltage below specified minimum levels. Wire type and sizing of conductors shall also be in accordance with

the manufacturer's wiring specifications. Conflicts with the manufacturer's requirements shall be brought to the attention of OSHM during the shop drawing phase, prior to product approval. Minimum wire size and type shall be as follows, unless manufacturer's requirements conflict. Conflicts shall be brought to the attention of the COTR and OSHM:

1. Signaling Line Circuits: 16AWG, Type FPLR/FPLP, solid copper
 2. Notification Appliance Circuits: 14AWG, Type FPLR/FPLP, solid copper, twisted pair
 3. 120VAC Circuits: 12AWG, Type THHN, solid copper
 4. Interfaced Circuits: 16AWG, Type FPLR/FPLP, solid copper
 5. VESDAnet: 16AWG, Type FPLR/FPLP, shielded, stranded
 6. Speaker Circuits: 16 AWG, Type FPLR/FPLP, solid copper, twisted pair
 7. Firefighter Telephone Circuits: 18 AWG, Type FPLR/FPLP, 2-hr rated, shielded, solid copper, twisted pair
 8. Battery Cable: 14AWG, solid copper
- C. Connectors: 1. Twist-type wire nuts are permitted at junction boxes 4" square or smaller. No more than two wires are permitted to be connected in this manner. 2. Stab-lock or similar connectors are not permitted. 3. Terminal blocks shall be utilized at terminal cabinets.
- D. Terminal Cabinets: Provide a terminal cabinet at the base of any circuit riser, on each floor at each riser, and where indicated on the drawings. Cabinet size shall be appropriate for the size of the wiring to be connected. All terminals shall be permanently labeled with circuit type, circuit number, and area served. Terminal strips shall be appropriately sized to accommodate the cables being connected.
- E. Conductor Numbering: All conductors installed in the system shall be numbered at every junction point and device. Use a numbered shrink-wrap label designed specifically for this purpose. Wire numbers shall be the same as those designated on the as-built drawings. Mark each terminal in accordance with the wiring chart and diagrams of the system.
- F. Conductor Color Coding: Color coded conductors shall be consistent for each type of circuit. When renovating or adding to an existing system, color-coding shall match the existing system. Wire color coding schedule shall follow the requirements specified in Section 1.9 of this specification.
- G. Junction Boxes: Any junction/gang box used in a circuit shall sole be dedicated to that circuit. A common junction/gang box is not permitted for routing multiple circuits. 6 inches of slack cable shall be provided in all pull-through boxes without devices.
- H. Signaling Line and Notification Appliance Circuits
1. Signaling Line, notification appliance, and power circuits shall each be in separate conduit. Circuits accessory to (e.g. controls wiring), or not associated with the fire alarm system shall be run in separate conduit until the point of termination to the fire alarm system. 2
 2. Strobes are to be connected to circuits separate from speakers. This includes strobes and speakers that are mounted as a unit.
 3. Provisions for tying in signaling line and notification appliance circuits directly to the FACU mother board (board containing CPU) shall not be used. Initiation and indicating circuits shall be tied to a separate electronic board before connection to the motherboard.

I. Circuit Loading:

1. Spare capacity shall be in accordance with Section 1.07 of this specification.
2. Circuits operating at 24 VDC shall not operate at less than 21.6volts. Circuits operating at any other voltage shall not have a voltage drop exceeding 10% of nominal voltage.

J. Spare Circuits: Provide one spare signaling line and notification appliance circuit for each terminal box placed at each floor on each riser. Spare capacity shall also be provided in the FACU for these circuits. FACU control boards shall be provided to permit two spare initiation circuits and two spare indicating circuits. Spare circuits are not to include those provided on the motherboard.

K. Conduit:

1. All conductors shall be in grounded metal conduit. Conduit shall be Rigid metal or EMT. Flexible metal conduit not exceeding six-foot lengths shall be permitted from junction box to initiating device. On flexible metal conduit, use only insulated throat connectors.
2. Run conduit or tubing concealed unless specifically shown otherwise on the drawings.
3. Minimum conduit size shall be 3/4-inch.
4. Steel compression-type fittings shall be used. Set-screw type fittings are not permitted.
5. All sharp edges, including excess all-thread rod, shall be removed from the interior of junction boxes before wire is pulled.
6. Provide insulated throat connectors or plastic bushings on all conduit

L. Labeling: Label all addressable input/output fire alarm cover plates and appliances (including smoke detectors, manual pull stations, addressable modules, and notification appliances) with their address utilizing a label machine. Addresses on SLC or NAC devices/appliances shall include panel number, circuit number, and appliance number. Associated end of line shall also be identified as such. Each SLC field device address shall match the fire alarm display address.

M. Circuits to Interfaced Equipment:

1. Circuits to smoke management systems, fan shutdown systems, door locking systems, A/V shutdown, fire door release, firefighter telephones in elevator cabs, etc. shall terminate in terminal cabinets within 914mm (3 feet) of the controllers for those systems. The completion of those circuits from the terminal cabinets to the appropriate system shall be provided under the appropriate division specification.
2. Devices shall have a separate raceway up until the point of termination at the fire alarm system. Wiring other than fire alarm cabling shall not be permitted in any fire alarm conduit.

N. Load Control Relays:

1. Where required by NFPA 101, relays shall be located within three feet of the device controlled.
2. Wiring between the fire alarm addressable module and the equipment load relay shall be supervised.

3.3 ACCESS PANELS

- A. All devices shall be easily accessible for testing, repairs, and maintenance. Provide access panel of sufficient size where devices and junction boxes are concealed to allow access. The Contractor shall provide documentation showing accessibility.
- B. The minimum access size shall be 12" x 12". Larger panels may be required at the discretion of the OSHM FPE if access is deemed insufficient for maintenance staff.

3.4 FIRESTOPPING

- A. Seal all holes caused by penetrating conduit, piping, or other penetrations that pass through floors, walls, or ceilings. Firestop penetrations through floor slabs, fire-rated walls, shafts, or any fire-rated assembly in accordance with Division 07 Section Penetration Firestopping.

3.5 MARKING

- A. All metal surfaces shall be painted. Metal conduit in finished areas shall be painted the color to match adjacent surfaces. Junction boxes in unfinished areas shall be painted full gloss enamel red. Painting shall be in accordance with Section 09 Painting.
- B. 2-inch-wide red bands shall be applied every 10 feet when not using red-colored conduit. Tape shall not be used for banding purposes.
- C. Prior to acceptance testing, label all addressable input/output fire alarm cover plates and appliances (including detectors, manual pull stations, addressable modules, and notification appliances) with their address utilizing a labeling machine. Handwritten labels are not acceptable. Addresses on SLC or NAC appliances shall include panel number, circuit number, and appliance number. Associated end of line appliances shall also be identified as such. Each SLC field appliance address shall match the fire alarm display address. Relays and monitors shall be labeled with their associated function or the equipment they control.
- D. All circuits entering and leaving a terminal cabinet or panel shall be marked with their designations. Where the circuits then travel to another panel or terminal cabinet, those locations and designations shall also be marked in the terminal cabinet.
- E. All duct smoke detectors shall be labeled with the designation of the air handling unit it serves and that unit's location.
- F. Monitor modules associated with sprinkler and fire pump equipment shall be labeled with the corresponding system number or valve designation.

3.6 DEVICE INSTALLATION

- A. Amplifiers: Install a backup amplifier in each fire alarm zone where one or more amplifiers are present.

- B. Strobes: Wall mount strobes shall be a minimum of 80 inches (2032 mm) above the finished floor or 6 inches (152 mm) below the ceiling, whichever is lower. The bottom of the strobe lens shall be above the 80-inch level.
- C. Speakers: Speakers should typically be set on the 1/2 watt tap. Speakers in areas with ceilings above 3048 mm (10 feet) shall be set on the 1-watt tap.
- D. Smoke Detectors:
 - 1. In raised floor spaces and high-velocity plenum areas, the smoke detectors shall be installed to protect 225 sq. ft per detector.
 - 2. Detectors shall be installed in elevator machine rooms and elevator hoistways.
 - 3. Install smoke detectors a minimum of 3 feet (914 mm) away from supply air vents.
 - 4. New smoke detectors shall be installed with dust covers. The dust covers shall be removed just prior to acceptance testing.
 - 5. The indicating LED on the smoke detector shall be visible from the floor. Where ceiling conditions prevent viewing of the LED from the floor, a remote indicating lamp shall be installed. Lamp shall be installed not more than 6 feet above finished floor.
- E. Duct Detectors:
 - 1. Installation of duct detector sampling tubes shall be verified by OSHM prior to final acceptance.
 - 2. Sampling holes shall be oriented facing into the airflow, or as manufacturer's requirements specify.
 - 3. Duct detector sampling tubes shall be verified by OSHM for proper orientation and airflow. Airflow shall be tested per manufacturer's instructions and shall be confirmed with a magnehelic/manometer (airflow/pressure) meter to ensure air is passing over/through the detector chamber. Testing shall be completed while the air handling units are active at their normal settings.
 - 4. Duct detector sampling tube length and installation shall be in accordance with manufacturer's requirements.
 - a. Where possible and where access would be available to the endcap, sampling tubes shall be installed extending out the opposite side of the duct.

3.7 INSPECTION & TESTING

- A. Test Equipment: The Contractor shall supply personnel, communication devices, and all equipment necessary for performance of the final test.
- B. Provide no less than 48-hours' notice for the inspections and testing below. Notice for final acceptance shall be in accordance with the Final Acceptance section.
- C. Conduit Rough-In Inspections:
 - 1. Prior to installation of any wire on a circuit, and/or prior to ceiling close-in, Contractor shall provide conduit plans for review. OSHM shall inspect and approve conduit layout and installation.

2. Installation shall be in accordance with NFPA 70, NFPA 72, Division 26 of this specification, and Section 283111.
3. If deviations from the approved shop drawings are present, red-line plans shall be available for review at the time of inspection.
4. Contractor shall correct all deficiencies before wire installation commences. OSHM shall verify all deficiencies

D. Wiring Tests:

1. Test equipment used for the test shall have been calibrated within one year of the test date.
2. Continuity Tests: Continuity of both wires on a circuit as well as the shield (if present) shall be demonstrated.
3. Megger Tests: After all wiring has been installed, and prior to making any connections to panels or devices, all wiring shall be megger tested for insulation resistance, grounds, and/or shorts. Conductors shall be tested to a minimum of 550 Mega-Ohms. The tests shall be witnessed by OSHM, SI Life Safety Group, and the Contracting Officer. Test results recorded for use at the final acceptance test. Results recorded shall be values between positive/negative, positive/ground, negative/ground, positive/shield, negative/shield, and shield/ground.
4. Loop Resistance Tests: Measure and record the resistance of each circuit with each pair of conductors in the circuit short-circuited at the farthest point from the circuit origin. The tests shall be witnessed by the Contracting Officer, OSHM and the SI Life Safety Group. Test results shall be recorded for use at the final acceptance test.
5. Verify the absence of unwanted voltages between circuit conductors and ground.
6. If partial circuit testing conducted, Contractor shall keep a written record of testing make it available to OSHM for tracking purposes.
7. Contractor shall correct all deficiencies before device installation commences. OSHM shall verify all deficiencies.

- E. Preliminary Testing: Conduct preliminary tests to ensure that all devices and circuits are functioning properly. After preliminary testing is complete, provide a letter to the COTR certifying that the installation is complete and fully operable. The letter shall state that each initiating and indicating device was tested in place and functioned properly, including proper device message descriptions and proper placement of graphic icons in fire alarm systems with graphic notification equipment. The letter shall also state that all panel functions were tested and operated properly. The Contractor and an authorized representative from each supplier of equipment shall be in attendance at the preliminary testing to make necessary adjustments.

- F. Final Acceptance Test: Notify the Contracting Officer in writing when the system is ready for final acceptance testing. Submit request for test at least 14 calendar days prior to the test date. A final acceptance test will not be scheduled until megger test results, the loop resistance test results, preliminary testing letter, and the submittals required in Part 1 are provided to the Contracting Officer. Test the system in accordance with the procedures outlined in the NFPA 72 and as follows:

1. Verify that the control unit is in the normal condition as detailed in the manufacturer's operating and maintenance manual.
2. Complete operational tests under emergency generator power.
3. Complete operational tests under battery power and as described above under battery power. Test the battery charger.

4. Test each initiating and indicating device and circuit for proper operation and response. Disconnect the confirmation feature for smoke detectors during tests to minimize the amount of smoke or test gas needed to activate the detector.
5. Test the airflow through all duct detectors to ensure flow is within the Manufacturer's requirements.
6. Commission all Air Aspirating Smoke Detectors in accordance with Manufacturer's requirements.
7. Test the system for all specified functions in accordance with the contract drawings and specifications and the manufacturers operating and maintenance manual.
8. Verify the audibility and intelligibility results that were recorded on the as-built drawings as part of the preliminary testing above.
9. Visually inspect all wiring at all panels and terminal cabinets. Up to 5% of all devices shall be disconnected and wiring visually inspected during the acceptance test. More devices may be inspected at the discretion of the OSHEM FPE if deficiencies are observed. A minimum of 2 devices per circuit shall be inspected.
10. Verify that all software control and data files have been entered or programmed in the FACU.
11. Verify that shop drawings reflecting as-built conditions are accurate.
12. Measure the current in circuits to assure that there is the calculated spare capacity for the circuits. Current should not exceed 80% of the capacity of the circuit.
13. Measure voltage readings for circuits to assure that voltage drop does not exceed manufacturer's requirements. Voltage drop shall not exceed manufacturer's requirements for the devices in question.
14. Measure the voltage drop at the most remote appliance on each notification appliance circuit. Voltage drop shall not exceed manufacturer's requirements for the devices in question.
15. All faults and trouble signals associated with the completed work shall be cleared from the panel.
16. All deficiencies shall be corrected by the Contractor.

G. Conduct Intelligibility Testing

1. Test intelligibility in identified Acoustically Distinguishable Spaces (ADS) in accordance with NFPA 72. Commercially available test equipment shall be provided by the installing contractor. The mean value of at least three readings shall be used to compute the intelligibility score at each test location. Each ADS shall be tested in at least one location.
2. The following areas are not required to meet a minimum intelligibility score:
 - a. Mechanical/Electrical/Plumbing Rooms
 - b. Private Bathrooms
 - c. Elevator Cars
 - d. Individual Offices
 - e. Closets
 - f. Rooms identified during design where intelligibility cannot be reasonably predicted
 - g. Ensure that the minimum required Common Intelligibility Scale (CIS) value of 0.7 (Speech Transmission Index (STI) 0.5) is obtained in normally occupied areas such as galleries, lobbies, offices, conference rooms, etc. Measurements should be taken near the head level applicable for the space under normal conditions (e.g., standing, sitting, as appropriate).

- h. Areas of the building identified during design and provided with hard wall and ceiling surfaces (such as metal, concrete, terrazzo, or glass) that are found to cause excessive sound reflections may be permitted to have a CIS score less than the minimum required value if approved by OSHEM. Building occupants in these areas must be able to determine that a voice signal is being broadcast by walking no more than 33 feet to find a location with at least the minimum required CIS score.
- i. Areas of the building identified during design where occupants are not normally expected to be present are permitted to have a CIS score less than the minimum required value if personnel can determine that a voice signal is being broadcast by walking no more than 50 feet to a location with at least the minimum required CIS score.
- j. Occasionally, large, cavernous-type open areas are present in SI buildings. In such buildings, hard wall/ceiling surfaces and building geometry may cause excessive sound reflections. Areas identified during design are permitted to have locations with a CIS value lower than the normal, minimum required CIS value when the following conditions are met:
 - 1) The CIS value is not less than 0.6 (STI 0.4) at any location within the cavernous area.
 - 2) Occupants in the cavernous area must walk no more than 98 feet to find another location having at least the normal, minimum required CIS value.
 - 3) Occupants located in the cavernous area can adequately understand the message content in the voice signal being broadcast. Whether the voice message is adequately understood shall be determined by OSHEM. 4) Justification for the deviation from the normal, minimum CIS criteria shall be provided to and approved by OSHEM. The justification shall address all factors relevant to the request and deviation from the normal, minimum CIS criteria, including but not limited to: the operational requirements that restrict the installation of acoustical wall and ceiling treatments; the potential use of special speaker technologies such as directional speakers or stacked speaker systems; and the availability of physically larger or higher-fidelity speakers even though such speakers might not be Listed for fire alarm use.

3.8 TRAINING

- A. Training sessions shall cover all aspects of system performance, including system architecture, signaling line circuit configurations, sensor and other initiating device types, locations, and addresses, fire alarm control unit function key operation, and other functions as designated by the COTR.
- B. Required Instruction Time: Provide 24 hours of instruction after final acceptance of the system. The instruction shall be given during regular working hours on such dates and times as are selected by the Contracting Officer. The instruction may be divided into two or more periods at the discretion of the Contracting Officer to account for the nature of shift work of SI employees. One training session shall be videotaped by the contractor. Video shall be delivered to the COTR.
- C. Provide a typeset printed or typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame. Install the frame in a conspicuous location observable from the FACU. The card shall show those steps to be taken by an operator when a

signal is received as well as the functional operation of the system under all conditions, normal, alarm, supervisory and trouble. The instructions shall be approved by OSHM.

- D. All training sessions shall be conducted following final system certification and acceptance. Three additional training sessions shall be provided for all security personnel on all shifts six months after final system certification.
- E. All training sessions shall be conducted by an authorized fire alarm system distributor representative, who has received specific training from the manufacturer for the training of other persons regarding the inspection, testing, and maintenance of the system provided. Manufacturer Factory Certifications for training personnel shall be provided to OSHM for review prior to scheduling of training.

3.9 KEYS

- A. Keys and locks for all equipment added as part of scope of work shall be identical. All panels and terminal cabinets provided shall be keyed the same. Air aspirating smoke detection control panels shall be keyed the same. Provide not less than five keys of each type required. Identify keys by an appropriate number stamped on each key or on a metal tag attached thereto. Provide a key numbering chart in each operation and maintenance manual furnished.

3.10 ON-SITE DOCUMENTATION

- A. Provide a document box in the same room as the main FACU. The box shall be red in color and shall be capable of holding a full set of as-built plans. Both electronic and hardcopy plans shall be provided.

END OF SECTION 28 31 11

SECTION 31 1 000 - SITE CLEARING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 010000 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

1. Protecting existing vegetation to remain.
2. Removing existing vegetation.
3. Clearing and grubbing.
4. Stripping and stockpiling topsoil.
5. Removing above- and below-grade site improvements.
6. Disconnecting, capping or sealing, and removing site utilities.
7. Temporary erosion- and sedimentation-control measures.

- B. Related Sections:

1. Division 010000 for temporary utility services, construction and support facilities, security and protection facilities, and temporary erosion- and sedimentation-control measures.

1.3 DEFINITIONS

- A. Subsoil: All soil beneath the topsoil layer of the soil profile, and typified by the lack of organic matter and soil organisms.
- B. Surface Soil: Soil that is present at the top layer of the existing soil profile at the Project site. In undisturbed areas, the surface soil is typically topsoil; but in disturbed areas such as urban environments, the surface soil can be subsoil.
- C. Topsoil: Top layer of the soil profile consisting of existing native surface topsoil or existing in-place surface soil and is the zone where plant roots grow. Its appearance is generally friable, pervious, and black or a darker shade of brown, gray, or red than underlying subsoil; reasonably free of subsoil, clay lumps, gravel, and other objects more than 2 inches in diameter; and free of subsoil and weeds, roots, toxic materials, or other nonsoil materials.
- D. Plant-Protection Zone: Area surrounding individual trees, groups of trees, shrubs, or other vegetation to be protected during construction, and indicated on Drawings.
- E. Tree-Protection Zone: Area surrounding individual trees or groups of trees to be protected during construction, and indicated in Division 010000.

- F. Vegetation: Trees, shrubs, groundcovers, grass, and other plants.

1.4 MATERIAL OWNERSHIP

- A. Except for stripped topsoil and other materials indicated in Division 010000 to be stockpiled or otherwise remain Owner's property, cleared materials shall become Contractor's property and shall be removed from Project site.

1.5 INFORMATIONAL SUBMITTALS

- A. Existing Conditions: Documentation of existing trees and plantings, adjoining construction, and site improvements that establishes preconstruction conditions that might be misconstrued as damage caused by site clearing.
 - 1. Use sufficiently detailed photographs or videotape.
 - 2. Include plans and notations to indicate specific wounds and damage conditions of each tree or other plants designated to remain.
- B. Record Drawings: Identifying and accurately showing locations of capped utilities and other subsurface structural, electrical, and mechanical conditions.

1.6 QUALITY ASSURANCE

- A. Preinstallation Conference: Conduct conference at Project site as part of Preconstruction.

1.7 PROJECT CONDITIONS

- A. Traffic: Minimize interference with adjoining roads, streets, walks, and other adjacent occupied or used facilities during site-clearing operations.
 - 1. Do not close or obstruct streets, walks, or other adjacent occupied or used facilities without permission from Owner.
 - 2. Provide alternate routes around closed or obstructed traffic ways as illustrated in the Maintenance of Traffic Plans, Demolition Plans, Erosion and Sediment Control Plans or as required by Owner.
- B. Salvable Improvements: Carefully remove items indicated to be salvaged and store on Owner's premises where indicated in Division 010000.
- C. Utility Locator Service: Notify utility locator service or Miss Utility for area where Project is located before site clearing.
- D. Do not commence site clearing operations until temporary erosion- and sedimentation-control and plant-protection measures are in place.
- E. The following practices are prohibited within protection zones:

1. Storage of construction materials, debris, or excavated material.
 2. Parking vehicles or equipment.
 3. Foot traffic.
 4. Erection of sheds or structures.
 5. Impoundment of water.
 6. Excavation or other digging unless otherwise indicated.
 7. Attachment of signs to or wrapping materials around trees or plants unless otherwise indicated.
- F. Do not direct vehicle or equipment exhaust towards protection zones.
- G. Prohibit heat sources, flames, ignition sources, and smoking within or near protection zones.
- H. Soil Stripping, Handling, and Stockpiling: Perform only when the topsoil is dry or slightly moist.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Satisfactory Soil Material: Requirements for satisfactory soil material are specified in Sections 312000 "Earth Moving" and 329113 "Soil Preparation."
1. Obtain approved borrow soil material off-site when satisfactory soil material is not available on-site.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Protect and maintain benchmarks and survey control points from disturbance during construction.
- B. Locate and clearly identify trees, shrubs, and other vegetation to remain. Wrap a 1-inch green vinyl tie tape flag around each tree trunk at 54 inches above the ground.
- C. Protect existing site improvements to remain from damage during construction.
1. Restore damaged improvements to their original condition, as acceptable to Owner.

3.2 TEMPORARY EROSION AND SEDIMENTATION CONTROL

- A. Provide temporary erosion- and sedimentation-control measures to prevent soil erosion and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways, according to erosion- and sedimentation-control Drawings and requirements of authorities having jurisdiction.

- B. Verify that flows of water redirected from construction areas or generated by construction activity do not enter or cross protection zones.
- C. Inspect, maintain, and repair erosion- and sedimentation-control measures during construction until permanent vegetation has been established.
- D. Remove erosion and sedimentation controls and restore and stabilize areas disturbed during removal.

3.3 TREE AND PLANT PROTECTION

- A. General: Protect trees and plants remaining on-site according to requirements in Division 010000.
- B. Repair or replace trees, shrubs, and other vegetation indicated to remain or be relocated that are damaged by construction operations, in a manner approved by Landscape Architect.

3.4 EXISTING UTILITIES

- A. Arrange for disconnecting and sealing indicated utilities that serve existing structures before site clearing, when requested by Contractor.
 - 1. Verify that utilities have been disconnected and capped before proceeding with site clearing.
- B. Locate, identify, disconnect, and seal or cap utilities indicated to be removed or abandoned in place.
 - 1. Arrange with utility companies to shut off indicated utilities.
- C. Locate, identify, and disconnect utilities indicated to be abandoned in place.
- D. Interrupting Existing Utilities: Do not interrupt utilities serving facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated:
 - 1. Notify COTR not less than five (5) days in advance of proposed utility interruptions.
 - 2. Do not proceed with utility interruptions without Owner's written permission.
- E. Excavate for and remove underground utilities indicated to be removed.
- F. Removal of underground utilities is included in earthwork sections.

3.5 CLEARING AND GRUBBING

- A. Remove obstructions, trees, shrubs, and other vegetation to permit installation of new construction.
 - 1. Do not remove trees, shrubs, and other vegetation indicated to remain or to be relocated.

2. Grind down stumps and remove roots, obstructions, and debris to a depth of 18 inches below exposed subgrade.
 3. Use only hand methods for grubbing within protection zones.
 4. Chip removed tree branches. Contractor may temporarily stockpile in areas approved by COTR until such time as the removal and proper disposal of the chipped mulch and associated debris is arranged by the Contractor.
- B. Fill depressions caused by clearing and grubbing operations with satisfactory soil material unless further excavation or earthwork is indicated.
1. Place fill material in horizontal layers not exceeding a loose depth of 8 inches, and compact each layer to a density equal to adjacent original ground.

3.6 TOPSOIL STRIPPING

- A. Remove sod and grass before stripping topsoil.
- B. Strip topsoil to depth indicated on Drawings or as noted in Section 329113 "Soil Preparation" in a manner to prevent intermingling with underlying subsoil or other waste materials.
1. Remove subsoil and nonsoil materials from topsoil, including clay lumps, gravel, and other objects more than 2 inches in diameter; trash, debris, weeds, roots, and other waste materials.
- C. Stockpile topsoil away from edge of excavations without intermixing with subsoil. Grade and shape stockpiles to drain surface water. Cover to prevent windblown dust and erosion by water.
1. Limit height of topsoil stockpiles to 72 inches.
 2. Do not stockpile topsoil within tree protection zones.
 3. Stockpile surplus topsoil to allow for respreading.

3.7 SITE IMPROVEMENTS

- A. Remove existing above- and below-grade improvements as indicated and necessary to facilitate new construction.
- B. Remove slabs, paving, curbs, gutters, and aggregate base as indicated.
1. Unless existing full-depth joints coincide with line of demolition, neatly saw-cut along line of existing pavement to remain before removing adjacent existing pavement. Saw-cut faces vertically.
 2. Paint cut ends of steel reinforcement in concrete to remain with two coats of antirust coating, following coating manufacturer's written instructions. Keep paint off surfaces that will remain exposed.

3.8 DISPOSAL OF SURPLUS AND WASTE MATERIALS

- A. Remove surplus soil material, unsuitable topsoil, obstructions, demolished materials, and waste materials including trash and debris, and legally dispose of them off Owner's property in accordance with the Contractor's Waste Management Plan.
- B. Separate recyclable materials produced during site clearing from other nonrecyclable materials. Store or stockpile without intermixing with other materials and transport them to recycling facilities in accordance with the Contractor's Waste Management Plan. Do not interfere with other Project work.

END OF SECTION 31 10 00

SECTION 31 20 00 - EARTH MOVING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 010000 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Preparing subgrades for trails, pavements, turf and grasses, and plants.
 - 2. Subbase course for concrete trails and pavements.
 - 3. Subbase course and base course for asphalt paving.
 - 4. Subsurface drainage backfill for walls and trenches.
 - 5. Excavating and backfilling trenches for utilities and pits for buried utility structures.

- B. Related Sections:

- 1. Section 010000 for temporary controls, utilities, and support facilities.
 - 2. Section 311000 "Site Clearing" for site stripping, grubbing, stripping and stockpiling topsoil, and removal of above- and below-grade improvements and utilities.

1.3 DEFINITIONS

- A. Backfill: Soil material or controlled low-strength material used to fill an excavation.

- 1. Initial Backfill: Backfill placed beside and over pipe in a trench, including haunches to support sides of pipe.
 - 2. Final Backfill: Backfill placed over initial backfill to fill a trench.

- B. Base Course: Aggregate layer placed between the subbase course and hot-mix asphalt paving.

- C. Bedding Course: Aggregate layer placed over the excavated subgrade in a trench before laying pipe.

- D. Borrow Soil: Satisfactory soil imported from off-site for use as fill or backfill.

- E. Drainage Course: Aggregate layer supporting the slab-on-grade that also minimizes upward capillary flow of pore water.

- F. Excavation: Removal of material encountered above subgrade elevations and to lines and dimensions indicated.

1. Authorized Additional Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions as directed by onsite Geotechnical Engineer to address unsuitable conditions. Authorized additional excavation and replacement material will be paid for according to Contract provisions for import, export and the unit prices for said items as agreed to in a formal change order or as approved by the COTR.
2. Bulk Excavation: Excavation more than 10 feet in width and more than 30 feet in length.
3. Unauthorized Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions without direction by COTR in consultation with the Geotechnical Engineer. Unauthorized excavation, as well as remedial work directed by COTR, shall be without additional compensation.

G. Fill: Soil materials used to raise existing grades.

H. Rock: Rock material in beds, ledges, unstratified masses, conglomerate deposits, and boulders of rock material 3/4 cu. yd or more in volume that exceed a standard penetration resistance of 100 blows/2 inches when tested by a geotechnical testing agency, according to ASTM D1586.

I. Structures: Buildings, footings, foundations, retaining walls, slabs, tanks, curbs, mechanical and electrical appurtenances, or other man-made stationary features constructed above or below the ground surface.

J. Subbase Course: Aggregate layer placed between the subgrade and base course for hot-mix asphalt pavement, or aggregate layer placed between the subgrade and a cement concrete pavement or a cement concrete or hot-mix asphalt walk.

K. Subgrade: Uppermost surface of an excavation or the top surface of a fill or backfill immediately below subbase, drainage fill, drainage course, or topsoil materials.

L. Utilities: On-site underground pipes, conduits, ducts, and cables, as well as underground services within buildings.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of the following manufactured products required:

1. Geotextiles.
2. Warning tapes.

B. Samples for Verification: For the following products, in sizes indicated below:

1. Geotextile: 12 by 12 inches.
2. Warning Tape: 12 inches long; of each color.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified testing (geotechnical engineer) agency.

- B. Material Test Reports: For each on-site and borrow soil material proposed for fill and backfill as follows:
 - 1. Classification according to ASTM D2487.
 - 2. Laboratory compaction curve according to ASTM D698.
- C. Preexcavation Photographs or Videotape: Show existing conditions of adjoining construction and site improvements, including finish surfaces that might be misconstrued as damage caused by earth moving operations. Submit before earth moving begins.

1.6 QUALITY ASSURANCE

- A. Geotechnical Testing Agency Qualifications: Qualified according to ASTM E329 and ASTM D3740 for testing indicated.
- B. Preexcavation Conference: Conduct conference at Project site as part of Preconstruction Meeting.

1.7 PROJECT CONDITIONS

- A. Traffic: Minimize interference with adjoining roads, streets, walks, and other adjacent occupied or used facilities during earth moving operations.
 - 1. Do not close or obstruct streets, walks, or other adjacent occupied or used facilities without permission from Owner.
 - 2. Provide alternate routes around closed or obstructed traffic ways in accordance with the Maintenance of Traffic Plans or as coordinated with and agreed to by Owner.
- B. Utility Locator Service: Notify utility locator service and/or "Miss Utility" for area where Project is located before beginning earth moving operations.
- C. Do not commence earth moving operations until temporary erosion- and sedimentation-control measures, specified on drawings are in place.
- D. The following practices are prohibited within protection zones:
 - 1. Storage of construction materials, debris, or excavated material.
 - 2. Parking vehicles or equipment.
 - 3. Foot traffic.
 - 4. Erection of sheds or structures.
 - 5. Impoundment of water or discharging of dewatering waste unless approved by landscape architect or Owner.
 - 6. Excavation or other digging unless otherwise indicated.
 - 7. Attachment of signs to or wrapping materials around trees or plants unless otherwise indicated.
- E. Do not direct vehicle or equipment exhaust towards protection zones.
- F. Prohibit heat sources, flames, ignition sources, and smoking within or near protection zones.

PART 2 - PRODUCTS

2.1 SOIL MATERIALS

- A. General: Provide borrow soil materials when sufficient satisfactory soil materials are not available from onsite excavations.
- B. Satisfactory Soils:
 - 1. For compacted fill to support pavements, soil Classification CL, ML, SC, SM, SP, SW, GC, GM, GP, or GW according to ASTM D2487 or a combination of these groups; free of rock or gravel larger than 3 inches in any dimension, debris, waste, frozen materials, vegetation, and other deleterious matter.
 - a. Maximum dry density greater than 105 pcf.
 - 2. For backfill against the retaining wall, soil classification SC, SM, SP, SW, GC, GM, GP, or GW or a combination of these groups; free of rock or gravel larger than 3 inches in any dimension, debris, waste, frozen materials, vegetation, and other deleterious matter.
 - a. Liquid limit: 40.
 - b. Plasticity index: 15.
- C. Unsatisfactory Soils: Soil Classification OL, CH, MH, OH, and PT according to ASTM D2487, or a combination of these groups.
 - 1. Unsatisfactory soils also include satisfactory soils not maintained within 2 percent of optimum moisture content at time of compaction.
- D. Subbase Material: Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D2940; with at least 90 percent passing a 1-1/2-inch sieve and not more than 12 percent passing a No. 200 sieve.
- E. Base Course: Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D2940; with at least 95 percent passing a 1-1/2-inch sieve and not more than 8 percent passing a No. 200 sieve.
- F. Engineered Fill: Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D2940; with at least 90 percent passing a 1-1/2-inch sieve and not more than 12 percent passing a No. 200 sieve.
- G. Bedding Course: Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D2940; except with 100 percent passing a 1-inch sieve and not more than 8 percent passing a No. 200 sieve.
- H. Drainage Course: Narrowly graded mixture of washed crushed stone, or crushed or uncrushed gravel; ASTM D448; coarse-aggregate grading Size 57; with 100 percent passing a 1-1/2-inch sieve and 0 to 5 percent passing a No. 8 sieve.

- I. Filter Material: Narrowly graded mixture of natural or crushed gravel, or crushed stone and natural sand; ASTM D448; coarse-aggregate grading Size 67; with 100 percent passing a 1-inch sieve and 0 to 5 percent passing a No. 4 sieve.
- J. Sand: ASTM C33; fine aggregate.
- K. Impervious Fill: Clayey gravel and sand mixture capable of compacting to a dense state.

2.2 GEOTEXTILES

- A. Refer to specifications included on Erosion and Sediment Control drawings.

2.3 ACCESSORIES

- A. Detectable Warning Tape: Acid- and alkali-resistant, polyethylene film warning tape manufactured for marking and identifying underground utilities, a minimum of 6 inches wide and 4 mils thick, continuously inscribed with a description of the utility, with metallic core encased in a protective jacket for corrosion protection, detectable by metal detector when tape is buried up to 30 inches deep; colored as follows:
 - 1. Red: Electric.
 - 2. Yellow: Gas, oil, steam, and dangerous materials.
 - 3. Orange: Telephone and other communications.
 - 4. Blue: Water systems.
 - 5. Green: Sewer systems.
 - 6. Purple: Reuse or Reclaimed Water Systems

PART 3 - EXECUTION

3.1 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earth moving operations.
- B. Protect and maintain erosion and sedimentation controls during earth moving operations.
- C. Protect subgrades and foundation soils from freezing temperatures and frost. Remove temporary protection before placing subsequent materials.

3.2 DEWATERING

- A. Prevent surface water and ground water from entering excavations, from ponding on prepared subgrades, and from flooding Project site and surrounding area.

- B. Protect subgrades from softening, undermining, washout, and damage by rain or water accumulation.

- 1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.

3.3 EXPLOSIVES

- A. Explosives: Do not use explosives.

3.4 EXCAVATION, GENERAL

- A. Unclassified Excavation: Excavate to subgrade elevations regardless of the character of surface and subsurface conditions encountered. Unclassified excavated materials may include rock, soil materials, and obstructions. No changes in the Contract Sum or the Contract Time will be authorized for rock excavation or removal of obstructions.

- 1. If excavated materials intended for fill and backfill include unsatisfactory soil materials and rock, replace with satisfactory soil materials.
 - 2. Remove rock to lines and grades indicated to permit installation of permanent construction without exceeding the following dimensions:
 - a. 24 inches outside of concrete forms other than at footings.
 - b. 12 inches outside of concrete forms at footings.
 - c. 6 inches outside of minimum required dimensions of concrete cast against grade.
 - d. 6 inches beneath pipe in trenches, and the greater of 24 inches wider than pipe or 42 inches wide.

- B. Classified Excavation: Excavate to subgrade elevations. Material to be excavated will be classified as earth and rock. Do not excavate rock until it has been classified and cross sectioned by Geotechnical Engineer. The Contract Sum will be adjusted for rock excavation according to unit prices included in the Contract Documents. Changes in the Contract Time may be authorized for rock excavation.

- 1. Earth excavation includes excavating pavements and obstructions visible on surface; underground structures, utilities, and other items indicated to be removed; together with soil, boulders, and other materials not classified as rock or unauthorized excavation.
 - a. Intermittent drilling; blasting, if permitted; ram hammering; or ripping of material not classified as rock excavation is earth excavation.
 - 2. Rock excavation includes removal and disposal of rock. Remove rock to lines and subgrade elevations indicated to permit installation of permanent construction without exceeding the following dimensions:
 - a. 24 inches outside of concrete forms other than at footings.
 - b. 12 inches outside of concrete forms at footings.
 - c. 6 inches outside of minimum required dimensions of concrete cast against grade.

- d. 6 inches beneath pipe in trenches, and the greater of 24 inches wider than pipe or 42 inches wide.

3.5 EXCAVATION FOR WALKS AND PAVEMENTS

- A. Excavate surfaces under walks and pavements to indicated lines, cross sections, elevations, and subgrades.
- B. Excavations at Edges of Tree- and Plant-Protection Zones:
 - 1. Excavate by hand to indicated lines, cross sections, elevations, and subgrades. Use narrow-tine spading forks to comb soil and expose roots. Do not break, tear, or chop exposed roots. Do not use mechanical equipment that rips, tears, or pulls roots.

3.6 EXCAVATION FOR UTILITY TRENCHES

- A. Excavate trenches to indicated gradients, lines, depths, and elevations.
 - 1. Excavate trenches to allow installation of top of pipe below frost line.
- B. Excavate trenches to uniform widths to provide the following clearance on each side of pipe or conduit. Excavate trench walls vertically from trench bottom to 12 inches higher than top of pipe or conduit unless otherwise indicated.
 - 1. Clearance: 12 inches each side of pipe or conduit.
- C. Trench Bottoms: Excavate and shape trench bottoms to provide uniform bearing and support of pipes and conduit. Shape subgrade to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits. Remove projecting stones and sharp objects along trench subgrade.
 - 1. For pipes and conduit less than 6 inches in nominal diameter, hand-excavate trench bottoms and support pipe and conduit on an undisturbed subgrade.
 - 2. For pipes and conduit 6 inches or larger in nominal diameter, shape bottom of trench to support bottom 90 degrees of pipe or conduit circumference. Fill depressions with tamped sand backfill.
 - 3. For flat-bottomed, multiple-duct conduit units, hand-excavate trench bottoms and support conduit on an undisturbed subgrade.
 - 4. Excavate trenches 6 inches deeper than elevation required in rock or other unyielding bearing material to allow for bedding course.
- D. Trench Bottoms: Excavate trenches 4 inches deeper than bottom of pipe and conduit elevations to allow for bedding course. Hand-excavate deeper for bells of pipe.
 - 1. Excavate trenches 6 inches deeper than elevation required in rock or other unyielding bearing material to allow for bedding course.
- E. Trenches in Tree- and Plant-Protection Zones:

1. Hand-excavate to indicated lines, cross sections, elevations, and subgrades. Use narrow-tine spading forks to comb soil and expose roots. Do not break, tear, or chop exposed roots. Do not use mechanical equipment that rips, tears, or pulls roots.
2. Do not cut main lateral roots or taproots; cut only smaller roots that interfere with installation of utilities.

3.7 SUBGRADE INSPECTION

- A. Notify COTR and Geotechnical Engineer when excavations have reached required subgrade.
- B. If Geotechnical Engineer determines that unsatisfactory soil is present, continue excavation and replace with compacted backfill or fill material as directed.
- C. Proof-roll subgrade below pavements with a pneumatic-tired and loaded dump truck weighing not less than 20 tons to identify soft pockets and areas of excess yielding. Do not proof-roll wet or saturated subgrades.
 1. Completely proof-roll subgrade in one direction. Limit vehicle speed to 3 mph.
 2. Excavate soft spots, unsatisfactory soils, and areas of excessive pumping or rutting, as determined by Geotechnical Engineer, and replace with compacted backfill or fill as directed.
- D. Authorized additional excavation and replacement material will be paid for according to Contract provisions for changes in the Work.
- E. Reconstruct subgrades damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, as directed by COTR or Geotechnical Engineer, without additional compensation.

3.8 UNAUTHORIZED EXCAVATION

- A. Fill unauthorized excavation under foundations or wall footings by extending bottom elevation of concrete foundation or footing to excavation bottom, without altering top elevation. Lean concrete fill, with 28-day compressive strength of 2500 psi, may be used when approved by Geotechnical Engineer.
 1. Fill unauthorized excavations under other construction, pipe, or conduit as directed by Geotechnical Engineer.

3.9 STORAGE OF SOIL MATERIALS

- A. Stockpile borrow soil materials and excavated satisfactory soil materials without intermixing. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
 1. Stockpile soil materials away from edge of excavations. Do not store within drip line of remaining trees.

3.10 BACKFILL

- A. Place and compact backfill in excavations promptly, but not before completing the following:
 - 1. Construction below finish grade including, where applicable, subdrainage, dampproofing, waterproofing, and perimeter insulation.
 - 2. Surveying locations of underground utilities for Record Documents.
 - 3. Testing and inspecting underground utilities.
 - 4. Removing concrete formwork.
 - 5. Removing trash and debris.
 - 6. Removing temporary shoring and bracing, and sheeting.
 - 7. Installing permanent or temporary horizontal bracing on horizontally supported walls.
- B. Place backfill on subgrades free of mud, frost, snow, or ice.

3.11 UTILITY TRENCH BACKFILL

- A. Place backfill on subgrades free of mud, frost, snow, or ice.
- B. Place and compact bedding course on trench bottoms and where indicated. Shape bedding course to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits.
- C. Trenches under Roadways: Provide 4-inch- thick, concrete-base slab support for piping or conduit less than 30 inches below surface of roadways. After installing and testing, completely encase piping or conduit in a minimum of 4 inches of concrete before backfilling or placing roadway subbase course.
- D. Backfill voids with satisfactory soil while removing shoring and bracing.
- E. Place and compact initial backfill of subbase material (in roadbeds) or satisfactory soil elsewhere than roadbeds, free of particles larger than 1 inch in any dimension, to a height of 12 inches over the pipe or conduit.
 - 1. Carefully compact initial backfill under pipe haunches and compact evenly up on both sides and along the full length of piping or conduit to avoid damage or displacement of piping or conduit. Coordinate backfilling with utilities testing.
- F. Place and compact final backfill of satisfactory soil to final subgrade elevation.
- G. Install detectable warning tape directly above utilities, 12 inches below finished grade, except 6 inches below subgrade under pavements and slabs.

3.12 SOIL FILL

- A. Plow, scarify, bench, or break up sloped surfaces steeper than 1 vertical to 4 horizontal so fill material will bond with existing material.

B. Place and compact fill material in layers to required elevations as follows:

1. Under grass and planted areas, use satisfactory soil material.
2. Under walks and pavements, use satisfactory soil material.
3. Under footings and foundations, use engineered fill.

C. Place soil fill on subgrades free of mud, frost, snow, or ice.

3.13 SOIL MOISTURE CONTROL

A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill soil layer before compaction to within 2 percent of optimum moisture content.

1. Do not place backfill or fill soil material on surfaces that are muddy, frozen, or contain frost or ice.
2. Remove and replace, or scarify and air dry, otherwise satisfactory soil material that exceeds optimum moisture content by 2 percent and is too wet to compact to specified dry unit weight.

3.14 COMPACTION OF SOIL BACKFILLS AND FILLS

A. Place backfill and fill soil materials in layers not more than 8 inches in loose depth for material compacted by heavy compaction equipment, and not more than 4 inches in loose depth for material compacted by hand-operated tampers.

B. Place backfill and fill soil materials evenly on all sides of structures to required elevations, and uniformly along the full length of each structure.

C. Compact soil materials to not less than the following percentages of maximum dry unit weight according to ASTM D698:

1. Under structures, building slabs, steps, and pavements, scarify and recompact top 12 inches of existing subgrade and each layer of backfill or fill soil material at 95 percent.
2. Under walkways, scarify and recompact top 6 inches below subgrade and compact each layer of backfill or fill soil material at 92 percent.
3. Under turf or unpaved areas, scarify top 6 inches below subgrade, provide final backfill in layers not more than 8 inches in loose depth and compact to 85%.
4. For utility trenches, compact each layer of initial and final backfill soil material at 85 percent.

3.15 GRADING

A. General: Uniformly grade areas to a smooth surface, free of irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.

1. Provide a smooth transition between adjacent existing grades and new grades.

2. Cut out soft spots, fill low spots, and trim high spots to comply with required surface tolerances.
- B. Site Rough Grading: Slope grades to direct water away from buildings and to prevent ponding. Finish subgrades to required elevations within the following tolerances:
 1. Turf or Unpaved Areas: Plus or minus 1 inch.
 2. Walks/Trails: Plus or minus 1 inch.
 3. Pavements: Plus or minus 1/2 inch.

3.16 SUBBASE AND BASE COURSES UNDER PAVEMENTS AND WALKS

- A. Place subbase course and base course on subgrades free of mud, frost, snow, or ice.
- B. On prepared subgrade, place subbase course and base course under pavements and walks as follows:
 1. Install separation geotextile on prepared subgrade according to manufacturer's written instructions, overlapping sides and ends.
 2. Place base course material over subbase course under hot-mix asphalt pavement.
 3. Shape subbase course and base course to required crown elevations and cross-slope grades.
 4. Place subbase course and base course 6 inches or less in compacted thickness in a single layer.
 5. Place subbase course and base course that exceeds 6 inches in compacted thickness in layers of equal thickness, with no compacted layer more than 6 inches thick or less than 3 inches thick.
 6. Compact subbase course and base course at optimum moisture content to required grades, lines, cross sections, and thickness to not less than 95 percent of maximum dry unit weight according to ASTM D698.
- C. Pavement Shoulders: Place shoulders along edges of subbase course and base course to prevent lateral movement. Construct shoulders, at least 24 inches wide, of satisfactory soil materials and compact simultaneously with each subbase and base layer to not less than 95 percent of maximum dry unit weight according to ASTM D698.

3.17 FIELD QUALITY CONTROL

- A. Special Inspections: Contractor will engage a qualified special inspector to perform the following special inspections:
 1. Determine prior to placement of fill that site has been prepared in compliance with requirements.
 2. Determine that fill material and maximum lift thickness comply with requirements.
 3. Determine, at the required frequency, that in-place density of compacted fill complies with requirements.
- B. Testing Agency: Owner will engage a qualified geotechnical engineering testing agency to perform tests and inspections.

- C. Allow testing agency to inspect and test subgrades and each fill or backfill layer. Proceed with subsequent earth moving only after test results for previously completed work comply with requirements.
- D. Footing Subgrade: At footing subgrades, at least one test of each soil stratum will be performed to verify design bearing capacities. Subsequent verification and approval of other footing subgrades may be based on a visual comparison of subgrade with tested subgrade when approved by Architect.
- E. Testing agency will test compaction of soils in place according to ASTM D1556, ASTM D2167, ASTM D2922, and ASTM D2937, as applicable. Tests will be performed at the following locations and frequencies:
 - 1. Paved Areas: At subgrade and at each compacted fill and backfill layer, at least one test for every 2000 sq. ft. or less of paved area, but in no case fewer than three tests.
 - 2. Retaining Wall Area: At subgrade and at each compacted fill and backfill layer, at least one test for every 2000 sq. ft. or less of wall system area, but in no case fewer than three tests.
 - 3. Trench Backfill: At each compacted initial and final backfill layer, at least one test for every 150 feet or less of trench length, but no fewer than two tests.
- F. When testing agency reports that subgrades, fills, or backfills have not achieved degree of compaction specified, scarify and moisten or aerate, or remove and replace soil materials to depth required; recompact and retest until specified compaction is obtained.

3.18 PROTECTION

- A. Protecting Graded Areas: Protect newly graded areas from traffic, freezing, and erosion. Keep free of trash and debris.
- B. Repair and reestablish grades to specified tolerances where completed or partially completed surfaces become eroded, rutted, settled, or where they lose compaction due to subsequent construction operations or weather conditions.
 - 1. Scarify or remove and replace soil material to depth as directed by Geotechnical Engineer; reshape and recompact.
- C. Where settling occurs before Project correction period elapses, remove finished surfacing, backfill with additional soil material, compact, and reconstruct surfacing.
 - 1. Restore appearance, quality, and condition of finished surfacing to match adjacent work, and eliminate evidence of restoration to greatest extent possible.

3.19 DISPOSAL OF SURPLUS AND WASTE MATERIALS

- A. Remove surplus satisfactory soil and waste materials, including unsatisfactory soil, trash, and debris, and legally dispose of them off Owner's property and in accordance with Contractor's Waste Management Plan.

- B. Transport surplus satisfactory soil to designated storage areas on Owner's property. Stockpile or spread soil as directed by COTR.
 - 1. Remove waste materials, including unsatisfactory soil, trash, and debris, and legally dispose of them off Owner's property and in accordance with Contractor's Waste Management Plan.

END OF SECTION 31 20 00

SECTION 32 12 16 - ASPHALT PAVING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 010000 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Cold milling of existing asphalt pavement.
 - 2. Hot-mix asphalt paving.
 - 3. Asphalt surface treatments.

- B. Related Requirements:

- 1. Section 312000 "Earth Moving" for subgrade preparation, fill material, unbound-aggregate subbase and base courses, and aggregate pavement shoulders.

1.3 UNIT PRICES

- A. Work of this Section is affected by the unit price listed in the bid for Hot Mix Asphalt Pavement.

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site as part of the Preconstruction Meeting.

- 1. Review methods and procedures related to hot-mix asphalt paving including, but not limited to, the following:
 - a. Review proposed sources of paving materials, including capabilities and location of plant that will manufacture hot-mix asphalt.
 - b. Review requirements for protecting paving work, including restriction of traffic during installation period and for remainder of construction period.

1.5 ACTION SUBMITTALS

- A. Product Data: For each type of product.

- 1. Include technical data and tested physical and performance properties.

2. Job-Mix Designs: Certification, by authorities having jurisdiction, of approval of each job mix proposed for the Work.
3. Job-Mix Designs: For each job mix proposed for the Work.

1.6 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For manufacturer and testing agency.
- B. Material Certificates: For each paving material.
- C. Material Test Reports: For each paving material, by a qualified testing agency.
- D. Field quality-control reports.

1.7 QUALITY ASSURANCE

- A. Manufacturer Qualifications: A paving-mix manufacturer registered with and approved by authorities having jurisdiction or the DOT of state in which Project is located.
- B. Testing Agency Qualifications: Qualified according to ASTM D3666 for testing indicated.
- C. Regulatory Requirements: Comply with materials, workmanship, and other applicable requirements of DC Department of Transportation (DDOT) for asphalt paving work.
 1. Measurement and payment provisions and safety program submittals included in standard specifications do not apply to this Section.

1.8 FIELD CONDITIONS

- A. Environmental Limitations: Do not apply asphalt materials if subgrade is wet or excessively damp, if rain is imminent or expected before time required for adequate cure, or if the following conditions are not met:
 1. Prime Coat: Minimum surface temperature of 60 deg F.
 2. Tack Coat: Minimum surface temperature of 60 deg F.
 3. Slurry Coat: Comply with weather limitations in ASTM D3910.
 4. Asphalt Base Course: Minimum surface temperature of 40 deg F and rising at time of placement.
 5. Asphalt Surface Course: Minimum surface temperature of 60 deg F at time of placement.

PART 2 - PRODUCTS

2.1 AGGREGATES

- A. General: Use materials and gradations that have performed satisfactorily in previous installations.

- B. Coarse Aggregate: ASTM D692, sound; angular crushed stone, crushed gravel, or cured, crushed blast-furnace slag.
- C. Fine Aggregate: ASTM D1073 or AASHTO M 29, sharp-edged natural sand or sand prepared from stone, gravel, cured blast-furnace slag, or combinations thereof.
 - 1. For hot-mix asphalt, limit natural sand to a maximum of 20 percent by weight of the total aggregate mass.

2.2 ASPHALT MATERIALS

- A. Asphalt Binder: AASHTO M 320, PG 64-22.
- B. Asphalt Cement: ASTM D3381 for viscosity-graded material or ASTM D946 for penetration-graded material.
- C. Cutback Prime Coat: ASTM D2027, medium-curing cutback asphalt, MC-30 or MC-70 MC-250.
- D. Emulsified Asphalt Prime Coat: ASTM D977 or AASHTO M 140 emulsified asphalt, or ASTM D2397 or AASHTO M 208 cationic emulsified asphalt, slow setting, diluted in water, of suitable grade and consistency for application.
- E. Tack Coat: ASTM D977 or AASHTO M 140 emulsified asphalt, or ASTM D2397 or AASHTO M 208 cationic emulsified asphalt, slow setting, diluted in water, of suitable grade and consistency for application.
- F. Fog Seal: ASTM D977 or AASHTO M 140 emulsified asphalt, or ASTM D 2397 or AASHTO M 208 cationic emulsified asphalt, slow setting, factory diluted in water, of suitable grade and consistency for application.
- G. Water: Potable.
- H. Undersealing Asphalt: ASTM D3141; pumping consistency.

2.3 AUXILIARY MATERIALS

- A. Recycled Materials for Hot-Mix Asphalt Mixes: Reclaimed asphalt pavement; reclaimed, unbound-aggregate base material; and recycled asphalt shingles from sources and gradations that have performed satisfactorily in previous installations, equal to performance of required hot-mix asphalt paving produced from all new materials.
- B. Herbicide: Commercial chemical for weed control, registered by the EPA, and not classified as "restricted use" for locations and conditions of application. Provide in granular, liquid, or wettable powder form.
- C. Sand: ASTM D1073 or AASHTO M 29, Grade No. 2 or No. 3.

- D. Joint Sealant: ASTM D6690 or AASHTO M 324, Type I Type II or III Type IV, hot-applied, single-component, polymer-modified bituminous sealant.
- E. Pavement Marking Paint: Latex, waterborne emulsion, lead and chromate free, ready mixed, complying with FS TT-P-1952, Type II, with drying time of less than 45 minutes.
 - 1. Color: as indicated on plans.
- F. Glass Beads: AASHTO M 247, Type 1 made of 100 percent recycled glass.

2.4 MIXES

- A. Recycled Content of Hot-Mix Asphalt: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 10 percent or more than **50** percent by weight.
 - 1. Surface Course Limit: Recycled content no more than 10 percent by weight.
- B. Hot-Mix Asphalt: Dense-graded, hot-laid, hot-mix asphalt plant mixes approved by authorities having jurisdiction; designed according to procedures in AI MS-2, "Mix Design Methods for Asphalt Concrete and Other Hot-Mix Types"; and complying with the following requirements:
 - 1. Provide mixes with a history of satisfactory performance in geographical area where Project is located.
 - 2. Base Course: 4.5 in. 19.0 mm PG64-22
 - 3. Surface Course: 1.5 in. 9.5 mm PG64-22
- C. Emulsified-Asphalt Slurry: ASTM D 3910, Type 1 Type 2 Type 3.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that subgrade is dry and in suitable condition to begin paving.
- B. Proof-roll subgrade below pavements with heavy pneumatic-tired equipment to identify soft pockets and areas of excess yielding. Do not proof-roll wet or saturated subgrades.
 - 1. Completely proof-roll subgrade in one direction. Limit vehicle speed to 3 mph.
 - 2. Proof roll with a loaded 10-wheel, tandem-axle dump truck weighing not less than 20 tons.
 - 3. Excavate soft spots, unsatisfactory soils, and areas of excessive pumping or rutting, as determined by Architect, and replace with compacted backfill or fill as directed.
- C. Proceed with paving only after unsatisfactory conditions have been corrected.

3.2 COLD MILLING

- A. Clean existing pavement surface of loose and deleterious material immediately before cold milling. Remove existing asphalt pavement by cold milling to grades and cross sections indicated.

1. Mill to a depth of 3 inches.
2. Mill to a uniform finished surface free of excessive gouges, grooves, and ridges.
3. Control rate of milling to prevent tearing of existing asphalt course.
4. Repair or replace curbs, manholes, and other construction damaged during cold milling.
5. Excavate and trim unbound-aggregate base course, if encountered, and keep material separate from milled hot-mix asphalt.
6. Patch surface depressions deeper than 1 inch after milling, before wearing course is laid.
7. Handle milled asphalt material according to approved waste management plan required in Section 017419 "Construction Waste Management and Disposal."
8. Keep milled pavement surface free of loose material and dust.
9. Do not allow milled materials to accumulate on-site.

3.3 PATCHING

- A. Asphalt Pavement: Saw cut perimeter of patch and excavate existing pavement section to sound base. Excavate rectangular or trapezoidal patches, extending 12 inches into perimeter of adjacent sound pavement, unless otherwise indicated. Cut excavation faces vertically. Remove excavated material. Recompact existing unbound-aggregate base course to form new subgrade.
- B. Portland Cement Concrete Pavement: Break cracked slabs and roll as required to reseal concrete pieces firmly.
 1. Pump hot undersealing asphalt under rocking slab until slab is stabilized or, if necessary, crack slab into pieces and roll to reseal pieces firmly.
 2. Remove disintegrated or badly cracked pavement. Excavate rectangular or trapezoidal patches, extending into perimeter of adjacent sound pavement, unless otherwise indicated. Cut excavation faces vertically. Recompact existing unbound-aggregate base course to form new subgrade.
- C. Tack Coat: Before placing patch material, apply tack coat uniformly to vertical asphalt surfaces abutting the patch. Apply at a rate of 0.05 to 0.15 gal./sq. yd.
 1. Allow tack coat to cure undisturbed before applying hot-mix asphalt paving.
 2. Avoid smearing or staining adjoining surfaces, appurtenances, and surroundings. Remove spillages and clean affected surfaces.
- D. Placing Patch Material: Fill excavated pavement areas with hot-mix asphalt base mix for full thickness of patch and, while still hot, compact flush with adjacent surface.
- E. Placing Patch Material: Partially fill excavated pavements with hot-mix asphalt base mix and, while still hot, compact. Cover asphalt base course with compacted, hot-mix surface layer finished flush with adjacent surfaces.

3.4 REPAIRS

- A. Leveling Course: Install and compact leveling course consisting of hot-mix asphalt surface course to level sags and fill depressions deeper than 1 inch in existing pavements.

1. Install leveling wedges in compacted lifts not exceeding 3 inches thick.
- B. Crack and Joint Filling: Remove existing joint filler material from cracks or joints to a depth of 1/4 inch.
 1. Clean cracks and joints in existing hot-mix asphalt pavement.
 2. Use emulsified-asphalt slurry to seal cracks and joints less than 1/4 inch wide. Fill flush with surface of existing pavement and remove excess.
 3. Use hot-applied joint sealant to seal cracks and joints more than 1/4 inch wide. Fill flush with surface of existing pavement and remove excess.

3.5 SURFACE PREPARATION

- A. General: Immediately before placing asphalt materials, remove loose and deleterious material from substrate surfaces. Ensure that prepared subgrade is ready to receive paving.
- B. Herbicide Treatment: Apply herbicide according to manufacturer's recommended rates and written application instructions. Apply to dry, prepared subgrade or surface of compacted-aggregate base before applying paving materials.
 1. Mix herbicide with prime coat if formulated by manufacturer for that purpose.
- C. Cutback Prime Coat: Apply uniformly over surface of compacted unbound-aggregate base course at a rate of 0.15 to 0.50 gal./sq. yd. Apply enough material to penetrate and seal, but not flood, surface. Allow prime coat to cure.
 1. If prime coat is not entirely absorbed within 24 hours after application, spread sand over surface to blot excess asphalt. Use enough sand to prevent pickup under traffic. Remove loose sand by sweeping before pavement is placed and after volatiles have evaporated.
 2. Protect primed substrate from damage until ready to receive paving.
- D. Emulsified Asphalt Prime Coat: Apply uniformly over surface of compacted unbound-aggregate base course at a rate of 0.10 to 0.30 gal./sq. yd. per inch depth. Apply enough material to penetrate and seal, but not flood, surface. Allow prime coat to cure.
 1. If prime coat is not entirely absorbed within 24 hours after application, spread sand over surface to blot excess asphalt. Use enough sand to prevent pickup under traffic. Remove loose sand by sweeping before pavement is placed and after volatiles have evaporated.
 2. Protect primed substrate from damage until ready to receive paving.
- E. Tack Coat: Apply uniformly to surfaces of existing pavement at a rate of 0.05 to 0.15 gal./sq. yd.
 1. Allow tack coat to cure undisturbed before applying hot-mix asphalt paving.
 2. Avoid smearing or staining adjoining surfaces, appurtenances, and surroundings. Remove spillages and clean affected surfaces.

3.6 PLACING HOT-MIX ASPHALT

- A. Machine place hot-mix asphalt on prepared surface, spread uniformly, and strike off. Place asphalt mix by hand in areas inaccessible to equipment in a manner that prevents segregation of mix. Place each course to required grade, cross section, and thickness when compacted.
 - 1. Place hot-mix asphalt base course in number of lifts and thicknesses indicated.
 - 2. Place hot-mix asphalt surface course in single lift.
 - 3. Spread mix at a minimum temperature of 250 deg F.
 - 4. Begin applying mix along centerline of crown for crowned sections and on high side of one-way slopes unless otherwise indicated.
 - 5. Regulate paver machine speed to obtain smooth, continuous surface free of pulls and tears in asphalt-paving mat.
- B. Place paving in consecutive strips not less than 10 feet wide unless infill edge strips of a lesser width are required.
 - 1. After first strip has been placed and rolled, place succeeding strips and extend rolling to overlap previous strips. Overlap mix placement about 1 to 1-1/2 inches from strip to strip to ensure proper compaction of mix along longitudinal joints.
 - 2. Complete a section of asphalt base course before placing asphalt surface course.
- C. Promptly correct surface irregularities in paving course behind paver. Use suitable hand tools to remove excess material forming high spots. Fill depressions with hot-mix asphalt to prevent segregation of mix; use suitable hand tools to smooth surface.

3.7 JOINTS

- A. Construct joints to ensure a continuous bond between adjoining paving sections. Construct joints free of depressions, with same texture and smoothness as other sections of hot-mix asphalt course.
 - 1. Clean contact surfaces and apply tack coat to joints.
 - 2. Offset longitudinal joints, in successive courses, a minimum of 6 inches.
 - 3. Offset transverse joints, in successive courses, a minimum of 24 inches.
 - 4. Construct transverse joints at each point where paver ends a day's work and resumes work at a subsequent time. Construct these joints using either "bulkhead" or "papered" method according to AI MS-22, for both "Ending a Lane" and "Resumption of Paving Operations."
 - 5. Compact joints as soon as hot-mix asphalt will bear roller weight without excessive displacement.
 - 6. Compact asphalt at joints to a density within 2 percent of specified course density.

3.8 COMPACTION

- A. General: Begin compaction as soon as placed hot-mix paving will bear roller weight without excessive displacement. Compact hot-mix paving with hot, hand tampers or with vibratory-plate compactors in areas inaccessible to rollers.
 - 1. Complete compaction before mix temperature cools to 185 deg F.

- B. Breakdown Rolling: Complete breakdown or initial rolling immediately after rolling joints and outside edge. Examine surface immediately after breakdown rolling for indicated crown, grade, and smoothness. Correct laydown and rolling operations to comply with requirements.
- C. Intermediate Rolling: Begin intermediate rolling immediately after breakdown rolling while hot-mix asphalt is still hot enough to achieve specified density. Continue rolling until hot-mix asphalt course has been uniformly compacted to the following density:
 - 1. Average Density: 96 percent of reference laboratory density according to ASTM D6927 or AASHTO T 245, but not less than 94 percent or greater than 100 percent.
 - 2. Average Density: 92 percent of reference maximum theoretical density according to ASTM D 2041, but not less than 90 percent or greater than 96 percent.
- D. Finish Rolling: Finish roll paved surfaces to remove roller marks while hot-mix asphalt is still warm.
- E. Edge Shaping: While surface is being compacted and finished, trim edges of pavement to proper alignment. Bevel edges while asphalt is still hot; compact thoroughly.
- F. Repairs: Remove paved areas that are defective or contaminated with foreign materials and replace with fresh, hot-mix asphalt. Compact by rolling to specified density and surface smoothness.
- G. Protection: After final rolling, do not permit vehicular traffic on pavement until it has cooled and hardened.
- H. Erect barricades to protect paving from traffic until mixture has cooled enough not to become marked.

3.9 INSTALLATION TOLERANCES

- A. Pavement Thickness: Compact each course to produce the thickness indicated within the following tolerances:
 - 1. Base Course: Plus or minus 1/2 inch.
 - 2. Surface Course: Plus 1/4 inch, no minus.
- B. Pavement Surface Smoothness: Compact each course to produce a surface smoothness within the following tolerances as determined by using a 10-foot straightedge applied transversely or longitudinally to paved areas:
 - 1. Base Course: 1/4 inch.
 - 2. Surface Course: 1/8 inch.

3.10 SURFACE TREATMENTS

- A. Fog Seals: Apply fog seal at a rate of 0.10 to 0.15 gal./sq. yd. to existing asphalt pavement and allow to cure. With fine sand, lightly dust areas receiving excess fog seal.

- B. Slurry Seals: Apply slurry coat in a uniform thickness according to ASTM D 3910 and allow to cure.

- 1. Roll slurry seal to remove ridges and provide a uniform, smooth surface.

3.11 PAVEMENT MARKING

- A. Do not apply pavement marking paint until layout, colors, and placement have been verified with Engineer.
- B. Allow paving to age for 30 days before starting pavement marking.
- C. Sweep and clean surface to eliminate loose material and dust.
- D. Apply paint with mechanical equipment to produce pavement markings, of dimensions indicated, with uniform, straight edges. Apply at manufacturer's recommended rates to provide a minimum wet film thickness of 15 mils.
 - 1. Broadcast glass beads uniformly into wet pavement markings at a rate of 6 lb/gal.

3.12 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.
- B. Thickness: In-place compacted thickness of hot-mix asphalt courses will be determined according to ASTM D3549.
- C. Surface Smoothness: Finished surface of each hot-mix asphalt course will be tested for compliance with smoothness tolerances.
- D. Asphalt Traffic-Calming Devices: Finished height of traffic-calming devices above pavement will be measured for compliance with tolerances.
- E. In-Place Density: Testing agency will take samples of uncompacted paving mixtures and compacted pavement according to ASTM D979 or AASHTO T 168.
 - 1. Reference maximum theoretical density will be determined by averaging results from four samples of hot-mix asphalt-paving mixture delivered daily to site, prepared according to ASTM D2041, and compacted according to job-mix specifications.
 - 2. In-place density of compacted pavement will be determined by testing core samples according to ASTM D 1188 or ASTM D2726.
 - a. One core sample will be taken for every 1000 sq. yd. or less of installed pavement, with no fewer than three cores taken.
 - b. Field density of in-place compacted pavement may also be determined by nuclear method according to ASTM D2950 and correlated with ASTM D1188 or ASTM D2726.
- F. Replace and compact hot-mix asphalt where core tests were taken.

- G. Remove and replace or install additional hot-mix asphalt where test results or measurements indicate that it does not comply with specified requirements.

3.13 WASTE HANDLING

- A. General: Handle asphalt-paving waste according to approved waste management plan required in Section 017419 "Construction Waste Management and Disposal."

END OF SECTION 32 12 16

SECTION 32 91 00 - PLANTING SOILS

PART 1 - GENERAL

1.1 SUMMARY

- A. The scope of work includes all labor, materials, tools, supplies, equipment, facilities, transportation and services necessary for, and incidental to performing all operations in connection with furnishing, delivery, and installation of Planting Soils and /or the modification of existing site soil for use as Planting Soils, complete as shown on the drawings and as specified herein.
- B. The scope of work in this section includes, but is not limited to, the following:
 - 1. Locate, purchase, deliver and install the components that make up the Planting Soils:
 - a. Base Soil
 - b. Coarse Sand
 - c. Compost
 - d. Lightweight Aggregate
 - e. No. 8 Pea Gravel
 - f. No. 57 Stone
 - 2. Prepare, deliver and install the Soil Mixes:
 - a. Planting Soil
 - b. Lawn Soil
 - c. High-Use Lawn Soil
 - d. Lightweight Extensive Soil
 - 3. Testing and analysis for specification conformance, prior to placement of soils.
 - 4. Finish Grading of Planting Soil area surfaces.
 - 5. Clean up and disposal of all excess and surplus material.

1.2 CONTRACT DOCUMENTS

- A. Shall consist of specifications, general conditions, and the drawings. The intent of these documents is to include all labor, materials, and services necessary for the proper execution of the work. The documents are to be considered as one. Whatever is called for by any parts shall be as binding as if called for in all parts.

1.3 RELATED DOCUMENTS AND REFERENCES

- A. Related Documents:

1. Drawings and general provisions of contract, including general and supplementary conditions and Division I specifications, apply to work of this section.
 2. Related Specification Section
 - a. Section 01 5639 – Site Protection
 - b. Section 31 1000 – Site Clearing
 - c. Section 31 2000 – Earth Moving
 - d. Section 32 9200 – Lawns and Grasses
 - e. Section 32 9300 – Exterior Plants
 - f. Section 33 4100 – Storm Utility Drainage Piping
- B. References: The following specifications and standards of the organizations and documents listed in this paragraph form a part of the Specification to the extent required by the references thereto. In the event that the requirements of any of the following referenced standards and specifications conflict with each other the more stringent requirement shall prevail.
1. ASTM: American Society of Testing Materials cited section numbers.
 2. U.S. Department of Agriculture, Natural Resources Conservation Service, 2003. National Soil Survey Handbook, title 430-VI. Available Online.
 3. US Composting Council www.compostingcouncil.org and the Digital Resource Center <https://www.compostingcouncil.org/page/DigitalResourceCenter>, including “Compost and its Benefits”.
 4. Methods of Soil Analysis, as published by the Soil Science Society of America (<http://www.soils.org/>).

1.4 PERMITS AND REGULATIONS

- A. The Contractor shall obtain and pay for all permits related to this section of the work unless previously excluded under provision of the contract or general conditions. The Contractor shall comply with all laws and ordinances bearing on the operation or conduct of the work as drawn and specified. If the Contractor observes that a conflict exists between permit requirements and the work outlined in the contract documents, the Contractor shall promptly notify the SMITHSONIAN Contracting Officer’s Technical Representative (COTR) in writing including a description of any necessary changes and changes to the contract price resulting from changes in the work.
- B. Wherever references are made to standards or codes in accordance with which work is to be performed or tested, the edition or revision of the standards and codes current on the effective date of this contract shall apply, unless otherwise expressly set forth.
- C. In case of conflict among any referenced standards or codes or among any referenced standards and codes and the specifications, the more restrictive standard shall apply, or the SMITHSONIAN COTR shall determine which shall govern.

1.5 PROTECTION OF WORK, PROPERTY AND PERSON

- A. The Contractor shall adequately protect the work, adjacent property, and the public, and shall be responsible for any damages or injury due to the Contractor's actions. Reference Specification **Section 01 5639 – Site Protection** for the specific requirements governing this work.

1.6 CHANGES IN WORK

- A. All changes in the work, notifications and contractor's request for information (RFI) shall conform to the contract general condition requirements.
- B. CORRECTION OF WORK
- C. The Contractor shall re-execute any work that fails to conform to the requirements of the contract and shall remedy defects due to faulty materials or workmanship upon written notice from the SMITHSONIAN COTR, at the soonest possible time that can be coordinated with other work and seasonal weather demands but not more than 180 (one hundred and eighty) days after notification.

1.7 DEFINITIONS

- A. Acceptance, Acceptable, or Accepted: Formal approval by the SMITHSONIAN COTR in writing.
- B. Aesthetic Acceptance of Grades: Formal approval by the SMITHSONIAN COTR in writing of the aesthetic correctness of the contours. Aesthetic acceptance does not address whether an area drains properly, whether the areas are at the correct elevations, or whether it has been compacted properly.
- C. Base Soil: naturally produced and harvested soil from the A, B and C horizons and that the soil as further defined in this specification.
- D. Compacted soil: soil where the density of soil is greater than the maximum allowable resistance to penetrometer (measured in psi) as defined later in this specification.
- E. Compaction: The process by which a force is applied to the soil to achieve a desired soil density as defined in this specification.
- F. Compost: well decomposed stable organic material as defined by the US Composting Council and further defined in this specification.
- G. Drainage: The rate at which soil water moves through the soil transitioning the soil from saturated condition to field capacity. Most often expressed as saturated hydraulic conductivity (Ksat; units are inches per hour).
- H. End of Warranty Acceptance: The date when the SMITHSONIAN COTR accepts that the plants and work in this section meet all the requirements of the warranty. It is intended that the materials and workmanship warranty for Planting, Planting Soil, and Irrigation (if applicable) work run concurrent with each other, and further defined in this specification.

- I. Final Acceptance: The date at the end of Planting Soils installation where the SMITHSONIAN COTR accepts that all work in these sections is complete and the Warranty period has begun. This date may be different than the date of substantial completion for the other sections of the project, and further defined in this specification.
- J. Fine grading: The final grading of the soil to achieve exact contours and positive drainage, often accomplished by hand rakes or drag rakes other suitable devices, and further defined in this specification, and further defined in this specification.
- K. Finished grade: surface or elevation of Planting Soils after final grading and 12 months of settlement of the soil, and further defined in this specification.
- L. Minor disturbance: Minor grading as part of agricultural work that only adjusts the A horizon soil.
- M. Ped: a clump or clod of soil held together by a combination of clay, organic matter, and fungal hyphae, retaining the original structure of the harvested soil.
- N. Scarify: Loosening and roughening the surface of soil and sub soil prior to adding additional soil on top, and further defined in this specification.
- O. Smithsonian Contracting Officer's Technical Representative (COTR): The person or entity, appointed by the Owner to represent their interest in the review and approval of the work and to serve as the contracting authority with the Contractor. The Smithsonian COTR may appoint other persons to review and approve any aspects of the work.
- P. Soil Horizons: distinct layers of the soil profile distinguished by differences in such features as color, texture, organic matter content, and other characteristics as defined in the USDA National Soil Survey Handbook
- Q. http://www.nrcs.usda.gov/wps/portal/nrcs/detail/soils/scientists/?cid=nrcs142p2_054242.
- R. Soil Mix: Specialized soil types produced as a result of the combination of Base Soil with soil components (such as Coarse Sand, Compost, or **Lightweight Aggregate**) as defined in **Part 2 – Section 2.4.A**
- S. Soil Tilling: Loosening the surface of the soil to the depths specified with acceptable mechanical equipment as further defined in this specification.
- T. Soil Organic Matter: natural occurring organic matter that is a stable part of the soil matrix. Compost is not considered soil organic matter.
- U. Subgrade: surface or elevation of subsoil remaining after completing excavation, or top surface of a fill or backfill, or the top elevation of required drainage layers above structure before placing planting soil.
- V. Weeds: Any plant that is not on the planting plan.

1.8 LONG LEAD ITEM

- A. The Contractor shall be advised that the sourcing, testing, procurement and installation of Planting Soil is a CRITICAL PATH item, requiring timely attention to meet the requirements of the Documents. Having been informed that all aspects of planting soil, the Contractor shall consider this to be A LONG LEAD TIME ITEM. Contractor's failure to heed this notice shall not be a reason for substitution of unacceptable material(s). The SMITHSONIAN COTR will not accept materials that do not meet requirements.

1.9 SUBMITTALS

- A. See the contract General Conditions for policy and procedures related to submittals.
- B. Planting Soils Installation Plan: Submit, a minimum of twelve (12) weeks prior to the anticipated date of the start of soil installation, a Planting Soils Installation Plan that includes a written narrative of soil procurement, testing, mixing, delivery, storage and handling; mock up preparation, locations and installation process.
- C. The plan shall include the projected timeline for soil work and describe all equipment to be used to mix, deliver, spread, grade and compact the soil. Particular attention should be paid to contingency plans and/or schedule modification options due to the impact of weather on soil moisture, delivery, storage, and handling.
- D. Base Soil Source:
 - 1. Submit source information of Base Soil including:
 - a. Soil supplier name
 - b. Name of contact
 - c. Mailing address
 - d. Physical address of soil harvesting site or soil stock pile
 - e. Phone number
- E. Submit all product submittals twelve (12) weeks prior to the start of the soil work.
- F. Soil Mix components' Product Data and Samples: For each type of manufactured product, submit data and certificates that the product meets the specification requirements, signed by the product manufacturer, and complying with the following:
 - 1. Submit manufacturers or supplier's product data and literature, and certified analysis for standard products and bulk materials, complying with testing requirements and referenced standards and specific requested testing.
 - 2. Submit soil analytical test results as described in **Section 1.10.F.2** below.
- G. Soil Testing for Base Soil, Planting Soil, Lawn Soil, High-Use Lawn Soil, and Lightweight Extensive Soil.

1. Submit soil test analysis report for each sample of Base Soil and the Soil Mixes from an approved soil-testing laboratory and where indicated in Part 2 of the specification as follows:
 - a. Submit Base Soil for testing at least twelve (12) weeks before scheduled installation of the Soil Mix.
 - b. Submit Soil Mix tests no more than two (2) weeks after the approval of the Base Soil, Compost, Coarse Sand, and Lightweight Aggregate. Do not send any Soil Mix to the testing laboratory for testing until the Base Soil, Compost, Coarse Sand, and Lightweight Aggregate have been approved.
 - c. If tests fail to meet the specifications, obtain other sources of material, retest and resubmit until accepted by the SMITHSONIAN COTR.
 - d. No soil components shall be used until certified test reports by an approved Testing Agency have been approved by the SMITHSONIAN COTR.
 - e. No Soil Mix shall be used until certified test reports by an approved testing laboratory have been received and approved by the SMITHSONIAN COTR.
 - f. If, at any time during the project, the Base Soil, soil components or Soil Mixes requires adjustment to meet the specifications and/or performance criteria, the Contractor shall submit the adjusted soil components and/or mixes for testing as specified herein.
 - g. All soil testing will be at the expense of the Contractor.
 - h. Failure to complete tests as specified will result in rejection of test results
2. For each type of test required, select and use the same testing laboratory throughout the work from the provided list below. Notify the SMITHSONIAN COTR of any change in laboratory and do not proceed until the SMITHSONIAN COTR has provided pre-approval of the change.
 - a. Laboratory shall be an independent, certified laboratory with experience and capability to conduct the testing indicated and that specializes in the types of tests to be performed. Tests shall be made in strict compliance with the standards of the Association of Official Analytical Chemists and ASTM.
 - b. The following table lists testing parameters, approved soil testing facilities, and specific laboratory testing packages that will provide required data. Testing agencies not listed below shall be approved by the SMITHSONIAN COTR.

Soil Test Requirements, Parameters, Approved Laboratories and Analytical Packages			
Soil Component	Test Parameter	Approved Labs	Laboratory Analytical Suite
Soil Compaction Curve	ASTM D698 - 12e2 Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort	Terracon 4545 42nd St., NW, Suite 307 Washington, DC 20016 Phone: (202) 375 7900	
		Or approved equal	

Soil Nutrient and Chemical Tests (Select 1 lab for testing for the duration of the project)	Soil Organic Matter Soil pH (active pH) Cation Exchange Capacity Electrical Conductivity (EC) Total Nitrogen (Kjeldahl N) Nitrate-N Available Phosphorus Potassium Magnesium Calcium	Ward Laboratories 4007 Cherry Ave, Kearney, NE 68847 Phone: (800) 887-7645	S-4 + Ammonium-N, Total N, Exchangeable Aluminum, Wet Aggregate Stability
		Waypoint Analytical Laboratories 7621 Whitepine Rd	S3M + Nitrate- Nitrogen, Ammonium-
	Sodium Manganese Zinc Lead Boron Ammonium-N	Richmond, VA 23237 Phone: 804.743.9401	Nitrogen, Electrical Conductivity (EC), and Exch. Aluminum (Waypoint does not offer Total-N)
		Or approved equivalent	
Compost Testing (Select 1 lab for testing for the duration of the project)	pH Boron Calcium Copper Iron Magnesium Manganese Total nitrogen Ammonium-nitrogen Nitrate-nitrogen Organic-nitrogen C:N Ratio Phosphorus Potassium Sulfur Zinc Soluble salts Moisture Dry matter Wet Density	Woods End Laboratory 290 Belgrade Road Mt. Vernon ME 04352 Phone: 207-293-2457	Premium Analysis
		Ward Laboratories 4007 Cherry Ave, Kearney, NE 68847 Phone: (800) 887-7645	Manure & Compost Complete
		Or approved equivalent	
Soil Biological Tests (Both labs required)	PLFA (phospholipid-fatty acid) test Mycorrhizal spore counts Mycorrhizal root colonization	Ward Laboratories 4007 Cherry Ave, Kearney, NE 68847	PLFA Wet Aggregate Stability
		MIDI Labs 125 Sandy Drive Newark, DE 19713 Mycoroots 1970 NW Lance Way, Corvallis, OR 97330	 Mycorrhizal spore count Mycorrhizal root colonization

H. Soil Samples for Base Soil, Planting Soil, Lawn Soil, High-Use Lawn Soil, and Lightweight Extensive Soil:

1. Submit samples of each product, concurrently with the submission of the laboratory test reports to the SMITHSONIAN COTR for approval. Samples shall be submitted in two-

gallon bags and labeled to indicate product, characteristics, and locations in the work. Samples arriving without labels will be rejected. Samples will be reviewed for consistency with specification requirements as can be reasonably ascertained and with certified data.

2. Submit samples a minimum of twelve (12) weeks prior to the anticipated date of the start of soil installation.
3. Samples of Base Soil and the Soil Mixes shall be submitted at the same time as the required soil testing analysis of that material, per **Section 1.10.F.2** above. Analysis data and Samples that don't arrive together, thereby verifying the sample matches the data, will be rejected.

I. Subgrade Infiltration and Density testing reports:

1. See requirements for testing in **Section 1.15B** below.

1.10 SOIL INSTALLATION MOCKUP

- A. Prior to installation of Planting Soils, construct at the site a mockup of each Soil Mix required and configuration using the means and methods and equipment proposed by the Contractor to complete the work. Installation of the mockup shall be in the presence of the SMITHSONIAN COTR. The purpose of the mockup is to test the methods of installation, finish grading and compaction of the soil and to serve as a benchmark for completed soil compaction. The mockup shall be as follows:

1. Mockup of the following Soil Mix to be provided:
 - a. Planting Soil
 - b. Lawn Soil
 - c. High-Use Lawn Soil
 - d. Lightweight Extensive Soil
2. The mockup area may remain as part of the installed work at the end of the project if protected from further compaction, contamination, or other disturbance.
 - a. Locate mock-up on site in a proposed planting area easily referenced by workers performing soil installation and finish grading operations.
3. Where soil mockups are intended to remain in place, complete subgrade soil infiltration and density testing per the requirements outlined in **Section 1.15B** below, before construction of the soil mockup.
4. Following acceptance of the soil submittals, in areas that can be protected from disturbance and further compaction, install mockups of each Soil Mix, configuration and soil modification, 3-meter X 6 meter (10-foot X 20-foot) X the full depth of the deepest installation, using the requirements of these specifications. Compaction methods, including the type of compaction equipment and number of passes required to achieve the required compaction shall be evaluated and results measured.
5. Compaction in the mockup soil shall be examined and tested as described below:
 - a. Soil density and moisture shall be tested using a nuclear soil density gauge (nuclear densometer) if required by the SMITHSONIAN COTR. Soil density

- (measured as the percent derived from the dry density divided by the maximum dry density) shall be tested to confirm that soil installation methods will achieve specified soil conditions (density and moisture). Results of nuclear soil density testing shall be used to compare to other soil density and compaction testing methods that are specified in this section.
- b. Cone penetrometer (see **Part 1.14.G** below). A cone penetrometer will be used to test for compacted layers or changes in soil density within an installed soil profile. A minimum of four readings from each Soil Mix shall be taken at the specified depths of the soil profile. Readings from the cone penetrometer will be compared to results of the nuclear soil density gauge to provide a relative basis of soil compaction measurements.
 - c. Cone penetrometer must only be used in dry to moist soils (described as the Acceptable Condition in **Section 1.16.A**). The Penetrometer cannot be used in excessively dry or very moist or wet soils. Refer to the section noted above for requirements concerning soil moisture levels.
 - d. In the event that the nuclear soil density gauge or the cone penetrometer readings exceed the specified densities, reconstruct the mockup, adjusting the soil compaction method achieve the desired results.
 - e. Conduct visual observations of soil structure. Record changes in soil particle configuration, such as platy (soil forms plate-like structure), massive (soil appears to have no form, with no observable soil pores), or granular (soil has distinct peds, or soil particles that are friable and break easily).
 - f. Where the modification requires ripping, tilling or fracturing soils that are over compacted, start the procedure in a new location so that the process is working on soil that is similar to the density of the expected soil.
6. Contractor shall submit a report of the final methods of soil installation to the SMITHSONIAN COTR. This shall include all penetrometer, nuclear densometer, and soil moisture readings that were catalogued during the mockup process.
 7. Provide a protective 1.2-meter-high (4-foot) fence around each mockup to keep all work and equipment from entering the surface of the mockup area.

1.11 OBSERVATION OF THE WORK

- A. The SMITHSONIAN COTR may observe the work at any time. They may remove samples of materials for conformity to specifications. Rejected materials shall be immediately removed from the site and replaced at the Contractor's expense. The cost of testing materials not meeting specifications shall be paid by the Contractor.
 1. Should the SMITHSONIAN COTR determine the need for nuclear densometer testing, he/she may request use of the Contractor's nuclear densometer (by a qualified professional) or cone penetrometer at any time to check soil compaction and moisture.
- B. The SMITHSONIAN COTR shall be informed of the progress of the work, so the work may be observed at the following key times in the construction process. The SMITHSONIAN COTR shall be afforded sufficient time to schedule visit to the site. Each required review may require multiple visits to reflect the phasing of work. Failure of the SMITHSONIAN COTR to make field observations shall not relieve the Contractor from meeting all the requirements of this specification.

1. SOIL MOCKUP REVIEW: At the time of construction of all soil mockups.
2. SUBGRADE REVIEW: Observe each area of soil installation including planter drainage and waterproofing work prior to the installation of any the Planting Soils.
3. COMPLETION of PLANTING SOILS INSTALLATION: Upon completion of all soil modification and installation of the Planting Soils.
4. COMPLETION OF FINE GRADING AND SURFACE SOIL MODIFICATIONS REVIEW: Upon completion of all surface soil modifications and fine grading but prior to the installation of shrubs, ground covers, or lawns.

1.12 PRE-CONSTRUCTION MEETING

- A. Schedule a pre-construction meeting with the SMITHSONIAN COTR at least seven (7) days before beginning work to review any questions the Contractor may have regarding the work, administrative procedures during construction and project work schedule.

1.13 QUALITY ASSURANCE

- A. Installer Qualifications: The installer shall be a firm having at least ten(10) years of experience of a scope similar to that required for the work, including the preparation, mixing and installation of soil mixes to support planting. The installer shall be experienced with the installation of plants and soil in on-grade and over-structure conditions.
 1. The bidders list for work under this section shall be approved by the SMITHSONIAN COTR.
 2. Installer Field Supervision: When any planting soil work is in progress, installer shall maintain, on site, an experienced full-time supervisor who can communicate in English with the SMITHSONIAN COTR.
 3. Installer's Field Supervisor shall have a minimum of five (5) years' experience as a field supervisor installing soil in on-grade and over-structure conditions, shall be trained and proficient in the use of field surveying equipment to establish grades and can communicate in English with the SMITHSONIAN COTR.
 4. The installer's crew shall be experienced in the installation of planting soil, plantings, and interpretation of planting plans, and soil installation plans.
 5. Submit references of past projects and employee training certifications that support that the Contractors meet all of the above installer qualifications and applicable licensures.
- B. Soil Supplier Qualifications: The soil supplier shall be a firm having at least five (5) years of experience of a scope similar to that required for the work, including the preparation, mixing and delivery of soil to support planting, including experience with the installation of plants and soil in Washington, D.C. ***It is the intent of this specification to use unscreened naturally formed loam soils sourced from the Piedmont region in Virginia and local parts of Maryland (generally west of I-95). These soils typically have characteristics (soil clay content, structural stability, and native organic matter content) that favor stable aggregate structure formation, drainage, and compaction resistance.*** The soil supplier shall be located in the area where these soils are found.

- C. Installer and Soil Supplier Bidders List: The bidders list for work under this section shall be approved by the SMITHSONIAN COTR prior to the start of the bidding process. Submit bidders list for approval.
1. Pre-approved soil supplier:
 - a. JK Enterprise Landscape Supply 15900 Lee Highway
Culpepper, VA 20121
P: 703-926-1967
 - b. Luck Ecosystems
PO Box 29682 Richmond, VA 23242
P: 877-904-5825
 - c. Or approved equal
 2. Pre-approved compost supplier:
 - a. Leafgro
Multiple bulk distributors available
 - b. WeCare Denali 7800 Kabik Ct.
Woodbine, MD 21797
P: 410-795-7666
- D. Soil testing laboratory qualifications: an independent, certified and accredited testing laboratory, with the experience and capability to conduct the testing indicated and that specializes in USDA agricultural soil testing, planting soil, and the types of tests to be performed. Geotechnical engineering testing labs shall not be used for chemical analyses.
- E. Base Soil source inspection: The SMITHSONIAN COTR may inspect the Base Soil source stockpiles as part of the approval process. The Soil Supplier shall accompany the SMITHSONIAN COTR.
1. If multiple sites or options are to be considered, arrange the inspection trip such that all optional locations can be inspected in one day. The SMITHSONIAN COTR may remove samples of the material as a record of the inspection and independent testing as needed.
- F. All delivered and installed Planting Soils shall conform to the approved submittals sample color, ped sizes and distribution, texture and approved test analysis.
1. The SMITHSONIAN COTR may request samples of the delivered or installed soil be tested for analysis to confirm the Soil Mix conforms to the approved material.
 2. All testing shall be performed by the same soil lab that performed the original Soil Mix testing.
 3. Testing results shall be within 10% plus or minus of the values measured in the approved Soil Mix.
 4. Any Soil Mix that fails to meet the above criteria, if requested by the SMITHSONIAN COTR, shall be removed and new soil installed.

G. Soil compaction testing: following installation or modification of soil, test soil compaction as follows:

1. Soil density shall be measured using a nuclear densometer (Troxler nuclear soil density gauge or equivalent) if required by the SMITHSONIAN COTR. Soil density shall be reported as a measured percentage derived from the dry density divided by the maximum dry density (See **Section 1.10.F.2**).
 - a. Soil density testing using a nuclear densometer must be completed by a certified operator.
 - b. One soil density measurement shall be completed after the second 12" lift of soil. This will be used to calibrate acceptable soil density and establish the approved cone penetrometer readings.
 - c. Soil density results must be within the range specified in Part 3 of this Section.
2. Soil compaction for detection of compacted layers shall be tested using a static cone penetrometer using method ASTM D3441 - 16 Standard Test Method for Mechanical Cone Penetration Testing of Soils. This test shall not be used on wet soils.
 - a. Maintain at the site, at all times, a soil cone penetrometer with pressure dial and a soil moisture meter to check soil compaction and soil moisture.
 - b. Penetrometer shall be AgraTronix Soil Compaction Meter distributed by Forestry Suppliers, www.forestry-suppliers.com or approved equal.
 - c. Prior to testing the soil with the penetrometer check the penetrometer readings in the mockup soils. Penetrometer readings are impacted by soil moisture and excessively wet or dry soils will read significantly lower or higher than soils at optimum moisture.
 - 1) Refer to **Section 1.16.A** for requirements concerning soil moisture levels.
3. Soil penetrometer readings shall be completed at 150mm (12-inch) intervals for the full depth of the soil profile. These shall be recorded for review by the SMITHSONIAN COTR.
4. The penetrometer readings shall be within 20% plus or minus of the readings in the approved mockup when at similar moisture levels.

H. Soil infiltration testing shall be completed for each Soil Mix or area installed. Soil infiltration testing shall be conducted using ASTM D3385 - 18: Standard Test Method for Infiltration Rate of Soils in Field Using Double-Ring Infiltrometer (Falling head method).

1. A 150mm (6") inner ring / 300mm (12") outer ring Double Ring Infiltrometer shall be used.
 - a. Turf-Tec IN7-W Infiltration Rings, 12" OD/6" ID by 4" Tall distributed by Forestry Suppliers, www.forestry-suppliers.com or approved equal.
 - b. Maintain the Double-Ring Infiltrometer at the site at all times.
2. One infiltration test per planting area shall be sufficient.

1.14 SITE CONDITIONS

- A. It is the responsibility of the Contractor to be aware of all surface and subsurface conditions, and to notify the SMITHSONIAN COTR, in writing, of any circumstances that would negatively impact the health of plantings. Do not proceed with work until unsatisfactory conditions have been corrected.
1. Should subsurface drainage or soil conditions be encountered which would be detrimental to growth or survival of plant material, the Contractor shall notify the SMITHSONIAN COTR in writing, stating the conditions and submit a proposal covering cost of corrections. If the Contractor fails to notify the SMITHSONIAN COTR of such conditions, they shall remain responsible for plant material under the warrantee clause of the specifications.
 2. This specification requires that all planting soil and irrigation (if applicable) work be completed and accepted prior to the installation of any plants.
- B. Subgrade (subsoil) drainage:
1. Perform infiltration and density tests on subsoil:
 - a. Infiltration rate for subgrade, Min. 0.25" per hour 1.) See **Section 1.14.H** for testing procedure.
 - b. Compaction rate for subgrade, static cone penetrometer method:
 - 1) 1.) 0 -150mm (0-6") depth: Range 120 to 180 pounds per square in (psi)
 - 2) 2.) 150mm - 300mm (6"-12") depth: Range 160 to 220 psi. 3.) See **Section 1.14.G.2** for testing procedure.
 - 3) Submit test results as part of the Submittals in **Section 1.10** above.

1.15 DELIVERY, STORAGE, AND HANDLING

- A. Weather: Do not mix, deliver, place or grade soils when frozen or with moisture greater than 70 percent of moisture level for optimum soil compaction as determined from Standard Proctor tests. Doing so is cause for the SMITHSONIAN COTR to reject the soil outright.
1. In addition to obtaining soil moisture levels from a nuclear soil density gauge, moisture can be determined by feel in the following manner:
 - a. With a handful of soil, form a ball by compressing the soil together.
 - 1) Above Field Capacity: If the ball glistens or is plastic (can form a ribbon or be molded), it is too wet.
 - 2) Nearing Field Capacity: If the ball breaks fractures into large pieces but not individual peds, it is too moist for handling.
 - 3) Acceptable Condition: If the ball breaks into individual soil peds with little hand pressure, it is then suitable for handling and spreading.
 - 4) At or Below Wilt Point: If the soil is unable to form a ball with hand pressure, it is unsuitable for handling and spreading.

- B. Protect soil and soil stockpiles, including the stockpiles at the soil blender's yard, from wind, rain and washing that can erode soil or separate fines and coarse material, and contamination by chemicals, dust and debris that may be detrimental to plants or soil drainage. Cover stockpiles with plastic sheeting or fabric at the end of each workday.
- C. Planting Soils shall be stockpiled, loaded, unloaded and transported using methods that protect the size percentage range and distribution of soil peds within the soil.
 - 1. Soil Mixes shall not be allowed to remain in stockpiles for longer than six (6) weeks, either at the soil source or the project site. If the soil will be stockpiled for longer than six (6) weeks, it must be planted to a cover crop or covered with mulch to protect it from sun/heat, rain, and wind erosion. ***The intent of this requirement is to harvest soil from the source only as it is needed and have it directly delivered to the project site, and then installed very soon after delivery. This minimizes soil handling and resulting soil degradation and saves storage space and maintenance. Careful planning and close coordination between the contractor and soil provider is essential.***
 - 2. Use of soil shooters, soil blowers, augers, and conveyors with drops greater than 1-meter (3 feet) or other soil conveyance devices that break up soil peds shall be prohibited. Craned soil in large bulk bags are permitted provided that the process to place the soil into the bags and empty them at the site respects the need to protect soil peds.
- D. All manufactured packaged products and material shall be delivered to the site in unopened containers and stored in a dry enclosed space suitable for the material and meeting all environmental regulations. All products shall be freshly manufactured and dated for the year in which the products are to be used.
- E. Bulk material: Coordinate delivery and storage with the SMITHSONIAN COTR and confine materials to neat piles in areas acceptable to the SMITHSONIAN COTR.

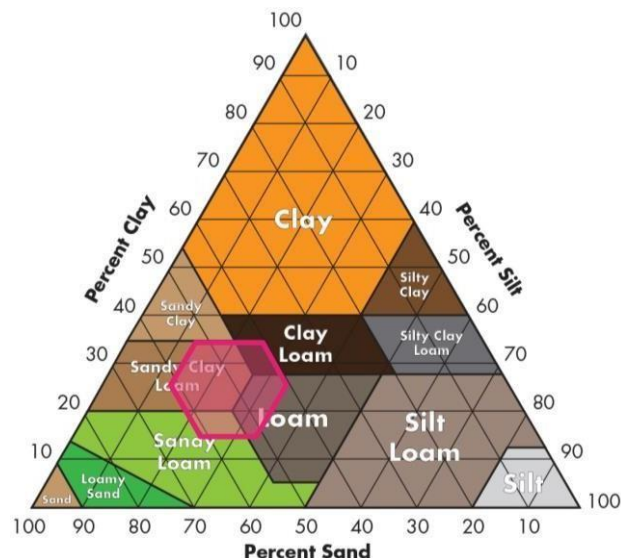
1.16 EXCAVATING AND GRADING AROUND UTILITIES

- A. Contractor shall carefully examine the civil, record, and survey drawings to become familiar with the existing underground conditions before digging.
- B. Determine location of underground utilities including irrigation and electrical systems and perform work in a manner that will avoid damage. Hand excavate as required. Maintain grade stakes set by others until parties concerned mutually agree upon removal.
 - 1. Prior to Work commencement review and clearly mark in field horizontal and vertical locations of existing public underground utilities and structures with respective utility companies.
 - 2. Prior to Work commencement review and clearly mark in field horizontal and vertical locations of existing private underground utilities and structures with the SMITHSONIAN COTR or the qualified party responsible for completing this work.

PART 2 - PRODUCTS

2.1 BASE SOIL

- A. Base Soil definition: Fertile, friable soil containing 15% or less of a total volume of the combination of subsoil, refuse, roots larger than 25mm (1-inch) diameter, clumps of heavy, sticky or stiff clay, stones larger than 75mm (3 inches) in diameter, noxious seeds, sticks, brush, litter, or any substances deleterious to plant growth. The percent (%) of the above objects shall be controlled by source selection, not by screening the soil. Base Soil shall be suitable for the germination of seeds and the support of vegetative growth. Base Soil shall not contain weed seeds in quantities that cause noticeable weed growth and hazardous weed species in the planting beds. *This can be controlled by selection of the soil source site and soil management/cover crops*. The contractor shall remove all weeds as they emerge.
- B. Base Soil shall be defined as a **Clay Loam, Loam, Sandy Loam, or Sandy Clay Loam** having between 17 to 37 percent clay, between 15 to 35 percent silt, and between 40 to 62 percent sand; that will be similar to or match the surrounding topsoil layer. For example, if the topsoil layer is loam soil, the imported soil must also be loam soil with similar organic matter content. Soils imported for subsoil layers, if necessary, must be similar to or match the surrounding subsoil. Soil with greater than 35 percent silt will be rejected.
1. If required, Base soil may be tested for wet aggregate stability, with results showing more than 60 percent stable soil aggregates greater than 0.20 mm
 2. Soil structure should have an identifiably strong aggregation of particles in the form of peds, or small clumps, commonly seen in soils characterized as granular or subangular blocky.



3. Chemical properties: Base Soil shall have the following soil chemical properties:

Item	Units	Range
Organic Matter Content	%	2.5 - 6
Total Nitrogen	mg kg ⁻¹	>200
Ammonium-nitrogen	mg kg ⁻¹	>10
Nitrate-nitrogen	mg kg ⁻¹	>5
Available Phosphorus	mg kg ⁻¹	>50

pH	Std Units	5.5 – 7.3
Cation Exchange Capacity (CEC)	meq/ 100 gm soil	>17
Soluble Salts/ Electrical Conductivity (EC)	dS.m- 1	<1
Potassium	mg kg ⁻¹	>90
Calcium	mg kg ⁻¹	>800
Magnesium	mg kg ⁻¹	>200
Sodium	mg kg ⁻¹	<180
Sulphate-S	mg kg ⁻¹	>20
Lead	mg kg ⁻¹	<50
Manganese	mg kg ⁻¹	<200
zinc	mg kg ⁻¹	<150
boron	mg kg ⁻¹	<3
Exch. Aluminum	mg kg ⁻¹	< 200

- C. Base Soil may be a harvested soil from fields or development sites. The organic content and particle size distribution shall be the result of natural soil formation. Manufactured soils where Coarse Sand, composted organic material or chemical additives has been added to the soil to meet the requirements of this specification section shall not be acceptable. Retained soil peds shall be the same color on the inside as is visible on the outside.
- D. Base Soil shall NOT have been screened. Soil may retain soil peds or clods larger than 2 inches in diameter throughout the stockpile after harvesting as determined by visual inspection by the SMITHSONIAN COTR.
- E. Base Soil Moisture: Protect the Base Soil from rain as required. Soil moisture shall be sufficient when it retains moisture, enabling a friable structure when squeezed, but not in such a wet condition as to leave mud on the hand (see field moisture determination procedure above). Cover the piles after harvesting and uncover during mid-summer/fall drought periods to dry out the soil if project schedule permits.
- F. Submit sample from the Base Soil source location per requirements outlined in **Section 1.10.G**. The sample shall be a mixture of the random samples taken around the source stockpile or field.

The soil sample shall be delivered with soil peds intact that represent the size and quantity of expected peds in the final delivered soil.

2.2 COARSE SAND

- A. Coarse Sand for amending the Base Soil (if required) shall be uniformly graded coarse sand consisting of clean, inert, rounded to sub-angular grains of quartz or other durable rock free from loam or clay, surface coatings, mica, and other deleterious materials. There shall be no coarse fragments over 1.0 cm in size or visible organic matter present. Particle size distribution for material passing a Number 4 Sieve shall be:

U.S. Sieve Size	% Passing Minimum	% Passing Maximum
4	90	100
10	75	95
18	48	75
35	20	45
60	0	20
140	0	5
270	0	3
0.002 mm	0	1

1. The pH shall not exceed 7.8 as determined from a 1:1 soil-distilled water suspension using a glass electrode pH meter American Society of Agronomy Methods of Soil Analysis, Part 2, 1986.
- B. Submit testing sample per requirements outlined in **Section 1.10.E**.

2.3 COMPOST

- A. Compost shall be a stable, humus-like material produced from the aerobic decomposition and curing of organic vegetative residues derived from feedstock consisting of woody stems, leaves, grass cuttings, and livestock manure (up to 10 percent of the compost mix by volume). No food products are acceptable as part of the compost feedstock. The compost shall be a dark brown to black color and be capable of supporting plant growth with appropriate management practices with no visible free water or dust, with no unpleasant odor. Compost shall contain no more than 1 percent foreign materials by weight, and be capable of passing through a one-half inch screen. Compost shall not have become anaerobic during the processing and storage process.
- B. Compost shall be commercially prepared Compost and meet US Compost Council STA/TMECC criteria or as modified in this section for “Compost as a Landscape Backfill Mix Component”.
- http://compostingcouncil.org/admin/wp-content/plugins/wp-pdfupload/pdf/191/LandscapeArch_Specs.pdf
- C. Chemical Properties - Compost shall conform to the following values:

Item	Units	Range
Organic Matter	Percent	>35%
Total Nitrogen	ppm	>1500
Carbon:Nitrogen Ratio	No units	Between 10:1 to 20:1
Extractable Nitrate	ppm	20-200
Total Phosphorus	ppm	5-2000
Available Phosphorus	ppm	5-200
pH	Std Units	5.5 - 8.0
Salt concentration	dS.m- 1	<6
Moisture	% wt	30-55

1. One hundred percent of the material shall pass a 1/2-inch (or smaller) screen. Debris such as metal, glass, plastic, wood (other than residual chips), asphalt or masonry shall not be visible and shall not exceed one percent dry weight.
2. The Compost shall be screened to ½-inch maximum particle size and shall contain no more than 3 percent material finer than 1.0mm (No. 18 sieve) as determined by sieve analysis.
 - a. Additional tests defined in Part I as Chemical Properties shall be performed and the results shall be utilized to evaluate amendments to the Soil Mixes that may be required.
3. Maturity:
 - a. CO2 test: Compost respiration shall be no more than 6 mg CO2-CIgBVS day.
 - b. Solvita test: The compost must achieve a maturity index of 6 or more.
4. Biological Values:
 - a. Active bacteria and fungi are not to be higher than 10 percent of total bacteria and fungi respectively.
 - b. Total bacteria to be a minimum of 1500 ng/g
 - c. Total fungi to be a minimum of 400 ng/g
 - d. Protozoa to be 100,000 (amoeba and flagellates with no more 3000 ciliates)
5. Pathogens/Metals/Vector Attraction reduction shall meet 40 CFR Part 503 rule, Table 3, page 9392, Vol. 58 No. 32.

D. Submit testing and sample per requirements outlined in **Section 1.10.E**.

2.4 LIGHTWEIGHT AGGREGATE

- A. Shall be a 5/8" lightweight expanded shale or clay, produced by the rotary kiln process and meeting the requirements of ASTM E2278-M18. Lightweight expanded shale shall have a proven record of durability and be non-corrosive, with the following properties:
 1. Soundness Loss: Max 30% with 4 cycles of Magnesium sulfate, in accordance with AASHTO T104
 2. Abrasion Resistance: Max 40% in accordance with ASTM C131.
 3. Chloride Content: Max. 100 ppm in accordance with AASHTO T 291.

2.5 NO. 8 PEA GRAVEL

- A. Shall be clean, dried and free of organic/deleterious materials, conforming to ASTM C-33 specifications for 3/8" aggregate.

2.6 NO. 57 STONE

- A. Shall be clean, coarse, open-graded, self-compacting aggregate blend of size 5, 6, & 7 stone.
- B. No. 57 Stone shall conform to the following particle size:

U.S. Sieve Size	% Passing Minimum	% Passing Maximum
1.5 inch	100	100
1 inch	95	100
0.5 inch	25	60
#4	0	10
#8	0	8
18	0	0
270	0	0
0.002 mm	0	0

2.7 SOIL MIXES

- A. All Soil Mixes shall be blended and prepared with the components (Base Soil, Compost, Coarse Sand, and Lightweight Aggregate) in the ratios summarized below. Preparation of Soil Mixes shall follow the requirement of Part 3 – Execution. Specific blending details are outlined per Soil Mix below.

SOIL MIX	Base Material	Second Component	Third Component	Ratio by Volume
Planting Soil	Base Soil	Compost <i>(if req'd)</i>		5 pts base soil to 1 pt compost <i>(if req'd)</i>
Lawn Soil	Base Soil	Compost <i>(if req'd)</i>		4 pts base soil to 1 part compost
High-Use Lawn Soil	Base Soil (Clay Loam or Sandy Clay Loam)	Coarse Sand	Compost	2 pts base soil to 3 pts coarse sand to 1 pt compost
	Base Soil (Loam or Sandy Loam)	Compost	Coarse Sand <i>(if req'd)</i>	4 pts base soil to 1 pts compost <i>(add 1 pt coarse sand if req'd)</i>

Biofiltration Soil	Base Soil (Clay Loam or Sandy Clay Loam)	Coarse Sand	Compost	2 pts base soil to 4 pts coarse sand to 1 pt compost
	Base Soil (Loam or Sandy Loam)	Compost	Coarse Sand (if req'd)	3 pts base soil to 2 pts compost (add 1 pt coarse sand if req'd)
Lightweight Extensive Soil	Base Soil	Lightweight Aggregate	Compost	4 pts base soil to 4 pts lightweight aggregate to 1 pt compost

- B. Estimated Soil Mix Densities for determining structural bearing capacity in applications over-structure are provided below. Estimated densities assume approximately 50 percent porosity and 2.5 percent organic matter in Planting Soil; 47 percent pore space for Lawn Soils, and 35 percent porosity and no organic matter in Coarse Sand.

SOIL MIX OR COMPONENTS	Dry Density	Saturated Density
Planting Soil	82 to 87 lbs/cu ft	110 to 114 lbs/cu ft
Lawn Soils	85 to 90 lbs/cu ft	113 to 118 lbs/cu ft
Biofiltration Soil	106 to 110 lbs/cu ft	125 to 132 lbs/cu ft
Lightweight Extensive Soil	69 to 75 lbs/cu ft	92 to 101 lbs/cu ft
Coarse Sand	109 to 115 lbs/cu ft	127 to 132 lbs/cu ft

2.8 PLANTING SOIL

- A. Planting Soil shall consist of Base Soil harvested from the soil source site. If the Base Soil has organic matter greater than 3.0 percent, and the soil is dark brown with strong aggregate structure (with discernable peds indicating granular or subangular blocky composition), no blending of amendments shall be done and the soil shall be used without further processing unless large rocks, sticks, or other debris greater than 75mm (3 inches) are present. Large items may be manually removed from the soil (preferred) or, if necessary, screened through a screen with 50mm – 75mm (2- to 3-inch) openings (only if large objects are greater than 10 percent of the harvested soil material). Soils must be examined and approved of by the SMITHSONIAN COTR.
1. If the soil organic matter content is less than 3.0 percent, the Planting Soil shall be blended with compost in a ratio of 10 parts Base Soil to 1 part Compost for each percentage point less than 3 percent. The Planting Soil shall be blended to be a uniform, homogenous mixture of the soil components. Additives to achieve specified criteria are not acceptable.
 2. Mix Compost into the soil with a loader bucket to loosely incorporate it into the Base Soil as follows.
 - a. Soil moisture shall not be greater than 17 percent by weight of the Base Soil mass. Wetter soil will remain cohesive and not blend with compost well.

- b. Spread the Base Soil out as a layer approximately 380mm - 500mm (15-20 inches) deep.
 - c. Place the Compost over the soil in a layer 75mm – 100mm (3- to 4-inches) thick.
 - d. Using the loader bucket, push and lift the soil and compost into a pile, back-drag the pile only once and lift the soil/compost blend in the bucket and allow it to roll [gently] out of the bucket back into the pile.
3. DO NOT OVER MIX! Do not mix with a soil blending machine. Do not screen the soil. This specification assumes that the various other operations of loading and delivery and final spreading of the soil at the site will further mix the two components to an acceptable amount.
4. Base Soil moisture: Prior to mixing Planting Soil (*if required*), protect the Base Soil required for this mix from rain so that the soil moisture remains above wilt point and below field capacity. Soil moisture shall be between 8 to 17 percent by weight of the soil mass when blending with amendments or when handling. Soil with greater than approximately 15 percent moisture may lose critical soil structure if handled too much. Soil with less than 8 percent moisture by weight may be too brittle and soil peds may shatter into structureless soil particles that will become cemented or compacted when rewetted.
 - a. Cover the piles after harvesting and uncover during summer/fall drought periods to dry out the soil required and if project schedule permits.
 - b. See additional requirements for soil moisture as defined in **Section 1.16.A**.
5. Planting Soil moisture: After mixing Planting Soil, protect from rain so that the soil moisture is sufficient to be friable and crumble when squeezed and not leave mud on the hand. Cover the piles after mixing and uncover during mid-summer/fall drought periods to dry out the soil if soil moisture requires and project schedule permits.
 - a. See additional requirements for soil moisture as defined in **Section 1.16.A**.
6. Submit Planting Soil sample per requirements outlined in **Section 1.10.G**. The sample should represent the size and distribution of the soil peds in the soil. Sample shall be marked with the proportion of Base soil to Compost. Submit testing data per requirements outlined in **Section 1.10.F**.
 - a. Planting Soil chemical qualities shall be tested and include:

Item	Units	Range
Total Organic Matter	Percent	3.0 – 5.0%
Total Nitrogen	mg kg ⁻¹	>750
Ammonium-nitrogen	mg kg ⁻¹	>20
Nitrate-nitrogen	mg kg ⁻¹	>25
Available Phosphorus	mg kg ⁻¹	>50
pH	Std Units	5.5 – 7.3
Cation Exchange Capacity (CEC)	meq/ 100 gm soil	>17
Soluble Salts/ Electrical Conductivity (EC)	dS.m ⁻¹	<1.0

Item	Units	Range
Potassium	mg kg ⁻¹	>90
Calcium	mg kg ⁻¹	>700
Magnesium	mg kg ⁻¹	>200
Sodium	mg kg ⁻¹	<100
Sulphate-S	mg kg ⁻¹	>20
Lead	mg kg ⁻¹	<50
Manganese	mg kg ⁻¹	<200
Exchangeable aluminum	mg kg ⁻¹	<180
zinc	mg kg ⁻¹	<50
boron	mg kg ⁻¹	<3
Exch. Aluminum	mg kg ⁻¹	<200

2.9 LAWN SOIL

- A. Lawn Soil shall consist of Base Soil harvested from the soil source site that will be blended with Compost in a ratio of 4 parts Base Soil to 1 part Compost. If the Base Soil has organic matter greater than 3.0 percent, and the soil is dark brown with strong aggregate structure, no blending of is necessary and the soil shall be used without further processing unless large rocks, sticks, or other debris greater than 4 inches are present. The soil components shall be blended to a homogenous, uniform mix free of large clay clods, rocks or sticks greater than 50mm (2 inches) in any dimension, or debris that could affect plant growth and development.
1. If the combined fine (no. 140 screen) and very fine (no. 270 screen) sand fraction of the soil particle size is greater than 30 percent of the total soil texture, then Coarse Sand shall be added to the soil blend with a final ratio of 3 parts Base Soil to 1 part Compost to 1 part Coarse Sand. Additives to achieve specified criteria are not acceptable.
 2. The soil components may first be blended using a loader bucket to loosely incorporate the Compost (and Coarse Sand, if required) into the Base Soil as follows.
 - a. Spread the Base Soil out as a layer approximately 380mm - 500mm (15-20 inches) deep.
 - b. Place the Compost over the soil in a layer 100mm (4 inches) thick.
 - c. If coarse sand is required to be added to the blended soil, the Base Soil shall be placed in a layer approximately 380mm (15 inches) thick, upon which a 100mm (4-inch) layer of Compost and a 100mm (4-inch) layer of Coarse Sand will be placed over the Base Soil.
 - d. Using the loader bucket, push and lift the soil and compost into a pile, back-drag the pile only once and lift the soil/compost blend in the bucket and allow it to roll [gently] out of the bucket back into the pile.
 3. Blending shall be completed by use of a rotating mixer or a 50mm (2-inch) screen to more evenly mix the soil components.
- B. Submit Lawn Soil sample per requirements outlined in **Section 1.10.G**. Submit testing data per requirements outlined in **Section 1.10.F**. Soil chemical qualities shall be tested and include:

Item	Units	Range
Total Organic Matter	Percent	>3.0%
Total Nitrogen	mg kg ⁻¹	>700
Ammonium-nitrogen	mg kg ⁻¹	>20
Nitrate-nitrogen	mg kg ⁻¹	>25
Available Phosphorus	mg kg ⁻¹	>50
pH	Std Units	6.5 – 7.8
Cation Exchange Capacity (CEC)	meq/ 100 gm soil	>14
Soluble Salts/ Electrical Conductivity (EC)	dS.m ⁻¹	<1.0
Potassium	mg kg ⁻¹	>90
Calcium	mg kg ⁻¹	>500
Magnesium	mg kg ⁻¹	>200
Sodium	mg kg ⁻¹	<100
Sulphate-S	mg kg ⁻¹	>20
Lead	mg kg ⁻¹	<50
Manganese	mg kg ⁻¹	<200
Exchangeable aluminum	mg kg ⁻¹	<180
Zinc	mg kg ⁻¹	<50
Boron	mg kg ⁻¹	<3
Exch. Aluminum	mg kg ⁻¹	<200

1. Soil moisture shall be as defined in Section **1.16.A**.
2. Protect prepared soils from rain during storage to prevent soils from becoming saturated during storage. Cover the piles after mixing and uncover during mid-summer/fall drought periods to dry out the soil if soil moisture requires and project schedule permits.

2.10 HIGH-USE LAWN SOIL

- A. High-Use Lawn Soil shall consist of Base Soil harvested from the soil source site that will be blended in a ratio per Soil Mix Table in **Section 2.7.A**. If the Base Soil has organic matter greater than 3.0 percent, and the soil is dark brown with strong aggregate structure, no blending of is necessary and the soil shall be used without further processing unless large rocks, sticks, or other debris greater than 4 inches are present. The soil components shall be blended to a homogenous, uniform mix free of large clay clods, rocks or sticks greater than 50mm (2 inches) in any dimension, or debris that could affect plant growth and development.
 1. The soil components may first be blended using a loader bucket to loosely incorporate the Coarse Sand and Compost into the Base Soil as follows.
 - a. Spread the Base Soil out as a layer approximately 380mm - 500mm (15-20 inches) deep.
 - b. Place the Compost over the soil in a layer 100mm (4 inches) thick.
 - c. Add the Coarse Sand over the soil in a layer 100mm (4-inch) thick.

- d. Using the loader bucket, push and lift the soil and compost into a pile, back-drag the pile only once and lift the soil/compost blend in the bucket and allow it to roll [gently] out of the bucket back into the pile.
 2. Blending shall be completed by use of a rotating mixer or a 50mm (2-inch) screen to more evenly mix the soil components.
- B. Submit High-Use Lawn Soil sample per requirements outlined in **Section 1.10.G**. Submit testing data per requirements outlined in **Section 1.10.F**. Soil chemical qualities shall be tested and include:

Item	Units	Range
Total Organic Matter	Percent	>3.0%
Total Nitrogen	mg kg ⁻¹	>700
Ammonium-nitrogen	mg kg ⁻¹	>20
Nitrate-nitrogen	mg kg ⁻¹	>25
Available Phosphorus	mg kg ⁻¹	>50
pH	Std Units	6.5 – 7.8
Cation Exchange Capacity (CEC)	meq/ 100 gm soil	>14
Soluble Salts/ Electrical Conductivity (EC)	dS.m ⁻¹	<1.0
Potassium	mg kg ⁻¹	>90
Calcium	mg kg ⁻¹	>500
Magnesium	mg kg ⁻¹	>200
Sodium	mg kg ⁻¹	<100
Sulphate-S	mg kg ⁻¹	>20
Lead	mg kg ⁻¹	<50
Manganese	mg kg ⁻¹	<200
Exchangeable aluminum	mg kg ⁻¹	<180
Zinc	mg kg ⁻¹	<50
Boron	mg kg ⁻¹	<3
Exch. Aluminum	mg kg ⁻¹	<200

1. Soil moisture shall be as defined in Section **1.16.A**.
2. Protect prepared soils from rain during storage to prevent soils from becoming saturated during storage. Cover the piles after mixing and uncover during mid-summer/fall drought periods to dry out the soil if soil moisture requires and project schedule permits.

2.11 BIOFILTRATION SOIL – Not Used

2.12 LIGHTWEIGHT EXTENSIVE SOIL

- A. Lightweight Extensive Soil shall be a blend of 50 percent Base Soil to 50 percent Lightweight Aggregate (1:1 blend ratio) to provide a uniform, homogenous blend of the soil components. The blended Soil/EC mix shall then be blended with compost in a ratio of 5 parts Soil/EC mix to 1 part compost. Blending using a rotating or tumbling drum or screen is recommended. The soil shall be tested for particle size distribution after blending to assure the proper ratio of expanded clay particles and soil particle sizes.
- B. Submit Lightweight Extensive Soil sample per requirements outlined in **Section 1.10.G**. Submit testing data per requirements outlined in **Section 1.10.F**. Soil chemical qualities shall be tested and include:

Item	Units	Range
Total Organic Matter	Percent	>2.0%
Total Nitrogen	mg kg ⁻¹	>500
Ammonium-nitrogen	mg kg ⁻¹	>20
Total Organic Matter	Percent	>2.0%
Total Nitrogen	mg kg ⁻¹	>500
Ammonium-nitrogen	mg kg ⁻¹	>20
Nitrate-nitrogen	mg kg ⁻¹	>25
Available Phosphorus	mg kg ⁻¹	>50
pH	Std Units	6.5 – 7.8
Cation Exchange Capacity (CEC)	meq/ 100 gm soil	>10
Soluble Salts/ Electrical Conductivity (EC)	dS.m ⁻¹	<1.0
Potassium	mg kg ⁻¹	>90
Calcium	mg kg ⁻¹	>800
Magnesium	mg kg ⁻¹	>200
Sodium	mg kg ⁻¹	<100
Sulphate-S	mg kg ⁻¹	>20
Lead	mg kg ⁻¹	<50
Manganese	mg kg ⁻¹	<200
Exchangeable aluminum	mg kg ⁻¹	<180
Zinc	mg kg ⁻¹	<50
Boron	mg kg ⁻¹	<3

- See additional requirements for soil moisture as defined in Section **1.16.A**.

PART 3 - EXECUTION

3.1 COORDINATION WITH PROJECT WORK

- A. The Contractor shall coordinate with all other work that may impact the completion of the work.
- B. Prior to the start of work, submit and obtain approval of the Planting Soils Installation Plan and prepare a detailed schedule of the work for coordination with other trades.
- C. Coordinate the relocation of any irrigation lines, heads or the conduits of other utility lines that are in conflict with tree locations. Root balls shall not be altered to fit around lines. Notify the SMITHSONIAN COTR of any conflicts encountered.

3.2 PRE-EXAMINATION, VERIFICATION AND ACCEPTANCE

- A. A Pre-installation Examination with the SMITHSONIAN COTR is required for the work of this section. Schedule the examination at least five days before the installation process begins.
 - 1. As the work proceeds, the Contractor shall schedule a pre-installation examination with the SMITHSONIAN COTR for each area of planting soil installation.
 - 2. Reference **Section 1.12** for further requirements governing pre-installation examinations.
- B. Upon receipt of delivery of the Soil Mixes, the Contractor shall visually inspect the soil for moisture content, non-aggregated soil particles, clumping, debris, deleterious or foreign materials, or any other physical conditions that could affect the quality of the soils and the Contractor's installation operations.
 - 1. The Contractor shall immediately notify the SMITHSONIAN COTR of any soil deliveries that exhibit any of the physical conditions noted above.
 - 2. The Contractor shall not accept or use soil that exhibits any of the physical conditions noted above.
- C. The Contractor shall be responsible for verification that all of the planting areas receiving Soil Mixes have been prepared in conformance with the Contract Documents.
 - 1. Verify that utilities have been installed and accepted.
 - 2. Verify that irrigation mainlines have been installed.
 - 3. Verify that there is a sufficient means for on-site watering of installed plants.
 - 4. Verify that the rough grading has been accepted by the SMITHSONIAN COTR.
- D. Examine subgrade for deficiencies including:
 - 1. Construction debris present within the area to receive Soil Mixes.
 - 2. Puddling of water, muddy soil conditions, or expressing of water from the subgrade or adjacent areas.
 - 3. The subgrade is not at the correct depths for installing the planting soil.
 - 4. Incomplete utility, irrigation and /or subsurface drainage installation.

5. Insufficient compaction of subgrade. Refer back to **Section 1.15.B** or the geotechnical subgrade requirements (*if applicable*) outlined in Specification **Section 31 2000 -- Earthwork**.
- E. Submit all noted deficiencies that will impact the proper installation or execution of the Work to the SMITHSONIAN COTR in writing prior to beginning soil installation operations. The Contractor assumes responsibility for all subgrade work and conditions upon beginning soil installation operations.
- F. **GRADE AND ELEVATION CONTROL**
- G. Provide grade and elevation control during installation of the Soil Mixes. Utilize grade stakes, surveying equipment, and other means and methods to assure that grades and contours conform to the grades indicated on the plans.
 1. Establish lines and levels, locate and lay out by instrumentation and similar appropriate means for planting area finish grades.
 2. Provide as many grade stakes and string lines as required to achieve smooth finish grades acceptable to the SMITHSONIAN COTR with positive surface drainage.
 3. High Points and Low Points: Provide grade stakes at high points and low points including top of berms, catch basin rims and area drain rims.

3.3 SUBGRADE PREPARATION FOR ON-GRADE CONDITIONS

- A. Protection of Existing Conditions:
 1. Refer to Specification **Section 01 5639 – Site Protection** for requirements governing this work.
 2. Submit written notification of conditions damaged during construction to the SMITHSONIAN COTR immediately.
- B. Excavate to the proposed subgrade where applicable. Maintain all required angles of repose of the adjacent materials as shown on the drawings or as required by this specification. Do not over excavate compacted subgrades of adjacent pavement or structures. Maintain a supporting 1:1 side slope of compacted subgrade material along the edges of all paving and structures where the bottom of the paving or structure is above the bottom elevation of the excavated planting area.
- C. Remove all construction debris and material including any construction materials from the subgrade.
- D. Confirm that the subgrade is at the proper elevation and compacted if required. Subgrade elevations shall slope approximately parallel to the finished grade and/or toward the subsurface drain lines as shown in the Drawings.
- E. Subgrade shall be scarified (roughened) to a depth of 75 – 150mm (3-6 inches) prior to placement of subsoil and Soil Mixes to create an uneven, broken surface in which the subgrade can be mixed with the first lift of Soil Mix placed. Scarification can be accomplished using a mini-disc, reverse tiller, or other suitable device as approved by the SMITHSONIAN COTR.

- F. Perform infiltration and density tests on subsoil as described in **Section 1.15.B**.
- G. Protect adjacent walls, walks and utilities from damage or staining by the soil. Use 12.5mm (1/2-inch) plywood as directed to cover existing concrete, metal and masonry work and other items as directed during the progress of the work.
 - 1. At the end of each working day, clean up any soil or dirt spilled on any paved surface.
 - 2. Any damage to the paving or site features or work shall be repaired at the Contractor's expense.

3.4 PREPARATION FOR OVER-STRUCTURE CONDITIONS

- A. Protection of Existing Conditions:
 - 1. Refer to Specification **Section 01 5639 – Site Protection** for requirements governing this work.
 - 2. Submit written notification of conditions damaged during construction to the SMITHSONIAN COTR immediately.
- B. In areas of work over structure, confirm that all waterproofing and drainage layers are complete. Remove any debris from the surface of the drainage layer.

3.5 SOIL MOISTURE

- A. Volumetric soil moisture level, in the Soil Mixes and the root balls of all plants, prior to, during and after soil installation and planting shall be above permanent wilt point and below field capacity. Reference **Section 1.16.A.1** for determining acceptable soil moisture levels for delivering, handling, placing, and grading activities. Failure to adhere to these requirements is cause for the SMITHSONIAN COTR to reject the soil outright.

3.6 SOIL INSTALLATION

- A. General
 - 1. Phase work such that equipment to deliver or grade soil and install Soil Mixes does not have to operate over previously installed soil work including lower lifts. Work in rows of lifts the width of the extension of the bucket on the loader. Install all lifts of the entire assembly in one row before proceeding to the next. Work out from the furthest part of each bed from the soil delivery point to the edge of each bed area.
 - 2. The contractor shall continuously check the compaction of the soil installation with a cone penetrometer (at 150mm (12-inch) intervals), as the work progresses, to assure that penetration resistance conforms to the values of the mock up. Final compaction readings with the cone penetrometer should be taken for each planting area at the completion of soil installation (see **Section 3.9**).
 - 3. Installing soil using soil or mulch blowers or soil slingers shall not be permitted for planting soil due to the over mixing and soil ped breakdown cause by this type of equipment. Use of soil "Gaylord" bags craned into the site is permitted

4. Where travel over installed soil is unavoidable, limit paths of traffic to reduce the impact of compaction in planting soil.
 - a. Wheel-driven vehicles are expressly forbidden to be allowed on installed soils. Only low- ground pressure (less than 6 psi), wide track 400mm to 600mm (18-24 inches) equipment may be allowed on installed soils and only along pre-approved corridors or lanes.
 - b. All grading and soil delivery equipment shall have buckets equipped with 100 mm (6-inch) long teeth to scarify any soil that becomes compacted.
 - c. Protect the surface of the soil in any areas that will receive repeated passes with motorized equipment with 19mm (3/4-inch) plywood matting.
 - 1) 3/4" plywood shall only be used when the duration of use does not exceed three days. This is considered for temporary use only.
 - 2) 2.) No part of any plywood matting shall be moved, altered, or changed in any way until access across, adjacent to, or through installed soil zones is no longer necessary or until the temporary use period has concluded.
 - d. Till the surface of the soil with a mini-disc or reverse tiller to a depth of 150mm to 200mm (6-8 inches) when the plywood is removed.
 - e. Each time equipment passes over the installed soil it shall reverse out of the area along the same path with the teeth of the bucket dropped to scarify the soil. In the event that the planting soil becomes over compacted, thereby failing to comply with **Section 3.9**, the soil shall be removed in 12" lifts and the compaction retested until specified compaction levels are observed before proceeding with new soil installation.
5. Prior to installing Planting Soils, the SMITHSONIAN COTR shall approve the condition of the subgrade and the previous scarification.
 - a. Immediately install the Soil Mix. Protect the loosened area from traffic. DO NOT allow the loosened subgrade to become compacted.
6. Compaction of soils to specified soil densities shall be accomplished using rollers, foot pressure, or manual tools.
 - a. Vibratory, plate and jumping jack compactors, or impact methods (using a backhoe bucket or similar to impact or "hammer" the soil) are strictly prohibited.
7. In the event that the loosened area becomes overly compacted, loosen the area again prior to installing planting soil.
8. Where possible place trees on soil pedestals such that the elevation of the tree root flare will be at the planned finished soil elevation. Tree pedestals will be compacted to 90- to 95% of the soil's maximum dry density, using the Standard Proctor method, to resist settling of the soil from the weight of the tree.

B. Planting Soil Installation

1. After inspecting the base grade of the soil, the Planting Soil shall be installed in lifts not to exceed 300mm (12-inch) to the required depths as shown in the Drawings.

- a. Compact each lift gently and evenly to approximately 80 to 84 percent of the soil's maximum dry density using the Standard Proctor method
 - b. After each lift has been successfully compacted, scarify the surface of the lift to a depth of 38mm (1.5 inches) and place the next soil lift. Repeat the placement of soil lifts for each soil layer until the Planting Soil grade elevation has been achieved.
 - c. Measure soil compaction using a nuclear soil density gauge to attain the required soil density approved in the soil mock up.
2. Where trees will be planted / installed, pack each lift of Planting Soil around the base and rootball of the tree using manual tools (shovels, rakes) to assure that no voids are present around the rootball and that soil is firmly in place.
 - a. Place the Planting soil in 300 mm (12-inch) lifts.
 - 1) Compact each lift gently and evenly to approximately 80 to 84 percent of the soil's maximum dry density using the Standard Proctor method.
 - 2) Scarify each lift to a depth of approximately 25mm (1 inch) prior to placement of each successive lift.
 - b. After the final Planting Soil lift has been installed, grade the surface of the Planting Soil smooth and even. Scarify the final soil lift to create a firm, friable planting surface.
- C. Planting Soil Installation in Soil Cells – Not Used
- D. Planting Soil Installation in Raised Planters – Not Used
- E. Lawn Soil Installation
 1. The Lawn Soil profile shall include the Planting Soil layer and the Lawn Soil.
 2. Install the Planting Soil in lifts not to exceed 300mm (12-inch) to the required depths as shown in the Drawings. Compact each lift gently and evenly compact soil to approximately 82 to 84 percent of the soil's maximum dry density using the Standard Proctor method. After each lift has been successfully compacted, scarify the surface of the lift to a depth of 38mm (1.5 inches) and place the next soil lift. Repeat the placement of soil lifts for each soil layer until the Planting Soil grade elevation has been achieved. Measure soil compaction using the cone penetrometer as required to attain the required soil density approved in the soil mock up.
 - a. Prior to placement of successive materials or lifts, determine soil density using the cone penetrometer. Resistance to penetrometer push shall be 110 to 170 psi.
 - b. Where trees will be planted / installed, pack each lift of Planting Soil around the base and rootball of the tree using manual tools (shovels, rakes) to assure that no voids are present around the rootball and that soil is firmly in place.
 3. Install the Lawn Soil in lifts not to exceed 300mm (12-inch) to the required depths as shown on the Plans. Compact each lift gently and evenly compact soil to approximately 82 to 84 percent of the soil's maximum dry density using the Standard Proctor method. After each lift has been successfully compacted, scarify the surface of the lift to a depth of 38mm (1.5 inches) and place the next soil lift. Repeat the placement of soil lifts for

each soil layer until the final grade of the soil profile has been achieved, as required to attain the required compaction approved in the soil mock up.

- a. Prior to placement of successive materials or lifts, determine soil density using the cone penetrometer. Resistance to penetrometer push shall be 110 to 170 psi.
- b. After placement of all Lawn Soil, the soil density shall be measure for the whole depth of the Lawn Soil layer to check for uniform density and that clay or compaction pans are not present.
- c. The Lawn Soil shall be finished with a firm, friable planting surface.

F. High-Use Lawn Soil Installation

1. The High Use Lawn Soil profile shall include the Planting Soil layer, a Coarse Sand drainage layer and a Lawn Soil layer.
2. Install the Planting Soil in lifts not to exceed 300mm (12-inch) to the required depths as shown on the Plans. Compact each lift gently and evenly compact soil to approximately 82 to 84 percent of the soil's maximum dry density using the Standard Proctor method. After each lift has been successfully compacted, scarify the surface of the lift to a depth of 38mm (1.5 inches) and place the next soil lift. Repeat the placement of soil lifts for each soil layer until the Planting Soil grade elevation has been achieved. Measure soil compaction using a nuclear soil density gauge to assure the required soil density approved in the soil mock up.
3. Install the 150mm (6-inch) Coarse Sand drainage layer over the Planting Soil, taking care to minimize damage, including excess compaction, to the Planting Soil layer.
 - a. The Coarse Sand drainage layer shall be installed evenly and raked smooth to a firm, smooth surface.
 - b. The Coarse Sand drainage layer shall not be compacted.
4. Install the High-Use Lawn Soil in lifts not to exceed 200mm (8-inch) to the required depths as shown on the Plans. Compact each lift gently and evenly compact soil to approximately 82 to 84 percent of the soil's maximum dry density using the Standard Proctor method. After each lift has been successfully compacted, scarify the surface of the lift to a depth of 38mm (1.5 inches) and place the next soil lift. Repeat the placement of soil lifts for each soil layer until the final grade of the soil profile has been achieved, as required to attain the required compaction approved in the soil mock up.
 - a. Prior to placement of successive materials or lifts, determine soil density using the cone penetrometer. Resistance to penetrometer push shall be 110 to 170 psi.
 - b. After placement of all High-Use Lawn Soil, the soil density shall be measure for the whole
 - c. depth of the Lawn Soil layer to check for uniform density and that clay or compaction pans are not present.
 - d. The Lawn Soil layer shall be graded smooth and even and compacted using a roller, or previously approved means. Any uneven surface areas shall be graded smooth.
 - e. The Lawn Soil shall be finished with a firm, friable planting surface.

G. Biofiltration Soil Installation

1. Biofiltration Soil profiles shall be installed as shown in the plans to comply with DOEE requirements.
2. Prior to installation of the Biofiltration soil profile, check that all subgrade preparation has been completed and accepted. The base of each biofiltration areas shall be scarified to a depth of 50mm (2-inches) and graded smooth with a rough, porous surface.
3. A rock drainage layer consisting of clean No. 57 stone (1" to 1.5" stone) shall be placed to the depth shown in the Drawings above the base grade concurrent with the installation of the bioretention drainage system. The stone drainage layer will be raked or graded smooth.
4. A layer of 3/8" No. 8 pea gravel shall be installed above the stone drainage layer. The pea gravel layer shall be placed as an even layer raked smooth.
5. The Biofiltration Soil shall be installed to the depths according to the plan details. For installation, the soil shall be installed in 200mm – 300mm (8- to 12-inch) lifts. Each lift shall be graded smooth and scarified prior to the placement of the next subsequent lift. Biofiltration Soil lifts shall not be compacted, but each lift must be raked to assure that soil voids have been eliminated.
6. After placement of the Biofiltration Soil, the full profile shall be wetted to allow for natural settling and compaction. This process shall be conducted a minimum of two times. The final grade elevation of the Biofiltration Soil shall be measured. Any resulting settling or voids shall be filled with added Biofiltration Soil to the final grade.
7. The final Biofiltration Soil shall be graded smooth and a firm, friable planting bed shall be prepared.

H. Lightweight Extensive Soil Installation

1. The structural base upon which the soil will be installed, including waterproofing and the drainage system, shall be provided by others.
2. After completion of the structural surface, waterproofing, and drainage layers, the installation of the protection layer, root barrier and separation geotextile should follow.
3. Next, the Lightweight Planting Soil shall be installed as shown in the Drawings.
 - a. The Lightweight Planting Soil shall be installed to the depth shown in the drawings in a single lift and raked and graded smooth. Compaction of this lift is not necessary.

3.7 INSTALLATION OF COMPOST TILL LAYER

- A. After Planting Soils are installed and just prior to the installation of shrub or groundcover plantings, spread 50mm (2 inches) of Compost over the beds and roto till into the top 100mm (4 inches) of the Planting Soil. This step will raise grades slightly, refer to **Section 3.10.A**.
 1. Planting Soils here refer to **Sections 3.7B** (Planting Soil).

3.8 POST SOIL INSTALLATION TESTING

- A. Soil density, soil compaction, and soil infiltration testing shall be completed as soil is installed and at completion of soil installation to assure that soils comply with specified requirements as established in this Section.

1. Soil density shall be measured using a nuclear soil density gauge as described in **Part 1.14.G** of this section.
 - a. Soil density measurements shall be completed for each planting area smaller than 100 square meters, or one test for each 100 square meters of a larger planting area.
 - b. Soil density shall be between 80 to 84 percent of the soil's maximum dry density using the Standard Proctor method in all areas. If soil density exceeds this range, additional soil density tests shall be conducted offset by 5 feet in each direction from the point of exceedance to determine extent of excessive soil density. The SMITHSONIAN COTR shall be consulted for corrective actions.
 - c. Soils with high compost or organic matter concentrations may not achieve the specified soil density range. Advise the SMITHSONIAN COTR if soil density measurements and the location of soil density measurements not meeting the specified range.
2. Compaction testing, including surface as well as subsurface compaction layers, shall be tested using a cone penetrometer as described in **Part 1.14G** of this Section.
 - a. Compaction testing shall be done for each 25 square meters of planting area at a minimum.
 - b. Compaction measured as resistance to penetration of the cone penetrometer in pounds per square inch (psi) shall not exceed 140 psi in the surface 150mm (6 inches), 180 psi in the 150 – 300 mm (6- to 12-inch) depth interval, or 200 psi at depths greater than 300 mm in the soil profile.
 - 1) If compaction testing detected compacted layers exceeding the allowances described above, soil shall be removed to the depth of compaction, and compacted soils shall be loosened with hand tools to break compacted layers and achieve appropriate soil density.
3. Infiltration testing shall be conducted as described in **Section 1.14H**. Infiltration testing shall be completed using the Double-Ring Infiltrometer.
 - a. Infiltration testing shall be done for each planting area of 100 square meters or less. Infiltration testing shall be done at a rate of one test per 100 square meters for planting areas greater than 100 square meters.
 - b. Infiltration rates for Planting Soil and Lawn Soil shall be greater than 15 mm per hour (0.6 inches/hour). If infiltration rates are not greater than 15 mm/hr, consult with the SMITHSONIAN COTR to determine corrective actions.
 - c. Infiltration rates for High-Use Lawn and Biofiltration Soils shall be greater than 25 mm per hour (1.0 inches/hour). If infiltration rates are not greater than 25 mm/hr, consult with the SMITHSONIAN COTR to determine corrective actions.

3.9 FINISH GRADING

- A. The finish grades of all planted areas shown in the Drawings are the elevations after settlement and shrinkage of the Compost Till Layer and planting soil. This settlement is anticipated to be within a few months after installation as the Compost breaks down. A minimum settlement of approximately 25mm (1-inch) of the soil depth is expected. The Contractor shall install the

planting soil at a higher level to anticipate this reduction of planting soil volume (approximately 25mm (1-inch)).

1. Grade the edges of shrub areas and ground cover areas soil surfaces to an elevation 50mm (2 inches) below the finished surface of adjacent paving and curbs, after initial soil settlement, unless indicated otherwise.
 2. The grades in bed areas shown in the Drawings is the soil line before mulch is added and after soil settlement. The grades in lawn areas shown in the Drawings in the thatch line of the sod after initial soil settlement, unless indicated otherwise.
- B. Utilize hand tools to keep surface rough without further compaction. Do not use the flat bottom of a loader bucket to fine grade, as it will cause the finished grade to become overly smooth and orslightly compressed.
- C. Inspect and survey finished soil grades for positive drainage from all areas toward the existing inlets, drainage structures and or the edges of planting beds consistent with soil grade designs unless indicated otherwise. Adjust grades as directed to reflect actual constructed field conditions of paving, wall and inlet elevations as shown in the Drawings. Notify the SMITHSONIAN COTR in the event that conditions make it impossible to achieve positive drainage.
1. Grade soil surface smooth to be free of high and low areas which will inhibit surface drainage.
 2. Provide smooth, rounded transitions between slopes of different gradients and direction.

3.10 TOLERANCES

- A. Grade soil surface to within 0.10-foot of grades indicated in the Drawings, except bring soil surface grades along headers, paving, curbs, and other structures to within 0.01-foot of grades indicated in the Drawings.
- B. Transition soil surface grades along paving, curbs and other structures to areas of less strict tolerance over a 5-foot distance.
- C. Fill all depressions and remove any rises or mounds in the overall plane of the slope. The tolerance for dips and bumps in shrub and ground cover planting areas shall be a 50mm (2-inch) deviation from the plane in 3-meters (10 feet). The tolerance for dips and bumps in lawn areas shall be a 25mm (1- inch) deviation from the plane in 3-meters (10 feet).

3.11 PROTECTION AND REPAIRS

- A. The Contractor shall take every precaution to ensure the integrity of the underdrainage, aeration and irrigation systems during and after soil placement. Any damage caused by the Contractor shall be repaired at no additional expense to the Owner.
- B. The Contractor shall be responsible to ensure that no soil disturbance will occur from construction traffic or other construction activities after placement of planting soil is complete. Disturbance shall be repaired by the Contractor at no additional expense to the Owner.

1. The Contractor shall place barricades to prevent compaction of planting soil from vehicles, equipment, or pedestrian traffic.
 2. Coordinate activities with other project contractors so that there is no soil disturbance from traffic or other construction activities subsequent to placement.
- C. Protect newly graded areas from traffic and erosion. Keep free of trash, debris or construction materials from other work.
- D. Repair and re-establish grades where completed or partially completed surfaces become eroded, rutted or compacted. Scarify, or, if directed by the SMITHSONIAN COTR, remove and replace soil to a depth as directed. Reshape and re-compact to the required density while soil is at a moisture content between permanent wilting point and field capacity.
- E. Where settling greater than 50mm (2-inches) occurs, before final acceptance or during the warranty period, remove finish surfacing, backfill with additional approved material, compact to specified rates, and restore any disturbed areas to a condition acceptable to the SMITHSONIAN COTR.
1. Repaired or restored areas shall follow the same procedures as specified for installation of new Soil Mixes.
- F. Any soil that becomes compacted to a density greater than 85 proctor density and/or the density in the approved mockup shall be dug up and reinstalled. This requirement includes compaction caused by other sub-contractors after the Soil Mix (Planting Soil) is installed and approved.
- G. Surface tilling shall not be considered adequate to reduce over compaction at levels 150mm (6 inches) or greater below finished grade. See **Section 3.7.A.4** for remediating over-compaction.

3.12 FINAL ACCEPTANCE / SOIL SETTLEMENT

- A. Upon written notice from the Contractor, the SMITHSONIAN COTR shall review the work and make a determination if the work is substantially complete.
- B. The date of substantial completion of the planting soil shall be the date when the SMITHSONIAN COTR accepts that all work in Planting, Planting Soil, and Irrigation installation sections is complete.
- C. Aesthetic Acceptance of Grades:
1. Upon completion of finish grading Work, schedule with the SMITHSONIAN COTR a review to obtain aesthetic acceptance.
 2. Provide three (3) days advance written notification.
 3. Do not commence planting or sodding Work until receiving aesthetic acceptance.
- D. At the end of the plant warranty and maintenance period, (see Specification Section 00 0000 -- Planting) the SMITHSONIAN COTR shall observe the soil installation work and establish that all provisions of the contract are complete, and the work is satisfactory.

1. Restore any soil settlement and or erosion areas to the grades shown in the Drawings. When restoring soil grades remove plants and mulch and add soil before restoring the planting. Do not add soil over the root balls of plants or on top of mulch.
- E. Failure to pass acceptance: If the work fails to pass final acceptance, any subsequent observations must be rescheduled as per above. The cost to the Owner for additional observations will be charged to the Contractor at the prevailing hourly rate of the SMITHSONIAN COTR.

3.13 EXCESS MATERIALS

- A. Excess Planting Soil: Remove the excess planting soil mixture and materials from the project area at no additional cost to the Owner.

END OF SECTION 32 91 00

SECTION 32 92 00 - LAWN AND GRASSES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Sodding.

- B. Related Sections:

- 1. Division 31 Section "Site Clearing" for removal limits of trees, shrubs, and other plantings affected by new construction.
 - 2. Division 31 Section "Earth Moving" for building and utility trench excavation, backfilling, compacting and grading requirements, and soil materials.
 - 3. Division 32 Section "Exterior Plants" for tree and shrub planting, and tree support systems.
 - 4. Division 32 Section 32 9100 "Planting Soil" for soil materials.

1.3 DESCRIPTION

- A. Sodding: This work shall consist of soil preparation, addition of soil amendments (as determined by the soil test results), watering, and placing grass sod on prepared areas, as specified in the Contract Documents or as directed by the Smithsonian or Smithsonian's representative.

1.4 DEFINITIONS

- A. Duff Layer: The surface layer of native topsoil that is composed of mostly decayed leaves, twigs, and detritus.
- B. Finish Grade: Elevation of finished surface of planting soil.
- C. Pesticide: A substance or mixture intended for preventing, destroying, repelling, or mitigating a pest. This includes insecticides, miticides, herbicides, fungicides, rodenticides, and molluscicides. It also includes substances or mixtures intended for use as a plant regulator, defoliant, or desiccant.
- D. Pests: Living organisms that occur where they are not desired or that cause damage to plants, animals, or people. These include insects, mites, grubs, mollusks (snails and slugs), rodents (gophers, moles, and mice), unwanted plants (weeds), fungi, bacteria, and viruses.

- E. Planting Soil: Material consisting of standardized topsoil; existing, native surface topsoil; existing, in-place surface soil; imported topsoil; or, manufactured topsoil; which is modified with soil amendments and perhaps fertilizers to produce a soil mixture best for plant growth.
- F. Subgrade: Surface or elevation of subsoil remaining after excavation is complete, or top surface of a fill or backfill before planting soil is placed.
- G. Subsoil: All soil beneath the topsoil layer of the soil profile and typified by the lack of organic matter and soil organisms.
- H. Surface Soil: Whatever soil is present at the top layer of the existing soil profile at the Project site. In undisturbed areas, the surface soil is typically topsoil, but in disturbed areas such as urban environments, the surface soil can be subsoil.

1.5 SUBMITTALS

- A. Product Data: For each type of product indicated.
 - 1. Pesticides and Herbicides: Include product label and manufacturer's application instructions specific to this Project.
- B. Submit certification of identification for the following:
 - 1. Source and name and telephone number of supplier.
 - 2. Certification of each seed mixture for turfgrass sod.
- C. Qualification Data: For qualified landscape Installer.
- D. Product certificates: For soil amendments and fertilizers, from manufacturer.
- E. Soils tests, analysis and recommendations: See requirements in section 32 1900 Planting Soils.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications: A qualified landscape Installer whose work has resulted in successful turf and meadow establishment.
 - 1. Professional Membership: Installer shall be a member in good standing of either the Professional Landcare Network or the American Nursery and Landscape Association.
 - 2. Experience: Five years experience in turf installation in addition to requirements in Division 1 Section "Quality Requirements."
 - 3. Installer's Field Supervision: Require Installer to maintain an experienced full-time supervisor on Project site when work is in progress.
 - 4. Personnel Certifications: Installer's field supervisor shall have certification in one of the following categories from the Professional Landcare Network:
 - a. Landscape Industry Certified Technician – Exterior.
 - b. Landscape Industry Certified Lawn Care Technician.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Packaged Materials: Deliver packaged materials in original, unopened containers showing weight, certified analysis, name and address of manufacturer, and indication of conformance with state and federal laws, as applicable. Store packaged materials in a manner that will prevent damage or intrusion of foreign matter. Any material that becomes contaminated will be removed from the job site.
- B. Sod: Harvest, deliver, store, and handle sod according to requirements in "Specifications for Turfgrass Sod Materials" and "Specifications for Turfgrass Sod Transplanting and Installation" in TPI's "Guideline Specifications to Turfgrass Sodding." Deliver sod in time for planting within 24 hours of harvesting. Protect sod from breakage and drying.
- C. Bulk Materials:
 - 1. Do not dump or store bulk materials near structures, utilities, walkways and pavements, or on existing turf areas or plants.
 - 2. Provide erosion-control measures to prevent erosion or displacement of bulk materials, discharge of soil-bearing water runoff, and airborne dust reaching adjacent properties, water conveyance systems, or walkways.
 - 3. Accompany each delivery of bulk fertilizers, lime and soil amendments with appropriate certificates.

1.8 PROJECT CONDITIONS

- A. Planting Restrictions: Plant during one of the following periods. Coordinate planting periods with initial maintenance periods to provide required maintenance from date of planting completion.
 - 1. Sod
 - a. Spring Planting: April 01 to June 15
 - b. Fall Planting: August 21 to November 01
- B. Weather Limitations: Proceed with planting only when existing and forecasted weather conditions permit planting to be performed when beneficial and optimum results may be obtained. Apply products during favorable weather conditions according to manufacturer's written instructions,

1.9 WARRANTY

- A. Special Warranty: Installer agrees to repair or replace plantings and accessories that fail in materials, workmanship, or growth within the specified warranty period.
 - 1. Failures include, but are not limited to, the following:
 - a. Death and unsatisfactory growth, except for defects resulting from abuse, lack of adequate maintenance, or neglect by Smithsonian, or incidents that are beyond Contractor's control.

- b. Structural failures including plantings falling or blowing over.
- 2. Warranty Periods from Date of Planting Completion:
 - a. Sodded Turf: 12 months, or completion of a single growing season, whichever is greater.

PART 2 - - PRODUCTS

2.1 TURFGRASS SOD

- A. Turfgrass Sod: Maryland Certified (labeled), inspected and approved by the Maryland Department of Agriculture, complying with Specifications for Turfgrass Sod Materials" in TPI's "Guideline Specifications to Turfgrass Sodding." Furnish viable sod of uniform density, color, and texture, strongly rooted, and capable of vigorous growth and development when planted. Sod shall be harvested during periods of active growth and irrigation. Dormant sod will not be acceptable.
- B. Turfgrass Species:
 - 1. Sod shall be a 90/10 mix with 90% of the mix consisting of a blend of at least three turf type tall fescues, and at least 10% Bluegrass. Approved varieties shall be selected "Recommended Turfgrass cultivars for Professional Seed Mixtures," University of Maryland Turfgrass Technical Update TT-77 most recent edition. A copy of this publication can be obtained by visiting the Maryland Turfgrass Council website <http://www.mdturf council.org> or by calling them at 410-836-2876.
 - a. Thickness of Cut: The thickness of the roots and soil should be ½" (12.7 mm) to ¾" (19.05 mm).
 - b. Pad Size: Individual pieces of sod shall be cut to the supplier's standard width and length. Maximum allowable deviation from standard widths and lengths shall be 5%. Broken pads and torn or uneven ends will not be acceptable.
 - c. Strength of Sod Sections: Under ideal conditions, standard size sections of sod shall be strong enough to support their own weight and retain their size and shape when suspended vertically from a firm grasp on the upper 10% of the section **without the use of netting**. Sod shall be a minimum of one year old.
 - d. Sod Viability: Sod shall not be harvested or transplanted under drought conditions.
- C. Subject to compliance with requirements, acceptable sources include but are not limited to:
 - 1. Oakwood Sod Farm,
Inc. 29307 Waller
Road Delmar, MD
21875 Phone: 410-
896-4009 Toll Free:
800-379-8488

2. Collins Wharf Sod
25361 Collins Wharf
Road Eden, MD 21822
Phone: 410-334-6676
Fax: 410-749-3815
cwsod@collinswharfsod.com
3. Summit Hall Sod Farm
21300 River Road
Poolesville, MD 20837-
9114 Phone: 301-948-
2900
Fax: 301-349-2668

2.2 INORGANIC SOIL AMENDMENTS

- A. Lime: ASTM C 602, agricultural liming material containing a minimum of 80 percent calcium carbonate equivalent and as follows:
 1. Class: O, with a minimum of 95 percent passing through No. 8 sieve and a minimum of 55 percent passing through No. 60 sieve.
 2. Provide lime in form of ground dolomitic limestone.
- B. Sulfur: Granular, biodegradable, containing a minimum of 90 percent sulfur, and with a minimum of 99 percent passing through No. 6 sieve and a maximum of 10 percent passing through No. 40 sieve.
- C. Iron Sulfate: Granulated ferrous sulfate containing a minimum of 20 percent iron and 10 percent sulfur.
- D. Aluminum Sulfate: Commercial grade, unadulterated.
- E. Perlite: Horticultural perlite, soil amendment grade.
- F. Agricultural Gypsum: Minimum 90 percent calcium sulfate, finely ground with 90 percent passing through No. 50 sieve.
- G. Sand: Clean, washed, natural or manufactured, and free of toxic materials.
- H. Diatomaceous Earth: Calcined, 90 percent silica, with approximately 140 percent water absorption capacity by weight.
- I. Zeolites: Mineral clinoptilolite with at least 60 percent water absorption by weight.

2.3 ORGANIC SOIL AMENDMENTS

- A. Compost: Well-composted, stable, and weed-free organic matter, pH range of 5.5 to 8; moisture content less than or equal to 30 percent by weight; 98 percent passing through 1-inch (25.4 mm)

sieve; soluble salt content of less than 5 decisiemens/m; not exceeding 0.5 percent inert contaminants and free of substances toxic to plantings.

- B. Wood Derivatives: Decomposed, nitrogen-treated sawdust, ground bark, or wood waste; of uniform texture and free of chips, stones, sticks, soil, or toxic materials.
- C. Manure: Well-rotted, unleached, stable or cattle manure containing not more than 25 percent by volume of straw, sawdust, or other bedding materials; free of toxic substances, stones, sticks, soil, weed seed, and material harmful to plant growth.

2.4 FERTILIZERS

- A. Bonemeal: Commercial, raw or steamed, finely ground; a minimum of 4 percent nitrogen and 10 percent phosphoric acid.
- B. Superphosphate: Commercial, phosphate mixture, soluble; a minimum of 20 percent available phosphoric acid.
- C. Commercial Fertilizer: Commercial-grade complete fertilizer of neutral character, consisting of fast- and slow-release nitrogen, 50 percent derived from natural organic sources of urea formaldehyde, phosphorous, and potassium in the following composition:
 - 1. Composition: 1 lb/1000 sq. ft. (0.45 kg/92.9 m²) of actual nitrogen, 4 percent phosphorous, and 2 percent potassium, by weight.
- D. Slow-Release Fertilizer: Granular or pelleted fertilizer consisting of 50 percent water-insoluble nitrogen, phosphorus, and potassium in the following composition:
 - 1. Composition: 20 percent nitrogen, 10 percent phosphorous, and 10 percent potassium, by weight.

2.5 SUBSOILS

- A. See requirements for Subsoils in specification 32 9100 Planting Soil.

2.6 PLANTING SOILS

- A. See requirements for Lawn Soil in specification 32 9100 Planting Soil.

2.7 PESTICIDES AND HERBICIDES

- A. General: Do not use pesticides and herbicides unless authorized by Smithsonian COTR.

PART 3 - EXECUTION

3.1 SOIL TESTING

- A. See requirements for Soil Testing in specification 32 9100 Planting Soil

3.2 EXAMINATION

- A. Examine areas to be planted for compliance with requirements and other conditions affecting performance.
 - 1. Verify that no foreign or deleterious material or liquid such as paint, paint washout, concrete slurry, concrete layers or chunks, cement, plaster, oils, gasoline, diesel fuel, paint thinner, turpentine, tar, roofing compound, or acid has been deposited in soil within a planting area.
 - 2. All areas to receive planting soil shall be free of construction debris, refuse, compressible or decayable materials, stones greater than two inches (50.8 mm) and standing water.
 - 3. Do not mix or place soils and soil amendments in frozen, excessively dry, wet, or muddy conditions.
 - 4. Suspend soil spreading, grading, and tilling operations during periods of excessive soil moisture until the moisture content reaches acceptable levels to attain the required results.
 - 5. Uniformly moisten excessively dry soil that is not workable and which is too dusty.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.
- C. If contamination by foreign or deleterious material or liquid is present in soil within a planting area, remove the soil and contamination as directed by Smithsonian COTR and replace with new planting soil.

3.3 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities, trees, shrubs, and plantings from damage caused by planting operations.
 - 1. Protect grade stakes set by others until directed to remove them.
- B. Install erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.

3.4 TURF AREA PREPARATION

- A. Limit turf subgrade preparation to areas to be planted.
- B. Subsoil areas which are intended to receive topsoil shall be maintained on a true and even grade, in compliance with plans, drawings or other applicable specifications pertaining to the site, with no significant depressions or elevations. Topsoil shall not be deposited or spread over

the subsoil until rough grading has been completed and all areas within the subgrade are within 2/10 of 1.0 foot (6/100 of 1 m) from the final subgrade. If the graded area develops weed or other plant growth, the plants shall be eliminated before topsoil is deposited or spread over the subsoil.

- C. For Newly Graded Subgrades refer to Smithsonian Gardens Planting Soil Specification 329100.
- D. Unchanged Subgrades Inside Tree Critical Root Zones: See Smithsonian Gardens Site Protection Specification 015639.
- E. Unchanged Subgrades Outside Tree Critical Root Zones: If turf is to be planted in areas unaltered or undisturbed by excavating, grading, or surface-soil stripping operations, prepare surface soil as follows:
 - 1. Remove existing grass, vegetation, and turf. Do not mix into surface soil.
 - 2. Loosen surface soil to a depth of at least 6 inches (152.4 mm). Apply soil amendments and fertilizers according to planting soil mix proportions and mix thoroughly into top 4 inches (101.6 mm) of soil. Till soil to a homogeneous mixture of fine texture.
 - a. Apply fertilizer directly to surface soil before loosening.
 - 3. Remove stones larger than ¾ inch (19.05 mm) in any dimension and sticks, roots, trash, brush, wire, grade stakes and other debris that would interfere with sodding.
 - 4. Legally dispose of waste material, including grass, vegetation, and turf, off Smithsonian's property.
- F. Finish Grading: Grade planting areas to a smooth, uniform surface plane with loose, uniformly fine texture. Grade to within plus or minus 1/2 inch (12.7 mm) of finish elevation. Roll and rake, remove ridges, and fill depressions to meet finish grades. Limit finish grading to areas that can be planted in the immediate future.
- G. Planting may be done immediately thereafter, provided the topsoil has remained in a good, friable condition and has not become muddy or hard. If it has become hard, it shall be tilled to a friable condition again before acceptance. Moisten prepared area before planting if soil is dry. Water thoroughly and allow surface to dry before planting. Do not create muddy soil.
- H. Before planting, obtain Smithsonian's COTR acceptance of finish grading; restore planting areas if eroded or otherwise disturbed after finish grading.

3.5 SODDING

- A. Consult with Smithsonian COTR regarding sodding boundaries. Sod layout shall utilize full panels of sod to create complete coverage of area to be sodded.
- B. Lay sod within 24 hours of harvesting. Do not lay sod if dormant or if ground is frozen or muddy.
- C. Lay sod to form a solid mass with tightly fitted joints. Butt ends and sides of sod; do not stretch or overlap. Stagger sod strips or pads to offset joints in adjacent courses. Avoid damage to subgrade or sod during installation. Tamp and roll lightly to ensure contact with subgrade,

eliminate air pockets, and form a smooth surface. Work sifted soil or fine sand into minor cracks between pieces of sod; remove excess to avoid smothering sod and adjacent grass.

1. Do not install sod on slopes greater than 3:1. Consult with Smithsonian COTR if any slopes exceed 3:1.
 2. Anchor sod on slopes exceeding 1:6 with biodegradable staples spaced as recommended by sod manufacturer but not less than 2 anchors per sod strip to prevent slippage.
- D. Saturate sod with fine water spray within two hours of planting. During first two weeks after planting, water daily or more frequently as necessary to maintain moist soil to a minimum depth of 1-1/2 inches (38.1 mm) below sod.

3.6 SATISFACTORY TURF

- A. Turf installations shall meet the following criteria as determined by Smithsonian COTR:
1. Satisfactory Sodded Turf: At end of maintenance period, a healthy, well-rooted, even-colored, viable turf has been established, free of weeds, open joints, bare areas, and surface irregularities.
- B. Use specified materials to reestablish turf that does not comply with requirements and continue maintenance until turf is satisfactory.
- C. Turf Grass Conditions will not be accepted if there is a poor or thin stand; improper application of sod, dead grass or sod, use of sod other than approved in specifications, improper fertilizer application either uneven spreading or insufficient amounts, or failure to re-fertilize during extended acceptance, and the presence of persistent weeds established in turf areas.

3.7 CLEANUP AND PROTECTION

- A. Promptly remove soil and debris created by turf work from paved areas. Clean wheels of vehicles before leaving site to avoid tracking soil onto roads, walks, or other paved areas.
- B. Remove nondegradable erosion-control measures after grass establishment period.

3.8 TIME RESTRICTIONS

- A. When permanent sodding is specified or directed, and is not allowed because of time restrictions specified above, utilize one or more of following methods to prevent erosion and sedimentation until permanent sodding is allowed.
1. Prepare soil as for permanent sodding.
 2. Apply temporary seeding, straw mulch, or shredded hardwood mulch in consultation with the Smithsonian COTR.
 3. Provide other erosion control measures acceptable to Engineer.

4. Remove straw or wood chips used as temporary mulch before initiation of permanent sodding application.

END OF SECTION 32 92 00

SECTION 32 93 00 - PLANTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.
- B. Section Includes:
 - 1. Trees.
 - 2. Shrubs.
 - 3. Tree stabilization.
 - 4. Fertilizing.
- C. Related Sections:
 - 1. Division 31 "Earthwork" for excavation, filling, and rough grading and for subsurface aggregate drainage and drainage backfill materials.
 - 2. Division 33 Sections "Storm Utility Drainage Piping" and "Subdrainage" for below-grade drainage of landscaped areas, paved areas, and wall perimeters.
 - 3. Division 32 Section 329200 "Grasses" for seeding and Section "Soil Preparation" for planting soils.

1.2 UNIT PRICES – NOT USED

1.3 DEFINITIONS

- A. Backfill: The earth used to replace or the act of replacing earth in an excavation.
- B. Balled and Burlapped Stock: Exterior plants dug with firm, natural balls of earth in which they are grown, with ball size not less than diameter and depth recommended by ANSI Z60.1 for type and size of tree or shrub required, wrapped, tied, rigidly supported, and drum laced as recommended by ANSI Z60.1.
- C. Balled and Potted Stock: Plants dug with firm, natural balls of earth in which they are grown and placed, unbroken, in a container. Ball size is not less than diameter and depth recommended by ANSI Z60.1 for type and size of plant required.
- D. Container-Grown Stock: Healthy, vigorous, well-rooted exterior plants grown in a container with well-established root system reaching sides of container and maintaining a firm ball when removed from container. Container shall be rigid enough to hold ball shape and protect root mass during shipping and be sized according to ANSI Z60.1 for type and size of exterior plant required.

- E. Finish/Finished Grade: Elevation of finished surface of planting soil.
- F. Manufactured Topsoil: Soil produced off-site by homogeneously blending mineral soils or sand with stabilized organic soil amendments to produce topsoil or planting soil.
- G. Multi-Stem: Where three or more main stems arise from the ground from a single root crown or at a point right above the root crown.
- H. Onsite topsoil: Topsoil removed from excavations onsite, stored, and reused for planting soil.
- I. Planting Soil: Native or imported topsoil, manufactured topsoil, or surface soil modified to become topsoil; mixed with soil amendments.
- J. Permanent Seeding: Final, desired construction conditions of areas designated as Native Seeding and Meadows. This planting is distinct from the Temporary Seeding which is constituted from annual grasses required for Erosion and Sedimentation Control Plans, as indicated on the Civil Engineering Drawings. Distinct from the annual grasses, the Permanent Seeding is composed of perennial grasses that will maintain permanent vegetative cover.
- K. Pesticide: A substance or mixture intended for preventing, destroying, repelling, or mitigating a pest. Pesticides include insecticides, miticides, herbicides, fungicides, rodenticides, and molluscicides. They also include substances or mixtures intended for use as a plant regulator, defoliant, or desiccant. Some sources classify herbicides separately from pesticides.
- L. Pests: Living organisms that occur where they are not desired or that cause damage to plants, animals, or people. Pests include insects, mites, grubs, mollusks (snails and slugs), rodents (gophers, moles, and mice), unwanted plants (weeds), fungi, bacteria, and viruses.
- M. Planting Area: Areas to be planted.
- N. Root Flare: Also called "trunk flare." The area at the base of the plant's stem or trunk where the stem or trunk broadens to form roots; the area of transition between the root system and the stem or trunk.
- O. Seed Mix: Specified mix of specific species of native grasses and herbs in seed form.
- P. Stem Girdling Roots: Roots that encircle the stems (trunks) of trees below the soil surface.
- Q. Subgrade: Surface or elevation of subsoil remaining after completing excavation, or top surface of a fill or backfill, before placing planting soil.
- R. Subsoil: All soil beneath the topsoil layer of the soil profile, and typified by the lack of organic matter and soil organisms.
- S. Topsoil: As defined by 329113 Soil Preparation.
- T. Temporary Seeding: Constituted from annual (and not permanent, perennial) grasses required for Erosion and Sedimentation Control Plans, as indicated on the Civil Engineering Drawings.

- U. Tree Stabilizing Elements: Materials, elements, and systems used to keep the tree or large shrub upright, plumb, and stable.

1.4 SUBMITTALS

- A. Qualification Data: For qualified landscape Installer, due within 30 days after signing of contract between Contractor and Installer and no less than ninety (90) days before planting. Include the following:
 - 1. Verification of successful establishment of exterior plants in commercial and/or substantial properties over a minimum of 10-year period.
 - 2. List of at least 10 exterior planting, noting project location, project size, project budget, photos, and owner contact information.
- B. Informational: For each schedule listed, due no fewer than one hundred-eighty (180) days before planting. Include the following:
 - 1. Replication of Plant Schedules indicating not only all categories listed on Drawings but also verified quantities and sources for each plant species. Quantities of trees when noted on the plant list are based upon the graphic symbols shown on the drawings. Plant Schedule quantities are included as a convenience only. Contractor is responsible for correct totals, and quantities on Drawings take precedence.
 - a. Contractor shall provide and install all trees based upon the graphic symbol quantities.
 - b. If a difference is found between the tree quantities noted on the plant list and the symbol count of trees shown on the drawings, notify the Owner a minimum of 3 days prior to the bid date.
 - 2. Schedule of Seeding quantities based upon the Seeding Rate noted in the Seeding Schedule and area takeoffs based upon the Landscape Plans.
 - 3. Schedule of Substitutions: Written requests for plant substitutions and documentation of suitability of substitution, what species it is replacing, and its compliance with the design intent of the project. Landscape Architect may reject the request for substitution. Approved substitutions shall be installed at no additional cost to the Owner.
 - 4. Plant Procurement Plan: List of locations and contact information of nurseries/sources of plants and verification that Contractor has secured acquisition of plants for appropriate planting period.
 - 5. Planting Timetable: Indicating anticipated planting dates for exterior plants, including preliminary staking schedule and dates on which Landscape Architect is requested to approve staking.
 - 6. (If applicable) Construction plan and schedule showing location(s) for storage of trees to be relocated and dates proposed for digging and replanting of trees to be relocated.
- C. Product Data and Samples: For each type of product indicated, due no fewer than ninety (90) days before planting. Include the following:

1. Verification of species variety for those plants specifically identified as requiring this verification as directed in the Plant Schedule's Notes. This provision is intended but not limited to those species whose requirements include dwarf varieties, disease resistance, and/or requirement of non-fruiting plants.
 2. List of plants that have particular planting needs and/or schedules that differ from the general norms of most plants, including notes indicating how these items will be delivered, installed and cared for.
 3. Fertilizer product data, method of application, and schedule of application—for each type of fertilizer and/or plant type (deciduous plants, evergreen plants, lawns, etc.).
 4. Product literature for slow release watering device, if applicable.
 5. Product literature for Tree Stabilizing Elements.
 6. Certification from qualified independent laboratory for Mulch for each different type to be applied on the project, including moisture content, pH range, sieve standards, soluble salt content, and color.
 7. Mulch sample for each type to be applied on the project: Three (3) quart-sized samples randomly selected from vendor.
 8. (If applicable) Product literature and documentation that Weed Barrier product is capable of:
 - a. Permeability rating of ten (10) to fifteen (15) gal/min/sq.ft. per ASTM D-4491
 - b. Fitness rating of at least a fifteen (15) to twenty (20) years.
- D. Product Certificates: To be filed with Contractor within five (5) business days of delivery, planting, or installation, for each type of manufactured product, from manufacturer, and complying with the following:
1. Delivery manifests for each load of plants delivered.
 2. Labels from seed bags for seed planted.
 3. Manufacturer's certified analysis for standard products.
 4. Analysis of other materials by a recognized laboratory made according to methods established by the Association of Official Analytical Chemists, where applicable.
- E. Supervision Certifications: Due no fewer than thirty (30) days before planting:
1. Name and phone number of Installer's Field Supervisor for this job, including certification by the Installer that they understand the requirement that Field Supervisor shall be on the Project site at all times that tasks relating to exterior planting or landscaping are in progress.
- F. Closeout Submittals:
1. Maintenance Plan, due within ninety (90) days of planting: Written, organized, and complete recommended procedures to be established by Owner for maintenance of exterior plants during a calendar year and procedures for grasses (meadow) for three (3) years. Must include all areas, plant/grass types, and tasks, a timetable for each task, weeding, herbiciding, fertilizing, pruning, watering, cleanup, mulching, mowing/burning, and any species-specific recommendations.

1.5 COORDINATION

- A. Coordination with Turf and Grasses: Plant trees, shrubs, and other plants after finish grades are established and before planting seeded areas unless otherwise indicated.
 - 1. (If applicable) When planting trees, shrubs, and other plants after planting turf areas, protect turf areas, and promptly repair damage caused by planting operations.

1.6 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at project site with Landscape Architect per coordination with COTR.

1.7 QUALITY ASSURANCE

- A. Installer Qualifications: A qualified landscape installer whose work has resulted in successful establishment of exterior plants, including both upland plants and wetland planting.
 - 1. Installer's Field Supervision: Require Installer to maintain an experienced full-time supervisor on Project site when planting is being delivered and when planting is in progress.
 - 2. Employee Training: Require Installer to employ workers trained in Native Grasses and Meadow planting/maintenance.
- B. Provide quality, size, genus, species, and variety of exterior plants indicated, complying with applicable requirements in ANSI Z60.1, "American Standard for Nursery Stock."
- C. Tree and Shrub Measurements: Measure according to ANSI Z60.1 with branches and trunks or canes in their normal position. Do not prune to obtain required sizes. Take caliper measurements 6 inches (150 mm) above the ground for trees up to 4-inch (100-mm) caliper size, and 12 inches (300 mm) above the ground for larger sizes. Measure main body of tree or shrub for height and spread; do not measure branches or roots tip-to-tip.
- D. Observation: Landscape Architect may observe trees and shrubs at site before planting for compliance with requirements for genus, species, variety, size, and quality. Landscape Architect retains right to observe trees and shrubs further for size and condition of balls and root systems, insects, injuries, and latent defects and to reject unsatisfactory or defective material at any time during progress of work. Remove rejected trees or shrubs immediately from Project site.
- E. Staking: No planting may occur until plants have been staked in the field and approved by the Landscape Architect via the COTR. All plants will be field-located using the Plan as an initial guide. Final staking will be accomplished on-site under supervision of Landscape Architect via the COTR. All plants will be either staked or placed in their containers/B&B in preliminary locations and approved before any planting pits are dug and plants removed from containers/B&B.

- F. Suitable Plant Materials: Prior to mobilization, the Contractor shall notify the Landscape Architect, in writing, if he/she believes any of the plant material identified on the Plan may not be suitable for the site or may fail to thrive or die. Substitution requests will be granted by the Landscape Architect prior to the start of construction activities. If notification is not given to the Landscape Architect, all planting that fails to thrive (except for defects resulting from lack of adequate maintenance as determined by the Owner as vandalism) shall be replaced by the Contractor at the Contractor's expense.
- G. Notification: Contractor shall notify the Landscape Architect in writing prior to planting when conditions detrimental to plant growth are encountered, such as rubble fill, poor planting soil, adverse drainage conditions, or obstructions. Failure to notify prior to planting will result in the responsibility/liability remaining with the Installer should plants fail to thrive per Warranty section following.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver exterior plants that are in balled and burlap form shall be freshly dug.
- B. Do not prune trees and shrubs before delivery except as approved by Landscape Architect. Protect bark, branches, and root systems from sun scald, drying, sweating, whipping, and other handling and tying damage. Do not bend or bind-tie trees or shrubs in such a manner as to destroy their natural shape. Provide protective covering of exterior plants during delivery. Do not drop exterior plants during delivery and handling.
- C. Handle planting stock by root ball.
- D. Do not bend or bind-tie trees in such a manner as to destroy natural shape of the trees.
- E. Provide protective covering to the trees during transport and delivery.
- F. Do not drop trees and shrubs at any time from digging through installation. Trees with damaged balls shall be rejected.
- G. Apply antidesiccant or antitranspirant to trees using a power spray to provide an adequate film over trunks, branches, stems, twigs, and foliage to protect them during digging, handling, transporting and immediately after planting.
 - 1. When deciduous trees are moved in leaf, spray with anti-desiccant or anti-transpirant at the nursery before digging or moving.
- H. Deliver exterior plants after preparations for planting have been completed and install immediately. If planting is delayed more than six hours after delivery, set exterior plants and trees in shade, protect from weather and mechanical damage, and keep roots moist.
 - 1. Set balled stock on ground and cover ball with soil, sawdust, or other acceptable material.
 - 2. Do not remove container-grown stock from containers before time of planting.
 - 3. Water root systems of exterior plants stored on-site with a fine-mist spray. Water as often as necessary to maintain root systems in a moist condition.

1.9 PROJECT CONDITIONS

- A. Field Measurements: Verify (as applicable) actual grade elevations, service and utility locations, irrigation system components, and dimensions of plantings and construction contiguous with new plantings by field measurements before proceeding with planting work.
- B. Planting Restrictions: Plant during periods that coordinate with construction schedule and appropriate to weather and climate conditions. In areas where planting will occur, do not undertake any activity, including planting, until soil has drained. Coordinate planting periods with maintenance periods to provide required maintenance from date of Substantial Completion.
- C. Schedule Restrictions: Plant during one of the following periods. Coordinate planting periods with maintenance periods to provide required maintenance from date of Substantial Completion.
 - 1. Woody Planting: a) Between April 15 and October 15 but not after first frost; and b) Within recommended planting period for specific species that may have more particular requirements than most plants.
 - 2. All trees dug in summer full foliage shall be handled in one of the following ways:
 - a. Seven to ten (7-10) days prior to digging by a gradual process of trenching and soaking, leaving the bottom surface of the fall attached to the soil until shipping.
 - b. Condition under irrigation in partial shade for at least four (4) days prior to shipping.
- D. Weather Limitations: Proceed with planting only when existing and forecasted weather conditions permit planting to be performed according to manufacturer's written instructions and warranty requirements.

1.10 WARRANTY

- A. Special Warranty: Installer's standard form in which Installer agrees to repair or replace plantings and accessories that fail in materials, workmanship, or growth within specified warranty period.
 - 1. Failures include, but are not limited to, the following:
 - a. Death and unsatisfactory growth, except for defects resulting from lack of adequate maintenance, neglect, abuse by Owner, or incidents that are beyond Contractor's control.
 - b. Structural failures including plantings falling or blowing over.
 - c. Faulty operation of tree stabilization and edgings.
 - d. Deterioration of metals, metal finishes, and other materials beyond normal weathering.
 - e. Contractor error in Execution.
 - f. Contractor failure to protect adjacent plants during herbicide treatments.
 - g. Plants that are growing from a deviant form from that specified, particularly cultivars that are not conforming to the standard of the variation.
 - 2. Warranty Periods from Date of Substantial Completion:

- a. Trees, Shrubs, Ground Cover, and Vines: Two years.
3. For Trees, Shrubs, Ground Cover, and Vines, include the following remedial actions as a minimum:
 - a. Remove dead exterior plants immediately. Replace immediately unless required to plant in the succeeding planting season.
 - b. Replace exterior plants that are more than 25 percent dead or in an unhealthy condition at end of warranty period.
 - c. Provide extended warranty for replaced plant materials; warranty period equal to original warranty period.
4. For Native Grasses and Meadows: Per Specification Section 329200 Grasses.

1.11 MAINTENANCE SERVICE

- A. Initial Maintenance Service for Trees and Shrubs: Provide full maintenance by skilled employees of landscape Installer. Maintain as required in Part 3. Begin maintenance immediately after each area is planted and continue until plantings are acceptably healthy and well established, but for not less than maintenance period below.
 1. Maintenance Period: 24 months from Substantial Completion.
 2. Contractor shall monitor the site on a quarterly basis in or on the following dates: January 15, April 15, July 15, and October 15 to assess the status of the plantings, watering, mulching, etc. and to avoid failure of revegetation efforts. Contractor shall remedy any minor conditions failures and correct any procedural problems. Contractor shall provide written documentation of plants that need to be placed. Copies shall be distributed to the Owner and Landscape Architect. Replacement plants shall be installed during appropriate planting windows as agreed upon between Installer and Landscape Architect.
- B. Continuing Maintenance Proposal: From Installer to Owner, in the form of a standard yearly (or other period) maintenance agreement, starting on date initial maintenance service is concluded. State services, obligations, conditions, and terms for agreement period and for future renewal options.

PART 2 - PRODUCTS

2.1 TREE AND SHRUB MATERIAL

- A. General: Furnish nursery-grown plants true to genus, species, variety, cultivar, stem form, shearing, and other features indicated in Plant List, Plant Schedule, or Plant Legend indicated on Drawings and complying with ANSI Z60.1; and with healthy root systems developed by transplanting or root pruning. Provide well-shaped, fully branched, healthy, vigorous stock, densely foliated when in leaf and free of disease, pests, eggs, larvae, and defects such as knots, sun scald, injuries, abrasions, and disfigurement.

1. Unacceptable trees include those with damaged, crooked, or multiple leaders; tight vertical branches where bark is squeezed between two branches or between branch and trunk ("included bark"); crossing trunks; cut-off limbs more than 3/4 inch (19 mm) in diameter; or with stem girdling roots.
 2. Trunk shall be free of unhealed branch removal wounds greater than one (1) inch in diameter.
 3. Collected Stock: Do not use plants harvested from the wild, from native stands, from an established landscape planting, or not grown in a nursery, unless otherwise indicated.
- B. Provide trees and shrubs of sizes, grades, and ball or container sizes complying with ANSI Z60.1 for types and form of trees and shrubs required. Trees and shrubs of a larger size may be used if acceptable to Landscape Architect, with a proportionate increase in size of roots or balls.
- C. Root-Ball Depth: Furnish trees and shrubs with root balls measured from top of root ball, which shall begin at root flare according to ANSI Z60.1.
- D. Provide plants from nurseries/sources of a similar climate and from no more than four (4) degree latitude or longitude difference. If proposed sources are from greater distances, submit for approval no less than 120 days in advance of planting.

2.2 SHADE AND FLOWERING TREES

1. Provide balled and burlapped or container-grown trees as appropriate.
 2. Branching Height: One-third to one-half of tree height or as noted on Plant Schedule, the latter being the overriding specification when present.
- B. Small Upright Trees: Branched or pruned naturally according to species and type, with relationship of caliper, height, and branching according to ANSI Z60.1; stem form as follows:
1. Stem Form: As noted in Plant Schedules.
 2. Provide balled and burlapped or container-grown trees as appropriate to the species.
- C. Small Spreading Trees: Branched or pruned naturally according to species and type, with relationship of caliper, height, and branching according to ANSI Z60.1; stem form as follows:
1. Stem Form: As noted on Planting Schedules.
 2. Provide balled and burlapped or container-grown trees as appropriate to the species.

2.3 DECIDUOUS SHRUBS

- A. Form and Size: Shrubs with not less than the minimum number of canes required by and measured according to ANSI Z60.1 for type, shape, and height of shrub.
1. Shrub sizes indicated are sizes after pruning.
 2. Provide balled and burlapped or container-grown trees as appropriate to the species.

2.4 CONIFEROUS EVERGREENS

- A. Form and Size: Normal-quality, well-balanced, coniferous evergreens, of type, height, spread, and shape required, complying with ANSI Z60.1.
- B. Form and Size: Specimen quality as described, symmetrically shaped coniferous evergreens.
 - 1. Shearing Designation: Never sheared.
 - 2. Provide balled and burlapped or container-grown trees as appropriate to the species.

2.5 BROADLEAF EVERGREENS

- A. Form and Size: Normal-quality, well-balanced, broadleaf evergreens, of type, height, spread, and shape required, complying with ANSI Z60.1.
- B. Form and Size: Specimen quality as described in Planting Schedules, symmetrically shaped broadleaf evergreens.
 - 1. Shearing Designation: Natural, never sheared.
 - 2. Provide balled and burlapped or container-grown trees as appropriate to the species.

2.6 MULCHES

- A. A. Organic Mulch: Suitable as a top dressing of trees and shrubs, well-composted, stable, and weed-free organic matter, pH range of 5.5 to 8; moisture content 35 to 55 percent by weight; 100 percent passing through 1-inch (25-mm) sieve; soluble salt content of 2 to 5 decisiemens/m; not exceeding 0.5 percent inert contaminants and free of substances toxic to plantings; and Organic Matter Content: 50 percent maximum dry weight. Consisting of one of the following:
 - 1. Use mulch generated by the project's clearing and grubbing activities and stored on site as noted on Construction Drawings or directed by the COTR.
 - 2. Double Shredded Hardwood Mulch: Double shredded hardwood mulch, free from deleterious materials and suitable as a top dress of plantings.
 - 3. For additional mulch, use ground or shredded bark, pine straw, or wood and bark chips appropriate to individual planting bed and plant species' needs.
 - 4. Mulch to be a natural brown to black mulch. Other colors are not permitted.
- B. Compost Mulch: Well-composted, stable, and weed-free organic matter, pH range of 5.5 to 8; moisture content 35 to 55 percent by weight; 100 percent passing through 1-inch (25-mm) sieve; soluble salt content of 2 to 5 decisiemens/m; not exceeding 0.5 percent inert contaminants and free of substances toxic to plantings; and as follows:
 - 1. Organic Matter Content: 50 percent maximum dry weight.
 - 2. Feedstock: Agricultural, food, or industrial residuals; biosolids; yard trimmings; or source-separated or compostable mixed solid waste.
- C. Stakes and Guys:

1. Upright and Guy Stakes: Rough-sawn, sound, new hardwood, redwood, or pressure-preservative-treated softwood, free of knots, holes, cross grain, and other defects, 2-by-2-inch nominal (38-by-38-mm actual) by length indicated, pointed at one end.
2. Flexible Ties: Wide rubber or elastic bands or
3. straps of length required to reach stakes or turnbuckles.
4. Guys and Tie Wires: ASTM A641/A641M, Class 1, galvanized-steel wire, 2-strand, twisted, 0.106 inch (2.7 mm) in diameter.
5. Hose Chafing Guards: Reinforced rubber or plastic hose at least 1/2 inch (13 mm) in diameter, black, cut to lengths required to protect tree trunks from damage.
6. Guy Cables: 5-strand, 3/16-inch- (4.8-mm-) diameter, galvanized-steel cable, with zinc-coated turnbuckles, a minimum of 3 inches (75 mm) long, with two 3/8-inch (10-mm) galvanized eyebolts.
7. Flags: Standard surveyor's plastic flagging tape, white, 6 inches (150 mm) long.

2.7 TREE-WATERING DEVICES

- A. Watering Pipe: PVC pipe 4 inches (100 mm) in diameter, site-cut to length as required, and with snug-fitting removable cap.
- B. Slow-Release Watering Device: Standard product manufactured for drip irrigation of plants and emptying its water contents over an extended time period, manufactured from UV-light-stabilized nylon-reinforced polyethylene sheet, PVC, or HDPE plastic. Color: dark chocolate, tan, green, or other color approved by the Landscape Architect through the submittals process.

2.8 WEED BARRIERS FOR TREES

- A. A. Nonwoven Fabric: Polypropylene or polyester fabric, three (3) oz. per sq. yd. minimum. Needle punched or otherwise treated for hydrolic effectiveness
- B. C. Composite Fabric: Woven, needle-punched Polypropylene substrate bonded to a nonwoven Polypropylene fabric, 4.8 oz. per sq. yd.

2.9 MISCELLANEOUS PRODUCTS

- A. Burlap: Non-synthetic, biodegradable.
- B. Antidesiccant: Water-insoluble emulsion, permeable moisture retarder, film forming, for trees and shrubs. Deliver in original, sealed, and fully labeled containers and mix according to manufacturer's written instructions.
- C. Trunk-Wrap Tape: Do not use.
- D. Planter Drainage Gravel: Washed, sound crushed stone or gravel complying with ASTM D 448 for Size No. 8.

- E. Weed Barrier: Woven or nonwoven geotextile manufactured for separation applications and made of polypropylene, polyolefin, or polyester fibers or combination of them.
 - 1. Permeability rating of ten (10) to fifteen (15) gal/min/sq.ft. per ASTM D4491
 - 2. Fitness rating of at least fifteen (15) to twenty (20) years.
- F. Mycorrhizal Fungi: Dry, granular inoculant containing at least 5300 spores per lb (0.45 kg) of vesicular-arbuscular mycorrhizal fungi and 95 million spores per lb (0.45 kg) of ectomycorrhizal fungi, 33 percent hydrogel, and a maximum of 5.5 percent inert material.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas to receive exterior plants for compliance with requirements and conditions affecting installation and performance.
 - 1. Verify that no foreign or deleterious material or liquid such as paint, paint washout, concrete slurry, concrete layers or chunks, cement, plaster, oils, gasoline, diesel fuel, paint thinner, turpentine, tar, roofing compound, or acid has been deposited in soil within a planting area.
 - 2. Verify that plants and vehicles loaded with plants can travel to planting locations with adequate overhead clearance.
 - 3. Suspend planting operations during periods of excessive soil moisture until the moisture content reaches acceptable levels to attain the required results. Uniformly moisten excessively dry soil that is not workable or which is dusty.
- B. Excavation: If contamination by foreign or deleterious material or liquid is present in soil within a planting area, remove the soil and contamination as directed by Landscape Architect via COTR and replace with new planting soil.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.
 - 1. Prior to beginning construction in each planting area, notify COTR and engage a private utility locating service at no cost to the Owner to locate, verify, and mark/flag the location of all underground utilities prior to planting operations beginning.
- D. Prior to planting, thoroughly remove weeds from planting areas.
- E. Maintain grade stakes until removal is mutually agreed upon by all parties concerned.

3.2 PREPARATION FOR PLANTING

- A. Protect structures, utilities, sidewalks, pavements, and other facilities, and lawns and existing exterior plants from damage caused by planting operations.
- B. Provide erosion-control measures per the Project's Erosion and Sediment Control Plan per the

- C. Remove all undesirable vegetation from planting areas through appropriate means: cutting, digging, herbicide, or other techniques.
- D. Layout exterior plants at locations directed by Landscape Architect. Stake locations of individual trees and shrubs and outline areas for multiple plantings. Do not plant until staking has been approved by Landscape Architect through COTR.
- E. Trunks: Inspect tree trunks for injury, improper pruning, and insect infestation. Reject if found. DO NOT WRAP TRUNKS WITH TREE TAPE.
- F. Apply antidessicant to trees and shrubs using power spray to provide an adequate film over trunks, branches, stems, twigs, and foliage to protect during digging, handling, and transportation.
 - 1. If deciduous trees or shrubs are moved in full leaf, spray with antidesiccant at nursery before moving and again two weeks after planting.
- G. Wrap trees and shrubs with burlap fabric over trunks, branches, stems, twigs, and foliage to protect from wind and other damage during digging, handling, and transportation.
- H. Determine soils needs per section 329113 Soil Preparation and execute soils modifications as necessary per Section Specifications.

3.3 PLANTING BED ESTABLISHMENT

- A. Spread planting soil mix to a depth of planting bed. Do not spread if planting soil or subgrade is frozen, muddy, or excessively wet.
- B. Finish Grading: Grade planting beds to a smooth, uniform surface plane with loose, uniformly fine texture. Roll and rake, remove ridges, and fill depressions to meet finish grades.
- C. Before planting, restore planting beds if eroded or otherwise disturbed after finish grading.

3.4 EXCAVATION FOR TREES AND SHRUBS

- A. Planting pits and excavated trenches: Excavate circular planting pits.
 - 1. Excavate approximately two times as wide as ball diameter for balled and burlapped and container-grown stock. Trim perimeter of bottom leaving center area of bottom raised slightly to support root ball and assist in drainage away from center. Do not further disturb base. Ensure that root ball will sit on undisturbed base soil to prevent settling. Scarify sides of planting pit smeared or smoothed during excavation.
 - 2. Excavate at least 12 inches (300 mm) wider than root spread and deep enough to accommodate vertical roots for bare-root stock.
 - 3. DO NOT EXCAVATE DEEPER THAN DEPTH OF THE ROOT BALL, measured from the root flare to the bottom of the root ball.
 - 4. If area under the plant was initially dug too deep, add soil to raise it to the correct level and thoroughly tamp the added soil to prevent settling.

5. Maintain angles of repose of adjacent materials to ensure stability. Do not excavate subgrades of adjacent paving, structures, hardscapes, or other new or existing improvements.
 6. Maintain supervision of excavations during working hours.
 7. Keep excavations covered or otherwise protected when unattended by Installer's personnel.
- B. Subsoil removed from excavations may be used as backfill if it is appropriate. Modify if necessary to support thriving plant(s).
- C. Obstructions: Notify Landscape Architect if unexpected rock or obstructions detrimental to trees or shrubs are encountered in excavations.
- D. Drainage: Notify Landscape Architect if subsoil conditions evidence unexpected water seepage or retention in tree or shrub pits.
- E. Fill excavations with water and allow to percolate away before positioning trees and shrubs.

3.5 TREE, SHRUB, AND VINE PLANTING

- A. Inspection: At time of planting, verify that root flare is visible at top of root ball according to ANSI Z60.1. If root flare is not visible, remove soil in a level manner from the root ball to where the top-most root emerges from the trunk. After soil removal to expose the root flare, verify that root ball still meets size requirements.
- B. Roots: Remove stem girdling roots and kinked roots. Remove injured roots by cutting cleanly; do not break.
- C. Balled and Burlapped Stock: Set each plant plumb and in center of planting pit or trench with root flare 1 inch (25 mm) above adjacent finished grades.
1. Backfill: Planting soil.
 2. After placing some backfill around root ball to stabilize plant, carefully cut and remove burlap, rope, and wire baskets from tops of root balls and from sides, but do not remove from under root balls. Remove pallets, if any, before setting. Do not use planting stock if root ball is cracked or broken before or during planting operation.
 3. Backfill around root ball in layers, tamping to settle soil and eliminate voids and air pockets. When planting pit is approximately one-half filled, water thoroughly before placing remainder of backfill. Repeat watering until no more water is absorbed.
 4. Continue backfilling process. Water again after placing and tamping final layer of soil.
- D. Balled and Potted and Container-Grown Stock: Set each plant plumb and in center of planting pit or trench with root flare 1 inch (25 mm) above adjacent finished grades.
1. Backfill: Planting soil.
 2. Carefully remove root ball from container without damaging root ball or plant.
 3. Backfill around root ball in layers, tamping to settle soil and eliminate voids and air pockets. When planting pit is approximately one-half filled, water thoroughly before placing remainder of backfill. Repeat watering until no more water is absorbed.

4. Continue backfilling process. Water again after placing and tamping final layer of soil.
- E. Slopes: When planting on slopes, set the plant so the root flare on the uphill side is flush with the surrounding soil on the slope; the edge of the root ball on the downhill side will be above the surrounding soil. Apply enough soil to cover the downhill side of the root ball.
- F. Before planting, verify that root flare is visible at top of root ball according to ANSI Z60.1.
- G. Set balled and burlapped stock plumb and in center of pit or trench with top of root ball flush with adjacent finish grades.
 1. Remove burlap and wire baskets from tops of root balls and partially from sides, but do not remove from under root balls. Remove pallets, if any, before setting. Do not use planting stock if root ball is cracked or broken before or during planting operation.
 2. Place planting soil mix around root ball in layers, tamping to settle mix and eliminate voids and air pockets. When pit is approximately one-half backfilled, water thoroughly before placing remainder of backfill. Repeat watering until no more water is absorbed. Water again after placing and tamping final layer of planting soil mix.
- H. Set container-grown stock plumb and in center of pit or trench with top of root ball flush with adjacent finish grades.
 1. Carefully remove root ball from container without damaging root ball or plant.
 2. Place planting soil mix around root ball in layers, tamping to settle mix and eliminate voids and air pockets. When pit is approximately one-half backfilled, water thoroughly before placing remainder of backfill. Repeat watering until no more water is absorbed. Water again after placing and tamping final layer of planting soil mix.
- I. Organic Mulching: Apply 3-inch (75-mm) average thickness of organic mulch extending 12 inches (300 mm) beyond edge of planting pit or trench. Do not place mulch within 3 inches (75 mm) of trunks or stems.
- J. Install trunk protection for trees per detail/s on Landscape Drawings. Take care not to damage root balls.

3.6 TREE AND SHRUB PRUNING

- A. Remove only dead, dying, or broken branches. Do not prune for shape.
- B. Prune, thin, and shape trees and shrubs as directed by Landscape Architect.
- C. Prune, thin, and shape trees and shrubs according to standard horticultural practice. Prune trees to retain required height and spread. Unless otherwise indicated by Landscape Architect, DO NOT CUT TREE LEADERS; remove only injured or dead branches from flowering trees. Prune shrubs to retain natural character.

3.7 TREE STABILIZATION

- A. Trunk Stabilization: Unless otherwise indicated, provide trunk stabilization as follows:
1. Upright Staking and Tying: Stake trees of 2- through 5-inch (50- through 125-mm) caliper. Stake trees of less than 2-inch (50-mm) caliper only as required to prevent wind tip-out. Use a minimum of 2 stakes of length required to penetrate at least 18 inches (450 mm) below bottom of backfilled excavation and to extend one third of trunk height above grade. Set vertical stakes and space to avoid penetrating root balls or root masses.
 2. Use 2 stakes for trees up to 12 feet (3.6 m) high and 2-1/2 inches (63 mm) or less in caliper; 3 stakes for trees less than 14 feet (4.2 m) high and up to 4 inches (100 mm) in caliper. Space stakes equally around trees.
 3. Support trees with bands of flexible ties at contact points with tree trunk. Allow enough slack to avoid rigid restraint of tree.
 4. Support trees with two strands of tie wire encased in hose sections at contact points with tree trunk. Allow enough slack to avoid rigid restraint of tree..
 5. Mark each guy wire with a six (6) inch piece of white plastic surveyors tape.
- B. Guying and Staking: Guy and stake trees exceeding 14 feet (4.2 m) in height and more than 3 inches (75 mm) in caliper unless otherwise indicated. Securely attach no fewer than 3 guys to stakes 30 inches (760 mm) long, driven to grade.
1. For trees more than 6 inches (150 mm) in caliper, anchor guys to pressure-preservative-treated deadmen 8 inches (200 mm) in diameter and 48 inches (1200 mm) long buried at least 36 inches (900 mm) below grade. Provide turnbuckle for each guy wire and tighten securely.
 2. Support trees with bands of flexible ties at contact points with tree trunk and reaching to turnbuckle. Allow enough slack to avoid rigid restraint of tree.
 3. Support trees with strands of cable or multiple strands of tie wire encased in hose sections at contact points with tree trunk and reaching to turnbuckle. Allow enough slack to avoid rigid restraint of tree.
 4. Attach flags to each guy wire, 30 inches (760 mm) above finish grade.

3.8 INSTALLING SLOW-RELEASE WATERING DEVICE

- A. Provide one device for each tree.
- B. Place device on top of the mulch at base of tree stem and fill with water according to manufacturer's written instructions.

3.9 PLANTING BED MULCHING

- A. Mulch backfilled surfaces of planting beds and other areas indicated. Provide mulch ring around trees in lawn areas.
1. Organic Mulch: Apply 3-inch (75-mm) average thickness of organic mulch, and finish level with adjacent finish grades. Do not place mulch against plant stems.

3.10 PLANT MAINTENANCE

- A. A. Tree and Shrub Maintenance: Maintain plantings by pruning, cultivating, watering, weeding, fertilizing, restoring planting saucers, adjusting and repairing stakes and guy supports, resetting to proper grades or vertical position, and other measures as required to establish healthy, viable plantings. Spray or treat as required to keep trees and shrubs free of insects and disease. Restore or replace damaged burlap tree wrappings.

3.11 PESTICIDE APPLICATION

- A. Apply pesticides and other chemical products and biological control agents according to authorities having jurisdiction and manufacturer's written recommendations. Coordinate applications with Owner's operations and others in proximity to the Work. Notify Owner before each application is performed.
- B. Pre-Emergent Herbicides (Selective and Nonselective): Apply to tree, shrub, and ground-cover areas according to manufacturer's written recommendations. Do not apply to seeded areas.
- C. Post-Emergent Herbicides (Selective and Nonselective): Apply only as necessary to treat already-germinated weeds and according to manufacturer's written recommendations.
- D. Protect adjacent areas from receiving treatment.

3.15 CLEANUP AND PROTECTION

- E. During planting, keep adjacent paving and construction clean and work area in an orderly condition.
- F. Protect exterior plants from damage due to landscape operations, operations by other contractors and trades, and others. Maintain protection during installation and maintenance periods. Treat, repair, or replace damaged plantings.

3.16 DISPOSAL

- G. Disposal: Remove surplus soil and waste material, including excess subsoil, unsuitable soil, trash, and debris, and legally dispose of them off Owner's property.

END OF SECTION 32 93 13

SECTION 334100 - STORM UTILITY DRAINAGE PIPING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 010000 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

1. Pipe and fittings.
2. Cleanouts.
3. Drains.
4. Manholes.
5. Catch Basins.
6. Stormwater Inlets.
7. Concrete.
8. Pipe outlets.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Shop Drawings:
 1. Drains: Include plans, elevations, sections, details, frames, covers, and grates.

1.4 INFORMATIONAL SUBMITTALS

- A. Product Certificates: For each type of pipe and fitting, from manufacturer.
- B. Field quality-control reports.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Do not store plastic pipe, and fittings in direct sunlight.
- B. Protect pipe, pipe fittings, and seals from dirt and damage.

1.6 PROJECT CONDITIONS

- A. Interruption of Existing Storm Drainage Service: Do not interrupt service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary service according to requirements indicated:
1. Notify Owner no fewer than two (2) days in advance of proposed interruption of service.
 2. Do not proceed with interruption of service without Owner's written permission.

PART 2 - PRODUCTS

2.1 PE PIPE AND FITTINGS

- A. Corrugated PE Drainage Pipe and Fittings NPS 3 to NPS 10: AASHTO M 252M, Type S, with smooth waterway for coupling joints.
1. Silttight Couplings: PE sleeve with ASTM D1056, Type 2, Class A, Grade 2 gasket material that mates with tube and fittings.
 2. Soiltight Couplings: AASHTO M 252M, corrugated, matching tube and fittings.
- B. Corrugated PE Pipe and Fittings NPS 12 to NPS 60: AASHTO M 294M, Type S, with smooth waterway for coupling joints.
1. Silttight Couplings: PE sleeve with ASTM D1056, Type 2, Class A, Grade 2 gasket material that mates with pipe and fittings.
 2. Soiltight Couplings: AASHTO M 294M, corrugated, matching pipe and fittings.

2.2 PVC PIPE AND FITTINGS

- A. PVC Profile Sewer Piping:
1. Pipe: ASTM F794, PVC profile, gravity sewer pipe with bell-and-spigot ends for gasketed joints.
 2. Fittings: ASTM D3034, PVC with bell ends.
 3. Gaskets: ASTM F477, elastomeric seals.

2.3 CONCRETE PIPE AND FITTINGS

- A. Reinforced-Concrete Sewer Pipe and Fittings: ASTM C76.
1. Bell-and-spigot ends and gasketed joints with ASTM C443, rubber gaskets.
 2. Class IV.

2.4 NONPRESSURE TRANSITION COUPLINGS

- A. Comply with ASTM C1173, elastomeric coupling for joining underground nonpressure piping. Include ends of same sizes as piping to be joined.
- B. Ring-Type, Flexible Couplings:
 - 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Fernco Inc.
 - 2. Description: Elastomeric compression seal with dimensions to fit inside bell of larger pipe and for spigot of smaller pipe to fit inside ring.

2.5 CLEANOUTS

- A. Plastic Cleanouts:
 - 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Jumbo Manufacturing
 - b. Circle Manufacturing Co.
 - c. Ferguson
 - 2. Description: PVC body with PVC threaded plug. Include PVC sewer pipe fitting and riser to cleanout of same material as sewer piping.

2.6 DRAINS

- A. Area or Inline Drains:
 - 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Nyloplast; ADS
 - b. Other
 - 2. Description: ASTM A536 Ductile Iron square secured grate and PVC inline drain. Include bottom outlet with inside calk or spigot connection, of sizes indicated.
 - 3. Top-Loading Classification(s): H-10 (Pedestrian) Medium Duty
 - 4. Color: all exposed surfaces to be dark in color (gray, black, or brown) including vertical stack of the inline drain- paint as needed.

2.7 MANHOLES

A. Standard Precast Concrete Manholes:

1. Description: ASTM C478, precast, reinforced concrete, of depth indicated, with provision for sealant joints.
2. Diameter: 48 inches minimum unless otherwise indicated.
3. Ballast: Increase thickness of precast concrete sections or add concrete to base section as required to prevent flotation.
4. Base Section: 6-inch minimum thickness for floor slab and 4-inch minimum thickness for walls and base riser section, and separate base slab or base section with integral floor.
5. Riser Sections: 4-inch minimum thickness, and lengths to provide depth indicated.
6. Top Section: Eccentric-cone type unless concentric-cone or flat-slab-top type is indicated, and top of cone of size that matches grade rings.
7. Joint Sealant: ASTM C990, bitumen or butyl rubber.
8. Resilient Pipe Connectors: ASTM C923, cast or fitted into manhole walls, for each pipe connection.
9. Steps: ASTM A615/A615M, deformed, 1/2-inch steel reinforcing rods encased in ASTM D4101, PP, wide enough to allow worker to place both feet on one step and designed to prevent lateral slippage off step. Cast or anchor steps into sidewalls at 12- to 16-inch intervals. Omit steps if total depth from floor of manhole to finished grade is less than 48 inches.
10. Adjusting Rings: Interlocking HDPE rings with level or sloped edge in thickness and diameter matching manhole frame and cover, and of height required to adjust manhole frame and cover to indicated elevation and slope. Include sealant recommended by ring manufacturer.
11. Grade Rings: Reinforced-concrete rings, 6- to 9-inch total thickness, to match diameter of manhole frame and cover, and height as required to adjust manhole frame and cover to indicated elevation and slope.

B. Manhole Frames and Covers:

1. Description: Ferrous; 24-inch ID minimum by 7- to 9-inch riser with 4-inch- minimum width flange and 26-inch- diameter minimum cover. Include indented top design with lettering cast into cover, using wording equivalent to "STORM SEWER."
2. Material: ASTM A536, Grade 60-40-18 ductile iron unless otherwise indicated.

2.8 CATCH BASINS

A. Standard Precast Concrete Catch Basins:

1. Description: ASTM C478, precast, reinforced concrete, of depth indicated, with provision for sealant joints.
2. Base Section: 6-inch minimum thickness for floor slab and 4-inch minimum thickness for walls and base riser section, and separate base slab or base section with integral floor.
3. Riser Sections: 4-inch minimum thickness, 48-inch diameter, and lengths to provide depth indicated.
4. Top Section: Eccentric-cone type unless concentric-cone or flat-slab-top type is indicated. Top of cone of size that matches grade rings.

5. Joint Sealant: ASTM C 990, bitumen or butyl rubber.
 6. Adjusting Rings: Interlocking rings with level or sloped edge in thickness and shape matching catch basin frame and grate. Include sealant recommended by ring manufacturer.
 7. Grade Rings: Include two or three reinforced-concrete rings, of 6- to 9-inch total thickness, that match 24-inch- diameter frame and grate.
 8. Steps: ASTM A615/A615M, deformed, 1/2-inch steel reinforcing rods encased in ASTM D4101, PP, wide enough to allow worker to place both feet on one step and designed to prevent lateral slippage off step. Cast or anchor steps into sidewalls at 12- to 16-inch intervals. Omit steps if total depth from floor of catch basin to finished grade is less than 60 inches.
 9. Pipe Connectors: ASTM C923, resilient, of size required, for each pipe connecting to base section.
- B. Frames and Grates: ASTM A536, Grade 60-40-18, ductile iron designed for A-16, structural loading. Include flat grate with small square or short-slotted drainage openings.
1. Size: 24 by 24 inches minimum unless otherwise indicated.
 2. Grate Free Area: Approximately 50 percent unless otherwise indicated.

2.9 STORMWATER INLETS

- A. Curb Inlets: Made with vertical curb opening according to jurisdictional details on the plans.
- B. Gutter Inlets: Made with horizontal gutter opening jurisdictional details on the plans. Include heavy-duty frames and grates.
- C. Frames and Grates: Heavy duty according to jurisdictional details on the plans.

2.10 CONCRETE

- A. General: Cast-in-place concrete according to ACI 318, ACI 350/350R and the following:
1. Cement: ASTM C150, Type II.
 2. Fine Aggregate: ASTM C33, sand.
 3. Coarse Aggregate: ASTM C33, crushed gravel.
 4. Water: Potable.
- B. Portland Cement Design Mix: 4000 psi minimum, with 0.45 maximum water/cementitious materials ratio.
1. Reinforcing Fabric: ASTM 185, steel, welded wire fabric, plain.
 2. Reinforcing Bars: ASTM A615, Grade 60 (420 MPa) deformed steel.
- C. Ballast and Pipe Supports: Portland cement design mix, 3000 psi minimum, with 0.58 maximum water/cementitious materials ratio.
1. Reinforcing Fabric: ASTM A185, steel, welded wire fabric, plain.
 2. Reinforcing Bars: ASTM A615, Grade 60 (420 MPa) deformed steel.

2.11 PIPE OUTLETS

- A. End walls or end sections: sized to fit pipe diameter and match pipe material.
- B. Riprap Basins: Broken, irregularly sized and shaped, graded stone according to MD Erosion and Sediment Control Manual Table H-2 "Stone Size", class as noted on drawings for each location.
 - 1. Color: Gray and black with no more than 30% bright or white coloring.

PART 3 - EXECUTION

3.1 EARTHWORK

- A. Excavation, trenching, and backfilling are specified in Section 312000 "Earth Moving."

3.2 PIPING INSTALLATION

- A. General Locations and Arrangements: Drawing plans and details indicate general location and arrangement of underground storm drainage piping. Location and arrangement of piping layout take into account design considerations. Install piping as indicated, to extent practical. Where specific installation is not indicated, follow piping manufacturer's written instructions.
- B. Install piping beginning at low point, true to grades and alignment indicated with unbroken continuity of invert. Place bell ends of piping facing upstream. Install gaskets, seals, sleeves, and couplings according to manufacturer's written instructions for use of lubricants, cements, and other installation requirements.
- C. Install manholes for changes in direction unless fittings are indicated. Use fittings for branch connections unless direct tap into existing sewer is indicated.
- D. Install proper size increasers, reducers, and couplings where different sizes or materials of pipes and fittings are connected. When installing pipe under streets or other obstructions that cannot be disturbed, use pipe-jacking process of microtunneling.
- E. Install gravity-flow, nonpressure drainage piping according to the following:
 - 1. Install piping pitched down in direction of flow.
 - 2. Install piping NPS 6 and larger with restrained joints at tee fittings and at changes in direction. Use corrosion-resistant rods, pipe or fitting manufacturer's proprietary restraint system, or cast-in-place concrete supports or anchors.
 - 3. Install piping with 12-inch minimum cover.
 - 4. Install PE corrugated sewer piping according to ASTM D2321.
 - 5. Install PVC profile gravity sewer piping according to ASTM D2321 and ASTM F1668.
 - 6. Install reinforced-concrete sewer piping according to ASTM C1479 and ACPA's "Concrete Pipe Installation Manual."

3.3 PIPE JOINT CONSTRUCTION

- A. Join gravity-flow, nonpressure drainage piping according to the following:
 - 1. Join corrugated PE piping according to ASTM D3212 for push-on joints.
 - 2. Join PVC profile gravity sewer piping according to ASTM D2321 for elastomeric-seal joints or ASTM F794 for gasketed joints.
 - 3. Join reinforced-concrete sewer piping according to ACPA's "Concrete Pipe Installation Manual" for rubber-gasketed joints.
 - 4. Join dissimilar pipe materials with nonpressure-type flexible couplings.

3.4 CLEANOUT INSTALLATION

- A. Install cleanouts and riser extensions from pipes to cleanouts at grade. Install piping so cleanouts open in direction of flow in pipe.
 - 1. Use Light-Duty, top-loading classification cleanouts in earth areas.
- B. Set cleanout frames and covers in earth with tops 6 inches above surrounding earth grade.

3.5 DRAIN INSTALLATION

- A. Install type of drains in locations indicated.
 - 1. Use Light-Duty, top-loading classification drains in earth areas.
 - 2. Use Heavy-Duty, H20 Traffic rated top-loading classification drains in paved areas.
- B. Fasten grates to drains.
- C. Set drain frames and covers per plan.

3.6 MANHOLE INSTALLATION

- A. General: Install manholes, complete with appurtenances and accessories indicated.
- B. Install precast concrete manhole sections with sealants according to ASTM C891.
- C. Where specific manhole construction is not indicated, follow manhole manufacturer's written instructions.
- D. Set tops of frames and covers flush with finished surface of manholes that occur in pavements. Set tops 3 inches above finished surface elsewhere unless otherwise indicated.

3.7 CATCH BASIN INSTALLATION

- A. Construct catch basins to sizes and shapes indicated.

- B. Set frames and grates to elevations indicated.

3.8 STORMWATER INLET AND OUTLET INSTALLATION

- A. Construct inlet head walls, aprons, and sides of reinforced concrete, as indicated.
- B. Construct riprap of broken stone, as indicated.
- C. Install outlets that spill onto grade, anchored with concrete, where indicated.
- D. Install outlets that spill onto grade, with flared end sections that match pipe, where indicated.

3.9 CONCRETE PLACEMENT

- A. Place cast-in-place concrete according to ACI 318.

3.10 CONNECTIONS

- A. Pipe couplings, expansion joints, and deflection fittings with pressure ratings at least equal to piping rating may be used in applications below unless otherwise indicated.
 - 1. Use nonpressure-type flexible couplings where required to join gravity-flow, nonpressure sewer piping unless otherwise indicated.
 - a. Ring-type flexible couplings for piping of different sizes where annular space between smaller piping's OD and larger piping's ID permits installation.

3.11 CLOSING ABANDONED STORM DRAINAGE SYSTEMS

- A. Abandoned Piping: Close open ends of abandoned underground piping indicated to remain in place. Include closures strong enough to withstand hydrostatic and earth pressures that may result after ends of abandoned piping have been closed. Use either procedure below:
 - 1. Close open ends of piping with at least 8-inch- thick, brick masonry bulkheads.
 - 2. Close open ends of piping with threaded metal caps, plastic plugs, or other acceptable methods suitable for size and type of material being closed. Do not use wood plugs.
- B. Abandoned Manholes and Structures: Excavate around manholes and structures as required and use one procedure below:
 - 1. Remove manhole or structure and close open ends of remaining piping.
 - 2. Remove top of manhole or structure down to at least 36 inches below final grade. Fill to within 12 inches of top with stone, rubble, gravel, or compacted dirt. Fill to top with concrete.
- C. Backfill to grade according to Section 312000 "Earth Moving."

3.12 IDENTIFICATION

- A. Materials and their installation are specified in Section 312000 "Earth Moving." Arrange for installation of green warning tape directly over piping and at outside edge of underground structures.
 - 1. Use detectable warning tape over ferrous piping.
 - 2. Use detectable warning tape over nonferrous piping and over edges of underground structures.

3.13 FIELD QUALITY CONTROL

- A. Inspect interior of piping to determine whether line displacement or other damage has occurred. Inspect after approximately 24 inches of backfill is in place, and again at completion of Project.
 - 1. Submit separate reports for each system inspection.
 - 2. Defects requiring correction include the following:
 - a. Alignment: Less than full diameter of inside of pipe is visible between structures.
 - b. Deflection: Flexible piping with deflection that prevents passage of ball or cylinder of size not less than 92.5 percent of piping diameter.
 - c. Damage: Crushed, broken, cracked, or otherwise damaged piping.
 - d. Infiltration: Water leakage into piping.
 - e. Exfiltration: Water leakage from or around piping.
 - 3. Replace defective piping using new materials, and repeat inspections until defects are within allowances specified.
 - 4. Reinspect and repeat procedure until results are satisfactory.
- B. Test new piping systems, and parts of existing systems that have been altered, extended, or repaired, for leaks and defects.
 - 1. Do not enclose, cover, or put into service before inspection and approval.
 - 2. Test completed piping systems according to requirements of authorities having jurisdiction.
 - 3. Schedule tests and inspections by authorities having jurisdiction with at least 24 hours' advance notice.
 - 4. Submit separate report for each test.
 - 5. Gravity-Flow Storm Drainage Piping: Test according to requirements of authorities having jurisdiction, UNI-B-6, and the following:
 - a. Exception: Piping with soiltight joints unless required by authorities having jurisdiction.
 - b. Test plastic piping according to ASTM F1417.
 - c. Test concrete piping according to ASTM C924.
- C. Leaks and loss in test pressure constitute defects that must be repaired.

- D. Replace leaking piping using new materials, and repeat testing until leakage is within allowances specified.

3.14 CLEANING

- A. Clean interior of piping of dirt and superfluous materials. Flush with water.

END OF SECTION 33 41 00